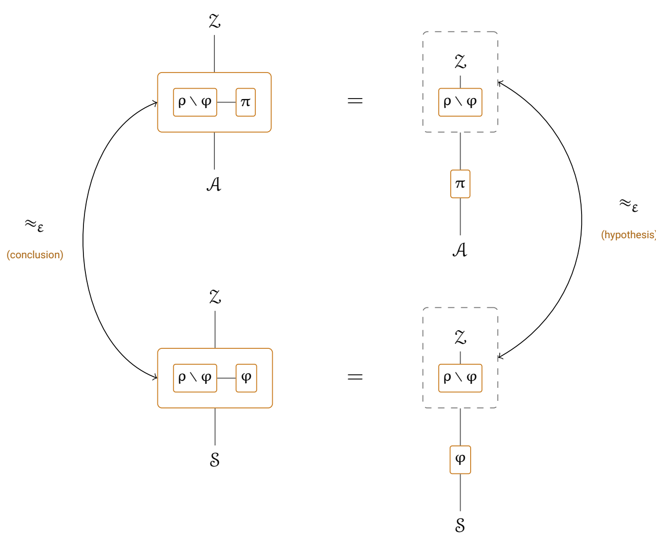
UC برای بازی‌بازان

UC for Gamers



هوش مصنوعی: Claude Opus 5 (1M context)
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ترکیب‌پذیری جهانی را، هر جا که نوشته می‌شود، به آمیزه‌ای از نوشتار و کد می‌نویسند؛ و همین است که پژوهشِ UC را سخت‌ابطال می‌کند و آزمودن یا وارسیِ صوری‌اش را سخت‌تر. ما، به سبکِ کدمحورِ امنیتِ ویژگی‌محور، زبانی دقیق برای برنامه‌نویسیِ کارکردهای UC می‌نشانیم: هویت و کنترلِ دسترسی داده‌اند، پوشانه‌ها آن‌ها را اعمال می‌کنند، فساد دفتری است که هر واسط می‌خواندش، و یک اجرا هندلری است که تنها یک ژتون را دست‌به‌دست می‌کند. آن‌گاه تقلید را روی رده‌های صریحِ ماشین و با کران‌های عینی تعریف می‌کنیم — بی‌پارامترِ امنیتی، بی‌مجانبی — و جعبه‌ابزارِ ترکیب از پیِ آن می‌آید: جانشانی، ترکیبِ موازی، و قضیهٔ تمامیت برای حریفِ بدل — هر سه نتیجهٔ یک لمِ جذب، لمی که پروتکل یا حریف را به درونِ محیط می‌بَرد و بازگروه‌بندیِ دقیقِ یک اجرای واحد است، نه مقایسهٔ دو اجرا — و ترایاییِ مفت، از خودِ متر. ویژگی‌های بازی‌محور نیز، چنان که خواهیم دید، خودشان کارکردند و در امتدادِ تقلید منتقل می‌شوند. یک کارکردِ امضا، یک ساعتِ جهانی و یک شبکهٔ با تأخیرِ کراندار سبک را در کار نشان می‌دهند. امیدمان این است که این کار به به‌کارگیریِ روش‌های صوری در UC کمک کند و سرانجام به اثباتی صوری از UC برسد.

فهرست مطالب

مقدمه

ترکیب‌پذیری جهانی را، تقریباً هر جا که نوشته می‌شود، به آمیزه‌ای از نوشتار و کد می‌نویسند. واسط‌های یک کارکرد کدند؛ اینکه چه کسی می‌تواند آن‌ها را فرا بخواند، یک جمله است. مدلِ فساد یک بند است؛ و اینکه واسطِ یک طرفِ فاسد در واقع چه برمی‌گرداند، به خواننده واگذار می‌شود. هر یک از این شکاف‌ها کوچک است، و داورِ انسانی بی‌آنکه بفهمد پُرش می‌کند. ماشین نمی‌تواند، و همین است که اثباتِ UC را سخت‌ابطال می‌کند و وارسیِ صوری‌اش را سخت‌تر: بسیاری از آنچه باید وارسی شود، هرگز نوشته نشده بود.

این مقاله آن بخش‌ها را به‌صورتِ کد می‌نویسد. هویت و کنترلِ دسترسی داده‌هایی می‌شوند که کارکرد در میدان‌های نام‌دار حمل می‌کند. پوشانه‌ها اعمالشان می‌کنند: نگهبان تصمیم می‌گیرد چه فراخوان‌هایی به هسته می‌رسند، خموش‌کننده تصمیم می‌گیرد هسته چه ادعایی می‌تواند بکند، و واسطه‌گر فراخوانِ یک طرفِ فاسد را به حریف می‌سپارد. فساد دفتری است که هر واسط می‌خواندش، دو بار، در نقاطی که خودِ کد نامشان می‌برد. یک اجرا هندلری است که تنها یک ژتون را دست‌به‌دست می‌کند. هیچ‌چیزِ اینجا مدلی تازه نیست — همان مدلِ کانتی [7] است، با چند قراردادش که صریح شده‌اند — اما بیانِ آن‌ها به کد و نه به نوشتار، همان چیزی است که می‌گذارد باقیِ مقاله اثبات شود و نه استدلال.

یک لم. آن‌گاه تقلید روی رده‌های صریحِ ماشین تعریف می‌شود، با کران‌های عینی: بی‌پارامترِ امنیتی، بی‌مجانبی، کرانی که یک عدد است. آنچه از پی می‌آید جعبه‌ابزارِ ترکیب است، و شکلِ آن ادعای اصلیِ ساختاریِ این مقاله است. جانشانی، ترکیبِ موازی و تمامیتِ حریفِ بدل، هر سه، نتیجهٔ یک لمِ جذب‌اند، لمی که پروتکل یا حریف را به درونِ محیط می‌بَرد. جذب مقایسهٔ دو اجرا نیست، بازگروه‌بندیِ دقیقِ یک اجراست: همان اجرا، با کمانک‌گذاریِ دیگر، پس مزیتِ دو سو یک عدد است و نه دو عدد با کرانی مشترک. ترایایی استثناست و از گونهٔ دیگری است، چون بر نامساویِ مثلثی روی متر تکیه دارد و نه بر هیچ بازگروه‌بندی؛ به همین سبب همراهِ متر بیان می‌شود، پیش از ماشین‌آلاتی که آن سه دیگر لازم دارند.

بازی‌ها کارکردند. یک ویژگی به سبکِ بازی‌محورِ بلر و راگوی [5] — جعل‌ناپذیری، زنده‌بودن، اصالت — چنان که پیدا می‌شود، خودش هم به‌صورتِ کارکرد بیان‌شدنی است، با مبارزطلبِ بازی به‌عنوانِ ماشینی معمولی و حکمش به‌عنوانِ واسطی معمولی.

ویژگی‌هایی که این‌گونه نوشته شوند در امتدادِ تقلید منتقل می‌شوند: اگر پروتکلی شیئی ایده‌آل را تقلید کند، ویژگی‌های آن شیء به آن فرود می‌آیند، زیرِ فرض‌هایی که مقاله بیانشان می‌کند و در هر کاربرد وارسی‌شان می‌کند و مفروضشان نمی‌گیرد.

نخست چه بخوانیم. دستگاه از آنچه شمارِ قضیه‌ها نشان می‌دهد بزرگ‌تر است، و همهٔ آن باربر نیست. پنج تعریف بیشترِ بارِ مقاله را حمل می‌کنند: یک کارکرد و پنج میدانش؛ یک اجرا و اینکه اشغالِ یک هویت چه معنایی دارد؛ یک دسته، که چند ماشین را چون یک اشغال‌گر می‌راند و جذب از آن ساخته شده است؛ خودِ جذب؛ و تناظرِ بازگروه‌بندی که دو اجرا را ماشین‌به‌ماشین جفت می‌کند. خواننده‌ای که این پنج را داشته باشد تیرکِ مقاله را دارد، و نتایجِ ترکیب به‌ترتیب از آن‌ها می‌آیند. کارکردهای عینی — یک طرحِ امضا، یک ساعتِ جهانی و یک شبکهٔ با تأخیرِ کراندار به‌پیرویِ کاتز، مائورر، تاکمان و زیکاس [11] و بادرچر، مائورر، چودی و زیکاس [3]، و یک کانالِ اصالت‌دار که روی آن‌ها محقق می‌شود — در پایان می‌آیند و هستند تا سبک را در کار نشان دهند، نه چون چیزی پیش از آن‌ها به آن‌ها نیاز داشته باشد.

این به چه کار می‌آید. پژوهشگرِ UC باید قراردادهای اینجا را همان قراردادهایی بیابد که خودش به کار می‌برد، اما نوشته‌شده. مخاطبِ دوم آن کسی است که این دقتِ افزوده برای اوست. هر کس که ابزاری روی UC می‌سازد — یک نوع‌سنج برای کدِ کارکرد، یک رمزگذاری در دستیارِ اثبات، یک داربستِ آزمون — مشخصه‌ای لازم دارد که پرسش‌هایی را حل کند که ارائهٔ غیررسمی می‌تواند به خوانندهٔ خود واگذارد، و حلِ همان‌ها بیشترِ کاری است که این مقاله می‌کند: نگهبان چه فراخوانی را می‌پذیرد، هستهٔ خموش‌شده چه ادعایی می‌تواند بکند، دفترِ فساد کِی خوانده می‌شود. اینکه این اندازه برای حملِ اثباتی ماشین‌وارسی‌شده از UC بس باشد یا نه، اینجا برقرار نشده است، و ما ادعایش نمی‌کنیم. همین است که چرا این چارچوب این‌گونه نوشته شده است.

فصل 1

کارکردها و سیستم‌ها

ما در چارچوبِ ترکیب‌پذیریِ جهانیِ کانتی [7] کار می‌کنیم، که چند قراردادش را در زیر صریح می‌کنیم.

شناسهٔ فرایند نامِ یک نمونهٔ در حالِ اجرای پروتکل است و سه بخش دارد:

$$\text{pid} = (\mathrm{F}, s, i),$$

یک نامِ کارکرد F، یک شناسهٔ نشست s، و یک شمارهٔ نمونه i. نام می‌گوید چه کدی اجرا می‌شود، نشست می‌گوید نمونه به کدام اجرای پروتکل تعلق دارد، و شماره چند نمونه از یک نام را در یک نشست از هم جدا می‌کند. برای سه بخشِ $id.\text{pid}$ می‌نویسیم $id.\mathrm{F}$، $id.s$ و $id.i$، و به همین شکل $\mathcal{F}.\mathrm{F}$، $\mathcal{F}.s$، $\mathcal{F}.i$ برای بخش‌های $\mathcal{F}.\text{pid}$؛ و $id.\text{pid} = \mathcal{F}.\text{pid}$ به‌تنهایی یعنی هر سه یکی‌اند.

نام‌ها عضوِ $\{0,1\}^* \cup \{A, Z, C\}$ و شناسه‌های طرف عضوِ $\{0,1\}^* \cup \{A, Z\}$اند، که در آن A، Z و C برای حریف، محیط و دفترِ فساد رزرو شده‌اند. چون رزروند، هیچ‌یک از این سه عضوِ $\{0,1\}^*$ نیست، پس میدانی که اعلام شده $\{0,1\}^*$ را در بر دارد هیچ‌یک از آن‌ها را در بر ندارد؛ و C نامِ هیچ طرفی نیست، چون دفتر ماشینی است که طرف‌ها فرایش می‌خوانند، نه ماشینی که طرفی آن را براند. هر سه به‌صورتِ یک نمونهٔ واحد در یک نشستِ واحد اجرا می‌شوند، که ثابتاً نشستِ 0 و نمونهٔ 0 است، پس شناسه‌های فرایندشان $(A, 0, 0)$، $(Z, 0, 0)$ و $(C, 0, 0)$ است. ما در تمامِ کتاب این‌ها را به همان نامِ خالیِ A، Z و C کوتاه می‌کنیم، چون آن دو مؤلفهٔ دیگر خبری حمل نمی‌کنند.

آزمون‌ها تقریباً همیشه فقط روی نام‌اند. هر جا می‌نویسیم $id.\mathrm{F} = A$ مقصودمان مؤلفهٔ نام است، هرگز کلِ سه‌گانه؛ و همین برای Z، برای C، و برای هر کارکردی که در آزمونِ عضویت نامش می‌آید.

دو تا از میدان‌های زیر نام‌هایی از این دست را جمع می‌کنند، و چیزهای متفاوتی جمع می‌کنند. مجموعهٔ طرف‌ها شناسه‌های طرف را در بر دارد، عضوهای $\{0,1\}^* \cup \{A, Z\}$؛ و مجموعهٔ فراخواننده‌ها کلِ شناسه‌های فرایند را، چون اینکه چه کسی می‌تواند فرا بخواند پرسشی است در بابِ نمونه‌ها و نه فقط در بابِ کد. پس مجموعهٔ فراخواننده‌ها را با این کوتاه‌نویسی‌ها می‌نویسیم:

$$\mathbf{A} := \{\text{pid} : \text{pid}.\mathrm{F} = A\}, \qquad \mathbf{Z} := \{\text{pid} : \text{pid}.\mathrm{F} = Z\},$$

$$\mathbf{Std} := \{\text{pid} : \text{pid}.\mathrm{F} \in \{0,1\}^*\}$$

برای شناسه‌های فرایندِ حریف، محیط، و هر چیزِ استاندارد. هر جا نمادگذاریِ قدیمی‌تر "A و Z" را می‌پذیرفت، اکنون $\mathbf{A} \cup \mathbf{Z}$ را می‌پذیرد، که از آن تنها $(A, 0, 0)$ و $(Z, 0, 0)$ هرگز رخ می‌دهند و اجرا از هر یک یکی فراهم می‌کند؛ و هر جا "$\{0, 1\}^*$" را می‌پذیرفت، اکنون **Std** را می‌پذیرد، و آنجا هر نشست و هر نمونه‌ای واقعاً رخ می‌دهد. مجموعه‌های طرف همان $\{A, Z\}$ خالی را نگه می‌دارند، چون آن‌ها شناسهٔ طرف‌اند و نه شناسهٔ فرایند، پس اعلامی که پیش‌ازاین $\mathbf{P} := \mathbf{N}$ خوانده می‌شد باید دو شقّه شود.

کارکرد هم به همین شکل نام می‌گیرد، با مؤلفهٔ نامش که به‌صورتِ زیرنویس نوشته می‌شود: $\mathcal{F}_{\mathsf{Sig}}$ کارکردی است با $\mathcal{F}_{\mathsf{Sig}}.\mathsf{F} = \mathsf{Sig}$، که نشست و نمونه‌اش ناگفته مانده‌اند، پس این نمادگذاری به‌جای شناسهٔ فرایندِ (Sig, s, i) می‌ایستد با s و iی نامعین. هر جا نمونهٔ خاصی مقصود باشد آن‌ها را جداگانه می‌دهیم؛ هر جا فقط کد مهم باشد، هیچ نوشته نمی‌شوند.

با ناگفته گذاشتنشان چیزِ کمی از دست می‌رود، چون کدِ خودِ کارکرد به‌سختی هرگز آن‌ها را می‌خواند. محلی یا جهانی، هسته برای طرفی که به آن خدمت می‌دهد و حالتی که نگه می‌دارد نوشته می‌شود؛ اینکه به کدام نشست تعلق دارد، و کدام نمونه از نامِ خودش است، کارِ اجراست و نه کارِ او. در این مقاله s تنها در سه جا رخ می‌دهد — سطرِ 3 از Guard، پوشانه‌هایی که آن را فرا می‌خوانند، و آزمونِ ایستای تعریفِ 1.3 — و i هرگز خوانده نمی‌شود. شمارهٔ نمونه جای خود را از راهی دیگر به‌دست می‌آورد: همان است که می‌گذارد دو نمونه از یک نام در یک نشست، آن شناسه‌های فرایندِ متمایزی را حمل کنند که تعریفِ 1.2 می‌خواهد. فصلِ 7 این مشاهده را به یک حکم بدل می‌کند: تغییرنامِ نشست‌ها و نمونه‌ها یک یکریختیِ اجراهاست، و تقلیدی که در یک شناسهٔ فرایند اثبات شود در هر شناسهٔ دیگری برقرار است، زیرِ یک شرطِ بهداشتی در بابِ اینکه چه کسی را می‌توان از میانِ نشست‌ها فرا خواند، شرطی که هر کارکردِ اینجا برآورده‌اش می‌کند.

تعریف 1.1 (هویت‌ها و کارکردها)**.** یک هویت زوجِ $\mathit{id} = (\mathsf{pid}, P)$ از یک شناسهٔ فرایند و یک شناسهٔ طرف است، و یک مجموعهٔ هویت هر مجموعهٔ $\mathcal{I}$ از این‌هاست — شناسهٔ فرایند سه‌گانه است، و دیگر مجموعه‌ای از زوج‌های نام نیست. یک کارکردِ $\mathcal{F}$ این پنج میدان را حمل می‌کند

$\mathcal{F}.\mathsf{pid}$	یک شناسهٔ فرایند	
$\mathcal{F}.\mathbf{P}$	مجموعهٔ طرف‌های خدمت‌گیرنده	واسط‌هایی که وجود دارند
$\mathcal{F}.\mathbf{N}$	شناسه‌های فرایندِ فراخواننده‌ای که می‌پذیرد	چه کسی می‌تواند آن واسط‌ها را فرا بخواند
$\mathcal{F}.\mathbf{U}$	کارکردهایی که به کار می‌برد	چه چیزی باید باشد تا او بتواند بدود
$\mathcal{F}.\mathsf{par}$	هر پارامترِ دیگر	کد جز موارد بالا چه می‌خواند

و این مجموعهٔ هویت را اشغال می‌کند

$$\mathsf{IDs}(\mathcal{F}) \;:=\; \big\{(\mathcal{F}.\mathsf{pid},\ P)\ :\ P \in \mathcal{F}.\mathbf{P}\big\}.$$

مجموعه‌های هویت دو چیز می‌گویند: اینکه گردایه‌ای از کارکردها چه هویت‌هایی را اشغال می‌کند، و اینکه یک ماشین چه هویت‌هایی را می‌تواند ادعا کند. این پنج میدان در سطرِ پارامترِ هر جعبهٔ زیر تکرار می‌شوند، بی نامِ صاحبشان، چون سرِ جعبه آن را می‌دهد. تنها دو تای اول $\mathsf{IDs}(\mathcal{F})$ را تعیین می‌کنند: شناسهٔ فرایند بینِ همهٔ هویت‌هایش مشترک است و طرف‌های خدمت‌گیرنده مؤلفهٔ دوم را می‌دهند.

میدانِ چهارم وابستگی را ثبت می‌کند. هر درایهٔ $\mathcal{F}.\mathbf{U}$ زوجی است $(\mathcal{F}', \mathcal{R})$: $\mathcal{F}'$ نامِ کارکردی است که باید حاضر باشد تا $\mathcal{F}$ بدود، و $\mathcal{R}$ محمولی است که می‌گوید $\mathcal{F}$ از هر چه آنجا یافت شود چه می‌خواهد. نام‌بردن از یک زیرروال به‌تنهایی به‌ندرت بس است — زیرروالی که به هیچ‌یک از طرف‌های فراخواننده‌اش خدمت نمی‌داد، یا هیچ‌یک از فراخواننده‌هایش را نمی‌پذیرفت، بی‌مصرف بود — پس می‌نویسیم

$$\mathsf{serves}(\mathcal{F}, \mathcal{G}) \;:\Longleftrightarrow\; \mathcal{G}.\mathbf{P} \supseteq \mathcal{F}.\mathbf{P} \;\wedge\; \mathcal{F}.\mathsf{pid} \in \mathcal{G}.\mathbf{N}$$

برای آن خواسته‌ای که بیش از همه پیش می‌آید: $\mathcal{G}$ به هر طرفی که $\mathcal{F}$ خدمت می‌دهد خدمت می‌دهد، و $\mathcal{F}$ را در میانِ فراخواننده‌هایش می‌پذیرد. بی عطفِ دوم، Guard هر فراخوان از $\mathcal{F}$ به $\mathcal{G}$ را رد می‌کرد.

آنچه $\mathbf{U}$ ثبت می‌کند همان است که کدِ $\mathcal{F}$ نامش را می‌برد. فراخوان‌هایی که پوشانه‌اش به‌جای او می‌گذارد — به $\mathcal{C}$ در سطرهای 2 و 6 از بخشِ 3.5، و به $\mathcal{A}$ از طریقِ Mediate — بینِ هر کارکردِ استاندارد مشترکند و کنار گذاشته می‌شوند، و این هیچ هزینه‌ای ندارد: هر دو هدف به $\{0,1\}^* \cup \{\mathrm{A}, \mathrm{Z}\}$ خدمت می‌دهند و $\mathbf{Std}$ را می‌پذیرند، پس $\mathsf{serves}(\mathcal{F}, \mathcal{C})$ و $\mathsf{serves}(\mathcal{F}, \mathcal{A})$ برای هر $\mathcal{F}$ِ استاندارد، هر میدانی هم که خودش داشته باشد، برقرارند و درایه‌ای برای هیچ‌یک هرگز نمی‌توانست شکست بخورد.

تعریف 1.2 (سیستم‌ها). یک کارکرد استاندارد است اگر $\mathcal{F}.\mathrm{F} \notin \{\mathrm{A}, \mathrm{Z}, \mathrm{C}\}$ و هر یک از هسته‌هایش چنان که در بخشِ 3.5 آمده پوشانده شده باشد — جز Leak، که چنان که در بخشِ 3.2 آمده پوشانده می‌شود، و Initialize، که یک رویه است و هیچ پوشانده نمی‌شود. یک سیستم (یا جهانِ) π مجموعه‌ای متناهی از کارکردهای استاندارد است که شناسه‌های فرایندِ متمایز حمل می‌کنند: اگر $\mathcal{F}, \mathcal{F}' \in \pi$ و $\mathcal{F} \neq \mathcal{F}'$ آن‌گاه $\mathcal{F}.\mathsf{pid} \neq \mathcal{F}'.\mathsf{pid}$. سیستم هر چه اعضایش اشغال کنند اشغال می‌کند،

$$\mathsf{IDs}(\pi) \;:=\; \bigcup_{\mathcal{F}\in\pi} \mathsf{IDs}(\mathcal{F}) \;=\; \big\{(\mathcal{F}.\mathsf{pid},\ \mathrm{P}) \;:\; \mathcal{F} \in \pi,\ \mathrm{P} \in \mathcal{F}.\mathbf{P}\big\}.$$

سیستم همان است که طراحِ پروتکل می‌نویسدش — ما پروتکل و سیستم را به‌جای هم به کار می‌بریم، و ایده‌آل و واقعی نامِ سبک‌های نوشتنِ یکی از آن‌هایند (در زیر)، نه دو ردهٔ جدا. سه ماشین هیچ بخشی از آن نیستند: محیطِ $\mathcal{Z}$، حریفِ $\mathcal{A}$، و دفترِ فسادِ $\mathcal{C}$ از بخشِ 3.1. هیچ‌یک استاندارد نیست، پس بنا بر تعریفِ 1.2 هیچ سیستمی هیچ‌یک از آن‌ها را در بر ندارد و $\mathsf{IDs}(\mathcal{Z})$، $\mathsf{IDs}(\mathcal{A})$ و $\mathsf{IDs}(\mathcal{C})$ بیرونِ هر سیستمی می‌مانند؛ به‌جایش اجرای بخشِ 3.6 هر سه را فراهم می‌کند. برای $\mathcal{Z}$ و $\mathcal{A}$ همین است که می‌گذارد

هر یک هویتِ خودش را ادعا کند و در همان حال روی سیستمِ زیرِ آزمون خموش باشد. برای $\mathcal{C}$ همین است که دفتر را از عملیاتِ مجموعه‌ایِ بخشِ 4.1 و فصلِ 6 برکنار نگه می‌دارد: هیچ مسدودگری برایش ساخته نمی‌شود، هیچ جانشانی‌ای جایش را نمی‌برد، و هر مقایسه روی یک و همان دفتر می‌دود. رزروکردنِ آن سه نام آن‌ها را از هر $\mathcal{F}.\mathbf{N}$ که به‌صورتِ **Std** اعلام شده هم بیرون نگه می‌دارد، پس هیچ کارکردی دفتر را در میانِ فراخواننده‌هایش نمی‌پذیرد — و این درست است، چون دفتر تنها فراخوانده است.

تمایزِ شناسه‌های فرایند کارِ واقعی می‌کند. سیستم با آنچه در هر شناسه‌ای که به کار می‌برد می‌نشاند تعیین می‌شود، پس عضویت را تنها شناسه حل می‌کند. و هر هویت آن شناسه را در مؤلفهٔ اولش ثبت می‌کند، پس اعضای متمایز $\mathsf{IDs}(\mathcal{F}) \cap \mathsf{IDs}(\mathcal{F}') = \emptyset$ دارند: اجتماعِ بالا مجزاست، و هیچ هویتی از یک سیستم دو بار اشغال نمی‌شود.

تعریف 1.3 (سیستم‌های خوش‌ساخت). بنویسیم $\pi^+ := \pi \cup \{\mathcal{Z}, \mathcal{A}, \mathcal{C}\}$ برای یک سیستم به همراهِ آن سه ماشینی که هر اجرا فراهم می‌کند. سیستمِ π خوش‌ساخت است، و می‌نویسیم $\mathsf{wf}(\pi)$، اگر

$$\forall \mathcal{F} \in \pi^+ \;\; \forall (\mathcal{F}', R) \in \mathcal{F}.\mathbf{U} \;\;\; \exists \mathcal{G} \in \pi^+ \;\; :$$
$$\mathcal{G}.F = \mathcal{F}'.F \;\; \wedge \;\; \big(\mathcal{G}.s = \mathcal{F}.s \;\vee\; \mathsf{global}(\mathcal{G})\big)$$
$$\wedge \;\; R(\mathcal{F}, \mathcal{G}).$$

سه چیز در بابِ این. شرط‌های روی $\mathcal{G}$ درست همان‌هایی هستند که Guard هنگامِ گذاشتنِ فراخوان تحمیل می‌کند — نامِ پذیرفته‌شده، نشستِ جور، مگر آنکه فراخوانده مشترک باشد، و میدان‌های برازنده — پس $\mathsf{wf}(\pi)$ درست همین را می‌گوید که فراخوان‌هایی که اعضای π برای گذاشتنشان ساخته شده‌اند می‌توانند موفق شوند. جورشدن با نام است، نه با خودِ کارکرد، و همین است که می‌گذارد یک وابستگی از جانشانیِ زیرسیستم جان سالم ببرد: درایه‌ای که نامِ φ_q را می‌برد با هر چه در آن نام بنشیند برآورده می‌شود، ایده‌آل یا واقعی، تا وقتی باقیِ شرط‌ها را برآورد. و نیمهٔ خواسته میدان‌های فراخواننده و فراخوانده را به هم گره می‌زند؛ wf بی آن تقریباً هیچ نمی‌گفت.

هر دو سور روی π^+ دامنه دارند، و هر بزرگ‌شدن جای خود را به‌دست می‌آورد: در سمتِ راست، درایه‌ای که نامِ یکی از آن سه ماشینِ فراهم‌شده را می‌برد اصلاً برآورده‌شدنی می‌شود، چون هیچ‌یک عضوِ هیچ سیستمی نیست؛ و در سمتِ چپ، درایه‌های خودِ آن‌ها وارسی می‌شود — بی‌درنگ همین‌جا، چون $\mathsf{serves}(\mathcal{Z}, \mathcal{A})$ و $\mathsf{serves}(\mathcal{A}, \mathcal{Z})$ برقرارند و $\mathcal{C}.\mathbf{U}$ تهی است. آنچه wf به‌تنهایی نمی‌کند، جان‌بردن از ساخت‌هایی است که در پی می‌آیند: افزودنِ عضو می‌تواند تعهد بیاورد، و جانشانیِ زیرسیستم هر خواسته را در اشغال‌گرِ تازه از نو می‌سنجد. گزارهٔ 6.4 (فصلِ 6) اولی را حل می‌کند و قضیهٔ 4.4 دومی را.

شرط‌ها، یک‌جا. خوش‌ساختی یکی از چند شرطِ ایستایی است که این مقاله بر سیستم‌ها، هسته‌ها و ماشین‌ها می‌گذارد، و هر یک همان جا که نخست لازم می‌شود

تعریف می‌شود — که خواننده را بی هیچ راهی می‌گذارد تا ببیند کدام نتیجه کدام شرط را می‌خواند. این جدول یک بار همه را جمع می‌کند. هیچ‌یک از آن‌ها در طولِ اجرا خوانده نمی‌شود: Exec میدان می‌خواند، نه شرط، چنان که گزارهٔ 7.7 به‌طورِ خاص برای **U** ثبت می‌کند.

شرط	چه چیزی را مقید می‌کند	چه کسی می‌خواندش
$\mathsf{wf}(\pi)$، تعریفِ 1.3	یک سیستم: هر درایهٔ **U** شاهدی دارد — فراخوان‌هایی که اعضایش برای گذاشتنشان ساخته شده‌اند می‌توانند موفق شوند	قضیهٔ 4.4، گزارهٔ 6.4، تعریفِ 5.1
جانشانیِ خوش‌ساخت، تعریفِ 4.2	سیستمِ ورودی در برابرِ آنچه می‌ماند: هیچ شناسهٔ فرایندِ مشترکی	گزارهٔ 4.3، قضیهٔ 4.4، گزارهٔ 4.16، قضیهٔ 4.20، قضیهٔ 4.24، گزارهٔ 5.10
سازگار، تعریفِ 6.1	دو سیستم: شناسهٔ فرایندِ مشترک یکسان خدمت می‌دهد و یکسان می‌پذیرد	قضیهٔ 6.3، ادعای دوم
نشست‌یکنوا، تعریفِ 7.2	یک سیستم یا محیط: اینکه چه کسی می‌تواند به عضوی برسد به نشستِ او وابسته نیست	لمِ 7.9، گزارهٔ 7.10، نتیجهٔ 7.12، نتیجهٔ 7.13
تنگ در I، نکتهٔ 6.5	یک سیستم: اعضای بیرونِ فراخواننده‌ها را با شناسهٔ I فرایند سنجاق می‌کنند	تعریفِ 5.1
هیچ فراخوانِ پاسخگو نمی‌گذارد، تعریفِ 9.2	هسته‌های یک سیستم	قضیهٔ 4.29، لمِ 4.18 (نیمهٔ واگذارنده)، گزارهٔ 5.9، نتیجهٔ 5.11
بی‌اعتنا به رد، تعریفِ 4.28	ماشینی در جایگاهِ حریف	همان چهار
هویت‌اتمی، تعریفِ 7.4	همهٔ کد؛ قراردادی جاری و نه آزمونی به‌ازای هر سیستم	لمِ 7.9، گزارهٔ 7.15
نشت‌خوش‌ساخت، بخشِ 3.2	یک تحقق	هیچ‌کس: ثبت شده، به کار نرفته

دو چیز که این جدول آشکار می‌کند. دو سطرِ اول یک نام دارند و دو شرطِ متفاوتند: wf در بابِ یک سیستم به‌تنهایی است، و جانشانیِ خوش‌ساخت در بابِ اینکه سیستمِ ورودی کجا فرود می‌آید. و wf و نشست‌یکنوایی دوگانِ یکدیگرند — فراخوان‌هایی که عضوی برای گذاشتنشان ساخته شده، در برابرِ اینکه چه کسی می‌تواند به او برسد — که وسوسه می‌کند آن‌ها را در یک شرط تا کنیم. جدا نگه داشته شده‌اند چون هیچ نتیجه‌ای هر دو را لازم ندارد، و چون یک تعریف در هر حال نمی‌توانست هر دو را حمل

کند: از بازیِ تعریفِ 5.1 خواسته می‌شود که خوش‌ساخت باشد و عامداً نشست‌یکنوا نیست، چنان که نکتهٔ 5.6 توضیح می‌دهد.

می‌گوییم $\mathcal{F}$ محلی است اگر $\mathcal{F}.\mathbf{N} = \{\mathsf{pid} : \mathsf{pid}.\mathsf{F} = \mathsf{F}^{\uparrow}\}$ برای نامِ والدِ (فراخوانندهٔ) معینِ $\mathsf{F}^{\uparrow}$. (این را به خودِ کارکرد وامی‌گذاریم که بگوید آیا **A** و **Z** می‌توانند فرایش بخوانند یا نه.) مفهوم‌های میانی ممکن‌اند: برپاسازی‌ای که از دو پروتکلِ نام‌دار در دسترس است $\{\mathsf{pid} : \mathsf{pid}.\mathsf{F} \in \{\mathsf{F}_1, \mathsf{F}_2\}\}$ را می‌پذیرد؛ و یک **N** می‌تواند کلِ شناسه‌های فرایند را سنجاق کند و نه نام‌ها را، و نمونه‌های اشغال‌شدهٔ فراخواننده‌هایش را رو نویسد — نکتهٔ 6.5 می‌گوید چرا اعضای خصوصیِ یک تحقق باید چنین کنند. در سرِ دیگرِ طیف می‌گوییم $\mathcal{F}$ جهانی است، به معنای برپاسازیِ مشترکِ UC تعمیم‌یافته [8]، اگر

$$\mathsf{global}(\mathcal{F}) \;:\Longleftrightarrow\; \mathbf{Std} \subseteq \mathcal{F}.\mathbf{N} \;\vee\; \mathcal{F}.\mathsf{F} \in \{\mathsf{A}, \mathsf{Z}, \mathsf{C}\},$$

که یک مفهوم است و نه دو: عطفِ اول حالتِ معمولی است، هر کارکردِ استانداردی می‌تواند فرایش بخواند، و عطفِ دوم آن سه ماشینی را می‌پوشاند که اجرا فراهم می‌کند — و هر چه در جایگاهشان بایستد — که به دلیلی دیگر جهانی‌اند و هر جا global آزموده شود همان‌گونه با آن‌ها رفتار می‌شود.

این تمایز همان چیزی است که مؤلفهٔ نشست برای آن است. کارکردِ محلی بخشی از یک اجرا از یک پروتکل است، پس فراخوان تنها از نشستِ خودش می‌تواند به آن برسد: سطرِ 3 از Guard می‌پرسد $\mathtt{id}'.\mathtt{s} = \mathtt{id}.\mathtt{s}$. شمارهٔ نمونه آزموده نمی‌شود، پس یک نشست می‌تواند هر چند نمونه از یک نام که بخواهد در بر داشته باشد و آن‌ها می‌توانند یکدیگر را فرا بخوانند. کارکردِ جهانی بینِ اجراها مشترک است — یک نشست از آن، در دسترسِ هر نشستی — پس آن آزمون برای او بخشیده می‌شود، و همهٔ نشست‌های $\mathcal{F}$ می‌توانند یک نمونهٔ مشترک را فرا بخوانند. کارکردِ ایده‌آلِ جهانی را با G در آغاز نام می‌دهیم و واقعیِ جهانی را با γ در آغاز، چنان که در $\mathcal{F}_{\mathsf{GSig}}$؛ و $\mathcal{G}$ خوش‌نویس همان می‌ماند که بوده، کارکردِ دومی در کنارِ $\mathcal{F}$. آن سه ماشینی که اجرا فراهم می‌کند جهانی‌اند: $\mathcal{A}$ و $\mathcal{C}$ همین حالا با بندِ اول، و $\mathcal{Z}$ با بندِ دوم، که آنجاست چون $\mathcal{Z}.\mathbf{N} = \mathbf{A} \cup \mathbf{Z}$ عامداً تنگ است و با این حال محیط باید از هر نشستی در دسترس باشد.

مشخصهٔ واقعی در برابرِ ایده‌آل. کارکردهای ایده‌آل و پروتکل‌های واقعی از چند جهت متفاوتند. اول، کارکردهای ایده‌آل وضعِ فسادِ طرف‌های موجود در مجموعهٔ طرف‌هایشان را می‌دانند. دوم، می‌توانند اجرا را به شبیه‌ساز بسپارند و می‌توانند واسط‌های ویژهٔ شبیه‌ساز عرضه کنند (مثلاً $\mathcal{F}_{\mathsf{Diffuse}}$ را ببینید). سرانجام، می‌توانند حالت را بینِ طرف‌ها مشترک کنند — و همین اشتراک نکتهٔ اصلی است: نگهبانِ فصلِ 2 هر فراخوان را به یک طرف گره می‌زند، پس طرف‌ها تنها از راهِ کارکردهایی با هم سخن می‌گویند که حالت را بینشان مشترک می‌کنند، و پروتکلِ واقعی با همتاهایش از راهِ زیرروال‌های ایده‌آل سخن می‌گوید و از هیچ راهِ دیگری. اینجا و در زیر، $\mathcal{F}_{\mathsf{Diffuse}}$ و $\mathcal{F}_{\mathsf{IO}}$ نامِ کارکردهایی است که نمی‌دهیمشان: یک کانالِ پخش و یک بازپخشِ ورودی–خروجی. هیچ‌چیزِ این مقاله به کدِ آن‌ها بند نیست. سرچشمهٔ تصادفِ $\mathcal{F}_{\mathsf{Rand}}$ و انبارِ $\mathcal{F}_{\mathsf{Store}}$ داده شده‌اند، در فصل‌های 13 و 14؛ آن‌ها کوچک‌ترین مثال‌های این مقاله‌اند و همان جایی هستند که خوانندهٔ ناآشنا با این سبک باید از آنجا شروع کند.

واسط‌های هستهٔ پروتکل‌های واقعیِ خوش‌مشخصه هیچ‌یک از این ویژگی‌ها را ندارند، جز آنچه در بلوک‌های ساختمانیِ ایده‌آل‌شده و پوشانه‌های کنترلِ دسترسی مستتر است؛ و تحقق‌های واقعی مشخص نمی‌کنند که نشت هنگامِ فساد چگونه حساب می‌شود، چون آن محتوای نوارهای کار است.[1]

مقدارِ آغازین. Initialize یک رویه است و نه یک واسط: یک بار می‌دود، وقتی کارکرد نخستین بار فعال می‌شود، و متغیرهایی را که باقیِ کد به کار می‌برد برپا می‌کند. هیچ فراخوانی نمی‌گذارد — اجرای بخشِ 3.6 هر ماشین را پیش از نصبِ توزیع‌گرش مقدارِ آغازین می‌دهد، پس چیزی نبود که فراخوانی را خدمت کند — هیچ فراخواننده‌ای نمی‌تواند به آن برسد، هیچ شناسهٔ فراخوانندهٔ ادعایی نمی‌گیرد، و کارکردی که هیچ اعلام نکند تهیِ آن را می‌گیرد. پروتکل‌های واقعی هیچ مقداردهِ جهانی از این دست حمل نمی‌کنند، تنها مقداردهِ ویژهٔ یک طرف در یک فرایند، که $(pid, P).\mathrm{Initialize}()$ نوشته می‌شود و در آن نمادگذاری را بدرفتاری می‌کنیم: آن‌ها خصوصی‌اند، نه واسط‌هایی که هر کس بتواند خطابشان کند.

[1] رهیافتی منظم‌تر ذخیره‌سازی را به‌صورتِ کارکردی ایده‌آل مدل می‌کرد که پاک‌کردنِ امن عرضه می‌کند؛ ما از آن لایهٔ افزوده می‌پرهیزیم تا نمادگذاری سبک بماند.

فصل 2

نگهبان‌ها، خموش‌کننده‌ها و واسطه‌گرها

این فصل سه تکه ماشین‌آلات را تعیین می‌کند. محمولِ Guard فراخوان‌هایی را که به یک ماشین می‌رسند مقید می‌کند؛ پوشانهٔ Silence فراخوان‌هایی را که او می‌گذارد؛ و پوشانهٔ Mediate فراخوانی را که طرفِ فاسد نمی‌تواند خودش پاسخش دهد به حریف می‌سپارد. واسطِ کاملِ بخشِ 3.5 ترکیبِ همین سه است.

بنویسیم Op برای واسطِ هستهٔ عمومیِ $\mathcal{F}$. ما تنها هسته‌ها را مشخص می‌کنیم؛ پوشانهٔ زیر هر یک را به واسطی کامل بدل می‌کند، و فراخواننده به FullOp می‌رسد، هرگز به Op. آزمون‌های کنترلِ دسترسی در محمولِ Guard جمع شده‌اند، که کارکرد نیست بلکه الگوریتم است: هویتِ هدفِ $\mathtt{id}$، هویتِ فراخوانندهٔ ادعاییِ $\mathtt{id}'$ و کارکردِ $\mathcal{F}$ را می‌گیرد و یک بولی برمی‌گرداند. دو قراردادِ ارائه. جعبه‌ها دو وزن دارند: پُر نشانهٔ کارکرد است، که هویت اشغال می‌کند، و تُهی نشانهٔ محمول یا پوشانه یا رویه، که هیچ اشغال نمی‌کند. و فراخوان چنین نوشته می‌شود: $\mathtt{id}.\mathsf{Op}(\mathtt{in})$ **as** $\mathtt{id}'$ که در آن $\mathtt{id}'$ هویتی است که در گذاشتنِ آن ادعا شده، و چنین دریافت می‌شود: $\mathtt{id}.\mathsf{Op}(\mathtt{in})$ **from** $\mathtt{id}'$ که در آن $\mathtt{id}'$ هویتی است که فراخوان زیرِ نامِ آن ادعا شده بود. اِعمال در پوشانه نشسته است، همان جا که سطرِ 1 آن را **require** می‌کند، پس فراخوانی که او ردش کند هرگز به هسته نمی‌رسد. سطرِ 1 آزمون می‌کند که واسط وجود دارد؛ سطرِ 2، که نامِ فراخواننده پذیرفته است و برای طرفِ درست عمل می‌کند؛ سطرِ 3، که فراخواننده و فراخوانده در یک نشستند، مگر آنکه فراخوانده بینِ نشست‌ها مشترک باشد.

محمولِ Guard

$\mathsf{Guard}_{\mathcal{F}}(\mathtt{id}, \mathtt{id}')$:

1: **return** $\mathtt{id}.\mathsf{pid} = \mathcal{F}.\mathsf{pid} \wedge \mathtt{id}.\mathtt{P} \in \mathcal{F}.\mathbf{P}$ — دارند وجود که واسطهایی /
2: $\wedge\ \mathtt{id}'.\mathsf{pid} \in \mathcal{F}.\mathbf{N} \wedge \mathtt{id}'.\mathtt{P} = \mathtt{id}.\mathtt{P}$ — پذیرفته فراخوانندههای /
3: $\wedge\ (\mathtt{id}'.s = \mathtt{id}.s \vee \mathsf{global}(\mathcal{F}))$ — باشد مشترک آنکه مگر نشست، یک /

Guard فراخوان‌هایی را که به ماشین می‌رسند مقید می‌کند؛ پوشانهٔ Silence در زیر، فراخوان‌هایی را که او می‌گذارد. این پوشانه یک مجموعهٔ هویتِ $\mathcal{I}$ و یک برنامه می‌گیرد، برنامه را می‌دواند، و هر فراخوانی که صادر کند را می‌رباید. فراخوانی که هویتی

در $\mathcal{J}$ ادعا کند با rej رد می‌شود، بی‌آنکه هرگز گذاشته شود؛ هر فراخوانِ دیگری گذاشته می‌شود و پاسخش پس داده می‌شود. پس $\mathcal{J}$ درست همان جایی است که برنامهٔ پوشانده‌شده در آن خموش است، و می‌تواند به‌عنوانِ هر چیزِ بیرونِ آن سخن بگوید.

دامنه، همان برنامهٔ پوشانده‌شده است و نه فراتر از آن. فراخوانی که او می‌گذارد ژتون را به ماشینی دیگر می‌سپارد، و آنچه آن ماشین می‌گوید با پوشانهٔ خودش حکم می‌شود، نه با این یکی. اگر جز این بود، کارکرد زیرروال‌هایش را ترایا خموش می‌کرد و هیچ‌یک نمی‌توانست به‌عنوانِ خودش سخن بگوید. هر کاربردِ Silence در زیر همین شکل را دارد: واسطی کامل که هستهٔ زیرِ خدمتش را می‌پوشاند.

خواندنِ این به‌عنوانِ هندلر، به معنای اثرهای جبری [12، 4]، به‌کار می‌آید. فراخوانی که برنامه صادر می‌کند عملی از اثرِ محیطی است، و k همان ادامه‌ای است که چشم‌به‌راهِ پاسخِ آن است. پوشانهٔ Silence تصمیم می‌گیرد با چه چیزی از سر گرفته شود: rej برای ادعای خموش‌شده، و در غیرِ این صورت پاسخِ فراخوانی که واقعاً گذاشته شد. بندِ **return** نتیجهٔ خودِ برنامه را دست‌نخورده رد می‌کند، پس خموش‌سازی آنچه برنامه می‌تواند بگوید را تغییر می‌دهد و هیچ‌چیزِ دیگر را. این قیاس تنها شکلِ پوشانه را تعیین می‌کند.

پوشانهٔ Silence

مجموعهٔ هویتِ $\mathcal{J}$، برنامهٔ $\mathrm{id.Op}(in)$ از id'

Silence$(\mathcal{J},\ \mathrm{id.Op}(in)\ \text{from}\ \mathrm{id}')$:

handle $\mathrm{id.Op}(in)$ from id' **with**		
return out	$\mapsto out$	
$\mathrm{id}''.\text{FullOp}(in'')$ as id'''; k	$\mapsto k(\text{rej})$	اگر $\mathrm{id}''' \in \mathcal{J}$
	$\mapsto k(\mathrm{id}''.\text{FullOp}(in'')\ \text{as}\ \mathrm{id}''')$	وگرنه

با داشتنِ Silence می‌توانیم بگوییم بر فراخوانی که طرفِ فاسد نمی‌تواند خودش پاسخش دهد چه می‌گذرد. هر واسطِ کاملِ زیر چنین فراخوانی را دو بار به حریف می‌سپارد، در راهِ ورود و در راهِ خروج، و هر دو بار در همان یک شکل: پوشانهٔ Mediate. این پوشانه یک مجموعهٔ فسادِ **C**، عملِ $\mathrm{id.Op}$ که در حالِ خدمت است، مقدارِ x که باید سپرده شود، و فراخوانندهٔ ادعاییِ id' عملی که به‌جایش می‌ایستد را می‌گیرد. اینکه پاسخ از کجا می‌آید در درونِ خودش تعیین شده است: هر عملی که باشد، پاسخ‌های طرفِ فاسد از $(\mathcal{A}, \mathrm{id.P})$ می‌آیند. یک چیز در بابِ آن آرگومانی معمولی نیست: هم **require** و هم **return**ِ Mediate از عملی برمی‌گردند که در کدِ آن ایستاده‌اند، و نه فقط از خودِ آن سطر. وقتی آزمونِ Mediate شکست بخورد او هیچ نمی‌کند و آن عمل ادامه می‌دهد.

این قلاب یک فراخوانِ معمولی روی واسطِ کاملِ $\mathcal{A}$ است، که هویتِ id واسطِ در حالِ خدمت را ادعا می‌کند. پس با $\text{Guard}_{\mathcal{A}}$ روبه‌رو می‌شود، و از آن می‌گذرد چون طرفِ ادعاییِ $\mathrm{id.P}$ همان طرفِ هدف است؛ و حریفی که پاسخ می‌دهد در بندِ خموش‌کنندهٔ خودِ $\mathcal{A}$ است، که تنها $(\mathcal{A}, \mathrm{id.P})$ را می‌پذیرد. پس حریف در حالِ پاسخ‌دادن برای $\mathrm{id.P}$ می‌تواند به‌عنوانِ خودش در $\mathrm{id.P}$ سخن بگوید و به‌عنوانِ هیچ‌چیزِ دیگر، و Mediate به خموش‌سازیِ خودش نیازی ندارد.

پوشانهٔ فسادِ Mediate

مجموعهٔ فسادِ **C**، عملِ $\mathrm{id}.\mathrm{Op}$، مقدارِ x، فراخوانندهٔ ادعاییِ id'

:Mediate$(\mathbf{C},\ \mathrm{id}.\mathrm{Op},\ x,\ \mathrm{id}')$

if $(\mathrm{id}'.\mathrm{F} \neq \mathrm{A}\ \wedge\ \mathrm{id}.\mathrm{P} \in \mathbf{C})$ **then**
    return $(\mathrm{A}, \mathrm{id}.\mathrm{P})$.FullA$(\mathrm{id}.\mathrm{Op},\ x)$ as id

فصل 3

مدلِ اجرا

3.1 فساد

فساد واسطی نیست که پروتکل‌های واقعی محققش کنند (پروتکلِ واقعی وقتی «دستور» بگیرد که فاسد شود چه می‌کند؟)، اما برای تقلید باید وجود داشته باشد. می‌توانیم فرض کنیم پروتکل‌های واقعی همیشه فساد را رد می‌کنند، اما آن‌گاه هرگز به‌عنوانِ فاسد ثبت نمی‌شوند.

ما فساد را به‌صورتِ کارکردی جهانی به نامِ $\mathcal{C}$ مدل می‌کنیم. چند ماشین واسطِ کاملشان یک‌جا داده شده و از هسته به‌دست نیامده — محیط، حریف، و عملِ نشتِ بخشِ 3.2 — اما $\mathcal{C}$ تنها ماشینی است که نه Silence را به کار می‌برد و نه Mediate، و گزارشِ وضع دلیلش است: آن پوشانه در هر فراخوان وضعِ فساد را می‌خواند، که همان چیزی است که این واسط پیاده‌اش می‌کند، پس پوشاندنش دوری بود. پس هر واسطِ زیر تنها آزمون‌هایی را که لازم دارد به‌دست حمل می‌کند، و FullStatus به‌طورِ خاص همیشه مجموعهٔ فاسدها را برمی‌گرداند، بی‌آنکه نخست بپرسد آیا آن طرف فاسد است.

نشتِ خودِ $\mathcal{C}$ ثابت است، پس آزمونِ فسادی که بخشِ 3.2 بر هر عملِ نشتِ دیگری تحمیل می‌کند اینجا چیزی برای محافظت نداشت؛ با این حال نگهبان نگه داشته می‌شود، تا حریف دفتر را در همان طرفی بخواند که برایش عمل می‌کند و در هیچ طرفِ دیگر.

خواندنِ دفتر. **C** حالتِ خودِ $\mathcal{C}$ است، پس هیچ ماشینِ دیگری رونوشتی از آن ندارد. هر جا **C** در کدِ کارکردی جز خودِ $\mathcal{C}$ ظاهر می‌شود، فهمِ مطلب این است که نخست واکشی شده است، با

$$\mathbf{C} \leftarrow (\mathcal{C}, \mathrm{id}.\mathrm{P}).\text{FullStatus}() \ \text{ as } \mathrm{id},$$

و ما آن سطر را هر جا که تنها چیزی را تکرار می‌کرد که کدِ پیرامون خودش روشن می‌کند حذف می‌کنیم. در درونِ $\mathcal{C}$ چیزی برای واکشی نیست: **C** مستقیم خوانده و نوشته می‌شود، و Corrupt در جا به‌روزش می‌کند.

نکته 3.1 (چرا FullStatus واسطه‌گری نمی‌شود). اگر مثلِ عمل‌های دیگر پوشانده می‌شد، FullStatus درست همان جا که مهم است بی‌مصرف می‌بود. فرض کنید

دفترِ فسادِ $\mathcal{C}$

$\mathsf{pid} := C$، $\mathbf{P} := \{0,1\}^* \cup \{A, Z\}$، $\mathbf{N} := \mathbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$، $\mathbf{U} := \emptyset$، $\mathsf{par} := \bot$

Initialize():

1: $\mathbf{C} \leftarrow \emptyset$

id.FullCorrupt() از id'

2: **require** Guard$_{\mathcal{C}}(\mathrm{id}, \mathrm{id}')$

3: **require** $\mathrm{id}'.F \in \{A, Z\}$ / زیررول هرگز بیرون، تنها

4: **require** $\mathrm{id}.P \in \{0,1\}^*$ / می‌مانند درستکار رزرو نام‌های

5: $\mathbf{C} \leftarrow \mathbf{C} \cup \{\mathrm{id}.P\}$

6: **return** ok

id.FullStatus() از id'

7: **require** Guard$_{\mathcal{C}}(\mathrm{id}, \mathrm{id}')$

8: **return** $\mathbf{C}$

id.FullLeak() از id'

9: **require** Guard$_{\mathcal{C}}(\mathrm{id}, \mathrm{id}')$

10: **require** $\mathrm{id}'.F = A$

11: **return** $\bot$ / نمی‌دارد نگه رازی هیچ دفتر ثابت:

$\mathcal{Z}$ به‌عنوانِ id' دست دراز کند تا وضعِ فسادِ $\mathrm{id}'.P$ را بپرسد. اگر $\mathrm{id}'.P$ فاسد باشد، دروازه فراخوان را رد می‌کند و Mediate آن را به حریف می‌سپارد — پس پاسخِ «آیا $\mathrm{id}'.P$ فاسد است؟» هر چیزی است که حریف بگوید، و می‌تواند بگوید نه. و چون هر فراخواننده‌ای دفتر را می‌خواند تا تصمیم بگیرد با طرفِ فاسد چگونه رفتار کند، پاسخی که حریف مهارش کند ماشین‌آلاتِ واسطه‌گری را به میلِ خود خاموش می‌کرد.

پس FullStatus هیچ واسطه‌گری و هیچ خموش‌سازی نمی‌کند: پس از نگهبان، $\mathbf{C}$ را برمی‌گرداند، حالتِ خودِ دفتر را. این همان یک عمل است که پاسخش باید از دفتر بیاید، و چون تنهاخواندنی است چیزی به حریف نمی‌دهد که پیش‌تر نتوانسته باشد استنتاج کند. هستهٔ Statusِ پیش‌نویس‌های قدیمی‌تر هم با آن می‌رود، چون پوشانه‌ای نمانده که از آن برکنار بماند.

فسادِ کارِ بیرون است. سطرِ 3 $\mathrm{id}'.F \in \{A, Z\}$ را می‌پذیرد، پس حریف و محیط هر یک می‌توانند طرفی را در $\mathbf{C}$ بگذارند و هیچ کارکردِ استانداردی اصلاً نمی‌تواند چیزی آنجا بگذارد. نگهبانِ بالای آن، واسطِ فاسدکننده را به همان طرفی که فاسدش می‌کند گره می‌زند و می‌پرسد $\mathrm{id}'.P = \mathrm{id}.P$: برای فاسدکردنِ P، حریف باید در (A, P) بدود، و محیط باید (Z, P) را ادعا کند، که سطرِ 3 می‌گذارد ریشه‌اش این کار را برای هر طرفی بکند. هر دو می‌توانند بخوانند هم: $\mathbf{A} \cup \mathbf{Z} \subseteq \mathcal{C}.\mathbf{N}$، و FullStatus کلِ $\mathbf{C}$ را برمی‌گرداند، پس هر یک می‌تواند در هر نقطه‌ای بپرسد و همیشه کلِ مجموعه را می‌آموزد. و تنها طرف‌های اصیل می‌توانند فاسد شوند: FullCorrupt می‌پرسد $\mathrm{id}.P \in \{0,1\}^*$، پس

نام‌های رزرو بنا بر ساخت درستکار می‌مانند.

هم‌اندازگی درست درمی‌آید، و آنچه تأمینش می‌کند خواندن است و نه هیچ تقسیمِ اینکه چه کسی می‌تواند بنویسد. ماشینِ نشسته در (A, P) در یک اجرای آزمایشِ تقلید $\mathcal{A}$ است و در اجرای دیگر $\mathcal{S}$، و هیچ‌چیز آن دو را به فاسدکردنِ همان طرف‌ها وادار نمی‌کند. اما $\mathcal{Z}$ در هر دو سو همان یک ماشین است و در هر دو **C** را می‌خواند، پس اشغال‌گرِ جایگاه اگر مجموعهٔ دیگری را فاسد کند، محیطی که تنها بپرسد او را می‌گیرد. پس هم‌اندازگی نتیجه است و نه فرض: هیچ محیطِ مجازی آن دو مجموعهٔ فساد را با مزیتی بیش از آنچه تعریفِ 4.11 به شبیه‌ساز می‌دهد از هم تشخیص نمی‌دهد. و اینکه محیط می‌تواند برای خودش فاسد کند این را تنها تیزتر می‌کند، چون فسادهایی که او می‌کند در هر دو سو از خودِ اوست؛ آنچه برای گرفتن می‌ماند ابتکارِ خودِ اشغال‌گر است.

فساد تنها با نامِ طرف اندیس‌گذاری می‌شود، پس P نمی‌تواند در یک شناسهٔ فرایند فاسد باشد و در دیگری درستکار.[1]

3.2 نشت

کارکردِ $\mathcal{F}$، عملِ نشت

pid، **P**، **N**، **U**، par

id.FullLeak() از id'

```
1: require id.pid = 𝓕.pid ∧ id.P ∈ 𝓕.P        / دارد وجود واسط
2: require id′.F = A ∧ id′.P = id.P            / طرف این برای 𝒜،
3: C ← (C, id.P).FullStatus() as id
4: require id.P ∈ C                            / می‌کند نشت فاسد طرفِ تنها
5: return Silence({id″ : id″ ≠ id}, id.Leak() from id′)
```

این سه خواسته همان‌هایی هستند که پوشانهٔ عمومیِ بخشِ 3.5 تحمیل می‌کند، و از نو بیان شده‌اند چون نشت به شیوهٔ معمول از هسته به‌دست نمی‌آید. سطرِ 1 نیمهٔ وجودِ Guard است؛ نیمهٔ فراخواننده عامداً ضعیف‌تر است، چون $\mathcal{A}$ باید بتواند از کارکردی نشت بگیرد که او را به‌عنوانِ فراخواننده نمی‌پذیرد، پس سطرِ 2 تنها می‌خواهد که حریف برای طرفی عمل کند که دارد می‌خواندش. سطرِ 4 دروازهٔ فساد است. بی آن، حالتِ طرفِ درستکار به میل خواندنی می‌شد، و همین است که سطرِ 3 از واسطِ کامل هر جای دیگری ردش می‌کند. خموش‌کننده همان خموش‌کنندهٔ سطرِ 5 است، پس هستهٔ نشتی که فراخوان می‌گذارد، مثلِ هر هستهٔ دیگری هویتِ خودش را ادعا می‌کند.

برای کارکردِ ایده‌آل، Leak بخشی از مشخصه است؛ پیام‌هایی که تا اینجا دریافت شده و محتوای فهرست‌های درونی نمونه‌های معمولند. برای پروتکل‌های واقعی

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[1] می‌شد به‌جایش یک هویت را فاسد کرد — **C** هویت‌ها را جمع می‌کرد و هر آزمونِ $id.P \in \mathbf{C}$ به $id \in \mathbf{C}$ بدل می‌شد — به بهای «پخش‌شدنِ» فسادها به زیرپروتکل‌ها. قراردادِ طرف‌گستر پاکیزه‌تر است، چون یک خانوادهٔ کامل را یک‌جا آلوده می‌کند.

می‌توان آن را چنان نشاند که هیچ نشت نکند، که یعنی مدلی از بی‌نشتیِ کامل؛ و چاره همان کارکردهای ایده‌آلی هستند که می‌گویند کدام بخش‌های حالت نشت می‌کنند.

پروتکلی نشت‌خوش‌ساخت است اگر بدنهٔ هر ماشین بینِ فراخوانی‌ها پاک باشد، هر تصادفِ نمونه‌برداری‌شده از راهِ $\mathcal{F}_{\mathsf{Rand}}$ کشیده شود و هر حالتی که از فراخوانی‌ای به فراخوانیِ دیگر می‌پاید در $\mathcal{F}_{\mathsf{Store}}$ نگه داشته شود و نه روی نوارِ ماشین — که هر یک آن را از راهِ Leak خودش رو می‌کند — و ورودی‌ها و خروجی‌ها تنها آنجا نشت کنند که پروتکل آن‌ها را از راهِ کارکردِ ورودی–خروجیِ $\mathcal{F}_{\mathsf{IO}}$ می‌گذراند، که پیش‌فرض نیست. آن‌گاه نوارهای کار هرگز نشت نمی‌کنند، چون پاکیِ بینِ فراخوانی‌ها چیزی رویشان نمی‌گذارد. هیچ نتیجهٔ زیر هیچ‌یک از این‌ها را مفروض نمی‌گیرد؛ این شرطی است که یک تحقق باید برآورده کند تا Leakِ آن معنایی داشته باشد، و ما ثبتش می‌کنیم و به کارش نمی‌بریم.

$\mathcal{Z}$ نمی‌تواند این واسط را فرا بخواند، پس برای تمایزناپذیری لازم نیست محقق شود. با این حال باید شبیه‌سازی‌شدنی باشد، چون حریف می‌تواند فرایش بخواند.

3.3 محیط

ما کارکردِ ویژه‌ای به نام $\mathcal{Z}$ تعریف می‌کنیم، با یک واسطِ یگانه به‌ازای هر $\mathtt{id}$، یعنی $\mathtt{id}$.FullZ، که هستهٔ $\mathtt{id}.\mathsf{Z}$ را نگهبانی و خموش می‌کند. میدانِ $\mathcal{Z}.\mathbf{N} = \mathbf{A} \cup \mathbf{Z}$ سزاوارِ تأکید است: حریف و خودِ محیط را می‌پذیرد و هیچ‌چیزِ دیگر را، پس هیچ کارکردِ استانداردی نمی‌تواند $\mathcal{Z}$ را فرا بخواند. هر چیزِ فراخ‌تری می‌گذاشت پروتکلی محیط را به‌عنوانِ یکی از زیرروال‌های خودش فرا بخواند — یک «پروتکلِ باز»، که زیرروال‌هایش نامعین می‌مانند. دو چیزِ متمایز این را ممنوع می‌کنند. تنگیِ $\mathcal{Z}.\mathbf{N}$ خودِ فراخوان را رد می‌کند: شناسهٔ فرایندِ فراخوانندهٔ استاندارد در **Std** است، که هیچ کجا با $\mathbf{A} \cup \mathbf{Z}$ برخورد نمی‌کند، پس سطرِ 2 از Guard، هر چه فراخواننده اعلام کند، ردش می‌کند. تعریفِ 1.3 نیمهٔ دیگرِ بازبودن را رد می‌کند، یعنی عضوی که نامِ وابستگی‌ای را می‌برد که هیچ عضوی فراهمش نمی‌کند. اینجا اولی است که کار می‌کند: نام‌بردن از $\mathcal{Z}$ wf را نمی‌لغزاند، چون $\mathcal{Z} \in \pi^+$ چنین درایه‌ای را برآورده‌شدنی می‌کند، پس اگر آن میدان فراخ‌تر بود، سیستمی می‌توانست خوش‌ساخت باشد و با این حال باز.

آن $\mathbf{Z}$ در $\mathcal{Z}.\mathbf{N}$ به دلیلی دیگر آنجاست: می‌گذارد $\mathcal{Z}$ خودش را فرا بخواند. واسط‌های $\mathcal{Z}$ِ محیط در طرف‌های مختلف جدایند، و پذیرفتنِ $\mathbf{Z}$ همان است که می‌گذارد آن‌ها به یکدیگر برسند.

اینکه کدامشان می‌تواند به کدام برسد را مجموعهٔ خموش‌شده حل می‌کند، و آنجا کاری بیش از دور نگه داشتنِ $\mathcal{Z}$ از سیستم‌ها می‌کند. هر ماشینِ دیگری به جایی که می‌دود گره خورده است: کارکردِ استاندارد تنها هویتِ خودش را می‌تواند ادعا کند، و $\mathcal{A}$ روی $\{\mathtt{id}'' \neq \mathtt{id}\}$ خموش است. هیچ‌چیز $\mathcal{Z}$ را به‌تنهایی گره نمی‌زد، چون Silence تنها ادعایی را آزمون می‌کند و Guard تنها هدف را، پس (Z, P) و $(\mathsf{Z}, \mathsf{P}')$ جایگزین‌پذیر می‌شدند و مؤلفهٔ طرفِ یک هویتِ $\mathcal{Z}$ هیچ خبری حمل نمی‌کرد. سطرِ 3 این گره را می‌دهد، و تا آنجا که کاربردها می‌گذارند شل: واسطی در $\mathtt{id}$ می‌تواند $\mathtt{id}'$ را ادعا کند هرگاه $\mathtt{id}'.\mathsf{P} = \mathtt{id}.\mathsf{P}$، یا $\mathtt{id}'.\mathsf{P} = \mathsf{Z}$، یا $\mathtt{id}.\mathsf{P} = \mathsf{Z}$. به زبانِ ساده، برای طرفِ خودت عمل کن، یا برای ریشه؛ و ریشه می‌تواند برای هر کسی عمل کند.

هر دو راهِ فرار لازمند. بی $\mathtt{id.P} = Z$ ریشه در تنگنا می‌ماند، و ریشه همان جایی است که Exec از آن آغاز می‌کند و تنها واسطی است که می‌تواند به دیگران برسد. بی $\mathtt{id}'.P = Z$ برگ‌ها در تنگنا می‌مانند: تنها راهِ بازگشتِ حریف به محیط $(A, P) \to (Z, P)$ است، و Mediate $\mathcal{A}$ را در $(A, \mathtt{id}.P)$ بیدار می‌کند، برای طرفی که پاسخش را می‌دهد؛ پس محیطی که نیاز داشته باشد آنچه حریف در P گفته را با آنچه در P' گفته بسنجد، هرگز نمی‌توانست آن دو را کنارِ هم بیاورد — این واسط‌ها جدایند، و نمی‌توانند از راهِ حالتِ مشترک هم آن را یک‌کاسه کنند — و ردهٔ تعریفِ 4.11 واقعاً کوچک می‌شد. با هر دو راهِ فرار، همبندی ستاره‌ای دوسویه است که در (Z, Z) ریشه دارد، و ما چیزی نمی‌شناسیم که محیطی نامقید بتواند بکند و این ستاره نتواند. سطرِ 2 در همین روح مقید می‌کند: جلوی ادعای (A, P) به‌دستِ $\mathcal{Z}$ را می‌گیرد، و از این راه جلوی گذشتنش از دروازهٔ فسادِ سطرِ 3 و راندنِ واسطِ طرفِ درستکار را — توانی که حریفی که او نقشش را بازی می‌کرد هم ندارد، چنان که نکتهٔ 4.8 در بابِ شبیه‌سازی که روی کمتری خموش شده ثبت می‌کند. آنچه آن سطر دیگر لازم نیست بکند، دور نگه داشتنِ $\mathcal{Z}$ از FullCorrupt است، که سطرِ 3 او را زیرِ نامِ خودش به آن می‌پذیرد.

A ∪ **Z** با هم همان است که کاربردها لازم دارند و نه بیشتر. پذیرفتنِ **A** ضروری است چون محیط هیچ راهِ دیگری به اجرا ندارد: وقتی $\mathcal{Z}$ ژتون را واگذارد، دوباره تنها زمانی می‌دود که چیزی فرایش بخواند، و تنها حریف می‌تواند؛ و همان کانال است که $\mathcal{Z}$ از آن هر چه حریف بخواهد بگویدش می‌شنود. پذیرفتنِ **Z** ضروری است چون محیط یک واسط به‌ازای هر طرف است و نه یک ماشین، و بی آن ریشه به هیچ‌یک از آن‌ها نمی‌رسید. جز این دو هیچ ضروری نیست: کارکردی که ناچار بود $\mathcal{Z}$ را فرا بخواند با آن مثلِ زیرروالی رفتار می‌کرد که خودش تعریفش نکرده، و $\mathcal{C}$ فسادها را به هیچ‌کس گزارش نمی‌دهد — حریف می‌داند چه فاسد کرده، چون خودش کرده — و همین است که چرا $\mathcal{Z}$ باید مجموعهٔ فساد را بپرسد. پس محیط تنها از آن دو ماشینی در دسترس است که بیرونِ هر سیستمی‌اند.

محیطِ $\mathcal{Z}$

pid := Z، **P** := $\{0,1\}^* \cup \{A, Z\}$، **N** := **A** ∪ **Z**، **U** := $\{(\mathcal{A}, \text{serves})\}$، par := $\mathcal{I}$

$\underline{\mathtt{id}.\text{FullZ}(\mathtt{in})}$ از $\mathtt{id}'$

1: **require** Guard$_{\mathcal{Z}}(\mathtt{id}, \mathtt{id}')$
2: $\bar{\mathcal{I}} \leftarrow \mathcal{I} \cup \{\mathtt{id}'' : \mathtt{id}''.F = A\}$ / دسترس از دور $\mathcal{A}$
3: $\quad \cup \ \{\mathtt{id}'' : \mathtt{id}.P \neq Z \ \wedge \ \mathtt{id}''.P \notin \{\mathtt{id}.P, Z\}\}$ / ریشه یا خود، طرفِ
4: $\mathtt{out} \leftarrow$ Silence$(\bar{\mathcal{I}},\ \mathtt{id}.Z(\mathtt{in})$ from $\mathtt{id}')$
5: **return** $\mathtt{out}$

3.4 حریف

حریفِ $\mathcal{A}$ همان شکل را دارد: یک واسطِ یگانه به‌ازای هر $\mathtt{id}$، یعنی $\mathtt{id}$.FullA، که هستهٔ $\mathtt{id}.A$ را نگهبانی و خموش می‌کند. آنچه او را از محیط جدا می‌کند این است که چه کسی می‌تواند به او برسد. آنجا که $\mathcal{Z}$.**N** تنها **A** ∪ **Z** است، میدانِ $\mathcal{A}$.**N** = **Std** ∪ **A** ∪ **Z** هر

شناسهٔ فرایندی را در بر دارد که هرگز فراخوانی می‌گذارد. تنها شناسهٔ $\mathcal{C}$ نیست، و $\mathcal{C}$ هیچ فراخوانی نمی‌گذارد. این فراخی یک خواسته است و نه یک راحتی: قلاب‌های واسطه‌گریِ فصلِ 2، و فراخوان‌هایی که کارکردهای ایده‌آل از راهِ کوتاه‌نویسیِ $\mathcal{A}(\cdot)$ می‌گذارند، هر دو فراخوان‌هایی هستند که ماشینی در حالِ عمل برای طرفی، روی حریف می‌گذارد.

این واسط‌ها همان چیزی هستند که دَرِ پشتی در این نمادگذاری چنین می‌نماید. حریف ژتونی از خودش ندارد و نمی‌تواند فراخوانی را آغاز کند، پس هر حمله‌ای که ترتیب می‌دهد از آنجا شروع می‌شود که ماشینِ دیگری مهار را به او می‌سپارد: واسطِ یک طرف از راهِ Mediate، یا مستقیماً محیط، که اصلاً همان راهی است که محیط به حریف دستور می‌دهد. آنچه او در پاسخ می‌تواند بگوید تنگ است: مجموعهٔ خموش‌شده $\{\mathrm{id}'' : \mathrm{id}'' \neq \mathrm{id}\}$ است، پس واسطی از $\mathcal{A}$ تنها می‌تواند همان هویتی را ادعا کند که در آن می‌دود.

در درونِ واسطی با هویتِ id، ما $\mathcal{A}(\mathrm{in})$ را برای فراخوانِ $(A, \mathrm{id}.P).\mathrm{FullA}(\mathrm{in})$ به‌عنوانِ id می‌نویسیم، و $\mathcal{Z}(\mathrm{in})$ را به همین شکل. ادعا باید id باشد و نه هیچ‌چیزِ دیگر، چون سطرِ 5 هسته را روی هر چیزی جز هویتِ خودش خموش می‌کند؛ پس حریفِ فراخواننده را به‌عنوانِ شناسهٔ ادعایی می‌آموزد و میدانِ جداگانه‌ای برایش لازم ندارد. هستهٔ او به‌صورتِ $\mathrm{id}.A(\mathrm{in})$ از id' وارد می‌شود، همان شکلی که Silence اعلام می‌کند و همان شکلی که هر هستهٔ دیگری در آن وارد می‌شود. مؤلفهٔ طرفِ هدف $\mathrm{id}.P$ است، پس فراخوان این را هم ثبت می‌کند که به‌نیابتِ چه کسی گذاشته شده است.

حریفِ $\mathcal{A}$

$\mathrm{pid} := A$، $\mathbf{P} := \{0,1\}^* \cup \{A, Z\}$، $\mathbf{N} := \mathbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$، $\mathbf{U} := \{(\mathcal{Z}, \mathrm{serves})\}$، $\mathrm{par} := \bot$

$\underline{\mathrm{id}.\mathrm{FullA}(\mathrm{in})}$ از id'

1: **require** $\mathrm{Guard}_{\mathcal{A}}(\mathrm{id}, \mathrm{id}')$
2: $\mathrm{out} \leftarrow \mathrm{Silence}(\{\mathrm{id}'' : \mathrm{id}'' \neq \mathrm{id}\},\ \mathrm{id}.A(\mathrm{in})\ \text{from}\ \mathrm{id}')$
3: **return** out

3.5 واسطِ کامل

هر چه پوشانه لازم دارد اکنون سرِ جایش است: Guard و Mediate و Silence از فصلِ 2، و دفترِ $\mathcal{C}$ از بخشِ 3.1 که **C** را می‌دهد. واسطِ کامل یک کارکردِ استاندارد ترکیبِ همین‌هاست.

سطرهای 2–4 و 6–7 فساد را می‌گردانند، و تنها جایی هستند که واسطِ کامل با دفتر مشورت می‌کند. دو بار خوانده می‌شود چون **C** با پیشرفتِ اجرا تغییر می‌کند و هر خواندن تنها یک عکسِ فوری است: آزمون‌های پیش از سطرِ 6 اولی را به کار می‌برند و آزمونِ سطرِ 7 دومی را. عضویت یکنواست، چون Corrupt هرگز جز افزودن نمی‌کند، پس طرفی که در خواندنِ اول درستکار است می‌تواند در خواندنِ دوم فاسد باشد، اما هرگز برعکس.

سطرِ 3 یک ترکیب را رد می‌کند: حریف در طرفِ درستکار، که آنجا کاری ندارد. هر چیزِ دیگری می‌گذرد، و Mediate باقی را حل می‌کند. پس طرفِ درستکار برای همه جز

کارکردِ $\mathcal{F}$
pid، **P**، **N**، **U**، par

$\mathtt{id.FullOp(in)}$ از id'

فاسد طرفِ در تنها $\mathcal{A}$ /

1: **require** Guard$_{\mathcal{F}}(\mathrm{id}, \mathrm{id}')$
2: **C** $\leftarrow (\mathrm{C}, \mathrm{id.P})$.FullStatus() as id
3: **require** $\neg(\mathrm{id}'.\mathrm{F} = \mathrm{A} \ \wedge\ \mathrm{id.P} \notin \mathbf{C})$
4: Mediate($\mathbf{C}$, id.Op, in, id')
5: $\mathrm{out} \leftarrow$ Silence($\{\mathrm{id}'' : \mathrm{id}'' \neq \mathrm{id}\}$, id.Op($\mathrm{in}$) from id')
6: **C** $\leftarrow (\mathrm{C}, \mathrm{id.P})$.FullStatus() as id
7: Mediate($\mathbf{C}$, id.Op, out, id')
8: **return** out

$\mathcal{A}$ خطاب‌پذیر است و هستهٔ خودش را می‌دواند؛ و طرفِ فاسد برای همه خطاب‌پذیر است، اما تنها $\mathcal{A}$ به هسته‌اش می‌رسد و هر فراخوانندهٔ دیگری را $\mathcal{A}$ به‌جای آن طرف پاسخ می‌دهد. محیط حالتِ ویژهٔ خودش را لازم ندارد: چه هویتِ خودش را ادعا کند و چه هویتِ بیرونیِ پذیرفته‌ای را، طرفِ فاسد در هر حال از راهِ $\mathcal{A}$ پاسخش می‌دهد.

می‌ماند طرفِ فاسدی که هر کسی جز $\mathcal{A}$ خطابش کند، که فراخوانش را نمی‌شود همین‌طور دواند، پس $\mathcal{A}$ به‌جای آن طرف پاسخ می‌دهد. سطرهای 4 و 7 همان یک پوشانه در دو لحظه‌اند. اولی طرفی را می‌گیرد که در لحظهٔ ورود از پیش فاسد است و in را واگذار می‌کند، پس هسته هرگز نمی‌دود. دومی طرفی را می‌گیرد که در حینِ دویدنِ هسته‌اش در سطرِ 5 فاسد شد و out را واگذار می‌کند، که دیگر مالِ آن طرف نیست که برگرداندش. یکنوایی باعث می‌شود آن دو، حالت‌ها را دقیقاً یک بار بپوشانند: طرفی که در اولی فاسد است پیش‌تر برگشته، و هیچ طرفی در اولی فاسد و در دومی درستکار نیست. هر دو قلاب $\mathcal{A}$ را در $(\mathrm{A}, \mathrm{id.P})$ از راهِ واسطِ کاملش خطاب می‌کنند، پس حریفی که پاسخ می‌دهد مثلِ هر فراخواندهٔ دیگری نگهبانی و خموش شده است. سطرِ 8 مقدارِ خودِ هسته را برمی‌گرداند، که در دو حالت رخ می‌دهد: برای طرفی که سرتاسر درستکار است، و برای طرفِ فاسدی که خودِ $\mathcal{A}$ خطابش می‌کند، که دروازه می‌پذیردش و هیچ‌یک از دو قلاب نمی‌گیردش، چون هر دو $\mathrm{id}'.\mathrm{F} \neq \mathrm{A}$ را آزمون می‌کنند.

نکته 3.2 (فسادِ میانِ فراخوان پیش‌دستانه نیست). هیچ‌چیز هسته‌ای را که پیش‌تر در حالِ دویدن است قطع نمی‌کند. فرض کنید هستهٔ سطرِ 5 حریف را فرا بخواند — از راهِ $\mathcal{A}(\cdot)$، یا از راهِ زیررَوالی که واسطه‌گری می‌کند — و حریف پیش از برگرداندنِ مقداری، $\mathrm{id.P}$ را فاسد کند. دفتر این را بی‌درنگ ثبت می‌کند، چون **C** حالتِ خودِ اوست، اما هسته نه خبر می‌شود و نه متوقف: با هر چه حریف برگردانده از سر می‌گیرد و تا پایان می‌دود. تنها در لحظهٔ بازگشتِ آن است که سطرِ 6 **C** را از نو می‌خواند، $\mathrm{id.P}$ را در آن می‌یابد، و out را در سطرِ 7 به Mediate می‌سپارد تا جایگزین شود.

پس حریف در طولِ چنین فراخوانی دو بار ژتون را در دست دارد، به این ترتیب: یک بار در درونِ هسته، برای فراخوانی که هسته گذاشت، و یک بار پس از بازگشتِ هسته،

برای واسطه‌گریِ خروجیِ آن. هیچ لحظه‌ای در میانه نمی‌گذارد آن فساد اثر کند، چون طرفی که در میانِ فراخوان فاسد می‌شود نمی‌تواند آنچه پیش‌تر کرده را نکرده کند. فراخوان از نو هم دروازه‌بانی نمی‌شود: سطرِ 3 در خواندنِ اول تصمیم گرفته شد، و فراخوانی که در درستکاریِ $\mathrm{id}.P$ پذیرفته شده پذیرفته می‌ماند. پس آنچه پوشانه تضمین می‌کند از سقط‌کردن ضعیف‌تر است، و بس است: هیچ‌چیزی که طرفِ تازه‌فاسد حساب کرده باشد بی‌واسطه‌گری به فراخوانندهٔ او نمی‌رسد. فراخوان‌هایی که پس از فساد می‌رسند، به‌جایش در راهِ ورود گرفته می‌شوند، با سطرِ 4.

3.6 اجراها

در یک اجرا، دقیقاً یک ماشین ژتونِ فعال‌سازی را در دست دارد. فراخوانِ $\mathrm{out} \leftarrow \mathrm{id}.\mathsf{Op}(\mathrm{in})$ کوتاه‌نویسیِ این است که فراخواننده با یک ادامه معلق شود و ژتون به واسطِ Op از $\mathcal{F}$ در هویتِ id سپرده شود. هر فراخوان میانِ ماشین‌ها یک واسطِ کامل را خطاب می‌کند، و FullOp همان است که Exec توزیعش می‌کند؛ هسته تنها به‌دستِ پوشانه‌ای که خدمتش می‌کند فراخوانده می‌شود، در درونِ ماشینی که حملش می‌کند، و هرگز از آن‌سوی یک سپردنِ ژتون. فراخوانی که به هویتی خطاب شود که هیچ ماشینی اشغالش نکرده با rej پاسخ می‌گیرد. یک نمونه می‌تواند چند ادامهٔ معلق داشته باشد و می‌تواند در حالِ تعلیق دوباره وارد شود، به‌ویژه به‌دستِ $\mathcal{A}$.

تعریف 3.3 (ماشین‌ها و اشغال). یک ماشین هر چیزی است که اجرا می‌دواندش: اعضای سیستم، و هر چه اجرا در کنارِ آن‌ها فراهم می‌کند. ماشین مجموعهٔ متناهی‌ای از هویت‌ها را اشغال می‌کند — برای کارکرد، همان $\mathsf{IDs}(\mathcal{F})$ از تعریفِ 1.1 — و Exec درست همان فراخوان‌هایی را به او توزیع می‌کند که به هویت‌های اشغال‌شده‌اش خطاب شده‌اند. اشغال‌گرانِ یک اجرا مجموعه‌های هویتِ دوبه‌دو مجزا حمل می‌کنند، و Exec برای هیچ خانوادهٔ دیگری تعریف نشده است. اشغال‌گرِ جایگاهِ حریف ماشینی است که هویت‌های شناسهٔ فرایندِ A را اشغال می‌کند و هیچ هویتِ دیگری را؛ حریفِ بخشِ 3.4 یکی از آن‌هاست، و هر شبیه‌سازِ فصلِ 4 نیز. یک فعال‌سازی بازهٔ ماکسیمالی است که در طولِ آن یک ماشین ژتون را در دست دارد.

هر فراخوان یک شناسهٔ ادعاییِ id' حمل می‌کند. کارکردهای استاندارد تنها هویتِ همان واسطی را می‌توانند ادعا کنند که فراخوان را می‌گذارد؛ $\mathcal{Z}$ و $\mathcal{A}$ می‌توانند هویت‌های دیگری را ادعا کنند، مشروط به آزمون‌های فصلِ 2. آن آزمون‌ها همان نگهبان‌اند، که id' را در برابرِ هویتِ id واسط و در برابرِ پارامترهای کارکرد می‌سنجد، و ما آن‌ها را با گزاره‌های **require** مشخص می‌کنیم. نگهبانی که شکست بخورد مقدارِ ممتازِ rej را به فرستنده برمی‌گرداند، و rej یک رد است و نه یک مقدار: هیچ ماشینی نمی‌تواند آن را به‌عنوانِ خروجیِ خودش برگرداند. این قاعده اِعمال می‌شود و مفروض گرفته نمی‌شود — هسته‌ای که کدش rej برمی‌گرداند، و پاسخِ واسطه‌گری‌شدهٔ rej از جایگاهِ

حریف، به‌جایش به‌صورتِ $\bot$ تحویل داده می‌شوند — پس فراخواننده‌ای که rej می‌گیرد می‌آموزد که نگهبانی فراخوانش را رد کرده، و هیچ‌چیزِ دیگری نمی‌تواند یکی بسازد. به‌ویژه $\bot \neq$ rej.

تصادف به‌ازای هر ماشین است: هر ماشین نوارِ سکهٔ خودش را حمل می‌کند، نامتناهی، یکنواخت، و مستقل از نوارِ هر ماشینِ دیگر، که تنها وقتی کدش نمونه‌برداری می‌کند خوانده می‌شود. احتمال‌ها روی یک اجرا، روی این نوارها به‌طورِ توأم‌اند. دسته‌کردنِ ماشین‌ها در یک اشغال‌گر — چنان که فصلِ 4 و بخشِ 4.4 می‌کنند — نوارها را یکی نمی‌کند: سکه‌های یک رونوشتِ میهمان همان‌اند که پیش از جذب بودند.

جز سیستمِ π، اجرا سه ماشین حمل می‌کند: محیطِ $\mathcal{Z}$، حریفِ $\mathcal{A}$، و دفترِ فسادِ $\mathcal{C}$، که حالتش **C** را هر پوشانه‌ای می‌خواند. بنا بر فصلِ 1 هیچ‌یک عضوِ π نیست، و همین سطرِ 1 را یک‌معنا می‌کند — دفتر یک بار مقدارِ آغازین می‌گیرد، به‌عنوانِ آرگومان، و نه بارِ دوم به‌عنوانِ عضو — و π را همان شیئی نگه می‌دارد که طراح نوشت. رویه آن سه را مقدارِ آغازین می‌دهد، و سپس محیط را در (Z, Z) فعال می‌کند، چنان که گویی خودش را فرا خوانده است. آن نخستین فعال‌سازی مثلِ هر فعال‌سازیِ دیگری از واسطِ کاملِ $\mathcal{Z}$ می‌گذرد، پس محیط از همان آغاز خموش است، و Guard$_{\mathcal{Z}}$ به‌سادگی می‌گذرد، چون هویتِ ادعایی و هدف یکی‌اند. هر چه محیط برگرداند همان برآمدِ کار است.

اجرای Exec
سیستمِ π، محیطِ $\mathcal{Z}$، حریفِ $\mathcal{A}$، دفترِ فسادِ $\mathcal{C}$

Exec($\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C}$):

1: $\mathcal{C}$.Initialize(); $\mathcal{F}$.Initialize() for every $\mathcal{F} \in \pi$
2: **handle** (Z, Z).FullZ() from (Z, Z) **with** ▷ $\mathcal{Z}$ نخست فعال می‌شود و ژتون را دارد
3: **return** $\mathtt{out} \mapsto \mathtt{out}$
4: $\mathtt{id}$.FullOp($\mathtt{in}$) from $\mathtt{id}'$; $\mathtt{k} \mapsto \mathtt{k}(\mathtt{id}$.FullOp($\mathtt{in}$) from $\mathtt{id}')$

اگر آن را به‌عنوانِ هندلر بخوانیم [12، 4]، اجرا دقیقاً یک عمل دارد: گذاشتنِ فراخوان. سطرِ 2 محیط را فعال می‌کند، و این تنها فعال‌سازی‌ای است که ماشینی آن را نگذاشته است. سطرِ 4 از آن پس هر فراخوانی را می‌گرداند، و ادامهٔ $\mathtt{k}$ همان جایی است که ژتون به آن بازمی‌گردد: معلق‌کردنِ فراخواننده، سپردنِ ژتون به $\mathtt{id}$، و از سر گرفتنِ فراخواننده با پاسخ، درست همان معنایی است که فراخواندنِ $\mathtt{k}$ روی آن پاسخ دارد. دقیقاً یک ماشین در سرتاسر ژتون را در دست دارد. بندِ **return** وقتی رسیده می‌شود که برنامهٔ محیط بایستد، و خروجیِ آن خروجیِ اجراست.

فصل 4

تقلید و ترکیبِ UC

4.1 جانشانیِ زیرسیستم

پروتکل را معمولاً در برابرِ زیرروالی می‌نویسند که بعداً پروتکلِ دیگری فراهمش می‌کند. جانشین‌کردنِ یکی به‌جای دیگری، عملی مجموعه‌ای روی سیستم‌هاست.

تعریف 4.1 (جانشانیِ زیرسیستم)**.** بگذارید ρ سیستمی باشد و $\varphi \subseteq \rho$ زیرسیستمی از آن. برای سیستمِ π، جانشانیِ φ با π در ρ این است:

$$\rho[\pi/\varphi] \;:=\; (\rho \setminus \varphi) \cup \pi.$$

تا اینجا هیچ‌چیز جلوی π ورودی را نمی‌گیرد که روی شناسهٔ فرایندی فرود بیاید که پروتکلِ پیرامون از پیش به کارش می‌برد، و این فراخوانی را با خطابی دوپهلو رها می‌کرد. ما این را ممنوع می‌کنیم و به تعمیرش نمی‌رویم.

تعریف 4.2 (جانشانیِ خوش‌ساخت)**.** جانشانیِ φ با π در ρ خوش‌ساخت است اگر هیچ شناسهٔ فرایندی هم در $\rho \setminus \varphi$ و هم در π رخ ندهد، یعنی اگر

$$\{\mathcal{F}.\mathsf{pid} : \mathcal{F} \in \rho \setminus \varphi\} \cap \{\mathcal{F}.\mathsf{pid} : \mathcal{F} \in \pi\} \;=\; \emptyset.$$

گزاره 4.3 (جانشانی)**.** بگذارید ρ و π سیستم باشند، $\varphi \subseteq \rho$، و جانشانیِ φ با π در ρ خوش‌ساخت باشد. آن‌گاه $\rho[\pi/\varphi]$ یک سیستم است. و اگر علاوه بر آن $\mathsf{IDs}(\pi) = \mathsf{IDs}(\varphi)$، آن‌گاه $\mathsf{IDs}(\rho[\pi/\varphi]) = \mathsf{IDs}(\rho)$.

اثبات. شناسه‌های فرایندِ $\rho[\pi/\varphi]$ همان‌های $\rho \setminus \varphi$ به همراهِ همان‌های π هستند. هر سیستم درونِ خودش شناسه‌های متمایز حمل می‌کند و خوش‌ساختی آن دو گردایه را مجزا می‌کند، پس هیچ شناسه‌ای تکرار نمی‌شود؛ اعضا استاندارد و به‌شمارِ متناهی‌اند، که هر دو از ρ و π به ارث رسیده، پس $\rho[\pi/\varphi]$ یک سیستم است. آن ادعای اول تنها بر خوش‌ساختیِ جانشانی تکیه دارد و $\mathsf{IDs}(\pi) = \mathsf{IDs}(\varphi)$ هیچ کجای آن وارد نمی‌شود.

هویت‌ها فرضِ جداگانه‌ای لازم ندارند: هویت شناسهٔ فرایندش را در مؤلفهٔ اول حمل می‌کند، پس اعضایی با شناسه‌های متمایز مجموعه‌های هویتِ مجزا دارند. و برای خودِ هویت‌ها،

$$\begin{aligned}\mathsf{IDs}(\rho[\pi/\varphi]) &= \mathsf{IDs}(\rho \setminus \varphi) \cup \mathsf{IDs}(\pi)\\ &= \mathsf{IDs}(\rho \setminus \varphi) \cup \mathsf{IDs}(\varphi) = \mathsf{IDs}(\rho),\end{aligned}$$

که در گامِ میانی $\mathsf{IDs}(\pi) = \mathsf{IDs}(\varphi)$ و در گامِ آخر $\varphi \subseteq \rho$ را به کار می‌برد. □

دو نیمهٔ این گزاره در زیر به دو شکلِ متفاوت به کار می‌روند. اینکه $\rho[\pi/\varphi]$ یک سیستم است می‌گذارد فصلِ 4 اصلاً از آن سخن بگوید، و تنها بر خوش‌ساختی تکیه دارد. اینکه هویت‌هایش همان‌های ρ است $\mathsf{IDs}(\pi) = \mathsf{IDs}(\varphi)$ را لازم دارد، و هیچ نتیجه‌ای در فصلِ 4 آن را مفروض نمی‌گیرد: مجازبودن در آنجا برای زوجی از سیستم‌ها و از راهِ اجتماعِ مجموعه‌های هویتشان بیان می‌شود، و گزارهٔ 4.16 و قضیهٔ 4.20 رو می‌گویند که هیچ نسبتی میانِ آن دو به کار نرفته است. هر جا آن تساوی خواسته باشد، مسدودسازی فراهمش می‌کند.

تعریفِ 4.2 شناسه‌های فرایند را حل می‌کند و گزارهٔ 4.3 هویت‌ها را. هیچ‌یک به وابستگی نمی‌پردازد، و وابستگی همان جایی است که جانشانی می‌تواند بد پیش برود: نام‌هایی که $\rho \setminus \varphi$ به آن‌ها تکیه دارد پس از آنکه φ جایش را به π بدهد هنوز اشغال‌شده‌اند، اما هر خواسته در اشغال‌گرِ تازه از نو سنجیده می‌شود. دو شرط این شکاف را می‌بندند.

به‌زبانِ غیررسمی، قضیهٔ 4.4ِ زیر می‌گوید جانشانیِ خوش‌ساخت wf را نگه می‌دارد، هرگاه درایه‌های πِ ورودی در جهانِ تازه برآورده‌شدنی باشند، و هر جا درایه‌ای از $\rho \setminus \varphi$ شاهدش را در درونِ φ می‌یافت، عضوی از π به‌جای φ بایستد. فرضِ (ii) همان معنای این است که π جانشینِ آمادهٔ φ باشد، و تقلید آن را فراهم نمی‌کند.

قضیه 4.4 (جانشانی خوش‌ساختی را نگه می‌دارد). بگذارید ρ و π سیستم باشند، $\varphi \subseteq \rho$، جانشانیِ φ با π در ρ به معنای تعریفِ 4.2 خوش‌ساخت باشد، و $\mathsf{wf}(\rho)$. فرض کنید

(i) هر درایهٔ هر $\mathcal{F} \in \pi$ شاهدی در $\left(\rho[\pi/\varphi]\right)^{+}$ دارد، به معنای تعریفِ 1.3؛ و

(ii) هرگاه φ درایه‌ای $\mathcal{F} \in \rho \setminus \varphi$ درایه‌ای $\mathcal{F}.\mathbf{U}$ $(\mathcal{F}', R) \in \mathcal{F}.\mathbf{U}$ داشته باشد که برخی شاهدش به معنای تعریفِ 1.3 در φ باشد، آن‌گاه برخی $\mathcal{G} \in \pi$ در $\mathcal{G}.F = \mathcal{F}'.F$، $\mathcal{G}.s = \mathcal{F}.s \vee \mathsf{global}(\mathcal{G})$ و $R(\mathcal{F}, \mathcal{G})$ صادق باشند.

آن‌گاه $\mathsf{wf}\big(\rho[\pi/\varphi]\big)$.

اثبات. بنویسیم $\rho' := \rho[\pi/\varphi] = (\rho \setminus \varphi) \cup \pi$، که بنا بر ادعای اولِ گزارهٔ 4.3 یک سیستم است — ادعای دوم اینجا در دست نیست، چون $\mathsf{IDs}(\pi) = \mathsf{IDs}(\varphi)$ فرضِ این قضیه نیست. $\mathcal{F} \in \rho'^{+}$ و درایهٔ $(\mathcal{F}', R) \in \mathcal{F}.\mathbf{U}$ را بگیرید؛ باید شاهدی در ρ'^{+} بنشانیم. بر حسبِ اینکه $\mathcal{F}$ از کجا می‌آید، سه حالت هست.

اگر $\mathcal{F} \in \pi$، فرضِ (i) درست همان ادعاست.

اگر $\mathcal{F}$ یکی از $\mathcal{Z}$، $\mathcal{A}$، $\mathcal{C}$ باشد، درایه‌هایش یک‌بار برای همیشه و مستقل از سیستم حل شده‌اند: $\mathcal{C}$.**U** تهی است، و درایه‌های $\mathcal{Z}$ و $\mathcal{A}$ نامِ یکدیگر را می‌برند، با $\mathcal{A}, \mathcal{Z} \in \rho'^{+}$ و serves که بینشان برقرار است، چنان که فصلِ 1 ثبت می‌کند. هر سه جهانی‌اند، پس عطفِ نشست بخشیده می‌شود.

اگر $\mathcal{F} \in \rho \setminus \varphi$، آن‌گاه $\mathcal{F} \in \rho^{+}$، پس $\mathsf{wf}(\rho)$ شاهدی $\mathcal{G} \in \rho^{+}$ برای این درایه فراهم می‌کند. یا $\mathcal{G} \notin \varphi$، که در آن صورت $\mathcal{G}$ در $(\rho \setminus \varphi) \cup \{\mathcal{Z}, \mathcal{A}, \mathcal{C}\} \subseteq \rho'^{+}$ است و دوباره خدمت می‌کند: همان ماشین است با همان میدان‌ها، پس $\mathcal{G}.\mathrm{F}$، $\mathcal{G}.s$ و هر چه R می‌خواند تغییر نکرده، و $\mathcal{F}$ هم تغییر نکرده است. یا $\mathcal{G} \in \varphi$، و فرضِ (ii) عضوی از π فراهم می‌کند که همان سه شرط را به‌جای او برمی‌آورد. در هر دو حالت درایه شاهدی در ρ'^{+} دارد. $\square$

فرضِ (ii) همان معنای این است که π جانشینِ آمادهٔ φ باشد، و تقلید آن را فراهم نمی‌کند. تعریفِ 4.11 رفتار را در مرز مقایسه می‌کند؛ در بابِ میدان‌های اعلام‌شدهٔ ماشین‌های پشتِ آن مرز هیچ نمی‌گوید، و πای که φ را بی‌نقص تقلید کند می‌تواند باز هم در پذیرفتنِ فراخوانندهای که φ می‌پذیرفت، یا خدمت‌دادن به طرفی که φ خدمت می‌داد، شکست بخورد. این آزمون ارزان است — روی درایه‌هایی از $\rho\setminus\varphi$ دامنه دارد که به درونِ φ اشاره می‌کردند و هیچ چیزِ دیگر — اما آزمون است، و جایش کنارِ خوش‌ساختیِ جانشانی است و نه در درونِ قضیهٔ ترکیب.

4.2 تقلید و جذب

هر چه اجرا لازم دارد اکنون تعیین شده است، پس می‌توانیم بگوییم یک سیستم کِی دستِ‌کم به‌اندازهٔ دیگری امن است. آن دو زیرِ یک و همان محیط نشانده می‌شوند، پس نخست باید تعیین شود که او چه چیزی را می‌تواند ادعا کند؛ و این نقشِ $\mathcal{Z}$.par در زیر است.

مقدارِ Exec$(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C})$ هر چیزی است که محیط برمی‌گرداند. ما محیط را چنان می‌گیریم که در ریشه یک بیت برمی‌گرداند — حدسش در بابِ اینکه در کدام سیستم نشانده شده — و اجرا را چنان می‌خوانیم که دقیقاً وقتی 1 برمی‌گرداند که محیط برگرداند. هیچ‌چیزِ دیگری در چارچوب آن مقدار را مقید نمی‌کند، پس این شرطی است بر محیط‌هایی که روی آن‌ها سور می‌زنیم، و نه نتیجه‌ای. آن‌گاه شکافِ میانِ دو اجرا یک عدد است، و نه یک شباهتِ غیررسمی.

تعریف 4.5 (مزیتِ UC). بگذارید π و φ سیستم باشند. برای محیطِ $\mathcal{Z}$، حریفِ $\mathcal{A}$ و شبیه‌سازِ $\mathcal{S}$،

$$\mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z},\mathcal{A},\mathcal{S}) \;:=\; \Big|\Pr\big[\mathsf{Exec}(\pi,\mathcal{Z},\mathcal{A},\mathcal{C})=1\big] - \Pr\big[\mathsf{Exec}(\varphi,\mathcal{Z},\mathcal{S},\mathcal{C})=1\big]\Big|,$$

که احتمال‌ها روی سکه‌های هر ماشینِ آن دو اجرا هستند، و $\mathsf{Exec} = 1$ کوتاه‌نویسیِ پیشامدی است که اجرا با برگرداندنِ 1 از سوی محیط می‌ایستد.

آن دو اجرا در سیستم و در ماشینی که هویت‌های حریف را اشغال می‌کند تفاوت دارند: $\mathcal{A}$ در یک سو، $\mathcal{S}$ در سوی دیگر. هر دو با همان یک جفت تعریف نوع‌بندی می‌شوند.

تعریف 4.6 (حریف‌ها). یک حریف ماشینی است که $\mathsf{IDs}(\mathcal{A})$ را اشغال می‌کند و هیچ هویتِ دیگری را، و واسط‌های حریف را حمل می‌کند — پوشانهٔ بخشِ 3.4، نگهبان و خموش‌کننده، و هر واسطِ دیگری که به ماشینِ آن بخش داده شده (فصلِ 9 یکی می‌افزاید) — به دورِ هستهٔ دلخواهی در هر یک از آن‌ها. ماشینِ بخشِ 3.4 یکی از آن‌هاست؛ بدلِ بخشِ 4.4 دیگری.

تعریف 4.7 (شبیه‌سازها). یک شبیه‌ساز حریفی است که دستِکم $\mathsf{Std} \cup \mathsf{Z}$ را می‌پذیرد — شناسه‌های فرایندِ هر کارکردِ استاندارد به همراهِ شناسهٔ محیط.

پس هر شبیه‌سازی حریف است، و همین است که می‌گذارد یکی از آن‌ها در ردهٔ حریف بایستد: گزارهٔ 4.12 و قضیهٔ 4.24 شبیه‌سازها را به‌عنوانِ حریف می‌خوانند، و شمول‌های بودجه‌ای بخشِ 4.5 آن دو رده را مستقیم می‌سنجند. اشغال و فراخواننده‌های پذیرفته همان معنای ایستادن در جایگاهِ حریف‌اند، پس هر چه بخشِ 3.4 در بابِ ماشینِ نشسته در (A, P) حل می‌کند بی‌تغییر برای شبیه‌ساز هم برقرار است: هیچ فراخوانی از خودش آغاز نمی‌کند، و خموش‌کننده‌اش تنها می‌گذارد هویتی را ادعا کند که در آن می‌دود. گزارهٔ 4.19 تنها اشغالِ هر چه به آن داده شود را می‌خواهد؛ قضیهٔ 4.29 پوشانه و فراخواننده‌های پذیرفته را هم به کار می‌برد، و نکتهٔ 4.8 توضیح می‌دهد چرا هیچ‌یک محدودیت نیست.

متر آن سه ماشین را به‌عنوانِ آرگومان می‌گیرد، پس تقلید را می‌شود با سورزدن روی رده‌هایی از آن‌ها و کران‌گذاشتن بر نتیجه بیان کرد. اینکه اصلاً چه محیط‌هایی به کار می‌آیند را همان زوجی که مقایسه می‌شود حل می‌کند.

نکته 4.8 (تقلید در بابِ جایگاهِ حریف چه چیزی را تعیین می‌کند). تعریفِ 4.10 زیر $\mathcal{Z}$.par را مقید می‌کند و در بابِ ماشین‌های جایگاهِ حریف هیچ نمی‌گوید؛ تعریف‌های 4.6 و 4.7 هر چه لازم است می‌گویند، و آن کمتر از پنج میدان است. میدان‌های pid و **P** با اشغال مفت می‌آیند: $\text{IDs}(\mathcal{F}) = \{(\mathcal{F}.\text{pid}, P) : P \in \mathcal{F}.\mathbf{P}\}$، پس اشغالِ $\text{IDs}(\mathcal{A})$ هر دو را سنجاق می‌کند، و $\mathcal{S}.\text{pid} = (A, 0, 0)$ در سطرِ 1 و $\text{global}(\mathcal{S})$ در سطرِ 3 را می‌دهد؛ و سنجاق‌کردنِ **P** همان است که خواسته است، چون اشغال‌گرِ جایگاهی که به طرف‌های کمتری خدمت دهد یک هویتِ A را تنها در یک سو اشغال‌نشده می‌گذارد، که در آن سو rej پاسخ می‌گیرد و در سوی دیگر پاسخی واقعی. میدانِ par هرگز خوانده نمی‌شود — مجموعهٔ خموش‌شده همان $\{\text{id}'' : \text{id}'' \neq \text{id}\}$ عینیِ سطرِ 2 است، که پوشانهٔ لازم آن را به‌عنوانِ کد حمل می‌کند — و این تشریفات نیست: شبیه‌سازی که روی کمتری خموش شده باشد می‌توانست هویتِ یک طرف را ادعا کند، از دروازهٔ فسادِ $\text{id}'.F = A$ در سطرِ 3 بگذرد، و واسطِ طرفِ درستکاری را براند، توانی که حریفی که او به‌جایش می‌ایستد ندارد. میدانِ **U** بی‌اثر است و تنها wf می‌خواندش، و آنجا π^{+} نامِ $\mathcal{A}$ رزرو را می‌برد و نه اشغال‌گرِ جایگاه را.
می‌ماند **N**، و همین است که چرا تعریفِ 4.7 آن را رو می‌خواهد. فراخ‌ترکردن ناممکن است، چون $\mathcal{A}.\mathbf{N}$ از پیش هر شناسهٔ فرایندی را در بر دارد که فراخوانی می‌گذارد؛ و تنگ‌ترکردن هیچ برای شبیه‌ساز نمی‌آورد و هر دو قضیهٔ زیر را می‌گیرد. حذفِ **Std** هر قلابِ واسطه‌گری در طرفِ فاسد را به همان $\perp$ محسوبِ بخشِ 3.6 بدل می‌کند — و قضیهٔ 4.20 روی هر ρ سور می‌زند، پس شبیه‌سازی که تنها شناسه‌های فرایندِ φ را بپذیرد در نخستین جهانِ بیرونی که شناسه‌های دیگری داشته باشد با آن‌ها روبه‌رو می‌شود. حذفِ **Z** محیط را تنها در سوی ایده‌آل بیرون می‌گذارد، و سطرِ 2 در جایگاه، درست همان دستورهایی را رد می‌کند که حریفِ واقعی پاسخشان می‌دهد.
و هیچ‌چیز در هیچ‌یک از این‌ها از زوجی که برایش شبیه‌سازی می‌شود نام نمی‌برد، و همین است که می‌گذارد قضیهٔ 4.20 شبیه‌سازِ فرض را بی‌تغییر تحویل دهد.

تعریف 4.9 (محیط‌ها). یک محیط ماشینی از اجراست که $\text{IDs}(\mathcal{Z})$ را اشغال می‌کند به همراهِ هویت‌های تعدادِ متناهی کارکردِ استاندارد که میهمان می‌کند — برای ماشینِ بخشِ 3.3 هیچ، و برای محیط‌های جذب‌شدهٔ تعریفِ 4.15 زیر، رونوشت‌هایی از پروتکلِ پیرامون. این ماشین واسطِ بخشِ 3.3 را، با خموش‌کننده و پارامترِ par، در هر هویتِ Z حمل می‌کند، و واسطِ کاملِ خودِ هر کارکردِ میهمان را در هویت‌های آن کارکرد؛ Exec در هر هویتی که او اشغال می‌کند به او توزیع می‌کند، و خروجی‌اش همان است که بخشِ $\mathcal{Z}$ او برمی‌گرداند.

تعریف 4.10 (محیط‌های مجاز). برای سیستمِ π، بنویسیم $\langle\pi\rangle := \{\mathrm{id} : \mathrm{id}.\mathsf{pid}.\mathsf{F} = \mathcal{F}.\mathsf{F} \text{ some for } \mathcal{F} \in \pi\}$ برای بستارِ نامیِ آن: هویت‌های هر شناسهٔ فرایندی که نامی را حمل می‌کند که π به کار می‌برد، اشغال‌شده یا نه، در هر نشست و هر نمونه. برای سیستم‌های π و φ، محیط‌های مجاز برای این زوج این‌هایند:

$$\mathbb{Z}_{\pi,\varphi} := \{\mathcal{Z} : \mathcal{Z}.\mathsf{par} \supseteq \langle\pi\rangle \cup \langle\varphi\rangle \text{ و } \mathcal{Z} \text{ از هیچ هویتی در } \mathsf{IDs}(\pi) \cup \mathsf{IDs}(\varphi) \text{میهمان نمی‌کند}\}.$$

شرطِ دوم بهداشتِ اشغال است: اجرا تنها برای اشغال‌گرانی با مجموعه‌های هویتِ دوبه‌دو مجزا تعریف شده است (تعریفِ 3.3)، پس محیطی که در هویتی میهمان کند که یکی از سیستم‌ها اشغالش کرده نمی‌توانست کنارِ آن نشانده شود، و مجازبودن آن را برای هر دو جهان یک‌جا ممنوع می‌کند.

شمول در زیر توجیه می‌شود. آن‌گاه یک حکمِ تقلید سه رده را تعیین می‌کند: ردهٔ $\mathbb{A}$ از حریف‌ها (تعریفِ 4.6)، ردهٔ $\mathbb{S}$ از شبیه‌سازها (تعریفِ 4.7)، و ردهٔ $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi}$ از محیط‌ها (تعریفِ 4.9). گرفتنِ $\mathbb{Z} = \mathbb{Z}_{\pi,\varphi}$ قوی‌ترین حکم از این سه را می‌دهد، و خوانشِ پیش‌فرض است.

تعریف 4.11 (تقلیدِ UC). بگذارید π و φ سیستم باشند، $\varepsilon \geqslant 0$، و $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi}$. می‌گوییم π کارکردِ φ را در ε و در برابرِ $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ تقلید-UC می‌کند اگر

$$\forall \mathcal{A} \in \mathbb{A}\ \exists \mathcal{S} \in \mathbb{S}\ \forall \mathcal{Z} \in \mathbb{Z} : \qquad \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) \leqslant \varepsilon.$$

ترتیبِ سورها جانِ این تعریف است: شبیه‌ساز می‌تواند به حریف وابسته باشد اما به محیط نه، پس یک $\mathcal{S}$ باید یک‌جا در برابرِ هر $\mathcal{Z} \in \mathbb{Z}$ کار کند. جز این، آن دو اجرا یک آزمایش‌اند که یک سیستم در آن عوض شده، و $\mathcal{Z}$ عیناً همان یک ماشین در هر دو سو است. ضعیف‌کردنش به $\forall\mathcal{A}\,\forall\mathcal{Z}\,\exists\mathcal{S}$ — شبیه‌سازی که برای محیطی که با آن روبه‌روست ساخته شده — مفهومی می‌دهد که تعریفِ 4.11 آن را نتیجه می‌دهد و عکسش را ما اثبات نمی‌کنیم، و آن دو در صورتِ عینیِ بخشِ 4.5 به‌صورتِ $\max_{\mathcal{A}} \min_{\mathcal{S}} \max_{\mathcal{Z}}$ و $\max_{\mathcal{A}} \max_{\mathcal{Z}} \min_{\mathcal{S}}$ از یک کمیت خوانده می‌شوند. نتایجِ ترکیب از این ضعیف‌سازی جان سالم می‌برند، چون هر اثباتِ زیر فرض را در محیطی می‌نشاند که نتیجه تعیینش کرده؛ پس آنچه این ترتیب می‌خرد معناست و نه ترکیب، چون تنها زیرِ آن است که $(\varphi, \mathcal{S})$ یک سیستم است که مثلِ $(\pi, \mathcal{A})$ رفتار می‌کند، هر که تماشا کند — جهانی ایده‌آل که می‌شود نامش برد و تحویلش داد.

تعریفِ 4.10 همان است که این مقایسه را معنادار می‌کند. چون $\mathcal{Z}.\mathsf{par}$ خودِ $\mathcal{Z}$ را خموش می‌کند، او هیچ هویتی از درونِ آن را نمی‌تواند ادعا کند، و سه دسته باید دور از دسترس بمانند. هویتی در $\mathsf{IDs}(\pi) \triangle \mathsf{IDs}(\varphi)$ در یک جهان اشغال است و در دیگری آزاد، پس به محیطی که اجازه یابد به‌عنوانِ آن سخن بگوید پیش از آنکه چیزی

رخ دهد گفته می‌شود در کدام جهان است، و هیچ شبیه‌سازی نمی‌تواند این را تعمیر کند. هویتی که درونیِ هر دو است چیزی را جدا نمی‌کند، اما محیطی که آن را ادعا کند به‌عنوانِ جزئی از سیستم سخن می‌گوید و نه از بیرون، و محیط برای این نیست. و هویتِ اشغال‌نشدهٔ یک نامِ به‌کاررفته به‌اندازهٔ اشغال‌شده کارساز است، چون Guard فراخواننده‌ها را با نام می‌پذیرد: فراخوانندهٔ محلی هر نمونه از نامِ والدش را می‌پذیرد و سطرِ 2 بیش از این نمی‌خواهد، پس محیطی که نمونهٔ تازه‌ای از آن نام را ادعا کند زیرروال‌های درونی را درست همان‌گونه می‌راند که فراخواننده‌هایشان — و در برابرِ زوجی مسدودشده (فصلِ 6) رو جهان‌ها را از هم تشخیص می‌دهد، چون مسدودگر فراخوانی را رد می‌کند که اشغال‌گرِ اصیل می‌پذیرد. خموش‌کردنِ بستار هر سه را یک‌جا ممنوع می‌کند. آنچه ادعاکردنی می‌ماند نام‌های تازه است، و آن‌ها تنها به چیزی می‌رسند که کلِ **Std** را می‌پذیرد: کارکردهای جهانی، که در هر مقایسه‌ای که خودشان محلِ نزاع نباشند بینِ دو جهان مشترکند.

خموش‌سازی حکم می‌کند که $\mathcal{Z}$ چه چیزی را می‌تواند ادعا کند، و فراخواندن کانالِ دومی است که به آن دست نمی‌زند. هویتی از $\mathsf{IDs}(\pi)\triangle\mathsf{IDs}(\varphi)$ در هر دو جهان می‌تواند فرا خوانده شود: آنجا که هیچ‌چیز اشغالش نکرده پاسخ rej است (بخشِ 3.6)، و آنجا که چیزی اشغالش کرده پاسخ باز هم rej است — به شرطی که آن چیز **Z** را در سطرِ 2 رد کند. این شرط، شرطی است بر سیستم‌هایی که مقایسه می‌شوند و نه بر ردهٔ محیط، و زوجی که آن را بشکند به‌سادگی تقلیدشدنی نیست، چون هیچ شبیه‌سازی نمی‌تواند پاسخی از خود بسازد که جهانِ ایده‌آل ماشینی برای دادنش ندارد. همین است که چرا زیرروال‌های محلیِ فصلِ 20 Z را از **N** خود بیرون نگه می‌دارند.

چون مجموعه‌های کوچک‌تر کم‌محدودکننده‌ترند، کم‌محدودترین محیط‌های مجاز آن‌هایی هستند که parشان درست همان بستار است — و مسدودسازیِ فصلِ 6 آن‌ها را برای زوجی سازگار تحویل می‌دهد:

$$\mathsf{IDs}(\overline{\pi}) \;=\; \mathsf{IDs}(\overline{\varphi}) \;=\; \mathsf{IDs}(\pi)\cup\mathsf{IDs}(\varphi)$$

بنا بر ادعای دومِ قضیهٔ 6.3، و مسدودگر نامش را از عضوی که به‌جایش می‌ایستد رونویسی می‌کند، پس

$$\langle\overline{\pi}\rangle \;=\; \langle\overline{\varphi}\rangle \;=\; \langle\pi\rangle\cup\langle\varphi\rangle.$$

پس $\mathcal{Z}.\mathsf{par} := \langle\overline{\pi}\rangle$ تنها مقداری نیست که آن دو سیستمِ مسدودشده می‌توانند مشترک داشته باشند، بلکه ضعیف‌ترین محدودیتی است که آن زوج می‌پذیرد.[1]

یک ویژگیِ این رابطه از پیش در دست است، و از گونه‌ای دیگر از هر چیزِ پس از آن است. نتایجِ ترکیبِ زیر — جانشانی، ترکیبِ موازی، تمامیتِ حریفِ بدل — نتیجه‌های یک لمِ جذبِ یگانه‌اند، که هر یک یک اجرا را بازگروه‌بندی می‌کند و همان یک مزیت را دو بار می‌خواند. ترایایی هیچ بازگروه‌بندی نمی‌کند. اثباتش نامساویِ مثلثی روی تعریفِ 4.5

[1] سورزدن روی همهٔ محیط‌ها بی‌اعتنا به par سازگار است اما برای به‌کارآمدن زیادی قوی است: شبیه‌ساز ناچار بود ترافیکِ درونیِ φ را فراخوان‌به‌فراخوان بازتولید کند. اگر می‌گذاشتیم شبیه‌ساز در هویت‌هایی که φ اشغال نمی‌کند کارکرد بزایاند، این شرط شل می‌شد و تنها $\mathsf{IDs}(\varphi)\setminus\mathsf{IDs}(\pi)$ برای دریغ‌کردن می‌ماند — که برای جفت‌شدنِ معمول تهی است؛ و اینکه زایاندن در ترکیب چه هزینه‌ای دارد را به کارِ آینده وامی‌گذاریم.

است، به همراهِ این آزمون که ردهٔ نتیجه برای زوجِ بیرونی مجاز است، پس به متر تعلق دارد و نه به ماشین‌آلات، و به هیچ‌یک از دستگاهِ باقیِ این بخش نیازی ندارد. به همین سبب همین‌جا بیان می‌شود.

گزاره 4.12 (ترایایی). بگذارید π، φ و ψ سیستم باشند و $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi} \cap \mathbb{Z}_{\varphi,\psi}$. اگر π کارکردِ φ را در ε_1 و در برابرِ $(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ تقلید-UC کند و φ کارکردِ ψ را در ε_2 و در برابرِ $(\mathbb{S}, \mathbb{S}', \mathbb{Z})$، آن‌گاه π کارکردِ ψ را در $\varepsilon_1 + \varepsilon_2$ و در برابرِ $(\mathbb{A}, \mathbb{S}', \mathbb{Z})$ تقلید-UC می‌کند.

اثبات. اول، نتیجه حکمِ مجازی از تقلید است. $\mathcal{Z}$ای که در هر دو رده باشد $\mathcal{Z}.\mathsf{par} \supseteq \langle\pi\rangle \cup \langle\varphi\rangle$ و $\mathcal{Z}.\mathsf{par} \supseteq \langle\varphi\rangle \cup \langle\psi\rangle$ دارد، پس $\mathcal{Z}.\mathsf{par} \supseteq \langle\pi\rangle \cup \langle\psi\rangle$؛ و در هیچ هویتی از $\mathsf{IDs}(\pi) \cup \mathsf{IDs}(\varphi)$ و نه از $\mathsf{IDs}(\varphi) \cup \mathsf{IDs}(\psi)$ میهمان نمی‌کند، پس در هیچ هویتی از $\mathsf{IDs}(\pi) \cup \mathsf{IDs}(\psi)$ هم نه — پس هر دو بندِ تعریفِ 4.10 برقرارند و $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\psi}$؛ و برای این، آن سه سیستم لازم نیست نسبتی با هم داشته باشند. اکنون بگذارید $\mathcal{A} \in \mathbb{A}$. تقلیدِ اول $\mathcal{S} \in \mathbb{S}$ را فراهم می‌کند، و چون تقلیدِ دوم حریف‌هایش را روی $\mathbb{S}$ سور می‌زند، خواندنِ $\mathcal{S}$ به‌عنوانِ حریفی در برابرِ φ به $\mathcal{S}' \in \mathbb{S}'$ می‌رسد. برای هر $\mathcal{Z} \in \mathbb{Z}$، نامساویِ مثلثی می‌دهد

$$\begin{aligned} \mathsf{Adv}^{\mathsf{uc}}_{\pi,\psi}(\mathcal{Z}, \mathcal{A}, \mathcal{S}') &\leqslant \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) + \mathsf{Adv}^{\mathsf{uc}}_{\varphi,\psi}(\mathcal{Z}, \mathcal{S}, \mathcal{S}') \\ &\leqslant \varepsilon_1 + \varepsilon_2, \end{aligned}$$

چون جملهٔ میانیِ $\Pr[\mathsf{Exec}(\varphi, \mathcal{Z}, \mathcal{S}, \mathcal{C}) = 1]$ حذف می‌شود. و چون $\mathcal{S}'$ تنها به $\mathcal{A}$ وابسته است، همان شبیه‌سازِ لازم است. □

سه ساختِ زیر — محیطِ جذب‌شدهٔ اینجا، و آن دوی بخشِ 4.4 — چند ماشین را در یک اشغال‌گرِ اجرا تا می‌کنند. آن‌ها یک شکلِ مشترک دارند، که یک بار نامش می‌بریم.

تعریف 4.13 (دسته). یک دسته چند ماشین را با هم به‌عنوانِ یک اشغال‌گرِ یگانهٔ اجرا می‌دواند، و ژتون را بینشان همان‌گونه که Exec می‌کند دست‌به‌دست می‌کند و هر یک را همان‌گونه که Exec می‌کرد مقدارِ آغازین می‌دهد. دسته مجموعهٔ اعلام‌شده‌ای از هویت‌ها را اشغال می‌کند، که هویت‌های خودِ اعضا باید بپوشانندش، و خروجی و پارامترهایش را از عضوی معین می‌گیرد. فراخوانِ میانِ دو ماشینِ میهمان در درون خدمت می‌شود، با مسیریابیِ خودِ دسته و بی‌آنکه روی اجرا گذاشته شود؛ فراخوانی که از بیرون می‌رسد به‌دستِ عضوی که هویتِ هدف را حمل می‌کند خدمت می‌شود، از راهِ واسطِ کاملِ همان عضو، با نگهبان و همه چیز؛ و فراخوانی که ماشینِ میهمان به بیرون می‌گذارد، هرگاه هدفش در مجموعهٔ اعلام‌شدهٔ بیرونی باشد روی اجرا گذاشته می‌شود و در غیرِ این صورت rej پاسخ می‌گیرد. یک نمونه می‌تواند علاوه بر این، فراخوان‌های خاصی را میانِ اعضایش به‌دست مسیریابی کند — یکی را پیش از دیگری بنشاند، حتی وقتی هدف

بیرونِ هویت‌های هر عضوی باشد — و چنین مسیریابیِ صریحی بر فراخوان‌هایی که نامشان می‌برد حکم می‌کند، و تنها باقی را به قاعدهٔ هدفِ بالا وامی‌گذارد. اینکه فراخوانِ بیرونیِ ماشینِ میهمان زیرِ چه ادعایی می‌رود بر اشغال می‌گردد، چون خموش‌کننده هستهٔ یک واسط را می‌پوشاند و فراتر نمی‌رود (فصلِ 2). عضوی که دسته هویت‌هایش را اشغال می‌کند فراخوان‌هایش را زیرِ هویتِ خودش می‌فرستد و تنها با خموش‌کنندهٔ خودش روبه‌رو می‌شود؛ عضوی که دسته هویت‌هایش را اشغال نمی‌کند هیچ هویتی برای ادعا ندارد، پس دسته هر یک از فراخوان‌های بیرونی‌اش را بازنویسی می‌کند تا زیرِ هویتی برود که خودش داردش — و اینکه کدام هویت، بخشی از همان چیزی است که نمونه اعلامش می‌کند، در کنارِ مجموعهٔ بیرونی‌اش — و ادعای بازنویسی‌شده با خموش‌کنندهٔ آن واسط روبه‌رو می‌شود. اینکه عضوی در کدام حالت است، همان است که دسته را به اجرایی که به‌جایش می‌ایستد وفادار نگه می‌دارد.

اینکه ترافیکِ درونی خدمت می‌شود و نه گذاشته، همان است که می‌گذارد دسته جایگاهِ حریفِ سیستمی را اشغال کند که هسته‌هایش او را پاسخگویانه فرا می‌خوانند، و ارزشِ گفتن دارد که چرا این تمایز مویِ دماغ نیست. پوشانهٔ Respond (تعریفِ 9.1) هر فراخوانی را که پاسخ‌دهنده می‌گذارد رد می‌کند، پس دسته‌ای که ناچار بود فراخوان‌های درونی‌اش را بگذارد در حالِ پاسخ‌دادن به هیچ‌یک از اعضای خودش نمی‌رسید — و شبیه‌سازِ فصلِ 20، که در درونِ هر یک از دو پاسخِ پاسخگو سه دورگهٔ میهمان را می‌راند، اصلاً نمی‌توانست بدود. خدمت‌کردنِ آن‌ها در درون، ژتون را همان جا که بود می‌گذارد، و این کلِ چیزی است که پاسخگویی می‌خواهد. آنچه دسته در درونِ یک پاسخِ پاسخگو باز هم نمی‌تواند بکند، رسیدن به چیزی است که میهمانش نکرده: دفتر و کارکردهای مشترک بیرونِ اویند، پس فراخوان‌هایشان گذاشته می‌شود، و رد می‌شود.

تعریفِ 4.13 پیش‌تر می‌گوید ماشینِ میهمان دو گونه دارد، بر حسبِ اینکه دسته هویت‌هایش را اشغال می‌کند یا نه، و همین تفاوت کلِ آنچه در پی می‌آید است. ماشینی که دسته هویت‌هایش را نگه می‌دارد از توزیعِ اجرا محو می‌شود و در درون خدمت می‌شود؛ و ماشینی که دسته هویت‌هایش را واگذار می‌کند باید آن‌ها را به کسی بسپارد، و فراخوان‌هایش را باید همان کس به‌نیابتِ او بگذارد. هر دو حرکت پیش می‌آید — اولی پروتکلی را جذب می‌کند و دومی حریفی را — پس با هم تعریفشان می‌کنیم و یک بار درستی‌شان را اثبات می‌کنیم.

تعریف 4.14 (جذب). بگذارید $\mathcal{Z}$ محیطی باشد و H مجموعهٔ متناهی‌ای از ماشین‌های یک اجرا، که به $\mathsf{H} = \mathsf{H}_{\mathsf{keep}} \uplus \mathsf{H}_{\mathsf{cede}}$ شقّه می‌شود و اعضای $\mathsf{H}_{\mathsf{cede}}$ اشغال‌گرانِ جایگاهِ حریف‌اند. جذبِ H در $\mathcal{Z}$ همان دستهٔ (تعریفِ 4.13) $\mathcal{Z}^{\mathsf{H}}$ از $\mathcal{Z}$ با یک رونوشت از هر عضوِ H است؛ این دسته $\mathsf{IDs}(\mathcal{Z}) \cup \mathsf{IDs}(\mathsf{H}_{\mathsf{keep}})$ را اشغال می‌کند، خروجی و par را از $\mathcal{Z}$ می‌گیرد، و مجموعهٔ بیرونی‌اش همان هویت‌هایی است که اجرای پیرامون هنوز خدمتشان می‌کند. عضوی از $\mathsf{H}_{\mathsf{keep}}$ هویت‌هایش را

نگه می‌دارد، پس فراخوان‌هایش زیرِ ادعای خودش می‌روند و فراخوان‌هایی که به او خطاب شوند در درونِ دسته خدمت می‌شوند. عضوی از H_{cede} هیچ هویتی نگه نمی‌دارد، پس دسته هر فراخوانِ بیرونیِ $id''.\text{FullOp}(x)$ را که او در حالِ دویدن در (A, P) می‌گذارد به این دستور بازنویسی می‌کند

$$(A, P).\text{FullA}\big(id''.\text{Op},\ x\big)\ \text{as}\ (Z, P),$$

و هویت‌هایی که او واگذار می‌کند در اجرای تازه به‌دستِ یک بازپخش برای او اشغال می‌شوند، که سرتاسر همان بدلِ $\mathcal{D}$ از بخشِ 4.4 است.

ما حالتِ واگذاری را تنها برای جایگاهِ حریف بیان می‌کنیم. تنها همین یکی در زیر به کار می‌رود، و تنها همین یکی است که واسطش شکلِ دستوری دارد که فراخوان‌ها را به آن بازنویسی کنیم – کارکردِ استاندارد آنچه کدِ خودش نامش می‌برد را خدمت می‌کند و نمی‌توانست به‌جای فراخوان‌های ماشینِ دیگری بایستد.

تعریف 4.15 (محیطِ جذب‌شده). بگذارید $\rho \subseteq \varphi$، و π سیستمی باشد که به‌جای φ نشانده می‌شود، و $\mathcal{Z}$ محیطی. محیطِ جذب‌شدهٔ $\mathcal{Z}^{\rho\setminus\varphi}$ همان جذبِ (تعریفِ 4.14) $\rho\setminus\varphi$ در $\mathcal{Z}$ است با هر عضو نگه‌داشته و هیچ عضو واگذارشده، پس دستهٔ $\mathcal{Z}$ است با یک رونوشت از هر $\mathcal{F} \in \rho \setminus \varphi$، که هر رونوشت هویت‌هایش را نگه می‌دارد؛ این دسته $\text{IDs}(\mathcal{Z}) \cup \text{IDs}(\rho \setminus \varphi)$ را اشغال می‌کند، خروجی و par را از $\mathcal{Z}$ می‌گیرد، و مجموعهٔ بیرونی‌اش $\text{IDs}(\pi) \cup \text{IDs}(\varphi) \cup \{id : id.F \in \{A, C\}\}$ است. این یک محیط است (تعریفِ 4.9) و نه یک کارکرد، چون روی چند شناسهٔ فرایند گسترده است. بالانویس تنها پوستهٔ میهمان‌شده را ثبت می‌کند – مجموعهٔ بیرونی π را هم می‌خواند – پس $\mathcal{Z}^{\rho\setminus\varphi}$ نسبت به جانشانی‌ای که بافت می‌دهد تعیین می‌شود، و دو جانشانی که $\rho \setminus \varphi$ مشترک دارند اما در π متفاوتند نامِ دو ماشینِ متفاوتند.

هر دو نیمهٔ مجموعهٔ بیرونی را همان چیزی وادار می‌کند که اجرای جذب‌شده باید بازتولیدش کند. اولی جایگاه است، که هویت‌هایش بر حسبِ اینکه محیطِ جذب‌شده در کدام جهان نشسته به π یا به φ می‌رسند. ما اجتماع را می‌گیریم و نه تنها $\text{IDs}(\varphi)$ را، تا همان یک ماشین در هر دو خدمت کند، و این حالا که آن دو لازم نیست همان هویت‌ها را اشغال کنند مهم است. دومی آن دو شناسهٔ رزروِ A و C است، که به‌عنوانِ شناسه نامیده شده‌اند و نه به‌عنوانِ ماشین، چون جایگاهِ حریف در سوی ایده‌آل $\mathcal{S}$ را در بر دارد و گزارهٔ 4.19 در هر دو سو یک و همان ماشینِ جذب‌شده را لازم دارد. آن‌ها آنجایند چون ماشین‌های میهمان واسط‌های کامل‌اند و نه هسته‌های خالی: هر یک دفتر را در سطرهای 2 و 6ِ بخشِ 3.5 می‌خواند، و هر یک فراخوانی در طرفِ فاسد را از راهِ Mediate به حریف می‌سپارد، که به $(A, id.P)$ قلاب می‌شود. هیچ‌یک از این دو شناسه به هیچ سیستمی تعلق ندارد. بی این نیمه، فراخوان به کارکردی میهمان در نخستین سطرش رد می‌شد، و قلابِ طرفِ فاسد به‌جای رسیدن به حریف رد می‌شد:

اجرای جذب‌شده درست همان جا که **C** به $\rho \setminus \varphi$ می‌رسید از بازگروه‌بندیِ اجرای بیرونی بودن دست می‌کشید. هر چیزِ دیگر با rej پاسخ می‌گیرد، چون در اجرای بیرونی هم چیزی خدمتش نمی‌کرد.

پارامتر دست‌نخورده منتقل می‌شود، و باید چنین باشد. آن مجموعهٔ خموش‌شدهٔ واسطِ خودِ $\mathcal{Z}$ است، پس هر تغییری در آن تغییر می‌دهد که $\mathcal{Z}$ چه فراخوان‌هایی می‌تواند بگذارد — و $\mathcal{Z}$ همان یک ماشینی است که باید در هر دو سو یکسان رفتار کند. کسی ممکن است گمان کند کارکردهای میهمان نیاز دارند $\mathsf{IDs}(\rho \setminus \varphi)$ از آن حذف شود تا بتوانند به‌عنوانِ خودشان سخن بگویند، اما ندارند: هر یک واسطی کامل است، و سطرِ 5 از پوشانهٔ خودش همان است که ادعایش را مجاز می‌کند، و پوشانهٔ $\mathcal{Z}$ فراتر از هستهٔ خودِ $\mathcal{Z}$ نمی‌رسد (تعریفِ 4.13). حذفِ آن هویت‌ها هیچ چیزِ تازه‌ای برایشان مجاز نمی‌کرد، و $\mathcal{Z}$ را درست روی همان هویت‌هایی ناخموش می‌کرد که پیش از جذب رویشان خموش بود.

پس جذب، محیطی برای زوجِ بیرونی را به محیطی برای زوجِ درونی بدل می‌کند و مجازبودن را نگه می‌دارد. دفترِ رفتارِ ویژه‌ای لازم ندارد: چون عضوِ هیچ سیستمی نیست، نه در φ است و نه در $\rho \setminus \varphi$، پس محیطِ جذب‌شده میهمانش نمی‌کند و مسدودسازی هیچ مسدودگری برایش نمی‌سازد. مثلِ $\mathcal{Z}$ و $\mathcal{A}$، Exec فراهمش می‌کند، و همان یک ماشینِ واحد در هر دو سوی هر مقایسه‌ای است. رونوشت‌های میهمانِ آن هویت‌هایشان را نگه می‌دارند، پس بنا بر تعریفِ 4.13 فراخوان‌هایشان را زیرِ ادعاهای خودشان می‌فرستند و با خموش‌کننده‌های خودشان روبه‌رو می‌شوند — نه با خموش‌کنندهٔ $\mathcal{Z}$ — و همین است که بازگروه‌بندیِ زیر لازم دارد؛ و ساخت‌های بخشِ 4.4 حالتِ مقابلند، چون حریفی را میهمان می‌کنند که دسته هویت‌هایش را اشغال نمی‌کند.

گزاره 4.16 (جذب مجازبودن را نگه می‌دارد). بگذارید ρ و π سیستم باشند، $\rho \subseteq \varphi$، و جانشانیِ φ با π در ρ خوش‌ساخت باشد. آن‌گاه

$$\mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi],\,\rho} \quad \Longrightarrow \quad \mathcal{Z}^{\rho\setminus\varphi} \in \mathbb{Z}_{\pi,\varphi}.$$

اثبات. برای نیمهٔ par، جذب این میدان را دست‌نخورده منتقل می‌کند، پس همین بس است که

$$\langle \rho[\pi/\varphi] \rangle \cup \langle \rho \rangle \;\supseteq\; \langle \pi \rangle \cup \langle \varphi \rangle.$$

هر نامی که φ به کار می‌برد ρ هم به کار می‌برد، چون $\rho \subseteq \varphi$، و هر نامی که π به کار می‌برد $\rho[\pi/\varphi]$ هم، چون $\rho[\pi/\varphi] \subseteq \pi$. برای نیمهٔ میهمانی، $\mathcal{Z}^{\rho\setminus\varphi}$ فراتر از هر چه $\mathcal{Z}$ میهمان کرده بود، در $\mathsf{IDs}(\rho\setminus\varphi)$ میهمان می‌کند: و آن دومی از $\mathsf{IDs}(\rho[\pi/\varphi]) \cup \mathsf{IDs}(\rho)$ می‌پرهیزد، که $\mathsf{IDs}(\pi) \cup \mathsf{IDs}(\varphi)$ را در بر دارد؛ و $\mathsf{IDs}(\rho \setminus \varphi)$ هیچ کجا با $\mathsf{IDs}(\varphi)$ برخورد نمی‌کند، چون آن دو زیرسیستم‌های مجزای ρاند، و هیچ کجا با $\mathsf{IDs}(\pi)$ برخورد نمی‌کند، بنا بر خوش‌ساختیِ جانشانی — تنها جایی که این فرض به کار می‌رود، و تنها برای نوع‌بندی به کار می‌رود. هیچ نسبتی میانِ $\mathsf{IDs}(\pi)$ و $\mathsf{IDs}(\varphi)$ وارد نمی‌شود. یک چیز که این حکم نمی‌گویدش: ماشینِ جذب‌شده هویت‌هایی را در درونِ par خودش

ادعا می‌کند، و هر رونوشتِ میهمان هویتِ خودش را. این تناقضی نیست — بنا بر تعریفِ 4.13 ادعاهای رونوشتِ میهمان با خموش‌کنندهٔ خودش روبه‌رو می‌شوند و هرگز با خموش‌کنندهٔ محیط، و مجازبودن شرطی است بر آن میدان، که هر جا آن میدان اِعمال شود خوانده می‌شود. □

آنچه این قضیه را کارآمد می‌کند این است که جذب یک اجرا را بازگروه‌بندی می‌کند بی‌آنکه تغییرش دهد. و چون همان تناظر به کارِ بخشِ 4.4 هم می‌آید، پیش از لم آن را می‌نشانیم.

پیکربندی‌ها. مقصودمان از پیکربندیِ یک اجرا حالتِ آن بینِ فعال‌سازی‌هاست: حالتِ هر ماشین، بخشِ ناخواندهٔ هر نوارِ سکه، اینکه کدام ماشین ژتون را در دست دارد و کجای کدِ خودش است، و فراخواننده‌هایی که معلق و چشم‌به‌راهِ پاسخند. فراخوان‌ها تودرتو می‌شوند، و بخشِ 3.6 می‌گذارد یک نمونه چند ادامهٔ معلق داشته باشد، پس آخری از این‌ها یک زنجیره است و نه یک مجموعه: یک درایه به‌ازای هر فراخوانی که گذاشته شده و هنوز پاسخ نگرفته، به همان ترتیبی که گذاشته شده، که فراخواننده و فراخوانی که چشم‌به‌راهش است را ثبت می‌کند.

آن دو اجرا همان ماشین‌ها را نمی‌دوانند — کارِ جذب همین است — اما وقتی $\mathcal{Z}^{H}$ را به‌جای یکی، همان ماشین‌هایی بشماریم که دسته‌شان می‌کند، همان‌ها را می‌دوانند، جز بازپخش‌هایی که هویتِ واگذارشده لازم دارد. تعریفِ 4.13 ژتون را بینِ آن‌ها همان‌گونه که Exec می‌کند دست‌به‌دست می‌کند، پس هر یک در هر دو سو به همان معنا آن را در دست دارد. با این شمارش، ماشین‌های آن دو اجرا جز بازپخش‌ها یک‌به‌یک متناظرند، و ما ι را برای این دوسویی می‌نویسیم: رونوشتِ میهمانِ هر عضوِ H به همان عضو، $\mathcal{Z}$ِ میهمان به $\mathcal{Z}$، و هر چه اجرا هنوز می‌دواندش، $\mathcal{C}$ هم در آن، به خودش.

تعریف 4.17 (تناظرِ بازگروه‌بندی). پیکربندیِ c از اجرای چپ و پیکربندیِ $\hat{c}$ از اجرای راست متناظرند، و می‌نویسیم $c \approx \hat{c}$، هرگاه

(C1) هر ماشینِ c و تصویرش زیرِ ι در همان حالت باشند — $\mathcal{C}$ هم در آن، پس آن دو بر **C** توافق دارند؛

(C2) هر ماشین و تصویرش همان نوارِ سکهٔ ناخوانده را داشته باشند؛

(C3) همان ماشین در هر دو، زیرِ ι، ژتون را در دست داشته باشد، در همان نقطه از همان عمل و با همان مقدارهای محلی؛ و

(C4) دو زنجیرهٔ فراخوانندهٔ معلق زیرِ ι توافق داشته باشند: طولِ برابر، و به‌ازای هر k، درایه‌های kاُم نامِ ماشین‌هایی را ببرند که ι یکی‌شان می‌کند و چشم‌به‌راهِ همان فراخوان باشند — همان هویتِ هدف، همان عمل، همان ورودی و همان هویتِ ادعایی.

هر جا خانوادهٔ راست بازپخشی حمل کند، ι دوسویی‌ای است میانِ ماشین‌های چپ و ماشین‌های راستِ جز بازپخش، و (C4) با حذفِ قاب‌های خودِ بازپخش خوانده

> می‌شود: فراخوانِ بازپخش‌شده یک قاب می‌افزاید که چپ ندارد، و دقیقاً یک بار با همان پاسخی که فراخوانِ چپ برگرداند از سر گرفته می‌شود — جز آنکه rejای که چپ می‌خواند می‌تواند در راست به‌صورتِ $\perp$ برسد، و این تنها جایی است که بی‌اعتنایی به رد خرج می‌شود.

تنها (C4) از این کمانک‌گذاری در خطر است. در چپ، هر تعلیقی را Exec ثبت می‌کند؛ در راست، تعلیق‌ها بینِ Exec و دسته تقسیم می‌شوند، و دسته آن‌هایی را ثبت می‌کند که به فراخوان‌های درونیِ خودش مربوطند. این بند می‌خواهد که آن دو زنجیره، وقتی زنجیره‌های خودِ دسته همان جا که گذاشته شده‌اند شمرده شوند، توافق داشته باشند. بازگروه‌بندی درست همین است: یک زنجیرهٔ تعلیق، که دو توزیع‌گر نگهش می‌دارند.

and correspondence، in begin executions two The **activation. first The** case lemma's the activation one the is this since why، saying worth is it and $\mathcal{C}$ initialises 1 Line machine. no by placed is it reach: not does analysis initialises 4.14 Definition sides، both on runs still execution the everything every so state، no carries relay a and would، Exec as copy hosted each tape No (C1). is which state، same the in start image its and machine so calls، no places Initialize operation The (C2). giving read، been has with and — (C4) giving empty، are chains both and suspended is nothing be cannot copies its initialises bundle the which in order the reason the it 2 Line configuration. the in nothing off read be can order that observed: ؛$\mathcal{Z}$ to goes token the left the On .(Z, Z) from (Z, Z).FullZ$()$ activates then which ،$\mathcal{Z}$ hosted the by carried is (Z, Z) where ،$\mathcal{Z}^{H}$ to goes it right the on same the at FullZ enter Both occupy. bundle the makes 4.14 Definition (C3). is which input، same the on identity

the and returned، value a or placed call a is activation every Thereafter on them of one survives correspondence that showing to comes lemma the with together halt configurations corresponding that and side، either case. by case activations those take paragraphs Its output. same

> **لم 4.18** (Absorption). Let an execution run the occupants $\mathrm{R} \cup \mathrm{H} \cup \{\mathcal{Z}\}$، with $\mathcal{C} \in \mathrm{R}$ and $\mathcal{Z}$ an environment، let $\mathrm{H} = \mathrm{H}_{\text{keep}} \uplus \mathrm{H}_{\text{cede}}$، and let $\mathcal{Z}^{\mathrm{H}}$ be the absorption of Definition 4.14. Ask of each member of H_{cede}، and of nothing in H_{keep}، that
>
> (آ) it be refusal-oblivious (Definition 4.28)؛
>
> (ب) no core of R place a responsive call on the identities it cedes (Definition 9.2)؛ and

(ج) $\mathcal{Z}$.par contain no Z.identity-

of sense the in families occupant legal being families two the Then،
،3.3 Definition

$$\mathsf{Exec}\big(\{\mathcal{Z}\} \cup \mathrm{H} \cup \mathrm{R}\big) \qquad \text{and}$$

$$\mathsf{Exec}\big(\{\mathcal{Z}^{\mathrm{H}}\} \cup \mathrm{R} \cup \{\text{identity ceded each at relay a}\}\big)$$

close. merely not distributions، as equal — distributed identically are

اثبات. The argument is one induction on activations، and the five para- graphs below discharge its step. Machines pairs the two occupant fami- lies and checks that every machine and its image start alike. Routing asks where each call lands and finds four cases — internal to the group، leaving it، entering at a kept identity، entering at a ceded one — of which only the last differs between the two families. Claims splits on which kind of mem- ber placed the call: one keeping its identities meets the same silencers on both sides، while a ceded member has none to claim، so its outward calls are rewritten، and the rewriting must survive three tests — where the bun- dle may speak from، whether the claim is silenced، and whether it passes the relay's guard. Refusals isolates the one value that can differ، a rej the left reads arriving as $\perp$ on the right، and spends hypothesis (a) on it؛ hy- pothesis (b) blocks the mirror problem، a responsive call on a relayed slot. Token checks that the two chains of suspensions agree once the bun- dle's own are counted where they were placed. Hypothesis (c) is spent in Claims، on the ceded member's right to claim (Z, P). Two conventions of Chapters 1 and 3.6 are used throughout and are worth restating. Dis- tinct machines draw from independent coins، and bundling machines into one does not merge their tapes، so a hosted copy's randomness is what it was before absorption. And Initialize places no calls، so the order in which the bundle initialises its hosted copies is unobservable. The proof is an induction on activations: the initial configurations correspond، as shown before the lemma؛ corresponding configurations produce the same next activation on both sides and step to corresponding configurations، each machine and its image reading the same coins in the same order، which is what (C2) of Definition 4.17 preserves؛ and corresponding halting con- figurations output alike، the output being the hosted $\mathcal{Z}$'s by Definition 4.14. The paragraphs below discharge the inductive step: which machines ex- ist and start alike، (C1)؛ where a call goes and what it claims، (C4)؛ and

both to common are paragraphs two first The (C3). moves، token the how
costs. member ceded a what are them after two the member؛ of kinds
together $\{\mathcal{Z}^{H}\} \cup R$ runs right the ؛$\{\mathcal{Z}\} \cup H \cup R$ runs left The Machines.
each of copy one with $\mathcal{Z}$ is $\mathcal{Z}^{H}$ and identity، ceded each at relay a with
collections two the bundles، it machines the as Counted .H of member
copy، its with H of member each pairs ι so relays، the for but coincide
is which over، relays the leaves and itself، with $\mathcal{C}$ and R of member each
sides، both on R initialises 1 Line them. gives 4.17 Definition reading the
relay a and would، Exec as copies hosted the initialises 4.14 Definition
is Occupancy alike. start image its and machine every so stateless، is
on target: unique a has call each so hypothesis، by sides both on disjoint
what hold relays the and $\mathsf{IDs}(\mathcal{Z}) \cup \mathsf{IDs}(H_{\mathsf{keep}})$ holds bundle the right the
go. let H_{cede}

in endpoints both With lands. it where ask and call a Take Routing.
the on $\mathcal{Z}^{H}$ inside served and left the on Exec by dispatched is it $\{\mathcal{Z}\} \cup H$
dispatched is it group that Leaving way. either machine same the by right،
the in lying target its right، the on outward placed and left the on Exec by
still execution the identities the makes 4.14 Definition which set، outward
Exec by dispatched is it ،H_{keep} of identity an at group the Entering serves.
identity، that carrying copy hosted the by right the on served and left the on
dispatched is it ،H_{cede} of identity an at Entering occupies. bundle the which
the is which right، the on relay the to and left the on member ceded the to
paragraphs two the of subject the and differ families two the place one
Exec side: either on machine no has it else anywhere Addressed below.
callee the at evaluated are Guards .rej answers $\mathcal{Z}^{H}$ and target no finds
do. claims the once agree they so id، claimed the from

car- call such Each identities. its keeps that member a for Claims،
could that silencers two the and sides، both on id claimed same the ries
gov- is interface hosted A both. on silencers same the are one change
:4.13 Definition by environment's، the by not and silencer own its by erned
inter- full a Being it. reach not does $\mathcal{Z}$ at Silence so identities، its keeps it
core its silencing 5 line side، either on identity own its exactly claims it face
،Mediate through and $\mathcal{C}$ on places، wrapper its calls the and ؛$\{\mathtt{id}'' \neq \mathtt{id}\}$ on
in- own environment's The sides. both on silencer that even outside stand
so unchanged، over carries absorption which ،par by governed is terface
exactly admits and identity same the at set same the from computed is $\bar{\mathcal{I}}$
claim. a loses nor gains machine Neither claims. same the

identity no has member a Such them. cedes that member a for Claims،
out- its rewrites 4.14 Definition so bundle، the inside claim to own its of

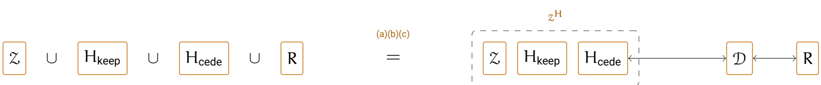
the where First، tests. three survive to has rewriting the and calls، ward
of behalf on runs member hosted the Whenever from. speak may bundle
is bundle the — (A, P) at running it has left the where is، that — P party
calling $\mathcal{Z}$ by either entered is it :$P = Z$ else or ،(Z, Z) at or (Z, P) at running
to party claimed the forcing Guard$_{\mathcal{A}}$ with internally served -identity،A an
only P party claim (Z, Q) at interface an letting 3 line and target's، the be
(Z, P) targets call inward whose relay، the by or ؛$P = Z$ or $Q \in \{P, Z\}$ when
party-tie the by not silenced: is (Z, P) claim the whether Second، exactly.
hy- which ،par by not and ،Z is name its since 2 line by not settled، just
at Guard passes it whether Third، -identities.Z of empties (c) pothesis
and ،2 line at equal parties the with $\mathbf{Z} \subseteq \mathcal{A}.\mathbf{N}$ ،1 line at $P \in \mathcal{A}.\mathbf{P}$ relay: the
under placed، left the call very the places then relay The .3 line at global
at conjuncts same the meet sides both so claimed، left the claim very the
way other the travels identity ceded the at arriving call A callee. same the
on passed it guard the under relay the reaches it tests: three same the by
admitted — payload the in caller its with (Z, P) at $\mathcal{Z}$ to handed is left، the
and — global($\mathcal{Z}$) and $\mathbf{A} \subseteq \mathcal{Z}.\mathbf{N}$ ،Guard$_{\mathcal{Z}}$ by and silencer own relay's the by
interface's that through (A, P) at interface member's hosted the reaches
left. the on passed that guard the guard،

places member ceded the call a Where differ. can value One Refusals.
relay the right the on while call، own its from rej reads left the refused، is
.$\perp$ as member hosted the to it delivers 3.6 Section and rej that returns
machine refusal-oblivious a this: closes what exactly is (a) Hypothesis
(b) Hypothesis unchanged. is distribution its so two، the to alike answers
on call responsive a problem، same the of direction other the stops what is
the and places، responder the call every refuses Respond identity: ceded a
slot relayed a on call responsive a so all، at answer to one place must relay
which — substance with answered left the where $\perp$ answered be would
visible: is does relay the else Nothing one. place may R of core no why is
claim، relay's the by met $\mathtt{id}'.F = A$ gate its detour، the survives Corrupt
the and -identity$A$ an under place must rewriting the call one the is which
cannot. bundle

is continuation its and call، a placing operation، one has Exec Token.
longer no $\mathcal{Z}^{H}$ to internal calls the right the On back. comes token the where
hosted the among token the passes 4.13 Definition but handler، that reach
own its on suspended is caller internal each so does، Exec as machines
call relayed each and answer؛ same the with resumed and continuation
elision the is which once، exactly resumed right، the on frame one adds
3.6 Section and used، are Exec of properties Two allows. 4.17 Definition

outstanding several hold may machine a instance: every to both grants
is second The suspended. while re-entered be may and continuations،
outward an on waiting itself is $\mathcal{Z}^{H}$ while needs R from arriving call a what
side. either on moment every at token the holds machine one Exactly call.
and order same the in inputs same the sees therefore machine Every
every at preserved is correspondence so tape، same the from answers
particular in — distributions as agree executions two the and activation
□ .1 returns $\mathcal{Z}$ that probability the on

side cases two its with equality one is lemma the Diagrammatically،
one ceded a outright، bundle the into disappears member kept a side: by
relay. a through out back bridged is



above figure 's4.20 Theorem in $\rho \setminus \varphi$ of all like — H_{keep} of member A
as only it reaches R so identity، its occupies $\mathcal{Z}^{H}$ outright: absorbed is —
same the on running too، hosted is H_{cede} of member A bundle. the of part
dummy (the $\mathcal{D}$ relay the identity: its occupy not does bundle the but tape،
their with tagged $\mathcal{Z}$ to calls 'sR handing place، its in stands (4.4 Section of
origi- its under calls outward member's hosted the re-emitting and caller
Hypotheses them. between sits relay a tell can side neither so claim، nal
is (a) indistinguishable: that keep what are ،H_{cede} of only asked (a)–(c)،
؛$\perp$ a into relays it rej every turns $\mathcal{D}$ since refusal-obliviousness، exactly
the claim، own 's$\mathcal{Z}$ on clash a and call responsive a out rule (c) and (b)
is figure 's4.20 Theorem show. could hop extra relay's a ways other two
the for standing occupant ceded one case، mirror the ؛$H_{cede} = \emptyset$ case the
dummy-adversary 's4.4 Section what is ،($H_{keep} = \emptyset$) slot adversary whole
on. runs construction

protocol a Absorbing apart. used are lemma the of halves two The
pure a is left is what and away fall hypotheses three all so nothing، cedes
stating worth is it and on، runs 4.20 Theorem case the is that rebracketing؛
all. at nothing costs it that emphasis the for separately

the let ،$\varphi \subseteq \rho$ let systems، be π and ρ Let .(Regrouping) **4.19 گزاره**
environment an be $\mathcal{Z}$ let and well-formed، be ρ in π by φ of replacement
and ،φ or π either σ for Then .IDs(ρ) ∪ IDs(π) of identity no at hosting

،(3.3 (Definition slot adversary the of $\mathcal{X}$ occupant any for

$$\mathsf{Exec}\big(\rho[\sigma/\varphi],\,\mathcal{Z},\,\mathcal{X},\,\mathcal{C}\big) \qquad \text{and} \qquad \mathsf{Exec}\big(\sigma,\,\mathcal{Z}^{\rho\setminus\varphi},\,\mathcal{X},\,\mathcal{C}\big)$$

close. merely not distributions، as equal — distributed identically are

اثبات. Lemma 4.18 at $H = \rho\setminus\varphi$ with $H_{\mathsf{cede}} = \emptyset$، so that $\mathcal{Z}^{H}$ is the absorbed
applies: lemma the of hypothesis no and 4.15 Definition of environment
Note none. are there and members ceded of asked are (c) and (b) (a)،
occupant two The side. ideal the is $\sigma = \varphi$ case the so ،$\rho[\varphi/\varphi] = \rho$
so cases، both in system a is collection left-hand the legal: are families
says well-formedness because $\sigma = \pi$ for — twice occupied is identity no
is collection that because $\sigma = \varphi$ for and id، process no share π and $\rho\setminus\varphi$
bundle the keeps $\mathcal{Z}$ on hypothesis hosting the right the on while — itself ρ
.IDs(σ) of clear ☐

another for subsystem one substituting that says now theorem The
advantage. in nothing costs protocol a inside

then ،ε within φ UC-emulates π if says: 4.20 Theorem Informally،
simulator and adversary The .ε same the within ρ UC-emulates $\rho[\pi/\varphi]$
are conclusion the of environments the unchanged؛ over carry classes
lies $\mathcal{Z}^{\rho\setminus\varphi}$ form absorbed whose $(\rho[\pi/\varphi],\rho)$ pair the for admissible those
hypothesis. the of class the in

قضیه 4.20 (Composition). Let ρ and π be systems، let $\varphi \subseteq \rho$، and let
the replacement of φ by π in ρ be well-formed in the sense of Defini-
tion 4.2. Suppose

$$\pi \text{ UC-emulates } \varphi \ \text{ within } \varepsilon \text{ against } (\mathbb{A},\mathbb{S},\mathbb{Z}).$$

Then

$$\rho[\pi/\varphi] \text{ UC-emulates } \rho \ \text{ within } \varepsilon \text{ against } (\mathbb{A}',\mathbb{S}',\mathbb{Z}'),$$

where

$$\mathbb{A}' = \mathbb{A}, \qquad \mathbb{S}' = \mathbb{S}, \qquad \mathbb{Z}' = \big\{\mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi],\,\rho} \;:\; \mathcal{Z}^{\rho\setminus\varphi} \in \mathbb{Z}\big\}.$$

Explicitly،

$$\forall\,\mathcal{A} \in \mathbb{A}' \;\; \exists\,\mathcal{S} \in \mathbb{S}' \;\; \forall\,\mathcal{Z} \in \mathbb{Z}' \;:$$

$$\mathsf{Adv}^{\mathsf{uc}}_{\rho[\pi/\varphi],\,\rho}(\mathcal{Z},\mathcal{A},\mathcal{S}) \;\leqslant\; \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}\big(\mathcal{Z}^{\rho\setminus\varphi},\mathcal{A},\mathcal{S}\big) \;\leqslant\; \varepsilon.$$

simulator and adversary The kinds. three of relations، Three اثبات. machine same the is adversary an copying: by over carry classes the mentions simulator a typing nothing and attacks، it system whichever computed thing one the is class environment The for. simulates it pair under class hypothesis's the of preimage the being copied، than rather once applied ،4.19 Proposition steps. two then is bound The absorption. the — exactly one inner the into advantage outer the turns side، each on outward its what is which sides، both serving machine absorbed same in- the bounds hypothesis emulation the and — allow to chosen was set an as shown equality، an is display the of relation first the So one. ner of are relations three The second. the with uniformity for only inequality reduction the because unchanged over carry adversaries The kinds. three as read is $\rho[\pi/\varphi]$ attacking $\mathcal{A} \in \mathbb{A}'$ an machine: same very the on hands largest the is equality and serve، would $\mathbb{A}' \subseteq \mathbb{A}$ Any .π attacking $\mathcal{A} \in \mathbb{A}$ an $\mathcal{S} \in \mathbb{S}$ the reason: mirror the for over carry simulators The choice. such conclusion، the for simulator the as returned is supplies hypothesis the be may it That smallest. the is equality and serve، would $\mathbb{S}' \supseteq \mathbb{S}$ any so pair the mentions simulator a typing nothing :4.8 Remark is all at returned .$(\rho[\pi/\varphi], \rho)$ for slot the in stands already (π, φ) for one so for، simulates it rather computed is something place one the are environments The systems، of pairs different two for classes are $\mathbb{Z}'$ and $\mathbb{Z}$ Here copied. than and $\mathbb{Z}'$ is the preimage of $\mathbb{Z}$ under absorption،

$$\mathbb{Z}' \;=\; \left((\cdot)^{\rho\setminus\varphi}\right)^{-1}(\mathbb{Z}) \;\cap\; \mathbb{Z}_{\rho[\pi/\varphi],\,\rho},$$

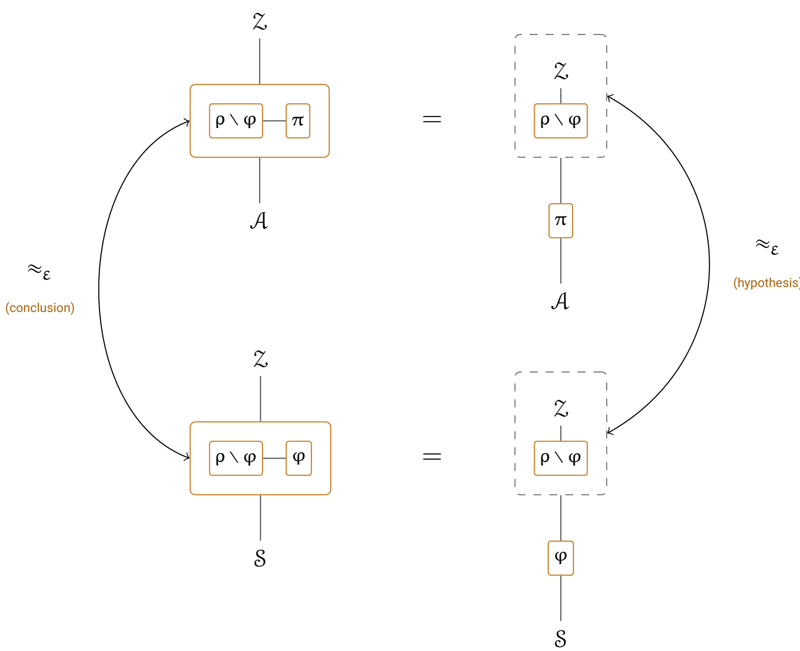
absorp- on becomes it what when exactly counts environment outer an so $\mathbb{Z}' \subseteq \mathbb{Z}_{\rho[\pi/\varphi],\rho}$ since class، genuine a is It covers. hypothesis the one is tion Proposi- ،$\mathbb{Z}_{\pi,\varphi}$ class admissible largest the is $\mathbb{Z}$ When construction. by all is preimage the and redundant condition second the makes 4.16 tion class largest the gives class largest the against emulation :$\mathbb{Z}_{\rho[\pi/\varphi],\rho}$ of again.

need- defined، advantage right-hand the makes also 4.16 Proposition between relation no and environments the on hypothesis closure no ing twice، applied 4.19 Proposition is inequality first The .IDs(φ) and IDs(π) ideal the on $\mathcal{X} = \mathcal{S}$ and $\sigma = \varphi$ and side real the on $\mathcal{X} = \mathcal{A}$ and $\sigma = \pi$ with Subtracting، one.

$$\mathsf{Adv}^{\mathsf{uc}}_{\rho[\pi/\varphi],\,\rho}(\mathcal{Z}, \mathcal{A}, \mathcal{S}) \;=\; \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}\left(\mathcal{Z}^{\rho\setminus\varphi}, \mathcal{A}, \mathcal{S}\right),$$

brack- experiment one are experiments two the equality: an fact in is it so ideal the because sides both on serves $\mathcal{S}$ same the and differently، eted inequal- second The absorption. same very the by regrouped is execution $\mathcal{Z}$ because $\mathbb{Z}$ in lies which ،$\mathcal{Z}^{\rho\setminus\varphi}$ to applied hypothesis emulation the is ity for only inequality an as relation first the displays theorem The .$\mathbb{Z}'$ in lies □ second. the with uniformity

،$\mathcal{Z} \in \mathbb{Z}'$ one and $\mathcal{S} \in \mathbb{S}'$ serving a ،$\mathcal{A} \in \mathbb{A}'$ one fixing Diagrammatically، string-diagram the in drawn below، square commuting the is proof the box the and interface، an is wire a system، a is box solid a :[9] of style occu- currently machine whichever is system a of bottom the off hanging slot. adversarial its pies



one is column each down Reading square. commuting a as Substitution، :1 شکل edges horizontal Both exchanged. being pair the marks box dotted the execution: the is edge right the and bounds، than rather equalities are both so absorption، are with one merely not quantity، same the then is left the on conclusion The hypothesis. (.4.20 (Theorem bound. same the

and $(\pi, \mathcal{A})$ to applied ،4.19 Proposition are edges bottom and top The mere not equalities، are both so free، is regrouping — respectively $(\varphi, \mathcal{S})$ to environ- absorbed (the right the on box dashed the why is which bounds، a between changes that thing only the is (4.19 Proposition of $\mathcal{Z}^{\rho\setminus\varphi}$ ment hypothesis، theorem's the is edge right The neighbour. its and column

con- theorem's the — edge left The environment. absorbed that to applied
not are 's$\approx_{\varepsilon}$ two the shows square the chasing forced: then is — clusion
the are equalities، two the by but، ε same the by bounded both merely
at hybrid one chain، a in squares such n Composing quantity. same very
below. 4.24 Theorem exactly is time، a

com- which in form the into combine now transitivity and Substitution
metric the with stated was 4.12 Proposition used. actually is position
does it where is here machinery؛ this of none needing proof its above،
work. its

sys- be ψ and ρ ،π Let .specification) a with (Composition **4.21 نتیجه**
and well-formed، be ρ in π by φ of replacement the let ،$\varphi \subseteq \rho$ let tems،
class the for $\mathbb{Z}'$ Write .$(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ against ε_1 within φ UC-emulate π let
then ،$(\mathbb{S}, \mathbb{S}', \mathbb{Z}')$ against ε_2 within ψ UC-emulates ρ If .4.20 Theorem of
.$(\mathbb{A}, \mathbb{S}', \mathbb{Z}')$ against $\varepsilon_1 + \varepsilon_2$ within ψ UC-emulates $\rho[\pi/\varphi]$

against ε_1 within ρ UC-emulates $\rho[\pi/\varphi]$ ،4.20 Theorem By .اثبات
statement a is hypothesis second The .$\mathbb{Z}' \subseteq \mathbb{Z}_{\rho[\pi/\varphi],\rho}$ and ،$(\mathbb{A}, \mathbb{S}, \mathbb{Z}')$
con- Both well. as $\mathbb{Z}' \subseteq \mathbb{Z}_{\rho,\psi}$ so class، same that against emulation of
the to applies it and hold، therefore 4.12 Proposition by needed tainments
middle. the in ρ with statements two □

with ρ analyzes One used. is composition which in form the is This
spec- whatever emulates it proving place، in φ subroutine idealized the
whose names، corollary the classes the against — wanted is ψ ification
con- that transfers corollary the simulators؛ step's first the are adversaries
emula- miss: to easy is hypothesis One runs. that system the to clusion
$\mathbb{Z}' \subseteq \mathbb{Z}_{\rho,\psi}$ that condition، side 's4.11 Definition as asserts، $\mathbb{Z}'$ against tion
class the environment every of par the under lie must names 'sψ so —
not. do they where vacuous is corollary the and retains،

4.3 ترکیبِ موازی

not is once at several Replacing subsystem. one replaces 4.20 Theorem
hybrids؛ of chain a along one that of applications n but argument new a
chain. the down classes three the of bookkeeping the is care needs what
satisfi- are and literally، meant are below hypotheses disjointness The
machine a is register the :1 Chapter of convention the of because only able
one ever only is there and system، no of member a and execution the of

once، at π_k every and φ_i every in lie would it member، a Counted it. of carry would π_k two no and disjoint be would ρ of subsystems two no so and $n \geqslant 2$ for unsatisfiable be would hypotheses the ids؛ process disjoint systems the of out register the Keeping .$\mathcal{C}$ modulo read be to have would written. as read be them lets what is

تعریف 4.22 let and system a be ρ Let .replacement) (Simultaneous systems For subsystems. disjoint pairwise be $\varphi_1, \dots, \varphi_n \subseteq \rho$ ،$\pi_1, \dots, \pi_n$

$$\rho[\pi_1/\varphi_1, \dots, \pi_n/\varphi_n] := \left(\rho \setminus \bigcup_{i=1}^{n} \varphi_i\right) \cup \bigcup_{i=1}^{n} \pi_i.$$

and $\rho_0 = \rho$ that so ،$0 \leqslant k \leqslant n$ for $\rho_k := \rho[\pi_1/\varphi_1, \dots, \pi_k/\varphi_k]$ Write first the by untouched is φ_k disjoint، pairwise are φ_i the Since .$\rho_n = \rho'$ and $\varphi_k \subseteq \rho_{k-1}$ so replacements، $k-1$

$$\rho_k = \rho_{k-1}[\pi_k/\varphi_k].$$

lets which replacement، single a by differ therefore hybrids Consecutive n as chain the walks below proof the step؛ each on act 4.20 Theorem needs also together steps the Fitting replacement. per one hops، game are. they and adjustable، be to classes the

لم 4.23 against ε within φ UC-emulates π Suppose .(Monotonicity) any ،$\mathbb{A}^- \subseteq \mathbb{A}$ any for $(\mathbb{A}^-, \mathbb{S}^+, \mathbb{Z}^-)$ against so does it Then .$(\mathbb{A}, \mathbb{S}, \mathbb{Z})$.$\mathbb{Z}^- \subseteq \mathbb{Z}$ any and $\mathbb{S}^+ \supseteq \mathbb{S}$ simulators of class

اثبات. Narrow- classes. three the over statement $\forall\exists\forall$ a is 4.11 Definition exis- the of that widening and quantifier universal either of range the ing legal a result the keeps $\mathbb{Z}^- \subseteq \mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi}$ and it، preserve each one tential emulation. of statement $\square$

sub- disjoint pairwise n replacing that says 4.24 Theorem Informally، 4.20 Theorem is it errors: individual the of sum the costs once at systems of those being adversaries the hybrids، of chain a along times n applied environ- the and first، the of those simulators the replacement، last the once. at step every for admissible ments

قضیه 4.24 $\varphi_1, \dots, \varphi_n \subseteq \rho$ let system، a be ρ Let .composition) (Parallel
π_k the that such systems be $\pi_1, \dots, \pi_n$ let and disjoint، pairwise be
pro- no shares π_k ،k each for and، ids process disjoint pairwise carry
k each for that Suppose .$\rho \setminus \varphi_k$ with id cess

$$\pi_k \text{ UC-emulates } \varphi_k \text{ within } \varepsilon_k \text{ against } (\mathbb{A}_k, \mathbb{S}_k, \mathbb{Z}_k),$$

step، -th$k$ the to attaches 4.20 Theorem class the for $\mathbb{Z}'_k$ write and

$$\mathbb{Z}'_k = \left\{ \mathcal{Z} \in \mathbb{Z}_{\rho_k, \rho_{k-1}} : \mathcal{Z}^{\rho_{k-1} \setminus \varphi_k} \in \mathbb{Z}_k \right\}.$$

that finally Assume

$$\mathbb{S}_{k+1} \subseteq \mathbb{A}_k \quad (1 \leqslant k < n), \qquad \mathbb{Z}^\star \subseteq \bigcap_{k=1}^{n} \mathbb{Z}'_k.$$

and ،$\mathbb{Z}^\star \subseteq \mathbb{Z}_{\rho', \rho}$ Then

$$\rho' := \rho[\pi_1/\varphi_1, \dots, \pi_n/\varphi_n] \text{ UC-emulates } \rho$$

$$\text{within } \sum_{k=1}^{n} \varepsilon_k \text{ against } (\mathbb{A}_n, \mathbb{S}_1, \mathbb{Z}^\star).$$

اثبات. A hybrid argument over n steps، and the work is in checking that
each step is a legal instance of Theorem 4.20 before any of them is taken.
Three preliminaries do that. The first shows each replacement is well-
formed and each ρ_k a system، by induction on k، so that the composition
theorem applies at every step. The second shows the class $\mathbb{Z}^\star$ is ad-
missible for every intermediate pair at once، and — taking the union over
k — for the outer pair too، so that the conclusion is itself a legal state-
ment of emulation. The third builds the chain of simulators downwards،
$\mathcal{S}_{n+1} := \mathcal{A}$ and each $\mathcal{S}_k$ supplied by the k-th step for the adversary $\mathcal{S}_{k+1}$،
which is what makes $\mathcal{S}_1$ depend on $\mathcal{A}$ alone. Only then is the hybrid taken:
consecutive games differ in one replaced subsystem and one adversary،
that difference is one application of Theorem 4.20، and summing the n
hops telescopes to $\sum_k \varepsilon_k$. First، each step is well-formed. The members
of $\rho_{k-1} \setminus \varphi_k$ are those of $\rho \setminus \varphi_k$ the first $k-1$ replacements leave in place،
together with $\pi_1, \dots, \pi_{k-1}$; by hypothesis π_k shares a process id with nei-
ther group. So the replacement of φ_k by π_k in ρ_{k-1} is well-formed in the
sense of Definition 4.2. Each ρ_k is a system، by induction on k: $\rho_0 = \rho$

well- that from 4.3 Proposition by one is $\rho_k = \rho_{k-1}[\pi_k/\varphi_k]$ and one، is ask which ،4.16 Proposition and 4.20 Theorem So replacement. formed step. -th$k$ the to apply systems، be to π_k and ρ_{k-1}
$\mathbb{Z}^\star \subseteq \mathbb{Z}'_k \subseteq \mathbb{Z}_{\rho_k,\rho_{k-1}}$ From .k fix and $\mathcal{Z} \in \mathbb{Z}^\star$ Let admissibility. Next، get we

$$\mathcal{Z}.\mathsf{par} \supseteq \langle \rho_k \rangle \cup \langle \rho_{k-1} \rangle,$$

$\mathsf{Adv}^{\mathsf{uc}}_{\rho_k,\rho_{k-1}}$ advantage the and hybrids of pair -th$k$ the for admissible is $\mathcal{Z}$ so by φ_k of replacement the to applied ،4.16 Proposition defined. is below parameter its :$\mathbb{Z}_{\pi_k,\varphi_k}$ in environment absorbed the places then ،ρ_{k-1} in π_k in- the so ،φ_k of and π_k of identities the contains already which ’s،$\mathcal{Z}$ is displayed the of union the Taking too. defined is $\mathsf{Adv}^{\mathsf{uc}}_{\pi_k,\varphi_k}$ advantage ner ،k all over containments

$$\mathcal{Z}.\mathsf{par} \supseteq \bigcup_{k=0}^{n} \langle \rho_k \rangle \supseteq \langle \rho' \rangle \cup \langle \rho \rangle,$$

the — $\bigcup_k \mathsf{IDs}(\rho_k) \supseteq \mathsf{IDs}(\rho') \cup \mathsf{IDs}(\rho)$ of identity no at hosts $\mathcal{Z}$ likewise and of clauses both so — $k = 1$ and $k = n$ at ،$\mathbb{Z}_{\rho_k,\rho_{k-1}}$ each of clause hosting state- legal a itself is conclusion the and $\mathbb{Z}^\star \subseteq \mathbb{Z}_{\rho',\rho}$ give 4.10 Definition emulation. of ment
$\mathcal{S}_k \in \mathbb{S}_k$ let $k = n, n-1, \ldots, 1$ For .$\mathcal{S}_{n+1} := \mathcal{A}$ put and $\mathcal{A} \in \mathbb{A}_n$ Fix is This .$\mathcal{S}_{k+1}$ adversary the for supplies step -th$k$ the simulator the be $k < n$ at and ،$\mathcal{S}_{n+1} = \mathcal{A} \in \mathbb{A}_n$ because $k = n$ at index: every at legitimate downwards، built is simulators of chain The .$\mathcal{S}_{k+1} \in \mathbb{S}_{k+1} \subseteq \mathbb{A}_k$ because alone. $\mathcal{A}$ on depends $\mathcal{S}_1$ and it، before one the from each games $n+1$ the consider and $\mathcal{Z} \in \mathbb{Z}^\star$ fix Now

$$G_k := \mathsf{Exec}\big(\rho_k, \mathcal{Z}, \mathcal{S}_{k+1}, \mathcal{C}\big), \qquad p_k := \Pr[\, G_k = 1 \,], \qquad 0 \leqslant k \leqslant n.$$

the so ،$\mathcal{A}$ under ρ' of that G_n and $\mathcal{S}_1$ under ρ of execution the is G_0 Here one in differ games Consecutive .ρ_n at ends and ρ_0 at starts sequence that and adversary، the playing machine the in and subsystem replaced ،$1 \leqslant k \leqslant n$ for :4.20 Theorem of application one exactly is difference

$$\begin{aligned} |\, p_k - p_{k-1} \,| &= \mathsf{Adv}^{\mathsf{uc}}_{\rho_k,\, \rho_{k-1}}\big(\mathcal{Z}, \mathcal{S}_{k+1}, \mathcal{S}_k\big) \\ &\leqslant \mathsf{Adv}^{\mathsf{uc}}_{\pi_k,\varphi_k}\big(\mathcal{Z}^{\rho_{k-1} \setminus \varphi_k}, \mathcal{S}_{k+1}, \mathcal{S}_k\big) \leqslant \varepsilon_k. \end{aligned}$$

the are G_{k-1} and G_k since backwards، read 4.5 Definition is equality The inequal- first The compares. $\mathsf{Adv}^{\mathsf{uc}}_{\rho_k,\rho_{k-1}}$ advantage the that executions two by $\mathcal{Z} \in \mathbb{Z}^\star \subseteq \mathbb{Z}'_k$ because available ،4.20 Theorem of reduction the is ity

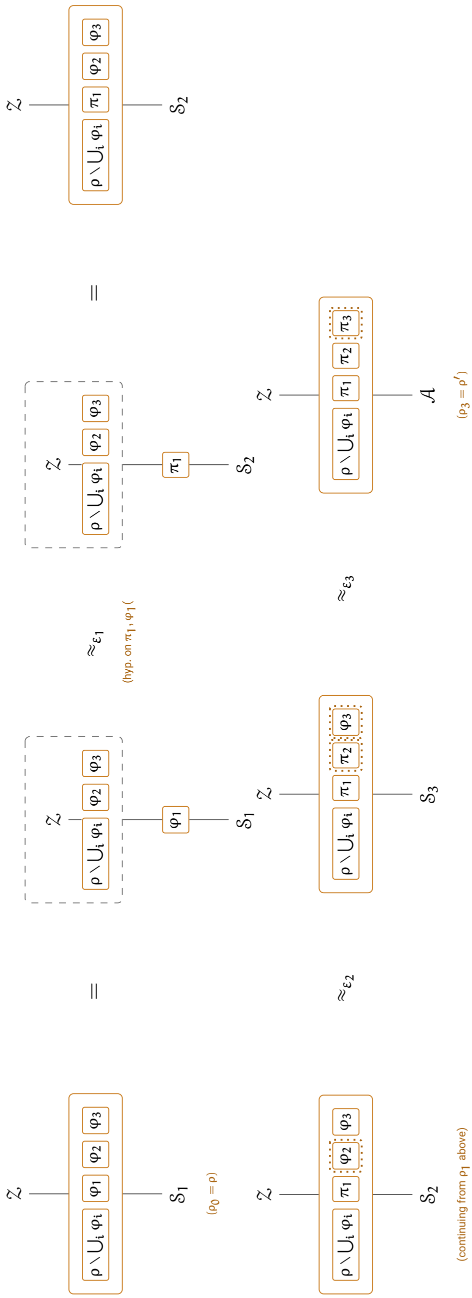
.$\mathbb{Z}^{\star}$ against statement a step -th$k$ the makes 4.23 Lemma so hypothesis، $\mathcal{S}_{k+1}$ adversary the to applied ،π_k by φ_k of emulation the is second The environment. absorbed the and

telescoping، and hops n the Summing

$$\mathsf{Adv}^{\mathsf{uc}}_{\rho',\rho}\big(\mathcal{Z},\mathcal{A},\mathcal{S}_1\big) = |p_n - p_0| \leqslant \sum_{k=1}^{n} |p_k - p_{k-1}| \leqslant \sum_{k=1}^{n} \varepsilon_k.$$

is this arbitrary، was $\mathcal{Z} \in \mathbb{Z}^{\star}$ and alone $\mathcal{A}$ from chosen was $\mathcal{S}_1 \in \mathbb{S}_1$ As theorem. the of statement the □

figure (the square own 's4.20 Theorem is k hop Diagrammatically، the and ،$\mathcal{S} \mapsto \mathcal{S}_k$ ،$\mathcal{A} \mapsto \mathcal{S}_{k+1}$ ،$\pi \mapsto \pi_k$ ،$\varphi \mapsto \varphi_k$ with proof) its above One ride. the for along slot untouched every carry to widened $\rho \setminus \varphi$ shell where from compactly continuing $2, 3$ hops and full in drawn 1 hop picture، :$\rho_3 = \rho'$ to $\rho_0 = \rho$ ends، it


(hyp. on π1, φ1)
(ρ0 = ρ)
(continuing from ρ1 above)
(ρ3 = ρ′)

of remainder true the — absorbed is 's)φ (all ρ_0 right: to left Reading
φ_1 only leaving it، with along $\mathcal{Z}$ into folding φ_2, φ_3 untouched the and ρ
handing ؛ε_1 within π_1 for it swaps hypothesis 's4.20 Theorem exposed.
match. to $\mathcal{S}_2$ to $\mathcal{S}_1$ from advanced $\mathcal{S}$ ،ρ_1 reassembles out back fold the
ends flat their to compressed only square، identical the are 3 and 2 Hops
dotted the with — add than rather repeat would more twice it redrawing —
and flip) to (about φ_k is slot which exactly hop، each at marking، boxes
last the in as telescoping، and hops three the Summing flipped). (just π_k
appear which of none 's،ε_k separate three turns above، proof the of lines
.$\rho \approx_{\varepsilon_1+\varepsilon_2+\varepsilon_3} \rho'$ bound one the into statement، own theorem's the in

it as chain the along behaves Corruption **chain. the along Corruption**
under- machine one is register the game a Within step. single a at does
corrupt cannot hop a so party، by keyed alike، φ_k and π_k ،$\rho \setminus \varphi_k$ neath
and instances separate are registers the games across halfway؛ party a
parties same the corrupts environment The hops. between nothing carry
؛3 line under calls same the placing machine same the being hop، every at
the — hop per change that machines two the so to، have not does slot the
corrupt not need — $\mathcal{S}_k \to \mathcal{S}_{k+1}$ occupant slot the and subsystem replaced
،(3.1 (Remark unmediated **C** reads environment the but parties؛ same the
commensu- and ،ε_k same the by caught is differently corrupting $\mathcal{S}_k$ a so
parties The assumption. an than rather consequence a again is rateness
absorbed the leave hooks Mediate their alike: handled are serves $\rho_{k-1} \setminus \varphi_k$
4.15 Definition of set outward the why is which slot، the for environment
environment no that only claims theorem the So .C and A ids the carries
.$\sum_k \varepsilon_k$ than more by apart patterns corruption the tells $\mathbb{Z}^\star$ in

has closures hybrid the of union The .classes) largest (The **نکته 4.25**
together ρ of part untouched the of consists ρ_k Each form. closed a
so ،$\pi_1, \dots, \pi_k$ with

$$\bigcup_{k=0}^{n} \langle \rho_k \rangle \;=\; \langle \rho \rangle \cup \bigcup_{i=1}^{n} \langle \pi_i \rangle \;=\; \langle \rho' \rangle \cup \langle \rho \rangle.$$

.$\mathbb{Z}_{\pi_k,\varphi_k}$ class admissible largest the is $\mathbb{Z}_k$ each that now Suppose
redun- $\mathbb{Z}'_k$ defining condition second the makes then 4.16 Proposition
is $\mathbb{Z}^\star$ of choice permissible largest the and ،$\mathbb{Z}'_k = \mathbb{Z}_{\rho_k,\rho_{k-1}}$ so dant،

$$\bigcap_{k=1}^{n} \mathbb{Z}_{\rho_k,\rho_{k-1}} \;=\; \left\{ \mathcal{Z} : \mathcal{Z}.\mathsf{par} \supseteq \textstyle\bigcup_{k=0}^{n} \langle \rho_k \rangle \right\} \;=\; \mathbb{Z}_{\rho',\rho}.$$

> the delivers therefore classes largest the against composition Parallel does. theorem single-step the as again، class largest

are adversaries The suggests. chain the as behave classes three The adversary an from starting argument the replacement، last the of those ending chain the first، the of those are simulators the ؛$\rho_n = \rho'$ against serves class intermediate Each .$\rho_0 = \rho$ against simulator a producing by which next، the of adversaries the and step one of simulators the as twice، for admissible be must environments The arranges. $\mathbb{S}_{k+1} \subseteq \mathbb{A}_k$ what is as additive، is advantage The intersection. the hence once، at step every it. makes always steps n over argument hybrid a

4.4 تمامیتِ حریفِ بدل

re- and it tells environment the whatever out carries adversary dummy The real a Anything own. its of strategy no has it told؛ is it whatever back ports dummy، the driving by achieve can environment an achieves، adversary quantifying by replaced be can adversaries over quantifying why is which sys- the for that، proves and it defines section This environments. over the records 9.4 Remark — precise makes theorem the classes and tems reach. not does claim the that calls، responsive placing cores regime، one

ma- of kind new a not is dummy The **fixes. already framework the What** carries it so core، particular a with 3.4 Section of adversary the is It chine. consequences Three wrapper. ’s$\mathcal{A}$ through pass calls its and fields ’s$\mathcal{A}$ written. is it of line a before code the of shape the settle

a silences box $\mathcal{A}$ the of 2 Line only. identity one claim may it First، identity the ،id claims places dummy the call every so ،$\{\mathrm{id}'' \neq \mathrm{id}\}$ on core cannot therefore ”id''' as id''.Op “call form the of instruction An at. runs it not or $(A, \mathrm{id}.P)$ as speaks dummy the :id''' arbitrary an for out carried be party، the but id claimed the not is chooses environment the What all. at may $\mathcal{Z}$ of root the ؛(A, Q) at than rather (A, P) at dummy the instructing by either. pick may so and party any for act

asks Guard predicate The forced. is target the of party the Second، at dummy the so ،$(A, \mathrm{id}.P)$ is claim the and callee، the of $\mathrm{id}'.P = \mathrm{id}.P$ does that target A other. no and P party of interfaces reaches (A, P) the ؛3 line or 2 line by call the refuses party، honest an or ،**A** admit not to it delivers 3.6 Section of rule the and back، refusal the passes dummy .$\perp$ as caller own dummy’s the

which ،$(A, P) \to (Z, P)$ too: fixed is environment the to route its Third،
agree. parties the and $\mathbf{A} \subseteq \mathcal{Z}.\mathbf{N}$ because admits Guard$_{\mathcal{Z}}$

environ- the from call A it. called who on dispatches core The **code. The**
mediation a — else anything from call a out؛ carry to instruction an is ment
In on. hand to message a is — core some inside placed $\mathcal{A}(\cdot)$ an or hook،
oper- an uses، already Mediate shape the has payload the directions both
value. a with together ation

$\mathcal{D}$ adversary Dummy

par := $\perp$ ،**U** := $\{(\mathcal{Z}, \text{serves})\}$ ،**N** := **Std** $\cup$ **A** $\cup$ **Z** ،**P** := $\{0,1\}^* \cup \{A, Z\}$ ،pid := A

id' from $\mathrm{id}.\mathsf{A}(\mathrm{in})$

:1 **if** $\mathrm{id}'.\mathrm{F} = \mathrm{Z}$ **then**
:2 parse in as $(\mathrm{id}''.\mathsf{Op},\ x)$ / unparseable: return $\perp$
:3 **return** $\mathrm{id}''.\mathsf{FullOp}(x)$ as id / carry out the instruction
:4 **else**
:5 **return** $(\mathrm{Z}, \mathrm{id}.\mathrm{P}).\mathsf{FullZ}(\mathrm{id}',\ \mathrm{in})$ as id / hand it to $\mathcal{Z}$، caller and all

in- target a names environment the direction: outward the is 3 Line
returns and call that exactly places dummy the and input، an and terface
the — remembered nothing and inspected is Nothing answer. its exactly
one: inward the is 5 Line .Initialize no has it so state، no keeps dummy
together environment the to handed is adversary the reaches whatever
is answers environment the whatever and from، came it identity the with
read relay same the are lines two The unchanged. caller that to returned
transparent. dummy the makes what is which directions، opposite in

an- guards the :$\perp$ with answered is parse cannot 2 line instruction An
names that payload a but ،rej with place cannot dummy the anything swer
noth- with machine a what is $\perp$ and own، its of value a needs all at call no
returns. say to ing

5 line on $\mathcal{Z}$ to passed is id' caller The on. pausing worth is detail One
which tell not could environment the it without payload: the merely not and
transparent. be cannot that loses that relay a and asking، is interface

moving about statement a is Completeness **adversary. an Interposing**
needs and it، around machines the into execution the from adversary an
sides: two from seen operation same the are Both move. each for name a
what so interface، dummy the and machine some between placed is $\mathcal{A}$
now place to used $\mathcal{A}$ what and inside، it reaches now $\mathcal{A}$ reach to used
.$\mathcal{D}$ to instruction an as leaves

an be $\mathcal{Z}$ Let .environment) the into absorbed (Adversary **4.26 تعریف** bundle the is $\mathcal{Z}^{\mathcal{A}}$ environment The adversary. an $\mathcal{A}$ and environment its keeping not copy the ،$\mathcal{A}$ of copy one with $\mathcal{Z}$ of (4.13 (Definition from par and output its takes and alone IDs$(\mathcal{Z})$ occupies it identities؛ interface 's$\mathcal{A}$ by internally served is -identity$A$ an on places $\mathcal{Z}$ call A .$\mathcal{Z}$ the call other Any 's.$\mathcal{Z}$ by -identityZ a on places $\mathcal{A}$ one and there، rewritten is ،(A, P) at running while $\mathtt{id}''$.FullOp(x) on places، $\mathcal{A}$ hosted instruction the to

$$(A, P).\text{FullA}\big(\mathtt{id}''.\text{Op},\ x\big)\ \text{ as } (Z, P),$$

un- the call a so target's، its never and at runs $\mathcal{A}$ party the to keyed party same that at relayed is (A, P) as only place could $\mathcal{A}$ absorbed an from arriving call A was. it where exactly refused or admitted and ،(A, P) at interface 's$\mathcal{A}$ to handed is $(\mathtt{id}', \mathtt{in})$ payload with -identity$A$.$\mathtt{id}'$ from call a as included، guard

sim- a be $\mathcal{S}$ Let .simulator) a before interposed (Adversary **4.27 تعریف** (Defini- bundle the is $\mathcal{S}[\mathcal{A}]$ simulator The adversary. an $\mathcal{A}$ and ulator 's$\mathcal{S}$ carries and IDs$(\mathcal{A})$ occupies it ؛$\mathcal{A}$ of copy one with $\mathcal{S}$ of (4.13 tion $\mathcal{S}[\mathcal{A}]$ so for، in stands it machine the being ،$\mathcal{A}$ not ،$\mathcal{S}$ — fields remaining the by served is -identityZ a from call A is. $\mathcal{S}$ whenever simulator a is partic- in hook mediation a id، process other any from one ؛$\mathcal{A}$ hosted to back route its — -identityZ a on calls 's$\mathcal{A}$ hosted The .$\mathcal{S}$ by ular، its ؛$\mathcal{Z}$ external the to (A, P) under bundle the leave — environment the served ،4.26 Definition of instruction the become calls outward other -identityZ a on places $\mathcal{S}$ call a and ؛(Z, P) from call a as (A, P) at $\mathcal{S}$ by .(A, P) at $\mathcal{A}$ to ،Guard$_{\mathcal{Z}}$ past handed، is

occu- $\mathcal{Z}^{\mathcal{A}}$ bundle The machine. of kind new a is construction Neither is it so claimed، $\mathcal{Z}$ what claims and occupies environment an what pies the being those -identity،A no occupies It is. $\mathcal{Z}$ when exactly admissible has $\mathcal{A}$ hosted the 4.13 Definition by so execution، right-hand the in relay's and above، rewriting the hence under: call a place to own its of identity no rewrit- the check to — below Siting by discharged — obligation the hence of case other the is 4 Chapter in Absorption silencer. 's$\mathcal{Z}$ against claim ten claim its keeps there functionality hosted a why is which ،4.13 Definition the in standing what is which ،IDs$(\mathcal{A})$ occupies $\mathcal{S}[\mathcal{A}]$ Meanwhile instead.

means. slot adversary

theorem. The Defini- is alone dummy the to respect with Emulation tion 4.11 with the adversary class cut down to the single machine $\mathcal{D}$. The claim is that nothing is lost.

One mild condition on the adversary is needed، and it is the price of the dummy being a relay.

> **تعریف 4.28** (Refusal-oblivious). An adversary is refusal-oblivious if its behaviour is unchanged when any rej it receives، as the answer to a call it placed، is replaced by $\perp$.

The dummy relays such an answer، and by Section 3.6 its own **return** of a rej reaches the caller as $\perp$؛ so where an unabsorbed adversary placing a refused call reads rej directly، the same adversary driving the dummy reads $\perp$. A refusal-oblivious adversary cannot tell the two apart، which is what the regrouping below needs. The dummy itself is refusal-oblivious — it relays its answer without branching on it — so the hypothesis costs the dummy class nothing، and a real adversary that reads a framework-internal refusal signal is in any case a modelling artefact rather than an attack.

Informally، Theorem 4.29 says that emulation with respect to the dummy alone loses nothing: anything a real adversary achieves، an environment can achieve by driving the dummy، and the simulator for $\mathcal{A}$ is $\mathbb{S}_{\mathcal{D}}[\mathcal{A}]$ — the one simulator the hypothesis supplies، with $\mathcal{A}$ interposed before it — which depends on $\mathcal{A}$ alone and not on $\mathcal{Z}$.

قضیه 4.29 (Completeness of the dummy adversary). Let π and φ be systems whose cores place no responsive calls (Definition 9.2)، let $\varepsilon \geqslant 0$، let $\mathbb{A}$ be a class of refusal-oblivious adversaries (Definition 4.28)، let $\mathbb{S}$ be a class of simulators، and let $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi}$. Suppose

$$\pi \text{ UC-emulates } \varphi \text{ within } \varepsilon \text{ against } (\{\mathcal{D}\}, \mathbb{S}, \mathbb{Z}).$$

Then

$$\pi \text{ UC-emulates } \varphi \text{ within } \varepsilon \text{ against } (\mathbb{A}, \mathbb{S}', \mathbb{Z}'),$$

where

$$\mathbb{S}' = \{\mathcal{S}[\mathcal{A}] : \mathcal{S} \in \mathbb{S},\ \mathcal{A} \in \mathbb{A}\},$$
$$\mathbb{Z}' = \{\mathcal{Z} \in \mathbb{Z}_{\pi,\varphi} : \mathcal{Z}.\mathsf{par} \cap \{\mathtt{id} : \mathtt{id}.\mathrm{F} = \mathrm{Z}\} = \emptyset \wedge \mathcal{Z}^{\mathcal{A}} \in \mathbb{Z} \text{ every for} \mathcal{A} \in \mathbb{A}\}.$$

.$\mathcal{D}$ for supplies hypothesis the simulator the be $\mathcal{S}_{\mathcal{D}} \in \mathbb{S}$ let Explicitly، Then

$$\forall \mathcal{A} \in \mathbb{A}\ \ \forall \mathcal{Z} \in \mathbb{Z}' :$$

$$\mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z}, \mathcal{A}, \mathcal{S}_{\mathcal{D}}[\mathcal{A}]) \leqslant \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z}^{\mathcal{A}}, \mathcal{D}, \mathcal{S}_{\mathcal{D}}) \leqslant \varepsilon,$$

on depends which ،$\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ is for asks 4.11 Definition simulator the so .$\mathcal{Z}$ on not and alone $\mathcal{A}$

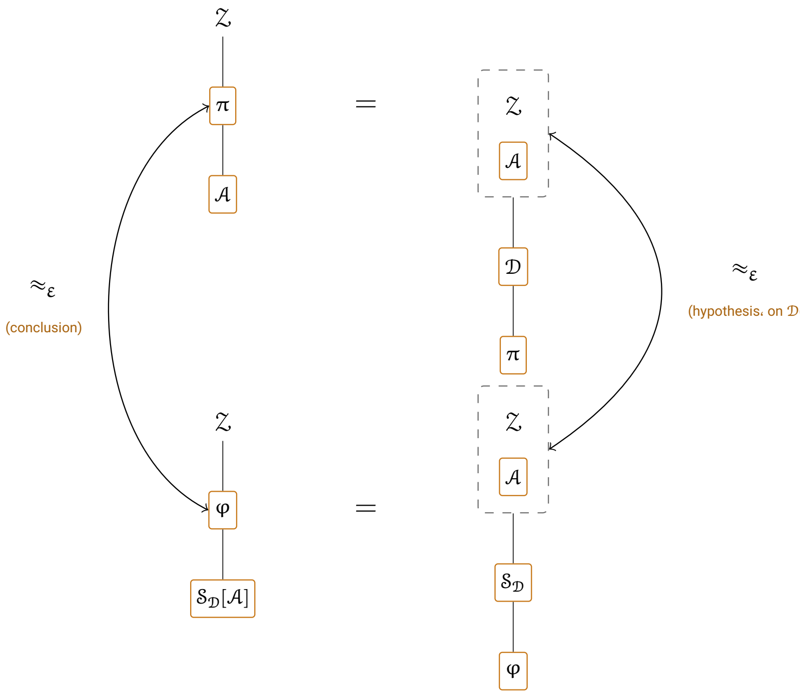
اثبات. The whole of the argument is two regroupings، one per world، labelled (R1) and (R2) below. Granting them، the two advantages of the display agree term by term and the hypothesis at $\mathcal{Z}^{\mathcal{A}}$ bounds the second، which is the theorem. Neither is a pure rebracketing — the right-hand executions carry a machine the left-hand ones do not، the relay standing at the ceded identities — so both are the ceding half of Lemma 4.18 rather than Proposition 4.19's. (R1) is that lemma applied directly، its three hypotheses being the theorem's three. (R2) is the same argument run again with φ for π and $\mathcal{S}_{\mathcal{D}}$ for $\mathcal{D}$، and it needs an argument of its own because there the adversary moves from one bundle to another rather than out of the execution: what the lemma's proof used of the machine in the slot is recounted and shown to hold of $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ as well. The closing paragraph checks the quantifier order — $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ depends on $\mathcal{A}$ alone، knowing nothing of $\mathcal{Z}$. Fix $\mathcal{A} \in \mathbb{A}$ and $\mathcal{Z} \in \mathbb{Z}'$، and let $\mathcal{S}_{\mathcal{D}} \in \mathbb{S}$ be the simulator the hypothesis supplies for $\mathcal{D}$. Membership in $\mathbb{Z}'$ gives $\mathcal{Z}^{\mathcal{A}} \in \mathbb{Z}$، so the hypothesis applies to it، and $\mathcal{Z}^{\mathcal{A}}.\mathsf{par} = \mathcal{Z}.\mathsf{par} \supseteq \langle\pi\rangle \cup \langle\varphi\rangle$، so the advantage on the right of the display is defined. The conventions of Proposition 4.19 — independent coins، unmerged call-free tapes، Initialize — are in force. The work is two regroupings، one per world:

$$\mathsf{Exec}(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C}) \equiv \mathsf{Exec}(\pi, \mathcal{Z}^{\mathcal{A}}, \mathcal{D}, \mathcal{C}), \tag{R1}$$
$$\mathsf{Exec}(\varphi, \mathcal{Z}, \mathcal{S}_{\mathcal{D}}[\mathcal{A}], \mathcal{C}) \equiv \mathsf{Exec}(\varphi, \mathcal{Z}^{\mathcal{A}}, \mathcal{S}_{\mathcal{D}}, \mathcal{C}), \tag{R2}$$

each an equality of distributions؛ granting them، the two advantages agree term by term and the hypothesis at $\mathcal{Z}^{\mathcal{A}}$ bounds the second by ε، which is executions right-hand the — rebracketing pure a is Neither display. the slot adversary the in relay the not، do ones left-hand the machine a carry 's.4.19 Proposition than rather 4.18 Lemma of half ceding the are both so — $\mathcal{Z}^{\mathrm{H}}$ that so ،$\mathrm{H}_{\mathsf{keep}} = \emptyset$ with $\mathrm{H} = \{\mathcal{A}\}$ at 4.18 Lemma (R1). of Proof Its .$\mathcal{D}$ is IDs$(\mathcal{A})$ ceded the at relay the and 4.26 Definition of $\mathcal{Z}^{\mathcal{A}}$ the is is؛ $\mathbb{A}$ because refusal-oblivious is $\mathcal{A}$ theorem's: the are hypotheses three sys- the on hypothesis the is which call، responsive a places π of core no legal، are families occupant The -identity.Z no silences $\mathcal{Z} \in \mathbb{Z}'$ and tems؛ (R2). of Proof go. let $\mathcal{A}$ identities the $\mathcal{D}$ and alone IDs$(\mathcal{Z})$ occupying $\mathcal{Z}^{\mathcal{A}}$ execu- the of out than rather another to bundle one from moves $\mathcal{A}$ Here its is argument the and face its on apply not does 4.18 Lemma so tion، a is transfers it That .$\mathcal{D}$ for $\mathcal{S}_{\mathcal{D}}$ and π for φ with again run half ceding which slot، the in machine the of used proof lemma's the what of matter dispatches it that verdicts، 'sGuard$_{\mathcal{A}}$ returns guard its that than more no is a reads $\mathcal{A}$ hosted the where — and ،$\mathtt{id}'.\mathrm{F} = \mathrm{Z}$ has claim the whether on the against left the on bundle the inside $\mathcal{S}_{\mathcal{D}}$ from rej instruction، refused ex- refusal-oblivious، is $\mathbb{A}$ that — right the on slot the across $\perp$ deemed left، the on $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ — $\mathcal{A}$ holds machine which is differs What (R1). in as actly the perform two the make 4.26 and 4.27 Definitions and — right the on $\mathcal{Z}^{\mathcal{A}}$ in- guarded machine's receiving the through routed each hand-offs، same across and side one on machine a inside travelling instruction the terface، in same the is $\mathcal{Z} \to \mathcal{A} \to \mathcal{S}_{\mathcal{D}} \to \varphi$ chain The other. the on execution the same the in instructions same the meets $\mathcal{S}_{\mathcal{D}}$ point: different a at cut both، is occupant the and verbatim، holds Siting claims، same the under order **N** and occupancy، by **P** and pid — way either fields same the by judged fields 's$\mathcal{S}_{\mathcal{D}}$ carrying inherits، $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ and meets $\mathcal{S}_{\mathcal{D}}$ which ،4.7 Definition by left- the routes 4.27 Definition rest. the reads guard no ؛4.27 Definition by on serves Exec that interface guarded same the through hand-off hand under bundle the leave $\mathcal{Z}$ to back calls own 's$\mathcal{A}$ hosted the and right، the right. the on Exec cross they as just (A, P)

are display the of advantages two the (R2) and (R1) By Conclusion. sim- The .ε by second the bounds $(\mathcal{D}, \mathcal{Z}^{\mathcal{A}})$ at hypothesis the and equal، knowing hypothesis the alone، $\mathcal{A}$ on depends and $\mathbb{S}'$ in lies $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ ulator was $\mathcal{Z} \in \mathbb{Z}'$ met؛ is 4.11 Definition of order quantifier the so ،$\mathcal{Z}$ of nothing arbitrary. □

Theo- as square commuting same the is proof the Diagrammatically، half ceding the from time this built proof)، its above figure (the 's4.20 rem part not is $\mathcal{Z}$ into moves what half: keeping the than rather 4.18 Lemma of stood. it where stands relay a and itself، adversary the but system the of



through- untouched φ and π starts: theorem the where is left Bottom before interposed $\mathcal{A}$ with $\mathcal{S}_\mathcal{D}$ — $\mathcal{S}_\mathcal{D}[\mathcal{A}]$ and occupant real-world the $\mathcal{A}$ out، and (R1) are edges bottom and top The one. ideal-world derived the — it again is read، is it side whichever on ،$\mathcal{Z}$ into $\mathcal{A}$ absorbing proof: the of (R2) this but — was 's4.19 Proposition as exactly distributions، of equality an (a)–(c) hypotheses it، for paying 4.18 Lemma of half ceding the is it time re- a and itself adversary the is moves what because nothing، of place in serves $\mathcal{Z}^\mathcal{A}$ absorbed same The place. its in stand to has ($\mathcal{S}_\mathcal{D}$ then ،$\mathcal{D}$) lay the is it all: at them compare can edge right the why is which sides، both exactly square the Chasing .$\mathcal{D}$ against only stated hypothesis، theorem's for now conclusion، theorem's the — edge left the ،4.20 Theorem in as the forced: is — machine fixed one than rather $\mathcal{A}$ refusal-oblivious every quantity. same the equalities، two the by are، 's$\approx_\varepsilon$ two

rea- same the for and ،4.20 Theorem in as move classes three The the and over، quantifies conclusion the those are adversaries The sons.

adversary، per one computed، are simulators the ؛$\mathcal{D}$ only had hypothesis
the and supplies؛ hypothesis the simulator the before it interposing by
it what when exactly counting $\mathcal{Z}$ outer an preimage، a are environments
$\mathbb{Z}$ Taking covers. hypothesis the one is $\mathcal{A}$ each absorbing on becomes
less — again one largest the $\mathbb{Z}'$ makes class admissible largest the be to
no condition a -identities،Z own their silence that environments the only
the because excluded wants، application no and imposes admissibility
en- an such which -claim،Z a under leaves 4.26 Definition of instruction
par leaves absorption since — alone side one on refuse would vironment
alone.

cost constructions two the ،4.5 Section of sense the in concretely Read
where $\mathbf{z} \oplus \mathbf{a}$ needs $\mathcal{Z}^{\mathcal{A}}$ then -bounded**a** is $\mathcal{A}$ if costs: adversary the what
invocation The .**s** needed $\mathcal{S}_{\mathcal{D}}$ where $\mathbf{s} \oplus \mathbf{a}$ needs $\mathcal{S}_{\mathcal{D}}[\mathcal{A}]$ and ،**z** needed $\mathcal{Z}$
indepen- both maximum، a take times per-invocation the and add counts
.φ and π of dent

4.5 امنیتِ عینی

about nothing says and classes arbitrary against stated is 4.11 Definition
turns resources bounded of machines with them Filling contain. they what
rela- the and one، concrete a into section previous the of statement each
budgets. on arithmetic into classes between tions

a $=$ pair a is budget A .classes) bounded and (Budgets **4.30 تعریف**
steps t most at takes it if -bounded**a** is machine A naturals. of (t, r)
at for runs it so all؛ in calls r most at answers and invocation each in
in bounds both meeting machine A execution. any in steps $t\,r$ most
rule halting the not or whether -bounded**a** counted is execution every
are below machines bounded from built bundles the — it into written is
of classes the for $\mathbb{Z}_{\pi,\varphi}(\mathbf{a})$ and $\mathbb{S}(\mathbf{a})$ ،$\mathbb{A}(\mathbf{a})$ Write sense. this in bounded
-bounded**a** of and simulators، -bounded**a** of adversaries، -bounded**a**
ma- componentwise؛ ordered are Budgets .$\mathbb{Z}_{\pi,\varphi}$ in lying environments
simulator a or several، of bundle a — execution an share that chines
by combine — adversary an running

$$(t, r) \oplus (t', r') \;:=\; \bigl(\max(t, t'),\; r + r'\bigr),$$

takes time per-invocation the so activation، per running machine one
add. counts invocation the while bound larger the

Bounding it. about promise a not machine، the of part is budget The
the — execution an bound to enough yet not is machines slot three the
it- on recurses that core a and too، machines are system a of members
token the carry would forever، another one calling members two or self،
Concrete it. reclaim to nothing with machine budgeted every from away
ev- of member every :systems budgeted over read therefore statements
def- this of sense the in own، its of budget a carries compared system ery
machines، many finitely runs then execution An verbatim. applied inition
any at steps machine one exactly and steps، many finitely taking each
of $\sum_i t_i r_i$ bound step total the within halts execution the so moment،
a given is budget a at Halting outright. defined is Exec and machines its
on suspended caller each resumes mid-call halts that machine a value:
read is return root its from away halts that environment an and ،$\perp$ with it
$\mathbb{A}(\mathbf{a}) \subseteq \mathbb{A}(\mathbf{a}')$ budget، the in monotone are classes The .0 returning as
4.23 Lemma makes That .$\mathbb{Z}_{\pi,\varphi}$ and $\mathbb{S}$ for likewise and ،$\mathbf{a} \leqslant \mathbf{a}'$ whenever
the growing budget، adversary the shrinking numbers: about statement a
em- preserve each budget environment the shrinking or budget، simulator
ulation.

تعريف 4.31 (Concrete UC advantage). Let π and φ be budgeted sys- tems، let $\mathbf{a}$ and $\mathbf{s}$ be budgets for the adversary and the simulator، and let $\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi}$ be a class of environments. Set

$$\mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}[\mathbf{a},\mathbf{s},\mathbb{Z}] := \max_{\mathcal{A}\in\mathbb{A}(\mathbf{a})} \min_{\mathcal{S}\in\mathbb{S}(\mathbf{s})} \max_{\mathcal{Z}\in\mathbb{Z}} \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}(\mathcal{Z},\mathcal{A},\mathcal{S}),$$

largest the ،$\mathbb{Z} = \mathbb{Z}_{\pi,\varphi}(\mathbf{z})$ case the for $\mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}[\mathbf{a},\mathbf{s},\mathbf{z}]$ abbreviate and
budget. that of class

largest the because budget a than rather class a is argument third The
compare below statements the and systems، of pair the on depends class
needed is class common a Where pairs. different over taken advantages
so. say to able be must notation the

read 4.11 Definition of quantifiers three the are operators three The
is simulator the reason: same the for and order same the in value، a as
are extrema The environment. the before and adversary the after chosen
rather finiteness is it and moment، a takes reason the though attained،
all، in steps $t\,r$ most at takes machine -bounded**a** An compactness. than
up coins، $t\,r$ most at and input any of symbols $t\,r$ most at reads it so
are there and did، it what to read it what from map a is it behaviour to

probabilities dyadic with them weight coins Its maps. such many finitely behaviour many finitely induces class bounded a so ،2^{tr} denominator of well، as budgeted systems the of members the With all. in distributions so behaviours، of choice each at number well-defined a is $\Pr[\mathsf{Exec} = 1]$ and values، many finitely of one takes 4.5 Definition of advantage each an (Over attained. is values many finitely over minimum or maximum a Therefore supremum.) a as read is max outer the $\mathbb{Z}$ class arbitrary

$$\pi \text{ UC-emulates } \varphi \text{ within } \varepsilon \text{ against } \big(\mathbb{A}(\mathbf{a}), \mathbb{S}(\mathbf{s}), \mathbb{Z}_{\pi,\varphi}(\mathbf{z})\big)$$

is minimum attained the — ε most at is quantity that when exactly holds se- concrete A boundary. the at exact direction reverse the makes what proba- a and budgets three relating inequality single a then is claim curity sight. in asymptotics no and parameter security no with bility،

فصل 5

ویژگی‌ها

one: of something asks property A systems. two compares Emulation
5.9 Proposition which — it against played game a win adversary no that
envi- the into absorbing slot the winning، not dummy's the to reduce will
func- a is game A either. for wanted is machinery new Little ronment.
and it، plays that adversary the is environment the other، any like tionality
what call may who settles already sections previous the of plumbing the
and event، an and class، environment an system، a is property a so —
on do rules game's a that work the of most do 1.1 Definition of fields the
hy- one comes: it as named is and small، is added be to has What paper.
4.4 Section conditions three the and system، game the on condition giene
moves. it anything of asks already

function- standard a is $\mathcal{F}$ for game A .system) game (Game، **تعریف 5.1**
with $\mathcal{G}\mathfrak{m}$ ality

$$\mathcal{G}\mathfrak{m}.\mathsf{pid} := (\mathcal{G}\mathfrak{m}.\mathsf{F}, 0, 0), \qquad \mathcal{G}\mathfrak{m}.\mathbf{P} := \mathcal{F}.\mathbf{P} \cup \{\mathsf{Z}\},$$
$$\mathcal{G}\mathfrak{m}.\mathbf{N} := \mathbf{Z}, \qquad \mathcal{G}\mathfrak{m}.\mathbf{U} := \{(\mathcal{F}, \mathsf{serves}_{\mathcal{G}\mathfrak{m}})\},$$

interfaces :finalizations more or one interfaces own its beside carrying
by borne and them for reserved prefix a ،Fin with begin names whose
and bit، a returning each root، the at served each interface، other no
one any once served — otherwise or finalization — $\mathcal{G}\mathfrak{m}$ of interface no
Here been. has them of

$$\mathsf{serves}_{\mathcal{G}\mathfrak{m}}(\mathcal{G}\mathfrak{m}, \mathcal{F}) \;:\Longleftrightarrow\; \mathcal{F}.\mathbf{P} \supseteq \mathcal{G}\mathfrak{m}.\mathbf{P} \setminus \{\mathsf{Z}\} \;\wedge\; \mathcal{G}\mathfrak{m}.\mathsf{pid} \in \mathcal{F}.\mathbf{N}$$

$\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \sigma$ is system game A exempted. root the with serves is
containing not and uses $\mathcal{F}$ whatever and $\mathcal{F}$ containing σ system a for
.6.5 Remark of sense the in $\{\mathcal{G}\mathfrak{m}\}$ at tight γ with and $\mathsf{wf}(\gamma)$ with ،$\mathcal{G}\mathfrak{m}$

over FinUF and FinCor properties: several carry may then، game، One was what record that oracles the sharing say، functionality، signature a en- The close. the at them off read verdict the in only differing and signed finalization which choosing by on judged be to which chooses vironment several with game a So does. it thing last the is choice that and call، to several، holding shell a but properties of conjunction a not is finalizations one maximizes that environment an below؛ advantage own its with each to. it asks here nothing and another، maximize not need

shell the :$\mathcal{F}$ for $\mathcal{G}\mathfrak{m}$ Game

par $:= \perp$ ،**U** $:= \{(\mathcal{F}, \text{serves}_{\mathcal{G}\mathfrak{m}})\}$ ،**N** $:=$ **Z** ،**P** $:= \mathcal{F}.$**P** $\cup \{Z\}$ ،pid $:= (\mathcal{G}\mathfrak{m}.\mathrm{F}, 0, 0)$

:Initialize()

:1 done $\leftarrow$ false
:2 … the game's own variables

$\mathtt{id}'$ from $\mathtt{id}$.Op($\mathtt{in}$)

:3 **require** $\neg$done / closed once finalized
:4 … the game's own code، which may call $\mathcal{F}$ at party $\mathtt{id}.\mathrm{P}$
:5 **return** out

$\mathtt{id}'$ from $\mathtt{id}$.Leak()

:6 **return** $\perp$ / a game keeps no party's secrets

shared is $\mathcal{F}$ for shell the property؛ per one ،$\mathcal{G}\mathfrak{m}$ of Finalizations

game the closing each root، the at served each ،Fin … named each

finalization) every of shape (the $\mathtt{id}'$ from $\mathtt{id}$.Fin()

:7 **require** $\neg$done / one finalization only
:8 **require** $\mathtt{id}.\mathrm{P} = Z$ / at the root، which stays honest
:9 done $\leftarrow$ true
:10 $w \leftarrow$ this property's verdict، read off the shell's state
:11 **return** w

$\mathtt{id}'$ from … ،$\mathtt{id}$.FinUF() ،$\mathtt{id}$.FinCor()

:12 … the same three lines، then each property's own verdict

what. by saying worth is it and forced، is field Each

and session the so experiment، per game one is There **id. process The** con- a not is 0 Session .0 at both fix we and information no carry instance its claiming by $\mathcal{G}\mathfrak{m}$ reaches environment the requirement: a but vention then 3 line and ،**Z** $\subseteq \mathcal{G}\mathfrak{m}.$**N** because admits 2 line which -identity،Z own obser- that is 7.17 Remark global. being not $\mathcal{G}\mathfrak{m} - \mathcal{G}\mathfrak{m}.s = \mathcal{Z}.s = 0$ asks session، caller's the shares $\mathcal{F}$ local a 1.3 Definition By general. in vation machines the where is which ،0 session in sits experiment whole the so too. live supplies execution the

Guard and (5 (line identity own its only claims core A **parties. served The** other: no at and P party at $\mathcal{F}$ reaches P party at $\mathcal{G}\mathfrak{m}$ so ،$\mathtt{id}'.\mathrm{P} = \mathtt{id}.\mathrm{P}$ asks chooses environment the and time، a at party one $\mathcal{F}$ exercises game the free being root its address، to $\mathcal{G}\mathfrak{m}$ of identity which choosing by party the finalizations، the for added is root The .$\mathcal{F}$.**P** Hence .3 line by any for act to why. says paragraph next the and

$\mathcal{G}\mathfrak{m}$ make would which ،**Std** Not else. nothing and **Z** **callers. admitted The** which ،**A** not game؛ the finalize functionality standard any let and global caller every refuses 2 line alone، **Z** With adversary. the to it hand would environment. the but

and game the renaming so: says 7.16 Corollary and Arbitrary، **name. The** game a against proved property a so probability، no changes together $\mathcal{F}$.**N** because matters This other. every under it against holds name one of the name to has $\mathcal{F}$.**N** so identity، own its only claim game the lets 5 line the protocol؛ that names protocol a inside ships that $\mathcal{F}$ the while game arbitrary not is What property. same the two the makes what is corollary touches. renaming no which code، caller's the is

the is root the because root the exempts serves$_{\mathcal{G}\mathfrak{m}}$ **dependence. The** game the party a serve not need $\mathcal{F}$ identity: bookkeeping own game's claimed a verifying — $\mathcal{F}$ query must verdict whose game A at. it calls never during record ones: usual the are they and options، two has — say forgery، as root the serve to $\mathcal{F}$ ask or needs، verdict the whatever calls oracle the .serves becomes serves$_{\mathcal{G}\mathfrak{m}}$ when well،

gap the and own، their on suffice not do above fields The **Tightness.** may it so ،$\langle\gamma\rangle$ on silenced is environment admissible An 's.6.5 Remark is from only it forbids 4.10 Definition but — uses γ name a of identity no claim inter- full a is functionality hosted a and occupies، γ identities at hosting .(4.13 (Definition silencer own its through identity own its claiming face of id process unoccupied an at machine a host may environment an So the of instance second invented an say، ،$(\mathcal{G}\mathfrak{m}.\mathrm{F}, 0, 1)$ at name: used a the — name by caller its admits that σ of member A name. own game's every pass calls its and machine، that admits — 1 Chapter of pattern local agree، sessions the agree، parties the ،**N** in lies name the :Guard of line then would environment The party. honest an at fire not does Mediate and $\Pr[\mathrm{Win}_{\mathrm{Fin}}] = 1$ and of، nothing records game the that $\mathcal{F}$ on oracle an hold it something produced adversary the that is verdict whose game any for given. not was

mem- Every completely. it refuses and that، refuses $\{\mathcal{G}m\}$ at Tightness in- an so occupies، system the ids process whole admits game the but ber every makes condition party 's6.5 Remark and ؛2 line fails instance vented 4.10 Definition where occupied، from speak could caller pinned a identity of member a to — open leaves tightness route one The hosting. forbids func- hosted a :$\mathcal{G}m.$**N** $=$ **Z** by closed is — game the is here which itself، I excludes، **Z** which id، process standard a claims and standard is tionality The own. its but hosted not are -identities$Z$ own environment's the while ids، process of class bare a admit may that member one the thus is game name. standard no is Z because exactly so doing in safe is it and

نکته 5.2 $\{0,1\}^*$ in ids Party .root) the at served is finalization a (Why party's corrupt a hands interface full the of 4 line and corrupted، be can such at served finalization A runs. core the before adversary the to call that whenever adversary the by answered be therefore would party a whatever be would won؟" I "have to answer the and corrupt، is party in- check no and exactly، problem 's3.1 Remark — says adversary the above. wrapper the in happening mediation it، repair can core the side FullCorrupt it: supplies already framework the and fix، the is root The $(\mathcal{G}m.\text{pid}, Z)$ at Mediate so ،**C** enters never Z so ،$\mathrm{id}.P \in \{0,1\}^*$ asks own. game's the always is computes 10 line bit the and fire never can ma- supplied three the keep to introduced were ids party reserved The 8 Line cost. extra no at honest game the keep they honest؛ chines place a is other no so identity، that to finalization every confines then win. to

نکته 5.3 finalization one exactly That .last) and once، finalization، (One enforced is answers، game the call last the is it that and answered، is a sets 9 Line jobs. both does flag one the and — assumed than rather finalization، later every refuses 7 line bypass، cannot wrapper the flag interface other every refuses 3 line and another's، or name own its environ- the so ،3.6 Section by rej returns **require** refused a thereafter؛ honest an at — else nothing and closed is game the that learns ment en- is core the before intercepts 4 line one corrupt a At least. at party، refused؛ than rather adversary the by answered is caller the so tered،

state game’s the write to unable being slot the nothing، changes that
environ- the stops Nothing already. fixed being event winning the and
— right is which either، afterwards functionality global a calling ment
won. it what change cannot then does it what and over، is game the
tables: its re-tests $\mathcal{F}_{\mathsf{Sig}}$ reason the for matters lines the of order The
instance suspended a lets 3.6 Section and calls، place may 10 line
rather computed is verdict the before set is flag the so re-entered، be
could finalizations nested two about، way other the it Were after. than
open. game the find each

safe finalizations several makes what is all them governs flag one That
environ- an own، its of flag a keep to each Were shell. one off hang to
run one in properties two on judged be and twice finalize could ment
ver- unforgeability its state a off verdict correctness a say، reading، —
them. between oracles the with from، computed been already had dict
the why is it and property، one to run a confines what is done Sharing
than rather flag shell’s the with own their of box a in sit finalizations
bookkeeping. own its carrying each

P $\in$ party a At .costs) it what and buys، corruption (What **5.4 نکته**
function- other any like wrapped are interfaces own game’s the $\{0,1\}^*$
does core the :P at $\mathcal{G}\mathrm{m}$ to call a intercepts 4 line $\mathrm{P} \in$ **C** once so ality’s،
the by answered is caller the and nothing، records game the run، not
an replaces Mediate — nothing environment an gains That adversary.
writ- never is state game’s the so core، a running than rather answer
Since party. that at oracle the it costs it and — adversary the by ten
party a corrupting ،(3.1 (Section alone party by indexed is corruption
wants that game A too. party that at oracle game’s the closes $\mathcal{F}$ of
the somewhere it serve therefore must party corrupt a at oracle an
outside own its of party a at or root، the at reach: not does corruption
.$\mathcal{F}$ on call no needs oracle that whenever do may it which ،$\mathcal{F}$.**P**
the of either not is cores game’s the of out adversary the keeps What
party، corrupt a at $\mathcal{A}$ admits 3 Line for. reach would one lines two
does it so ،$\mathrm{id}'.\mathrm{F} \neq \mathrm{A}$ tests Mediate and ones؛ honest at only it refusing
at slot the from call A places. itself adversary the call a intercept not
3.5 Section as exactly core، ’s$\mathcal{G}\mathrm{m}$ run therefore would party corrupt a
$\mathcal{G}\mathrm{m}$.**N** $=$ **Z** is it refuses What .$\mathcal{A}$ by addressed party corrupt a of says
not is field that of narrowness The does. else nothing and ،2 line at

own game's the and adversary the between thing only the but tidiness code.
Guard through go not does leakage because $\perp$ returns Leak 's$\mathcal{G}\mathfrak{m}$ And whatever party corrupt a at it read can adversary the :(3.2 (Section to willing not is it there nothing keep must game a so says، $\mathcal{G}\mathfrak{m}$.**N** over. hand

نکته 5.5 (Several instances of $\mathcal{F}$). A game may address as many in- stances of $\mathcal{F}$ as it likes، and the two ways it can differ. Instances in the game's own session need nothing extra، comparing 3 line sessions and ignoring the instance number، so $(\mathcal{F}.\mathrm{F}, 0, i)$ is reachable for every i. Instances in other sessions need global($\mathcal{F}$)، which is what waives that line. Either way the claim is $\mathcal{G}\mathfrak{m}$'s own identity، line 5 silencing its cores on everything else، so every instance sees one and the same caller and access control is decided once. And the instances really are copies: by Chapter 7 they run the same code and their fields differ in the process id alone، while Section 3.6 gives each its own state and its own coins. So a multi-instance property — n keys، n sessions — is written by having the game count them، with no extra convention about what a second instance means.
What counting them does not do is make transfer cheaper. Proposi- tion 5.10 replaces one subsystem، so a game over n instances every one of which is to be replaced costs n applications and $n\varepsilon$، by Theo- rem 4.24 along a chain of hybrids — exactly as Remark 7.18 records on the relocation side. One application transfers the property for the instances φ contains and says nothing of the others.

نکته 5.6 (A game is not session-uniform، and need not be). $\mathcal{G}\mathfrak{m}.\mathbf{N} = \mathbf{Z}$ meets **Z** while $\mathcal{G}\mathfrak{m}$ is not global، so Definition 7.2 fails of $\mathcal{G}\mathfrak{m}$ and the results of Chapter 7 do not apply to a game system. That is as it should be. A game is pinned to session 0 by the paragraph above، there is one of it per experiment، and relocating it is not something anyone wants: what the chapter applies to is the $\mathcal{F}$ under test، whose instances Remark 5.5 counts.

already 4 Chapter class the by settled is play may environments Which system. single a at taken defines،

for admissible environments The .environments) (Property **5.7 تعریف**
$\mathcal{Z}$.par $\supseteq \langle\gamma\rangle$ with those :4.10 Definition of $\mathbb{Z}_{\gamma,\gamma}$ are γ system game a
that them among those for $\mathbb{Z}^{\circ}_{\gamma}$ Write .IDs(γ) of identity no at host that
can 5.9 Proposition of absorption the ones the -identity،Z no silence
move.

the on Silenced say. to them needs game a what say clauses two The so uses، γ name a of identity no claim may environment an closure، name instance further a as or subroutines، 's$\mathcal{F}$ of one as ،$\mathcal{F}$ as speak cannot it ،$\mathcal{G}\mathfrak{m}$ reach which -identities،Z own its is claim may it what them؛ of any of envi- the So .**Std** of all admits what only reach which names، fresh and directly $\mathcal{F}$ touches and interfaces 's$\mathcal{G}\mathfrak{m}$ through game the plays ronment is which may، anyone as it touches it then and — global is $\mathcal{F}$ when only the in grants already 4.10 Definition what and means global being what against property a therefore is $\mathcal{F}$ global a of property A setting. emulation that. knowing written be must game the and it، shares that adversary an also it and ،3.3 Definition of hygiene occupancy the is clause hosting The game. second a installing from environment the stops

game a be $\gamma = \{\mathcal{G}\mathfrak{m}\} \cup \sigma$ Let .advantage) property (Win، **5.8 تعریف**
admissible be $\mathcal{Z}$ let finalizations، 's$\mathcal{G}\mathfrak{m}$ of one be Fin let system،
Write slot. adversary the occupy $\mathcal{X}$ let and ،(5.7 (Definition γ for
in 1 returns ($\mathcal{G}\mathfrak{m}$.pid, Z).FullFin interface the that event the for $\mathsf{Win}_{\mathsf{Fin}}$
of one is σ of Fin property the in $\mathcal{Z}$ of advantage The .Exec($\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C}$)

$$\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{X}} := 2\Pr[\,\mathsf{Win}_{\mathsf{Fin}}\,] - 1,$$

$$\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{X}} := \Pr[\,\mathsf{Win}_{\mathsf{Fin}}\,],$$

over being probability the reading، search the and reading decision the meant is two the of Which execution. the in machine every of coins the this by not and given، is game the where finalization each for fixed is choose. to how says below calibration on paragraph the definition؛ when σ for $\mathcal{F}$ write and ،$\mathcal{D}$ is it when subscript the from $\mathcal{X}$ drop We class a and environments of $\mathbb{Z} \subseteq \mathbb{Z}_{\gamma,\gamma}$ class a ،$\varepsilon \geqslant 0$ For .$\sigma = \{\mathcal{F}\}$ against ε within Fin property the has σ say we occupants، slot of $\mathbb{X}$ within and ،$\mathcal{X} \in \mathbb{X}$ every and $\mathcal{Z} \in \mathbb{Z}$ every for $\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{X}} \leqslant \varepsilon$ if $(\mathbb{Z}, \mathbb{X})$
against ε when $\mathbb{Z}$ $\{\mathcal{D}\} = \mathbb{X}$.

oc- Win$_{\mathsf{Fin}}$ one most at ،(5.3 (Remark game the closes flag one Since game a and disjoint، are finalizations distinct of events the run: any in curs executions several in measured properties several is them of several with one. off read numbers several than rather

Def- of that where signed، is advantage the reading decision the Under cannot environment an point: the is asymmetry the and not، is 4.5 inition rather verdict game's the being event the guess، a flip can it as Win$_{\mathsf{Fin}}$ flip to nothing is there and -1 scores win cannot it game a so own، its than of. value absolute an take

that and output، its than rather execution the of event an is Win$_{\mathsf{Fin}}$ returns environment the whatever is output execution's an deliberate: is that thing a but reported be to guess a not is property a and ،(4 (Chapter makes 5.2 Remark because distinction، the by lost is Nothing happened. the bit the is computes 10 line bit The matters. it where agree two the en- the name may we so unmediated، environment، the hands finalization at return its with $\mathcal{Z}$ for $\mathcal{Z}^\bigstar$ write ،Fin fixed a for it: reports that vironment $(\mathcal{G}\mathfrak{m}.\mathsf{pid}, \mathsf{Z}).$FullFin on placed I call any if 1 "return by replaced root the place. its earns that of part Every otherwise". 0 and ،1 with answered was party another at addressed finalization a because named is identity The adversary the by answered is and honest is party that where 8 line fails game's؛ the be not need back coming value the so corrupt، is it where be- named is operation The .5.2 Remark by happens، neither root the at the root، the at $\mathcal{G}\mathfrak{m}$ of interface every serves 1 line so ،$\mathsf{Z} \in \mathcal{G}\mathfrak{m}.\mathbf{P}$ cause finalizations several with because and — included oracles own game's per $\mathcal{Z}^\bigstar$ one on، reporting is it property which say must form reporting the be- bit، a as answer the of reading a than rather 1 against is test The .Fin bit no is rej and ،5.3 Remark by rej back comes finalization later a cause one first the because first، the than rather call any is it And .(3.6 (Section fails $\mathsf{P} \neq \mathsf{Z}$ for (Z, P) claiming environment an — refused be may placed noth- served؛ is one later a while — unset done leaving test، party 's2 line answered ever is call one most at since quantifying، by weakened is ing Exec So rest. the refusing and finalization one serving game the ،1 with on call every it، compute can $\mathcal{Z}^\bigstar$ and occurs، Win$_{\mathsf{Fin}}$ when exactly 1 returns caller، other no admits $\mathcal{G}\mathfrak{m}.\mathbf{N} = \mathbf{Z}$ itself: placed it one being interface that $\mathbf{Z}$ which id، standard a claims hosts environment the functionality a and Then excludes.

$$\Pr\big[\mathsf{Exec}(\gamma, \mathcal{Z}^\bigstar, \mathcal{X}, \mathcal{C}) = 1\big] \;=\; \Pr\big[\mathsf{Win}_{\mathsf{Fin}} \text{ in } \mathsf{Exec}(\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C})\big],$$

value the in only differing and calls same the placing machines two the membership whose in is $\mathcal{Z}$ class every in $\mathcal{Z}^{\star}$ puts also That with. halt they budgeted the and them، among $\mathbb{Z}_{\gamma,\gamma}$ — alone hosting and par on turns back. bit a reading of step further one the to up 4.30 Definition of classes used. is this of all where is 5.10 Proposition

،4.4 Section of dummy the is occupant slot default The **dummy. the Why** environment. the is it and adversary one has game a that is reason the and envi- an so strategy، no keeps and directions both in relays dummy The own ’s$\mathcal{F}$ where and do؛ can slot the everything holds it driving ronment ideal an how is which ،3.4 Section of $\mathcal{A}(\cdot)$ the — adversary the reach cores an- that environment the then is it — unpinned value a leaves functionality $\mathcal{X}$ Dropping it. give to means game a freedom the exactly is which swers، but generality of loss no thus is case dummy’s the for subscript the from 5.10 Proposition because kept is form longer the plays؛ who of choice a not is nothing costs abbreviation the That slot. that in simulator a needs the :4.4 Section of construction the is reason the and taste، of matter a environment. the into absorbs slot

be $\gamma = \{\mathcal{G}\mathfrak{m}\} \cup \sigma$ Let .environment) the into absorbs slot (The **5.9 گزاره** ،(9.2 (Definition calls responsive no place cores whose system game a (Defini- slot adversary the of occupant refusal-oblivious a be $\mathcal{X}$ let in lies again 4.26 Definition of $\mathcal{Z}^{\mathcal{X}}$ Then .$\mathcal{Z} \in \mathbb{Z}^{\circ}_{\gamma}$ let and ،(4.28 tion $\mathcal{G}\mathfrak{m}$ of Fin finalization every for and ،$\mathbb{Z}^{\circ}_{\gamma}$

$$\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{X}} \;=\; \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z}^{\mathcal{X}}}.$$

ε' within it has it ،$\mathbb{Z}^{\circ}_{\gamma}$ against ε' within Fin property the has σ if So Read occupants. refusal-oblivious of $\mathbb{X}$ class every for $\left(\mathbb{Z}^{\circ}_{\gamma}, \mathbb{X}\right)$ against occupant -bounded**x** an and environment -bounded**z** an concretely، -bounded.**z** $\oplus$ **x** is that environment absorbed an give

gives $\mathsf{H}_{\mathsf{keep}} = \emptyset$ with $\mathsf{H} = \{\mathcal{X}\}$ at 4.18 Lemma .اثبات

$$\mathsf{Exec}(\gamma, \mathcal{Z}, \mathcal{X}, \mathcal{C}) \;\equiv\; \mathsf{Exec}(\gamma, \mathcal{Z}^{\mathcal{X}}, \mathcal{D}, \mathcal{C}),$$

Now one. for one here، assumed three the being hypotheses three its and ،1 returns $(\mathcal{G}\mathfrak{m}.\mathsf{pid}, Z)$ at FullFin interface the that event the is $\mathsf{Win}_{\mathsf{Fin}}$ identity that occupies $\mathcal{G}\mathfrak{m}$ — alone slot adversary the touches absorption and alike، Fin each for probability one has event the so — sides both on Sec- is Admissibility reading. either under number one gives 5.8 Definition occupies environment an what occupies $\mathcal{Z}^{\mathcal{X}}$ that observation ’s4.4 tion

is budget the is؛ $\mathcal{Z}$ when exactly admissible is it so ،par 's$\mathcal{Z}$ carries and
in $\mathcal{X}$ each at read display the is sentence final The .8 Chapter of $\mathbf{z} \oplus \mathbf{a}$ the
$\square$ turn.

plac- system game a more: no and 's4.29 Theorem are hypotheses The
environ- an is so and ،9.4 Remark by this، outside is calls responsive ing
4.26 Definition of instruction the -identities،Z own its silences that ment
occupant the what on condition a is Neither -claim.Z a under leaving
in machine the whether care not need game a — point the is which does،
moved. be can it that only fairly، plays slot the

one same the is it and this، of all with comes condition scope One
Theo- of regrouping the on rests above proposition The throughout.
respon- no place cores whose systems for only holds which ،4.29 rem
same the otherwise؛ fails relay the where showing 9.4 Remark calls، sive
emula- the supplies what is which itself، 4.29 Theorem governs condition
and class، adversary wider a from wants 5.10 Proposition hypothesis tion
whose system game A once. at sides both on it inherits 5.11 Corollary
must and three all outside sits therefore $\mathcal{A}^!$ through slot the reaches $\mathcal{F}$
responsive a What directly. established hypothesis dummy-only its have
it. leaves 9.4 Remark where leave we like look would games of theory

takes finalization a readings two 's5.8 Definition of Which **calibration. The**
reads $2\,\Pr[\mathsf{Win}_{\mathsf{Fin}}]-1$ factor The this. is rule the and taste، of matter a not is
prob- with wins guesses that environment an :decision a as property the
right is That .1 scores wins always that one and ،0 scores and $\frac{1}{2}$ ability
environment the bit a is verdict the where — strategy a is guessing where
unforge- — property search a for wrong is It for. coin a flipped have could
there because — outright hard be to meant is winning where say، ability،
on ε bound a so ،$\Pr[\mathsf{Win}_{\mathsf{Fin}}] = (1 + \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z}})/2$ reads quantity same the
for which probability، winning the on $(1+\varepsilon)/2$ bound a is advantage the
half succeeding forger a permit would 0 of ε an scale: wrong the is ε small
the Declaring itself. $\Pr[\mathsf{Win}_{\mathsf{Fin}}]$ take therefore properties Search time. the
single a lets what is globally one fixing than rather finalization per reading
second the of is paper this in property Every once. at kinds both carry shell
out- impossible be to meant is below games the of each winning — kind
search the take them of all so — guess a than better no merely not right،
the is guessing where properties for there is reading decision the reading؛
them. among game a as written indistinguishability an strategy،
Its for. paid is it where is 5.10 Proposition and free، not is choice The
decision the for 2ε — scales two the at bound same the are displays two

transfer inherits finalization each so — reading search the for ε reading، at inherits two the mixing game a and fixes، reading own its scale the at both.

the beside than rather way this properties define to reason the Finally، them. carries emulation framework:

گزاره 5.10 (Properties transfer along emulation). Let $\gamma = \{\mathcal{G}\mathfrak{m}\} \cup \sigma$ be a game system، let $\varphi \subseteq \sigma$، let the replacement of φ by π in γ be well- formed in the sense of Definition 4.2، and let $\gamma[\pi/\varphi]$ be a game system as well. Suppose

$$\pi \text{ UC-emulates } \varphi \text{ within } \varepsilon \text{ against } (\{\mathcal{D}\}, \mathbb{S}, \mathbb{Z}).$$

Then there is $\mathcal{S} \in \mathbb{S}$ such that for every finalization Fin of $\mathcal{G}\mathfrak{m}$ and every $\mathcal{Z} \in \mathbb{Z}_{\gamma[\pi/\varphi],\gamma}$ whose reporting form has $\mathcal{Z}^{\star\,\gamma\setminus\varphi} \in \mathbb{Z}$ — a condition Proposition 4.16 makes vacuous whenever $\mathbb{Z}$ is the largest admissible class، though not when it is the largest class at a budget، absorption costing $\mathbf{c}_{\gamma\setminus\varphi}$ and the reporting form one step more —

$$\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma[\pi/\varphi],\mathcal{Z}} \leqslant \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{S}} + 2\varepsilon \qquad \text{(decision reading)،}$$

and، writing $\mathsf{Win}_{\mathsf{Fin}}$ and $\mathsf{Win}'_{\mathsf{Fin}}$ for the winning events of the two exe- cutions the display above compares — the real one and the ideal one —

$$\Pr[\,\mathsf{Win}_{\mathsf{Fin}}\,] \leqslant \Pr[\,\mathsf{Win}'_{\mathsf{Fin}}\,] + \varepsilon \qquad \text{(search reading).}$$

اثبات. Three moves and one bound. First the statement is checked to be about the systems it names، which is where the hypothesis on $\gamma[\pi/\varphi]$ is spent: emulation supplies neither the game's party set at the new oc- cupant nor its process id in that occupant's $\mathbf{N}$، and tightness is part of Definition 5.1 besides. Then the reporting form of Definition 5.8 turns the winning event into an output bit، rewriting what $\mathcal{Z}$ returns at the root and changing no run. Then absorption moves the game itself inside the envi- ronment، so that an environment playing the game becomes one distin- guishing the two systems. What is left is Theorem 4.20 applied at that absorbed environment. The search reading is the bound it gives; the deci- sion reading is that bound doubled، and nothing in the argument reads Fin، so it holds of each finalization separately. First، the two advantages are defined and are about the systems the statement names. Since $\mathcal{G}\mathfrak{m} \notin \sigma$ we have $\mathcal{G}\mathfrak{m} \notin \varphi$، so $\gamma[\pi/\varphi] = \{\mathcal{G}\mathfrak{m}\} \cup \sigma[\pi/\varphi]$; and $\mathbb{Z}_{\gamma[\pi/\varphi],\gamma}$ lies in both

a when weakening 4.10 Definition of clause each ،$\mathbb{Z}_{\gamma[\pi/\varphi],\gamma[\pi/\varphi]}$ and $\mathbb{Z}_{\gamma,\gamma}$
property a is side left-hand the That one. to shrinks closures two of union
not is it and spent، is $\gamma[\pi/\varphi]$ on hypothesis the where is all at advantage
occupant new the at $\mathcal{G}\mathfrak{m}.\mathbf{P} = \mathcal{F}'.\mathbf{P} \cup \{Z\}$ neither supplies emulation free:
4.4 Theorem of (ii) hypothesis being second the ،$\mathcal{G}\mathfrak{m}.\mathrm{pid} \in \mathcal{F}'.\mathbf{N}$ nor $\mathcal{F}'$
of part is tightness buys: it all that is Nor .4.20 Theorem of part no and
pin members 'sπ that addition in asks hypothesis the so ،5.1 Definition
least that supplies Emulation did. 'sφ where id process by callers their
the of nothing saying and boundary the at behaviour comparing all، of
it. behind fields declared

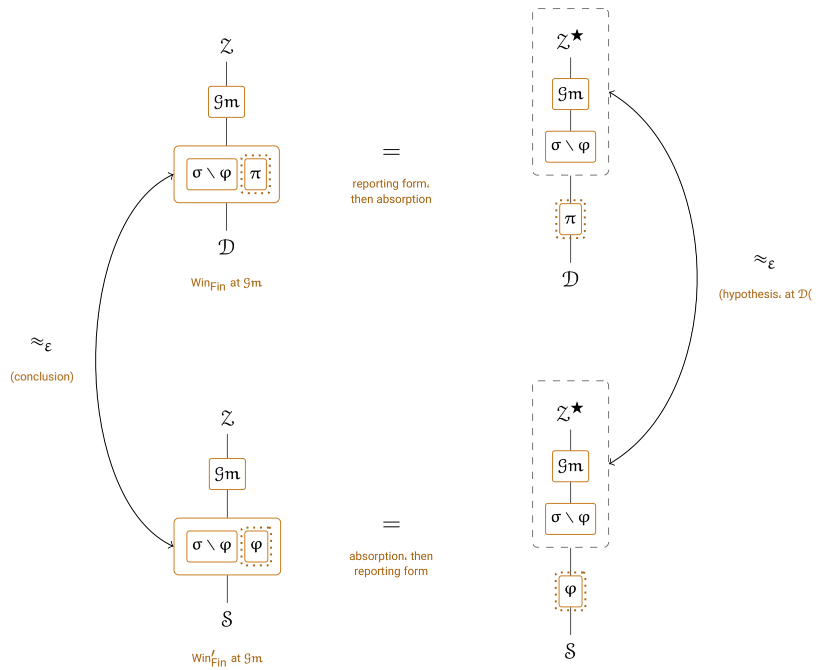
depends it :$\mathcal{D}$ for supplies hypothesis the simulator the be $\mathcal{S} \in \mathbb{S}$ Let
the take and ،Fin fix ،$\mathcal{Z}$ a such Fix .$\mathcal{Z}$ every serves $\mathcal{S}$ one so alone، $\mathcal{D}$ on
،5.8 Definition after paragraph the from Fin that for $\mathcal{Z}^\star$ form reporting
gives which

$$\Pr\left[\mathsf{Exec}(\mathcal{S}, \mathcal{Z}^\star, \mathcal{X}, \mathcal{C}) = 1\right] = \Pr\left[\mathsf{Win}_{\mathsf{Fin}} \text{ in } \mathsf{Exec}(\mathcal{S}, \mathcal{Z}, \mathcal{X}, \mathcal{C})\right]$$

$\mathcal{Z}^\star$ places hypothesis the :$\mathcal{X}$ occupant either and δ system game either for
$\mathcal{Z}^\star$ $\in$ and replacement، this to attaches 4.20 Theorem that $\mathbb{Z}'$ class the in
host. they what on and par on agree two the and is $\mathcal{Z}$ because $\mathbb{Z}_{\gamma[\pi/\varphi],\gamma}$
adversary the at ،γ in π by φ of replacement the to applied 4.20 Theorem
left-hand two the of difference the bounds ،$\mathcal{Z}^\star$ environment the and $\mathcal{D}$
$\mathcal{X} = \mathcal{S}$ and $\delta = \gamma$ against $\mathcal{X} = \mathcal{D}$ and $\delta = \gamma[\pi/\varphi]$ with — probabilities
which two، the between ε most at by differs $\Pr[\mathsf{Win}_{\mathsf{Fin}}]$ Hence .ε by —
being reading، decision the under advantage the display: second the is
order The first. the is which ،2ε most at by differs then ،$2\Pr[\mathsf{Win}_{\mathsf{Fin}}] - 1$
the what is bound probability the proof: for than rather use for matters
the Under doubled. it is bound decision-reading the and gives، argument
is display second the so probability، the is advantage the reading search
Nothing arises. never two of factor the and statement transfer the already
□ separately. finalization each of holds it so ،Fin reads argument the in

and one، textbook the is reduction The proof. that about things Two
Defi- of $\mathcal{Z}^{\star\,\gamma\setminus\varphi}$ environment absorbed the so: it makes what is absorption
environ- "an so ،$\gamma\setminus\varphi$ of rest the with along itself game the hosts 4.15 nition
distinguish- environment "an becomes "π against game the playing ment
4.20 Theorem what is which it، inside game the moving by "φ from π ing
the in $\mathcal{S}$ against stated is right the on property the And about. proved was
the keeps 5.8 Definition why is which dummy، the against than rather slot
meet need anyone obligation an not is it but — argument an as occupant
well. as environment the into absorbs slot the because form، that in

move the and move one is it because drawing، worth is reduction The
the in again، 1 Figure of square commuting the is It past. read to easy is
being pair the marking again box dotted the with and conventions same
moves: that game the is it time this but — exchanged



the of instead moving game the with square same the as transfer، Property :2 شکل
in belongs؛ game a where system، the inside sits $\mathcal{G}\mathfrak{m}$ column left the In protocol.
distinguishing. the doing machine the of part box، dashed the inside sits it right the
(.5.10 (Proposition reduction. whole the is boundary that Crossing

system، the inside sits it column left the In .$\mathcal{G}\mathfrak{m}$ watching by it Read
verdict the is $\mathrm{Win}_{\mathrm{Fin}}$ and it، plays environment the belongs: game a where
the of part box، dashed the inside sits it column right the In returns. it
whole the is boundary that Crossing distinguishing. the doing machine
en- an becomes π against game the playing environment an — reduction
a is left is what across، is game the once and — φ from π telling vironment
about. proved was 4.20 Theorem what is which question، emulation bare
rewrites only form reporting The free. are edges horizontal two The
it so untouched، places it call every leaving root، the at returns $\mathcal{Z}$ what
executions of identity an ،4.18 Lemma is absorption and run؛ no changes

is it and for، paid thing only the is edge right the So bound. a than rather
in $\mathcal{S}$ carries row bottom the why is which — $\mathcal{D}$ at applied hypothesis the
occupant the keeps 5.8 Definition why so and dummy، the not and slot the
:1 Figure in as exactly forced، then is edge left The all. at argument an as
two than rather quantity same the 's$\approx_\varepsilon$ two the makes square the chasing
.ε an share to happen that bounds

double. its is other the proposition's، the is quantity that of reading One
left the so ،$\Pr[\mathsf{Win}_{\mathsf{Fin}}]$ is advantage property the reading search the Under
under appears؛ two of factor no and statement transfer the already is edge
it. doubles which ،$2\Pr[\mathsf{Win}_{\mathsf{Fin}}] - 1$ is advantage the reading decision the
this of instance an is paper the in proof property per-functionality Every
not. do instances the where diagram a earns it why is which square، one

نتیجه 5.11 (Transfer، against every adversary). In the setting of Propo- sition 5.10، suppose in addition that the cores of γ and of $\gamma[\pi/\varphi]$ place no responsive calls (Definition 9.2)، that the simulator $\mathcal{S}$ that proposi- tion supplies and the occupant $\mathcal{A}$ below are refusal-oblivious (Defini- tion 4.28)، and that the environments named in lie in $\mathbb{Z}^\circ_\gamma$. Fix a finalization Fin of $\mathcal{G}m$. If σ has the property Fin within ε' against $\mathbb{Z}_{\gamma,\gamma}$ in the sense of Definition 5.8، then

$$\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}m,\sigma[\pi/\varphi],\mathcal{Z},\mathcal{A}} \leqslant \varepsilon' + 2\varepsilon$$

for every occupant $\mathcal{A}$ of the adversary slot and every $\mathcal{Z}$ the proposition admits with $\mathcal{Z}^{\mathcal{A}}$ among them. If in place of the hypothesis on advan- tages the ideal side satisfies $\Pr[\,\mathsf{Win}'_{\mathsf{Fin}}\,] \leqslant p$ under the same quantifi- cation، then $\Pr[\,\mathsf{Win}_{\mathsf{Fin}}\,] \leqslant p+\varepsilon$، with $\mathsf{Win}_{\mathsf{Fin}}$ and $\mathsf{Win}'_{\mathsf{Fin}}$ naming the real and ideal events as in Proposition 5.10 — which is the form a search property inherits، in the first display being the decision reading's.

اثبات. Three steps، of which the outer two are absorption. By Proposi- tion 5.9 at the real game system، $\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}m,\sigma[\pi/\varphi],\mathcal{Z},\mathcal{A}} = \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}m,\sigma[\pi/\varphi],\mathcal{Z}^{\mathcal{A}}}$، the right-hand side being the dummy's case. Proposition 5.10 at $\mathcal{Z}^{\mathcal{A}}$ bounds it by $\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}m,\sigma,\mathcal{Z}^{\mathcal{A}},\mathcal{S}} + 2\varepsilon$. And Proposition 5.9 again، now at the ideal game system، rewrites the first term as $\mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}m,\sigma,(\mathcal{Z}^{\mathcal{A}})^{\mathcal{S}}}$، which the hypothesis on σ bounds by ε'، the doubly absorbed machine being an admissible environ- ment by two applications of the same proposition. The probability form is the same three steps read on the second display of Proposition 5.10 rather than the first، absorption changing no probability at all. □

prop- the now is side ideal the on hypothesis The buys. this things Two over quantified ،σ about statement one — alone dummy the against erty asked own its on 5.10 Proposition where — else nothing and environments supplied. emulation the simulator particular the against property the for — leaks or corrupts simulator that how about assumed be need Nothing Sec- which refusal، framework-internal a on branch not does it that only absorb- does، it else whatever moves: it machine every of asks 4.4 tion ideal-side the machine a of part into it turns environment the into it ing real the of property the is conclusion the And covers. already hypothesis same the by dummy، the against only not adversary، every against system are absorptions two the concretely، Read side. other the on applied move innermost the $\mathcal{S}$ -bounded**s** an and $\mathcal{A}$ -bounded**a** an for price: whole the cost hosting one ،**z** $\oplus$ **a** $\oplus$ **s** budget environment at reached is statement per machine moved، and Chapter 8 charges nothing else.

نکته 5.12 Noth- .side) ideal the on does slot the of occupant the (What the whatever absorbs which above، proof the for needed is here ing instead is stake at is What supplies. 4.20 Theorem ε the into does slot propo- the executions two the right: the on hypothesis the of cost the Corruption system. the in as well as slot the in differ compares sition is not is that half The issue. at is it of half only and two، in divides ،$\mathrm{id}'.\mathrm{F} \in \{\mathrm{A}, \mathrm{Z}\}$ admits 3 line itself: for does environment the what it corruptions the for cooperation nobody's needs environment an so on identically them makes it root its from (Z, P) claiming and wants، initiative: own occupant's the is issue at is that half The sides. both asked is it what decline or to، it asked nobody parties corrupt may $\mathcal{S}$ right-hand the why is That relay. would $\mathcal{D}$ where slot، the through the against not and $\mathcal{S}$ against property the is 5.10 Proposition of side argument. an as occupant the keeps 5.8 Definition why and dummy، every against game the is form that in satisfy to has σ ideal an What the to it reduces 5.11 Corollary produce؛ may emulation the simulator environ- the into simulator the moving by so does and again، dummy it. of anything asking by than rather ment

reads environment The why. is register the and ،ε the escapes Nothing whole the returns and nothing mediates which ،FullStatus through **C** com- and point any at **C** read may it so ،(3.1 (Remark set corrupted whatever with together itself corrupted it parties the against it pare ad- class the environment an be $\mathcal{Z}$ Let add. to slot the instructed it

0 of return a and added comparison that with $\mathcal{Z}$ be $\mathcal{Z}'$ let and mits، is it what corrupting $\mathcal{D}$ fails، never it side real the On fails. it where prob- whatever with fails it side ideal the on besides؛ nothing and told what declining — direction either in with، strays simulator the δ ability advan- with distinguishes $\mathcal{Z}'$ So unbidden. corrupting or asked، is it therefore may One .$\delta \leqslant \varepsilon$ gives hypothesis emulation the and ،δ tage make، to instructed is it corruptions the echoes $\mathcal{S}$ though as reason commensurateness، 's3.1 Section is which — paid already price a at the discharges what not is it but having، worth is It properties. for read does، slot the absorbing ؛5.10 Proposition of right the on hypothesis conduct. simulator's the of all at nothing needs it and

the between divides Leakage .side) ideal the on (Leakage **5.13 نکته** because own its of check one needs and does، corruption as sides 2 line :Leak a read can slot the Only .Guard through go not does it identity. an such claim cannot environment the and ،$\mathrm{id}'.\mathsf{F} = A$ asks must: else something and there، game the protect not does $\mathcal{G}\mathfrak{m}$.**N** So read- is leakage game's the since party، the with together gate 's4 line other every at and corrupt، never is root the party، corrupt a at only able therefore state game's The .5.4 Remark by $\perp$ returns Leak 's$\mathcal{G}\mathfrak{m}$ party differ does What side. neither on route that by adversary the reaches produce؛ must $\mathcal{S}$ which 's،π against leakage 'sφ is sides the between a and ،ε same the by bounded emulation، of part ordinary an is that check. to further nothing with bound the inherits property

فصل 6

مسدودگرها

same the and one on environment an silences 4 Chapter of 4.10 Definition on depend not do it to open claims the so worlds، both in closure name re- no use 4.20 Theorem and 4.16 Proposition and — faces it world which narrower: is buys blocking What all. at sets identity two the between lation delivers so and set، silenced exact an as attainable closure that makes it 4.10 Definition of discussion the as class، admissible restricted least the It else. nothing and identities occupies blocker A records. 4 Chapter in for false is Guard of 2 line on conjunct the so ،**N** $:= \emptyset$ caller، no admits en- never therefore is core Block Its .rej receives caller the and call every so ،Guard through go not does Leakage formality. a is body its and tered with machine a all ،$\perp$ returns it reachable؛ is core Leak party's corrupt a requires 1.2 Definition as wrapped cores، are Both give. to has state no first the makes that wrapper the is it and functionality، standard any of unreachable.

$\mathcal{B}[\mathcal{F}]$ Blocker

par $:= \perp$ ،**U** $:= \emptyset$ ،**N** $:= \emptyset$ ،**P** $:= \mathcal{F}$.**P** ،pid $:= \mathcal{F}$.pid

$\mathtt{id}'$ from $\mathtt{id}$.Block()

/ unreachable due to Guard

:1 **return** $\perp$

$\mathtt{id}'$ from $\mathtt{id}$.Leak()

/ reachable، but there is no state

:2 **return** $\perp$

from for، in stands it functionality the by indexed is $\mathcal{B}[\mathcal{F}]$ blocker The By else. nothing and set identity an fix that fields two the copies it which ،$\mathcal{F}$ of identities the precisely occupies it :IDs$(\mathcal{B}[\mathcal{F}])$ = IDs$(\mathcal{F})$ construction it unreachable، Being besides. nothing and included، instance and session respect. other no in and identities those of occupant an as $\mathcal{F}$ for in stands By uses. system a ids process the for $\mathrm{I}_\pi := \{\mathcal{F}.\text{pid} : \mathcal{F} \in \pi\}$ Write we which member، one exactly by carried is $q \in \mathrm{I}_\pi$ each 1.2 Definition

Blockers $.\mathsf{IDs}(\pi) = \bigcup_{q\in I_\pi} \mathsf{IDs}(\pi_q)$ and $\pi = \{\pi_q : q \in I_\pi\}$ so ،π_q write a matters: distinction the and members، by not and ids by indexed are case which in code، different carrying worlds both in occur may id process q a such for blocker a and ،q uses π although π of member a not is φ_q id. same the at machines two put would

already is what of clear blocker a keeps instead $I_\varphi \setminus I_\pi$ by Indexing q has $\mathsf{IDs}(\pi)$ of identity no hence ،q a such carries π of member No there. φ_q from built blocker A $.\mathsf{IDs}(\varphi_q) \cap \mathsf{IDs}(\pi) = \emptyset$ and component first its in exist- an of id the on nor serves already π identity an on neither lands thus them of all and blockers، disjoint pairwise give ids Distinct member. ing throughout. ids distinct has again system enlarged the so ،π to new are

wherever agree addition in must compared being systems of pair The overlap. they

> when- if، compatible are φ and π Systems .(Compatibility) **6.1 تعریف** it at sitting functionalities two the both، in occurs id process a ever remaining the admit؛ they whom on and parties served their on agree may — run they code the and parameters، their use، they what — fields differ.

both to common ids the point: one exactly at used is Compatibility contributes it whatever so id، an such for added is blocker No worlds. serve could id shared a it Without sides. two the on agree already must uncovered. go would difference the and ،φ in and π in parties different only since needs، argument the than more is well as agree to **N** Requiring nothing، costs it but fixed؛ is id process the once set identity an enters **P** every in sides both on functionality same the by carried being id shared a use. we case

> ،φ and π systems For .systems) blocked system، (Blocker **6.2 تعریف** is $\mathsf{Blockers}(\varphi \setminus \pi)$ system blocker the
>
> $$\mathsf{Blockers}(\varphi\setminus\pi) := \left\{\mathcal{B}[\varphi_q] : q \in I_\varphi \setminus I_\pi\right\},$$
>
> are pair the of systems blocked the and
>
> $$\overline{\pi} := \pi \cup \mathsf{Blockers}(\varphi\setminus\pi), \qquad \overline{\varphi} := \varphi \cup \mathsf{Blockers}(\pi\setminus\varphi).$$

The argument of Blockers is notation rather than an input: the con-
struction reads the ids $I_\varphi \setminus I_\pi$ and φ's occupants there، not the mem-
ber set the difference denotes — at a shared id carrying different code،
$\varphi_q \in \varphi \setminus \pi$ and yet no blocker is built. Each id in $I_\varphi \setminus I_\pi$ is thus covered by
the blocker of the functionality living there، with that functionality's party
set. Each world gains exactly the process ids the other has and it lacks،
all blocking has to achieve. Ids common to both worlds get no blocker،
which is where Definition 6.1 does its work.

Informally، Theorem 6.3 says blocking equalises the two worlds: each
blocked system is again a system، and for a compatible pair the two oc-
cupy one and the same identity set، $\mathrm{IDs}(\pi) \cup \mathrm{IDs}(\varphi)$ — each world having
gained exactly the process ids the other has and it lacks.

قضیه 6.3 (Equalisation). Let π and φ be systems. Then the blocked systems $\overline{\pi}$ and $\overline{\varphi}$ of Definition 6.2 are systems. If π and φ are moreover compatible، then

$$\mathrm{IDs}(\overline{\pi}) = \mathrm{IDs}(\overline{\varphi}) = \mathrm{IDs}(\pi) \cup \mathrm{IDs}(\varphi).$$

اثبات. Two claims، and only the second needs compatibility. The first asks
the three things Definition 1.2 asks of a system — finiteness، standard-
ness، distinct process ids — of each blocked system، and gets them from
the indexing: blockers are indexed by the ids the other world has and this
one lacks، so they land on no existing member. The second compares
the two identity sets index by index over $I_\pi \cup I_\varphi$. At an id only one world
uses، both sides contribute the same set، a blocker copying the two fields
that fix an identity set. At a shared id no blocker is built، so the two con-
tribute $\mathrm{IDs}(\pi_q)$ and $\mathrm{IDs}(\varphi_q)$ as they stand — and compatibility supplies the
one thing the indexing does not، that the served party sets agree. That is
the only place it is used. For the first claim، being a system asks three
things of $\overline{\pi}$: that it be finite، that its members be standard، and that they
carry distinct process ids. Finiteness is immediate، the added members
being indexed by $I_\varphi \setminus I_\pi \subseteq I_\varphi$. Standardness is inherited: a blocker car-
ries the wrappers Definition 1.2 asks of a standard functionality، Block
as in Section 3.5 and Leak as in Section 3.2، and its process id is copied
from a member of φ، so its name is none of A، Z، C. For distinctness،
the members of Blockers$(\varphi \setminus \pi)$ are the $\mathcal{B}[\varphi_q]$ with $q \in I_\varphi \setminus I_\pi$، and
$\mathcal{B}[\varphi_q].\mathrm{pid} = \varphi_q.\mathrm{pid} = q$. These ids are pairwise distinct and none lies in

The .$I_{\overline{\pi}} = I_\pi \cup I_\varphi$ and ids process distinct carry $\overline{\pi}$ of members the so ،I_π
.$\overline{\varphi}$ to applies argument same

blocker a since ،$\mathcal{F}$ every for holds $\mathsf{IDs}(\mathcal{B}[\mathcal{F}]) = \mathsf{IDs}(\mathcal{F})$ second، the For
Hence set. identity an determine that fields two the copies

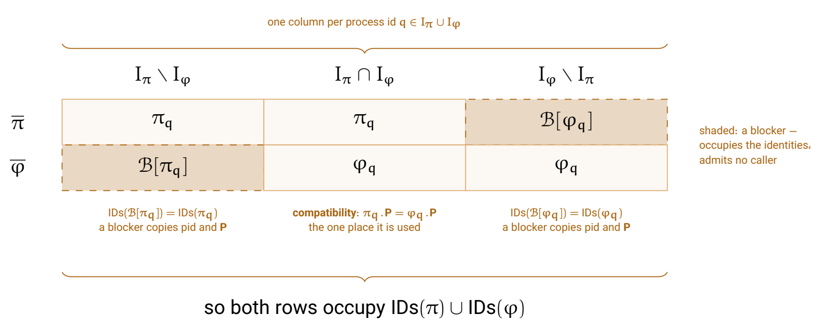
$$\begin{aligned} \mathsf{IDs}(\overline{\pi}) &= \bigcup_{q \in I_\pi} \mathsf{IDs}(\pi_q) \cup \bigcup_{q \in I_\varphi \setminus I_\pi} \mathsf{IDs}(\varphi_q), \\ \mathsf{IDs}(\overline{\varphi}) &= \bigcup_{q \in I_\varphi} \mathsf{IDs}(\varphi_q) \cup \bigcup_{q \in I_\pi \setminus I_\varphi} \mathsf{IDs}(\pi_q), \end{aligned}$$

that in q each at them Compare .$I_\pi \cup I_\varphi$ by indexed are unions both and both then $q \in I_\varphi \setminus I_\pi$ if and ،$\mathsf{IDs}(\pi_q)$ contribute both then $q \in I_\pi \setminus I_\varphi$ If set. the and $\mathsf{IDs}(\pi_q)$ contributes first the both، in lies q If .$\mathsf{IDs}(\varphi_q)$ contribute $\pi_q.\mathsf{pid} = \varphi_q.\mathsf{pid} =$ so ،q index the at carried are Both .$\mathsf{IDs}(\varphi_q)$ second follow not does that thing one the supplies compatibility and outright، q two those by fixed is set identity An .$\pi_q.\mathbf{P} = \varphi_q.\mathbf{P}$ indexing، the from by term agree therefore unions two The .$\mathsf{IDs}(\pi_q) = \mathsf{IDs}(\varphi_q)$ so fields،
equals each and term،

$$\bigcup_{q \in I_\pi} \mathsf{IDs}(\pi_q) \cup \bigcup_{q \in I_\varphi} \mathsf{IDs}(\varphi_q) = \mathsf{IDs}(\pi) \cup \mathsf{IDs}(\varphi).$$

□

places، three of one in falls id process A table. a is claim second The
each: at sits what is theorem the and



three The place. each at does blocking what and fall، can id process a Where :3 شکل
.6.3 Theorem of claim second the of cases three the are columns

a and blocker، a gains id the lacks that world the columns outer the In
for، in stands it functionality the of identities the exactly occupies blocker

rows two the So set. identity an fix that fields two only the copied having
working a contributes one although — there identities same the contribute
the is which nothing، answers that machine a other the and functionality
blocking all is sets identity Equalising differently. them shading of point
answered that blocker a achieves: it all be better had it and achieve، to has
compar- the and learn، could environment an what change would anything
worthless. be would enable to built was it ison

built is blocker No lives. hypothesis the where is column middle The
they as $\mathrm{IDs}(\varphi_q)$ and $\mathrm{IDs}(\pi_q)$ contribute rows two the so id، shared a for
carried are both — equal those makes indexing the in nothing and stand،
not. need sets party served the but outright، agree ids process the so ،q at
is column this and do، they that assumption the exactly is Compatibility
assump- the what also is it way، other the Read used. is it place only the
a is worlds two the in parties different serving id shared a costs: tion
adding — has it device one the because cover، cannot blocking difference
present. already is occupant an where unavailable is — occupant an

at systems are $\overline{\varphi}$ and $\overline{\pi}$ That this. of any needs claim second the Only
two the states theorem the why is which compatibility، of nothing asks all
alone. first the on lean may below 6.4 Proposition why and separately

two the of one is system a enlarging and system، a enlarges Blocking
undone. come could 1.3 Definition ways

sys- be φ and π Let .well-formedness) preserves (Blocking **6.4 گزاره**
.$\mathrm{wf}(\overline{\varphi})$ and $\mathrm{wf}(\overline{\pi})$ Then .$\mathrm{wf}(\varphi)$ and $\mathrm{wf}(\pi)$ with tems

com- no asks which ،6.3 Theorem of claim first the by system a is $\overline{\pi}$.اثبات
$\overline{\varphi}$ for argument the ؛$\overline{\pi}$ for argue We question. legal a is $\mathrm{wf}(\overline{\pi})$ so patibility،
.$(\mathcal{F}', R) \in \mathcal{F}.\mathbf{U}$ and $\mathcal{F} \in \overline{\pi}^+$ Take .$\mathrm{wf}(\pi)$ of place in $\mathrm{wf}(\varphi)$ with same the is
blocker a check: to nothing is there blockers added the of one is $\mathcal{F}$ If
in lies $\mathcal{F}$ Otherwise entry. no contributes it so construction، by $\mathbf{U} = \emptyset$ has
is witness That entry. that for $\mathcal{G} \in \pi^+$ witness a supplies $\mathrm{wf}(\pi)$ and ،π^+
so fields، member's no alters blocking and ،$\pi^+ \subseteq \overline{\pi}^+$ since present، still
conditions three The were. they what are reads R everything and $\mathcal{G}.s$ ،$\mathcal{G}.F$
.$\mathcal{G}$ same the of hold still therefore

en- :**U** empty their beyond blockers about nothing uses argument The
whenever well-formedness preserves system another to system a larging
once witness a because own، their of entries no carry added members the
Nei- change. never on judged is it fields the and removed never is present
☐ used. is claim second 's6.3 Theorem nor compatibility ther

own environment’s the keeps closure The .realizations) (Tight **6.5 نکته**
ad- An it. around route one the is hosting names؛ used the off claims
identity unoccupied an at machine a host may environment missible
and — say caller، a of instance second invented an — name used a of
through identity own its claiming interface، full a is machine hosted a
a Against .(4.13 (Definition par through than rather silencer own its
an- then can worlds two the name، that admit members whose pair
con- no and occupies، them of one only identity an at differently swer
absorp- refusing without intruder the refuse can class the on dition
of machines hosts 4.15 Definition of environment absorbed the tion:
— names admitted at occupants outside — shape same the exactly
admissibil- which runs، it code the in only intruder the from differs and
realization. the to belongs repair the of Part read. not does rightly ity
out- member every if members its of I subset a at tight system a Call
than rather system، the by occupied ids process whole admits I side
— serves callers admitted its of each parties only serves and names،
callers’، its exactly set party member’s private a makes this serves with
route host the closes Tightness is. subroutine private a what is which
every outside lies id process unoccupied an :I outside members the to
identity every makes condition party the and ،2 line fails and **N** pinned
for- 4.10 Definition where occupied، from speak could caller pinned a
I outside member a to addresses intruder an call a so hosting؛ bids
Exec by or **N** empty blocker’s the by too، side ideal the on refused، is
to route the is close not does tightness What occupant. no finding
used bare a admits that member interface an itself: I of member a
fake hosted a by reachable is — 1 Chapter of pattern local the — name
con- tightness and be، would member private a as exactly instance
Z ∪ **A** only admitting or too، those Pinning .I of member no strains
do we it؛ close would directly، them reaches environment the where
hygiene a is Tightness here. condition sufficient general a pursue not
necessary protocol، subroutine-respecting a of spirit the in condition
when of characterization full a leak؛ to not members private the for
open. leave we faithful are pairs blocked against comparisons

فصل 7

جابه‌جایی و تغییرنام

ses- own its reads hardly code functionality's a that records 1 Chapter the in ،Guard of 3 line at occurs s number: instance its never and sion This nowhere. occurs i and ،1.3 Definition in and it، invoke that wrappers and sessions Renaming statement. a into observation that turns chapter at proved emulation an so executions، of isomorphism an is instances same the at and ε same the within other، every at holds id process one func- a that — more no and needed is condition hygiene One budgets. every from reachable be session every outside from reachable tionality isolat- and — satisfies specifies paper the machine every which session، single a for written be realization a lets what is That work. the half is it ing rest. the covering as read and is، 15 Chapter of $\mathcal{F}_{\mathrm{Sig}}$ as session،

تعریف 7.1 (Relocation). A relocation is a permutation θ of the process ids such that

(i) $\theta(q).F = q.F$ for every q؛

(ii) $\theta(q).s = \theta(q').s \iff q.s = q'.s$ for all q, q' with $q.F, q'.F \in \{0,1\}^*$؛

(iii) $\theta(q) = q$ for each of the three ids $q \in \{(A,0,0),(Z,0,0),(C,0,0)\}$ the execution supplies.

names standard the on that exactly say (ii) and (i) Conditions

$$\theta(F, s, i) \;=\; \big(F,\ \hat{\theta}(s),\ \theta_{F,s}(i)\big)$$

per- a session، and name each for and، ids session of $\hat{\theta}$ permutation a for a of components two the write we — numbers instance of $\theta_{F,s}$ mutation spo- being ι and σ own، their of letters with than rather way this relocation proofs the notation whose ،4.17 Definition and 4.19 Proposition by for ken name by match wf and Guard because kept، are Names import. below

permuted are sessions functionality؛ different a become may nothing and
not and session a share ids two whether asks 3 line because blocks، as
component that block، a within freely permuted are instances and which؛
nowhere. read being

code: in literals as occur that ids process three the pins (iii) Condition
of $(C, \mathrm{id}.P)$ read register the in ،C ؛Mediate of $(A, \mathrm{id}.P)$ hook the in ،A
identity Every .5 line of relay the in and 2 line at ،Z and ؛6 and 2 lines
in party a with paired three these of one is paper the in outright named
no and three those pins it and — all them pins three the pinning so scope،
the and name، reserved a of id every pinning as read be not must It more.
(A, s, i) triple a holds **A** set the 1 Chapter by cosmetic: not is difference
of all fixing condition a so occurring، ever $(A, 0, 0)$ only ،i and s every for
every at $\hat{\theta}(s) = s$ give would unrestricted (ii) with together read and them
to (ii) Restricting all. at session a moves that relocation no leave and s
determined is $\hat{\theta}$ care: same the of half other the is names standard the
move to left are occur never that name reserved a of ids the and there،
where is shows 7.5 Proposition which class، own their inside harmlessly
stay. they

freedom That it. move to free is $\hat{\theta}$ and ،0 session pins here Nothing
only the is this since where، exactly seeing worth is it and price، a has
com- 3 Line another. from session one tells framework the which at point
three the and callee، the of tested is global while session caller's the pares
to $\mathcal{Z}$ or $\mathcal{A}$ from call a so ؛0 session in sit supplies execution the machines
own callee's the when exactly line that passes global not is that callee a
session moving relocation a tell would shape that of callee A .0 is session
the callee other every for can: else Nothing alone. it leaving one from 0
before 2 line at refused or global by waived either are machines supplied
gap، the closes callee the on condition one So at. looked is session the
.θ on restriction a than rather condition hygiene a is it and

تعریف 7.2 (Session-uniform). A functionality $\mathcal{F}$ is session-uniform if

$$\mathcal{F}.\mathbf{N} \cap (\mathbf{A} \cup \mathbf{Z}) \neq \emptyset \quad \Longrightarrow \quad \mathrm{global}(\mathcal{F}),$$

reach- is it if only session every outside from caller a admits it that so
its of each if session-uniform is system A session. every from able
is. hosts it functionality each if environment an and is، members

do from reached be can functionality session-uniform a sessions The it. of needs relocation what is which in، sits it session the on depend not being three all outright، condition the meet machines supplied three The global — $\mathcal{A}$ and $\mathcal{C}$ by the first disjunct of global، $\mathcal{Z}$ by the second — and so does every occupant of the adversary slot، global by that same sec- ond disjunct، and so does a blocker، whose **N** is empty and which line 2 therefore refuses whatever the session. Those are every **N** the paper de- clares؛ $\mathcal{F}_{\mathsf{Sig}}$ takes its as a parameter، where the condition is one line to check. What the condition excludes is the shape Chapter 1 leaves open in saying that a local $\mathcal{F}$ may itself declare whether **A** and **Z** may call it: a local functionality admitting the adversary is reachable from the slot in session 0 and in no other، and that is the one thing a relocation cannot carry. Remark 7.17 records the alternative، a one-line widening of line 3 that would dispense with the condition altogether.

تعریف 7.3 (Action of a relocation). A relocation θ acts on identities by $\theta(\mathsf{pid}, P) := (\theta(\mathsf{pid}), P)$، leaving the party component alone، and on identity sets pointwise. It sends a functionality $\mathcal{F}$ to $\theta\mathcal{F}$ with

$$\theta\mathcal{F}.\mathsf{pid} := \theta(\mathcal{F}.\mathsf{pid}), \quad \theta\mathcal{F}.\mathbf{P} := \mathcal{F}.\mathbf{P}, \quad \theta\mathcal{F}.\mathbf{N} := \theta(\mathcal{F}.\mathbf{N}),$$

$$\theta\mathcal{F}.\mathbf{U} := \theta(\mathcal{F}.\mathbf{U}), \quad \theta\mathcal{F}.\mathsf{par} := \theta(\mathcal{F}.\mathsf{par}),$$

where θ acts on a **U** entry by $\theta(\mathcal{F}', R) := (\theta\mathcal{F}', R)$ and on par through whatever identities and process ids it contains؛ the code of $\theta\mathcal{F}$ is the code of $\mathcal{F}$ with every process id occurring in it — in a call، a target، a claimed identity، a test، or a stored value — replaced by its image؛ and every identity set a machine declares rather than computes is re- placed by its image too، for which a bundle (Definition 4.13) means its occupied set، its outward set and any routing it fixes by hand. It acts on a system memberwise، and on any other machine of the execution the same way، a bundle's hosted members included، so that $\theta\mathcal{Z}$ hosts the images of what $\mathcal{Z}$ hosts. We require the requirement predicates to be equivariant، $R(\mathcal{F}, \mathcal{G}) \iff R(\theta\mathcal{F}, \theta\mathcal{G})$، as serves it is، reading only **P** and a membership of **N**. Every clause above relabels pointwise، so the action is one: id acts as the identity and $(\theta'\theta)\mathcal{F} = \theta'(\theta\mathcal{F})$. In particular θ^{-1} undoes θ on machines as it does on process ids، which is what lets the results below argue in both directions.

Relabelling the process ids a program names is not by itself relabelling what it does، and the gap between the two is a condition on the machine it. without false is 7.9 Lemma because it، state We model.

handles it if identity-atomic is Code .code) (Identity-atomic **7.4 تعریف**
to one project may It tokens. opaque as ids process and identities
name، their of two or them، of two test component؛ party or name its
mem- for one test equality؛ for components، instance or session party،
$\mathcal{F}$.**N** tests 2 line as given، been has code the them of set a in bership
3 and 2 lines as tests، those by them of set a form ؛$\mathcal{I}$ tests Silence and
destruc- and outright؛ one name ؛$\{\mathrm{id}'' \neq \mathrm{id}\}$ forms 5 line and $\bar{\mathcal{I}}$ form
or session a with compute not may It one. carries that value a ture
or session no and ordering، no arithmetic، no number: instance an
pro- or identity an inside from apart own، its on down written instance
whatever step، one as counts these of Each outright. named id cess
answer. one as call a counts 4.30 Definition as involved، identities the

Guard predicate The identity-atomic. is paper this in machine Every
all، at i reads core no else، nothing and equality for sessions compares
We (iii). condition of three the are outright named ids process only the and
7.3 Definition makes what is it and throughout، this be to model the take
merely not does there substitution the code identity-atomic on sound:
op- permitted each behaviour، relabelled the computes but literals rewrite
par- and names leaves θ because projections — equivariant being eration
sets the and injective is θ because membership and equality alone، ties
because identity named a code، the with relabelled are formed and tested
and machine a makes what also is It image. its by it replaces substitution
the in operations same the perform they steps: same the take image its
clause last the and both، on same the being flow control the order، same
have it inside ids the however each for step one charges definition the of
— unchanged over carry 4.30 Definition of budgets the So renamed. been
relocation a grant، not would symbol the by charging model cost a which
session. a of encoding the lengthen to free being

names $\mathrm{id}.i \bmod 2$ returning core A fail. both restriction the Without
in- another to carried untouched؛ it leaves substitution so id، process no
not is machine relabelled the and differently، answers it number stance
in target its misses $(F, \mathrm{id}.s, \mathrm{id}.i+1)$ calling core A relabelled. original the
4.6 Definition — write would anyone machine a is Neither way. same the
stated be to has exclusion the why is which core، arbitrary an permits
assumed. than rather

functionality a $\mathcal{F}$ relocation، a be θ Let .invariances) (Basic **7.5 گزاره**
Then system. a π and

(i) disjoint pairwise carrying occupants and ،$\mathsf{IDs}(\theta\mathcal{F}) = \theta(\mathsf{IDs}(\mathcal{F}))$
them؛ keep sets identity

(ii) $\theta(\mathbf{A}) = \mathbf{A}$ ،$\theta(\mathbf{Std}) = \mathbf{Std}$ so itself، onto class name each carries θ
$\{$pid : local a — names by declared **N** every and ،$\theta(\mathbf{Z}) = \mathbf{Z}$ and
image؛ own its is — them among pid.$F = F^{\uparrow}\}$

(iii) ؛$\mathsf{global}(\theta\mathcal{F}) \iff \mathsf{global}(\mathcal{F})$

(iv) ؛$\langle\theta\pi\rangle = \theta(\langle\pi\rangle) = \langle\pi\rangle$

(v) and ؛$\mathsf{wf}(\pi) \iff \mathsf{wf}(\theta\pi)$ and system، a is $\theta\pi$

(vi) relocation. a again is θ^{-1}

's7.1 Definition of which is see to thing useful the and items، Six .اثبات
an injectivity: and (i) conditions is (i) spends. one each conditions three
(ii) untouched. one and relabelled one fields، two by fixed is set identity
a being and themselves، into map classes name — alone (i) condition is
three the makes what is which well، as them onto partition، a on bijection
only reading global (ii)، from follows then (iii) invariant. sets id reserved
closures name again، (i) condition is (iv) .**Std** of subset a and name a
well-formedness one: substantial the is (v) names. by specified being
waived either is conjunct session the match، names because transports
preserves (ii) condition where ids standard two between read or global by
the runs converse the ؛7.3 Definition by equivariant is R relation the and it،
and (i) conditions — legitimate is that why is (vi) .θ^{-1} on argument same
(iii) and inverse، the of hold they so equivalence، an and identity an are (ii)
pid by fixed is set identity An (i) ids. three those fixes θ because holds
alone؛ second the leaving while first the relabels relocation a and ،**P** and
identities. on injective is θ because survives disjointness

The itself. into maps class name each 7.1 Definition of (i) By (ii)
$\theta^{-1}(x)$ point the F of class the in x for so ids، process the partition classes
؛F of class the is therefore and itself into maps which class، some in lies
unions are sets displayed three The well. as class that onto is map the so
them. of union a is name a by specified ids of set a and classes، name of
is second The .$\mathcal{F}.F \in \{A, Z, C\}$ or $\mathbf{Std} \subseteq \mathcal{F}.\mathbf{N}$ reads $\mathsf{global}(\mathcal{F})$ (iii)
first، the for ؛7.1 Definition of (i) by untouched

$$\mathbf{Std} \subseteq \theta(\mathcal{F}.\mathbf{N}) \iff \theta^{-1}(\mathbf{Std}) \subseteq \mathcal{F}.\mathbf{N},$$

bijection. a being θ above، (ii) item by $\theta^{-1}(\mathbf{Std}) = \mathbf{Std}$ and

keeps θ uses؛ system its names the by specified is closure name A (iv)
union a is $\langle\pi\rangle$ And .$\langle\theta\pi\rangle = \langle\pi\rangle$ and does π names the uses $\theta\pi$ so those،
as $\theta(\langle\pi\rangle) = \langle\pi\rangle$ gives (ii) item so ids، party with crossed classes name of
well.

injective، being θ ids، process distinct carry $\theta\pi$ of members The (v)
un- wrappers their and unchanged being names their standard، are and
an and $\mathcal{F} \in \pi^+$ take ،wf For finite. is family the relabelling؛ by touched
session the For sides. both on match Names .$\mathcal{G}$ witness with $(\mathcal{F}', R)$ entry
con- the waives above (iii) item and global$(\mathcal{G})$ either ،$\mathcal{G}.s = \mathcal{F}.s$ conjunct
the of entries the ،$\mathcal{F}$ is so then and — standard is $\mathcal{G}$ or sides، both on junct
— empty being $\mathcal{C}$.**U** and other each only naming machines supplied three
test. the preserves 7.1 Definition of (ii) and standard are ids both that so
requires. 7.3 Definition equivariance the by $R(\mathcal{F}, \mathcal{G}) \iff R(\theta\mathcal{F}, \theta\mathcal{G})$ Finally
both on identically settled are machines supplied three the of entries The
and above (ii) items by fixed being them of reads serves field every sides،
a ،θ^{-1} on argument same the run converse، the For .7.1 Definition of (iii)
.7.3 Definition from $\theta^{-1}(\theta\pi) = \pi$ using (vi)، item by relocation
of identity an as stated are 7.1 Definition of (ii) and (i) Conditions (vi)
does (iii) inverse؛ the of hold so and equivalence، an as and components
$\square$ ids. three those fixes θ because

declarations **N** pinned the only (ii)، By word. a deserve these of Two
pin. they members the with move they and all، at move 6.5 Remark of
Proposi- makes what is which invariant، are closures name (iv) by And
as exactly constrained class environment an against out come 7.10 tion
it. to corresponding merely one than rather before

تعریف 7.6 (Pid-blind core). A core is pid-blind if $\theta Op = Op$ for every
relocation θ: it names no process id that a relocation moves. It may
compare the sessions of two identities in scope، as Guard does، and it
may name one of the three process ids that (iii) of Definition 7.1 fixes
— but not some other id of a reserved name، which (iii) leaves free to
move.

read 2 Chapter of wrappers The pid-blind. is paper this in core Every
to 6 and 2 lines and $(A, \mathrm{id}.P)$ to hooks Mediate only؛ parties and names
names on tests by par from $\bar{\mathcal{I}}$ computes box $\mathcal{Z}$ the reserved؛ both ،$(C, \mathrm{id}.P)$
and $\mathrm{id}.P$ reads $\mathcal{F}_{\mathsf{Sig}}$ and names؛ payload its call the places $\mathcal{D}$ parties؛ and
identity. an of else nothing and **P** own its

different a and needs، 7.9 Lemma than property stronger a is That
execu- an of machine every which ،7.4 Definition needs lemma The one.
slot adversary the of occupant the and environment the meet، must tion
the name to having each pid-blind، be can those of neither and — included
con- the of reading the is buys pid-blindness What drives. it ids process
relabelled fields five with π is $\theta\pi$ cores، pid-blind of system a for clusion:
protocol، the is protocol relocated a so rewritten، code of line a not and
elsewhere. run

answers: definition the question the of half other the lemma، the Before
all. at reads execution an fields which

agreeing functionalities standard be $\mathcal{F}'$ and $\mathcal{F}$ Let .fields) (Inert **7.7 گزاره**
code. and par ،**P** ،pid on

(i) no then alone، **U** in differ they that so well، as **N** on agree they If
putting by got systems two the in them: distinguishes execution
the produces activation every id، process that at other the or one
configuration. same

(ii) same the on and state same the from then alone، **N** in differ they If
1 line at refuse wrappers both either ،id' from id.FullOp(in) call
point same the at core same the enter and admit both or ،rej with
state. same the in

اثبات. By inspection of where the two fields are read during an execution.
The field **U** is read only by the static check of Definition 1.3 and the results
about it، never by Exec or by any interface، which gives (i). The readers of
N inside an execution are two: line 2 of Guard، and the global predicate
and hence line 3. Both sit inside Guard$_{\mathcal{F}}$، which line 1 of the full interface
requires before anything else. (The field is read outside an execution too
— by Definition 1.3 through those two، by Definition 6.1، and by the typing
conditions of Definition 4.7 and Remark 6.5 — but those are conditions on
a system، not steps of one.) So the field decides that verdict and is read
nowhere else in the wrapper: the lines between line 1 and the core — the
register read، the corruption gate، the first Mediate hook، the silenced set
$\{\mathrm{id}'' \neq \mathrm{id}\}$ — are functions of pid، **P**، par، the code and the call، so two
wrappers that both admit reach line 5 together and enter the same core
at the same point in the same state. That is (ii). ☐

So **N** is access control and nothing else، and **U** is inert in every execu-
tion، as Remark 4.8 already records for the adversary slot. Inertness past

to strengthened be cannot it and step، one about statement a is guard the admits one call a that is reason obvious The returned. value the about one is one obvious less The apart. states two the sends refuses other the and entered core the because differently، end can admit both call a even that instance an permits 3.6 Section — wrapper very this re-enter may 5 line at the and — identity own its claim core a lets 5 line and re-entered، be to places that core A differs. field the where again، $\text{Guard}_{\mathcal{F}}$ meets call inner $\mathcal{F}$ an separates rej not is answer the if just 1 returns and $\mathtt{id}$ as $\mathtt{id}.\text{FullOp}'()$ ad- both call a on not، does that $\mathcal{F}'$ an from id process own its admitting verdicts guard in differ two the claim: exact the is survives What mitted. separate to enough is inner، or outer verdict، differing single a and alone، them.

> **نکته 7.8** (The two fields that are read). Neither **P** nor par admits a statement of this kind، and for different reasons. The field **P** fixes $\text{IDs}(\mathcal{F})$، hence which calls Exec dispatches to $\mathcal{F}$ at all and the second conjunct of line 1؛ and cores read it directly — Initialize of $\mathcal{F}_{\text{Sig}}$ types $\text{VK} : \mathcal{F}_{\text{Sig}}.\mathbf{P} \to \mathcal{K} \cup \{\square\}$، and the test $\exists\, P' \notin \mathbf{C}$ of Verify quantifies over that domain. So enlarging or shrinking a party set is not a symmetry. What is one is renaming: a permutation β of $\{0,1\}^*$ extended to fix $\mathcal{A}$ and $\mathcal{Z}$، acting on identities، on **P**، on **C** and on the party-indexed state، gives an isomorphism of executions by the argument of Lemma 7.9، provided cores read party ids only through equality and indexing — as $\mathcal{F}_{\text{Sig}}$ does، its $\exists P' \notin \mathbf{C} : \text{vk} = \text{VK}[P']$ being equivariant. The two places a party id enters otherwise survive it as well: line 3 names the party $\mathcal{Z}$، which β fixes، and FullCorrupt tests $\mathtt{id}.P \in \{0,1\}^*$، a set β permutes. The field par is read by the code by definition of the field، and $\mathcal{Z}$.par is read by Definition 4.10 besides؛ there is nothing to be invariant in. A relocation moves it and moves it to itself whenever it is declared by names، which is $\mathcal{Z}.\text{par} := \langle\overline{\pi}\rangle$.

instances and sessions relabelling that says 7.9 Lemma Informally، Proposi- is it observable: nothing changes execution an throughout tion 4.19's induction with θ in place of the absorption bijection، and every check is the equivariance of one line of one wrapper.

> **لم 7.9** (Relocation). Let θ be a relocation، π a session-uniform system، $\mathcal{X}$ an occupant of the adversary slot (Definition 3.3)، and $\mathcal{Z}$ a session-

is Exec that so ،IDs(π) of identity no at hosting environment uniform
Then four. the at defined

$$\mathsf{Exec}(\theta\pi, \theta\mathcal{Z}, \theta\mathcal{X}, \mathcal{C}) \qquad \text{and} \qquad \mathsf{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})$$

distributed. identically are

two :θ to up read ،$\iota := \theta$ at 4.17 Definition of correspondence The .اثبات
other's the is one when correspond chains caller or points program states،
is proof the and symmetric، is relation the bijection a is θ Since image.
below paragraphs four the step whose activations، on induction an again
so bijectively، relabelled is occupancy that checks Machines discharge.
takes Guards correspondence. in start and defined are executions both
the and injectivity، are two first the turn؛ in Guard of conjuncts three the
place only the is 3 line since work، real does that paragraph one the is third
session- where is this — another from session one tells framework the
would 0 session moving relocation a it without and spent، is uniformity
wrap- the checks register the mediation، Silencers، verdict. the change
corruption all، at image no needs register the that observes and pers،
step، the closes token claims، Routing، alone. party by indexed being
cor- The sides. both on branches same the taking code identity-atomic
paired machine each — $\iota := \theta$ with 4.17 Definition of that is respondence
to up read (C4) and (C3) (C1)، with and — itself with $\mathcal{C}$ image، its with
cor- callers suspended of chains two points، program two states، two :θ
on bijection a is θ Since other. the of -imageθ the is one when respond
induction an is proof the before as and symmetric، is relation the identities
step. the discharge below paragraphs The activations. on

(i) by bijectively، relabelled is occupancy and system a is $\theta\pi$ Machines.
the at hosts $\theta\mathcal{Z}$ and IDs(π) of clear hosts $\mathcal{Z}$ since ؛7.5 Proposition of (v) and
right the on disjoint pairwise are occupants the hosts، $\mathcal{Z}$ what of images
1 Line defined. are executions both so left، the on are they where exactly
state initial member's a and sides، both on member every and $\mathcal{C}$ initialises
outset. the at (C1) giving relabelled، image's its is

.$\theta\mathcal{F}$ at image its and $\mathcal{F}$ at $\mathtt{id}'$ from $\mathtt{id}.\mathsf{FullOp}(\mathtt{in})$ call a Take Guards.
$\mathtt{id}.\mathrm{P} \in$ **P** and injectivity، by $\theta(\mathtt{id}.\mathsf{pid}) = \theta(\mathcal{F}.\mathsf{pid})$ iff $\mathtt{id}.\mathsf{pid} = \mathcal{F}.\mathsf{pid}$:1 Line
by again ،$\theta(\mathtt{id}'.\mathsf{pid}) \in \theta(\mathcal{F}.\mathbf{N})$ iff $\mathtt{id}'.\mathsf{pid} \in \mathcal{F}.\mathbf{N}$:2 Line untouched. is
3 Line alone. leaves θ components reads test party the and injectivity،
of (iii) by sides two the on agrees disjunct second whose disjunction a is
together sides both on fails it — fails global$(\mathcal{F})$ suppose so ،7.5 Proposition

is execution the of machine Every first. the consider and — neither on or
and $\mathcal{C}$ ،$\mathcal{X}$ and hypothesis، by copies hosted 's$\mathcal{Z}$ and π session-uniform:
global which ،Z and C ،$\mathcal{A}$ are ids process their because interfaces own 's$\mathcal{Z}$
claim a and ،**Z** nor **A** neither meets $\mathcal{F}$.**N** So disjunct. second its by admits
fixing (iii) — sides both on 2 line at refused is $\mathcal{Z}$ of or $\mathcal{A}$ of id the carrying
sets two those does، $\mathcal{F}$.**N** as just **Z** and **A** missing $\theta(\mathcal{F}.\mathbf{N})$ and ids، those
cannot id supplied third The .7.5 Proposition of (ii) by -invariantθ being
carry therefore 3 line reach that claims The calls. no placing $\mathcal{C}$ all، at arrive
once $\{0,1\}^*$ in lying $\mathcal{F}$ of name the ،$\mathtt{id}$ does so and ids، process standard
holds it when exactly images the of holds $\mathtt{id}'.s = \mathtt{id}.s$ so fails؛ global$(\mathcal{F})$
session- paragraph one the is This .7.1 Definition of (ii) by originals، the of
change would 0 session moving relocation a it without for؛ is uniformity
callee، every at sides، both on same the is verdict the So here. verdict the
requirements three whose ،3.2 Section of interface leakage the including
read. lines these what only read

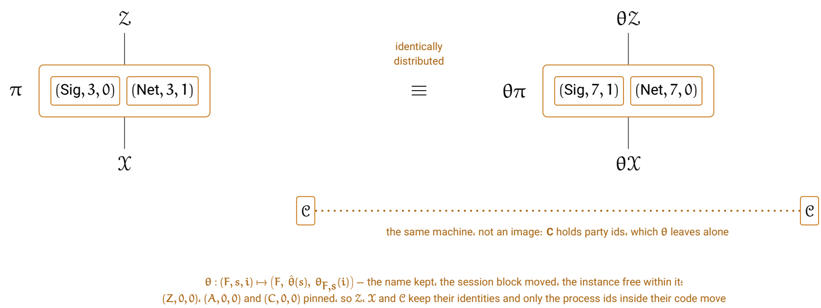
$\{\mathtt{id}'' \neq$ is 5 line of set silenced The register. the mediation، Silencers،
both on identity own its claims core a so ،$\{\mathtt{id}'' \neq \theta\mathtt{id}\}$ is image whose ،$\mathtt{id}\}$
$\theta(\mathrm{par})$ from computed is box $\mathcal{Z}$ the of $\bar{\mathcal{I}}$ set The else. nothing and sides
on computed set the of image the is it so parties، and names on tests by
wrapper The admits. that what of image the exactly admits and left، the
ad- 6 and 2 lines ؛$(A, \mathtt{id}.P)$ to hooks and $\mathtt{id}'.F$ and $\mathtt{id}.P$ ،**C** reads Mediate
(iii) that three the of one is here outright named id Every .$(C, \mathtt{id}.P)$ dress
collects **C** image: no needs itself register the And fixes. 7.1 Definition of
ma- same the literally is $\mathcal{C}$ so touch، not does relocation a which ids، party
Corruption configurations. corresponding at set same the holding chine
this. buys what is (3.1 (Section alone party by indexed being

its of occupant the to call a dispatches Exec token. claims، Routing،
by does occupancy while θ under correspond targets and identity، target
answered is identity unoccupied an to addressed call a paragraph؛ first the
by correspond Claims right. the on is image its when exactly left the on rej
a ،(7.4 (Definition identity-atomic being code The paragraph. previous the
-θ are that calls place and branches same the take image its and machine
values with point program same the reach they so another، one of images
is which order، same the in coins same the reads each and ؛θ by related
on operation one having Exec unchanged، is discipline token The (C2).
and (C3) is which arising، suspensions of chain same the and sides both
(C4).

being output the alike، output configurations halting Corresponding
the So fixes. 7.1 Definition of (iii) identity an ،(Z, Z) at return 's$\theta\mathcal{Z}$ and 's$\mathcal{Z}$

of probability the on particular in distributions، as agree executions two .1 returning □

same the and execution، One state. to than see to easier is lemma The side: by side renamed، id process every with execution



شکل 4: One execution beside the same execution with every process id renamed. Lemma 7.9 is that the two agree as distributions.

name The systems. two the in legible is do may relocation a Everything quietly may nothing and name by match wf and Guard because kept، is both – block a as moves session The functionality. different a become and caller whether asks 3 line because – together 7 to 3 from go members instance the And in. are they one which never and session a share callee because places، changing here 1 and 0 block، that within move to free is pinned، are literals as occur that ids three The it. reads code any of line no process the only and sides both on identities same the at sit $\mathcal{C}$ and $\mathcal{X}$ ،$\mathcal{Z}$ so rewritten. are code their inside ids

be- set، same the holding machine same the is it that: even not is $\mathcal{C}$ Index- party. a touch not does relocation a and ids party collects **C** cause worth is it and this، buys what is (3.1 (Section alone party by corruption ing have would id process by indexed register a – buys it much how noticing pattern corruption the and else، everything like θ by across carried be to disturb. could relocation a thing a be then would

looks lemma the way this Drawn give. cannot picture the warning One mediation every silencer، every conjunct، guard every is: nearly it and free، either which of all ids، pinned three the and parties names، only reads hook and test، session single the is exception The it. by fixed are or θ survive – remark a than rather hypothesis a is session-uniformity reason the is it them of one in sitting but session every outside from reachable callee a

that θ a after and before differently machines supplied the answer would
one that at apart come would executions two the and ،0 session moves
is rest the work؛ the of whole the is proof the of paragraph That line.
shows. already picture the bookkeeping

sys- φ ،π and relocation a be θ Let .transports) (Emulation **7.10 گزاره**
the of environment every and them of two the let (iii) and (ii) for tems؛
Then need. not does (i) which session-uniform، be named classes

and adversaries to adversaries carries θ and ،$\mathbb{Z}_{\theta\pi,\theta\varphi} = \theta\big(\mathbb{Z}_{\pi,\varphi}\big)$ (i)
؛4.4 Section of dummy the fixing simulators، to simulators

UC- $\theta\pi$ then ،$(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ against ε within φ UC-emulates π if (ii)
and ؛$(\theta\mathbb{A}, \theta\mathbb{S}, \theta\mathbb{Z})$ against ε within $\theta\varphi$ emulates

$\mathsf{Adv}^{\mathsf{uc}}_{\theta\pi,\theta\varphi}[\mathbf{a}, \mathbf{s}, \mathbf{z}] \;=$ ،$\mathbf{a}, \mathbf{s}, \mathbf{z}$ every and systems budgeted for (iii)
.$\mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}[\mathbf{a}, \mathbf{s}, \mathbf{z}]$

.7.5 Proposition of part different a on rests each and claims، Three .اثبات
trans- 4.10 Definition of clauses both classes: the about bookkeeping is (i)
host- the and -invariantθ are closures name because clause par the port،
because ways both going containment the with injectivity، by clause ing
to simulators and adversaries to go adversaries too؛ relocation a is θ^{-1}
fixes، θ ones the are read definitions those fields the because simulators
im- then is (ii) pid-blind. being core its outright، fixed is dummy the and
simulator returned the world، each to applied 7.9 Lemma from mediate
a is θ form: concrete the is (iii) alone. $\theta\mathcal{A}$ on depends which ،$\theta\mathcal{S}$ being
conclu- the class the onto classes budgeted three the of each of bijection
changes bijections along function a reindexing and over، quantifies sion
preserved is Session-uniformity infimum. an nor supremum a neither
Def- (i)، For part. each in than rather once checked is which throughout،
outright: -invariantθ is clause par The .$\mathcal{Z}$ of things two asks 4.10 inition
$\mathcal{Z}$.par when exactly $\theta(\langle\pi\rangle\cup\langle\varphi\rangle) = \langle\theta\pi\rangle\cup\langle\theta\varphi\rangle$ contains $\theta\mathcal{Z}$.par $= \theta(\mathcal{Z}$.par$)$
.7.5 Proposition of (iv) by invariant being closures name ،$\langle\pi\rangle\cup\langle\varphi\rangle$ contains
of images the at hosting $\theta\mathcal{Z}$ injectivity، by transports clause hosting The
$\mathsf{IDs}(\pi) \cup \mathsf{IDs}(\varphi)$ of image the being $\mathsf{IDs}(\theta\pi) \cup \mathsf{IDs}(\theta\varphi)$ and hosts $\mathcal{Z}$ what
a being θ^{-1} ways، both goes containment The proposition. that of (i) by
iden- whose ،$\mathsf{IDs}(\mathcal{A})$ occupies adversary an slot، the For (vi). by relocation
components party and fixes 7.1 Definition of (iii) id process the carry tities
relabels θ so pid-blind؛ is which wrapper، 's$\mathcal{A}$ carries it and alone، leaves θ
4.7 Definition shape. same the of adversary an is $\theta\mathcal{A}$ and alone core the
The simulators. to go simulators so (ii)، by fixes θ set a ،$\mathbf{N} \supseteq \mathbf{Std} \cup \mathbf{Z}$ asks

so — names payload its call the places 3 line — pid-blind is core dummy's
whose ،3.4 Section of $\mathcal{A}$ the of claimed is kind the of Nothing .$\theta\mathcal{D} = \mathcal{D}$
it. needs below nothing and arbitrary، leaves 4.6 Definition core

hypothesis the simulator the be $\mathcal{S} \in \mathbb{S}$ let and $\mathcal{A} \in \mathbb{A}$ fix (ii)، For
so alone، $\theta\mathcal{A}$ on depends it ؛$\theta\mathcal{S} \in \theta\mathbb{S}$ return ،$\theta\mathcal{A} \in \theta\mathbb{A}$ Given it. for supplies
7.9 Lemma ،$\theta\mathcal{Z} \in \theta\mathbb{Z}$ any For met. is 4.11 Definition of order quantifier the
gives world each to applied

$$\mathsf{Adv}^{\mathsf{uc}}_{\theta\pi,\theta\varphi}\big(\theta\mathcal{Z}, \theta\mathcal{A}, \theta\mathcal{S}\big) = \mathsf{Adv}^{\mathsf{uc}}_{\pi,\varphi}\big(\mathcal{Z}, \mathcal{A}, \mathcal{S}\big) \leqslant \varepsilon,$$

right. the on two the of -imagesθ the being left the on executions two the
.$\theta\mathbb{Z} \subseteq \mathbb{Z}_{\theta\pi,\theta\varphi}$ emulation، of statement legal a is conclusion the (i) By

of (ii) by **Z** and **A** fixes θ throughout: preserved is Session-uniformity
exactly session-uniform is $\theta\mathcal{F}$ so (iii)، its by global and 7.5 Proposition
memberwise. environment an of and system a of same the and is، $\mathcal{F}$ when
well. as respect this in θ under closed therefore is below class Every

the answer and steps same the take image its and machine a (iii)، For
therefore flow control its and identity-atomic being code the calls، same
and system budgeted a of member a ؛(7.4 (Definition both on same the
bijec- a is θ So reason. same the for budget same the carry image its
quantifies conclusion the class the onto classes three the of each of tion
(i)، by $\mathbb{Z}_{\theta\pi,\theta\varphi}(\mathbf{z})$ onto $\mathbb{Z}_{\pi,\varphi}(\mathbf{z})$ and themselves، onto $\mathbb{S}(\mathbf{s})$ and $\mathbb{A}(\mathbf{a})$ over:
of advantage the makes 7.9 Lemma .θ^{-1} by induced inverse with each
extrema nested two the so triples، corresponding at equal 4.5 Definition
function one to applied operators three same the are 4.31 Definition of
supremum a neither changes reindexing and bijections، along reindexed
Def- of finiteness the what here؛ needed is finiteness No infimum. an nor
separate a is which attained، are extrema the that is settles 4.31 inition
alike. sides both on holds and matter □

proof a and relocation، to up system a of property a thus is Emulation
family a for so say To outset. the at instance the and session the fix may
related are ids process of lists two when know to needs one instances، of
conditions. three is answer the and all، at relocation a by

$\vec{q} = (q_1, \dots, q_m)$ Let .another) to relocates tuple one (When **7.11 گزاره**
name. standard of ids process of tuples be $\vec{q}\,' = (q'_1, \dots, q'_m)$ and
for if، only and if j every for $\theta(q_j) = q'_j$ with θ relocation a is There

،k and j all

$$q_j.F = q'_j.F, \qquad \big(q_j.s = q_k.s \iff q'_j.s = q'_k.s\big),$$
$$\big(q_j = q_k \iff q'_j = q'_k\big).$$

reloca- a by related are name one of ids process two any particular In
tion.

or- in off read 7.1 Definition of conditions three the is Necessity .اثبات
assignment، session the from $\hat{\theta}$ build construction: a is Sufficiency der.
the between bijection well-defined a makes condition second the which
be- permutation a to it extend and sides، two the on occurring sessions
name each at then، side؛ each on size one of is over left is what cause
and third the which assignment، instance the from $\theta_{F,s}$ build session، and
the Taking extensible. again hence and injective make conditions fourth
(iii) and ،$\vec{q}\,'$ to $\vec{q}$ carrying relocation a assembles else everywhere identity
con- the is Necessity touched. never were ids reserved the because holds
shared sessions (i)، by names order: in off read 7.1 Definition of ditions
injective. being θ by equality and (ii)، by not or

occurring sessions many finitely the be S' and S let sufficiency، For
well-defined a $q_j.s \mapsto q'_j.s$ makes condition second The .$\vec{q}\,'$ in and $\vec{q}$ in
over left is what so — image an being $q'_j.s$ every onto، — $S \to S'$ bijection
to extends it and alike، infinite or finite size، one of is sides two the on
$s \in S$ session a and F name a fix Now ids. session of $\hat{\theta}$ permutation a
instance Their .$q_j.s = s$ and $q_j.F = F$ with j indices the consider and
then $q_j.i = q_k.i$ if injective: and defined well is $q_j.i \mapsto q'_j.i$ assignment
and agree؛ instances the and condition fourth the by $q'_j = q'_k$ so ،$q_j = q_k$
sessions and names their while $q'_j \neq q'_k$ so ،$q_j \neq q_k$ then $q_j.i \neq q_k.i$ if
in- the between bijection a again is Each differ. instances the and agree،
permutation a to extends it so sides، two the on occurring numbers stance
θ let and session، and name other every at identity the take did؛ $\hat{\theta}$ as $\theta_{F,s}$
iden- the as and names standard the on $(F,s,i) \mapsto (F,\hat{\theta}(s),\theta_{F,s}(i))$ as act
the because (iii) ،7.1 Definition meets θ Then ones. reserved the on tity
a claim، last the For .$\vec{q}\,'$ to $\vec{q}$ carries it and outright، fixed are ids reserved
□ vacuously. conditions three the meets name one of q' ،q pair single

and $\vec{q} \mapsto \pi(\vec{q})$ maps of pair a Call .suffices) instance (One **7.12 نتیجه**
$\varphi(\theta\vec{q}) = \theta\big(\varphi(\vec{q})\big)$ and $\pi(\theta\vec{q}) = \theta\big(\pi(\vec{q})\big)$ if family instance an $\vec{q} \mapsto \varphi(\vec{q})$

$\pi(\vec{q})$ by occupied ids process the lists $\vec{q}$ where ،θ relocation every for
speci- the and subroutines private its protocol، a — together $\varphi(\vec{q})$ and
with moving and tuple one by indexed against measured is it fication
UC- $\pi(\vec{q})$ that and session-uniform are $\varphi(\vec{q})$ and $\pi(\vec{q})$ Suppose it.
to down cut $\big(\mathbb{A}(\mathbf{a}), \mathbb{S}(\mathbf{s}), \mathbb{Z}_{\pi(\vec{q}),\varphi(\vec{q})}(\mathbf{z})\big)$ against ε within $\varphi(\vec{q})$ emulates
$\vec{q}\,'$ every at Then .$\vec{q}$ single a for environments، session-uniform the
UC-emulates $\pi(\vec{q}\,')$ ،7.11 Proposition of conditions three the meeting
cut-down same the and $\mathbb{S}(\mathbf{s})$ ،$\mathbb{A}(\mathbf{a})$ against ε same the within $\varphi(\vec{q}\,')$
id، process one occupying family a for particular، In pair. that for class
name. that of id every at emulation gives id one at emulation

equivari- ،$\vec{q}\,'$ to $\vec{q}$ carrying relocation a supplies 7.11 Proposition .اثبات
and ،$\vec{q}\,'$ at pair the to systems of pair the carrying one into it turns ance
re- the of session-uniformity — statement the transports 7.10 Proposition
proposi- that of proof the as original، the of that from coming pair located
one-id a :7.11 Proposition of clause final the is claim last The records. tion
□ name. its of id other any against conditions three the meets tuple

satisfiable، condition the makes what is tuple whole the by Indexing
id process principal the by indexing since recording، worth is point the and
given a fix may relocation A not. does — writes one thing first the — alone
in equivariance so id، another at sitting member private a move still and q
whole the by fixed id an have to $\pi(q)$ of member every force would alone q
three، reserved the and itself q are ids such only the and ؛q of stabiliser
Equivariance .q fixes that relocation some by movable being other every
all، at members private no with systems only admits therefore alone q in
wants. anyone case the not is which

to has else what asking and instead، systems of pair a for Written
be can name a of instance new A two. in splits statement the move،
(ii) condition is reason the and cannot، session new a itself؛ by up taken
nothing so blocks، as sessions permutes relocation a :7.1 Definition of
it. in behind another leaving while s session of out member one carry can

φ and π Let .session) one moving instance، one (Moving **7.13 نتیجه**
the and one at member a carrying each systems session-uniform be
occu- $q.\mathrm{F} = q_0.\mathrm{F}$ with id process a be q let and ،q_0 id process same
against ε within φ UC-emulates π Suppose system. neither by pied
the that is conclusion the below case each In .$\big(\mathbb{A}(\mathbf{a}), \mathbb{S}(\mathbf{s}), \mathbb{Z}_{\pi,\varphi}(\mathbf{z})\big)$

the and $\mathbb{S}(\mathbf{s})$ ،$\mathbb{A}(\mathbf{a})$ against ε same the within UC-emulates pair moved
.**z** at pair that of class admissible largest

(i) If $q_0.s = q.s$، the transposition $\theta := (q_0\ q)$ of instance numbers
the at name $q_0.F$ in that session is a relocation، and it fixes the
$\theta\varphi$ and $\theta\pi$ So system. either of member other every of id process
else nothing and q to carried q_0 at instance the with φ and π are
a in instance moved the naming member a that save — disturbed
6.5 Remark of manner the in callers its among it pinning by field،
it. with along carried mention that has otherwise، or

(ii) If $q_0.s \neq q.s$، no relocation carries q_0 to q while fixing a process
system a of member no so — $q_0.s$ session in name standard of id
q_0 carrying One entire. moves block the and fixed، is there lying
ses- in member a has system neither addition in if and exists؛ q to
and π of member each carries that chosen be may one ،$q.s$ sion
process the fixes and $q.s$ session into $q_0.s$ session in lying φ of
hypoth- further that Without elsewhere. lying member every of id
same the by — too move $q.s$ session in already members the esis
one fix can q to q_0 carrying relocation no ،q at applied argument
blocks two with pair a about then is conclusion the so — them of
one. than rather moved

اثبات. Both parts are Proposition 7.10 once a relocation has been exhib-
ited، so the work is exhibiting one. For (i) the transposition of two ids of
a common session does it، and the check is that it moves nothing else
occupied. For (ii) the negative half comes first — a relocation carrying q_0
to an id of a different session moves every standard id of that session،
by condition (ii) of Definition 7.1 — and the positive half then constructs
the relocation that moves session $q_0.s$ and fixes every third session. The
last sentence applies the negative half to θ^{-1}، which is a relocation by (vi)
of Proposition 7.5. Both parts are Proposition 7.10 once the relocation is
exhibited، its part (iii) supplying the classes. For (i)، the transposition has
$\hat{\theta} = \mathrm{id}$، so conditions (i) and (ii) of Definition 7.1 hold، and (iii) does be-
cause $q_0.F$ is the name of a member of a system and so none of A، Z، C.
It moves only the two ids q_0 and q، and q is occupied by neither system،
so every other occupied id is fixed؛ what is not fixed is a field naming q_0،
which Definition 7.3 relabels.

For (ii)، suppose $\theta(q_0) = q$ and let p be any process id of standard
name with $q_0.s = p.s$؛ by condition (ii) of Definition 7.1، $\theta(p).s = \theta(q_0).s =$
$p.s \neq q.s$، so $\theta(p) \neq p$. For existence، take $\hat{\theta}$ the transposition of the

name the at $q.i$ to $q_0.i$ carrying permutations instance and sessions two $q_0.s$ session carries θ That elsewhere. identity the ،$q_0.s$ session in $q_0.F$ lies member no if session؛ third a of id every fixes and $q.s$ session into .$q_0.s$ session of those exactly are moves it members the ،$q.s$ session in in lying member a be m let and q to q_0 carry θ let sentence، final the For carries and 7.5 Proposition of (vi) by relocation a is θ^{-1} Then .$q.s$ session every moves it to applied claim negative the so ،$q_0.s \neq q.s$ with q_0 to q give would $\theta(m) = m$ and them؛ among m ،$q.s$ session of id standard
□ too. m moves θ so ،$\theta^{-1}(m) = m$

permutation a components only the not are instances and Sessions a raises: 5 Chapter question a settles what is third the and move، can ،(5 (line name own game's the carrying claim a under $\mathcal{F}$ exercises game protocol a inside ships that $\mathcal{F}$ the while game، the name must $\mathcal{F}$.**N** so field the is it and field، one in differ two The instead. protocol that admits reads. Guard

> **تعریف 7.14** (Renaming). A renaming is a permutation ν of the process ids of the form
>
> $$\nu(F, s, i) = \big(\bar{\nu}(F),\ s,\ i\big)$$
>
> identi- on acts It .C and Z ،A fixing names the of $\bar{\nu}$ permutation a for in and prescribes، 7.3 Definition as sets declared and code fields، ties، image. its by code in outright occurring name every replaces addition

substi- 7.3 Definition before. not was and needed is clause extra The a what being names enough، is relocation a for which ids، process tutes name bare a against test a so them، moves renaming a keeps؛ relocation dis- 4.4 Section of dummy the $id'.F = Z$ the or ،Mediate in $id'.F \neq A$ — travels it paper this of machines the For too. travel to has — on patches so fixes، $\bar{\nu}$ three the are outright mention they names only the nowhere: fields. only all، at code of line no changes renaming a

> **گزاره 7.15** (Renaming). Let ν be a renaming. Then
>
> (i) ؛$\nu(\mathbf{Z}) = \mathbf{Z}$ and $\nu(\mathbf{A}) = \mathbf{A}$ ،$\nu(\mathbf{Std}) = \mathbf{Std}$ ؛$\mathsf{IDs}(\nu\mathcal{F}) = \nu(\mathsf{IDs}(\mathcal{F}))$ when exactly session-uniform is $\nu\mathcal{F}$ ؛$\mathsf{global}(\nu\mathcal{F}) \iff \mathsf{global}(\mathcal{F})$ and ؛$\mathsf{wf}(\pi) \iff \mathsf{wf}(\nu\pi)$ with system a is $\nu\pi$ ؛$\langle\nu\pi\rangle = \nu(\langle\pi\rangle)$ is؛ $\mathcal{F}$ renaming؛ a is ν^{-1}
>
> (ii) and slot adversary the of $\mathcal{X}$ occupant every ،π system every for

no with — IDs(π) of identity no at hosting $\mathcal{Z}$ environment every
— them of any of asked session-uniformity

$$\mathsf{Exec}(\nu\pi,\ \nu\mathcal{Z},\ \nu\mathcal{X},\ \mathcal{C}) \quad \text{and} \quad \mathsf{Exec}(\pi,\ \mathcal{Z},\ \mathcal{X},\ \mathcal{C})$$

and distributed؛ identically are
$(\mathbb{A}, \mathbb{S}, \mathbb{Z})$ against ε within φ UC-emulates π if ؛$\mathbb{Z}_{\nu\pi,\nu\varphi} = \nu(\mathbb{Z}_{\pi,\varphi})$ (iii)
the and ؛$(\nu\mathbb{A}, \nu\mathbb{S}, \nu\mathbb{Z})$ against ε within $\nu\varphi$ UC-emulates $\nu\pi$ then
budget. every at agree 4.31 Definition of advantages concrete

and re-run، proofs relocation corresponding the are parts three All .اثبات the overview is what changes. Names are now permuted rather than car- ried onto themselves، which is the point of the definition؛ the reserved classes and **Std** stay fixed، and with them global and session-uniformity، while a name closure is carried rather than fixed. In (ii) the one hard paragraph of Lemma 7.9 — the session conjunct — becomes trivial، a re- naming moving no session، which is why no session-uniformity hypoth- esis appears here at all. In (iii) what is lost is the reading، not the result: the environment class of the conclusion corresponds to the hypothesis's rather than being it. (i) is the proof of Proposition 7.5 with two changes of bookkeeping. Name classes are now permuted rather than carried onto themselves، so an **N** declared by names becomes the one declared by the images — which is the whole point of the definition — while **Std**، being the union of every standard class، and **A** and **Z**، being single reserved classes، are fixed، and with them global، Definition 7.2 and the reserved ids of (iii) of Definition 7.1. And a name closure is carried rather than fixed، $\langle\rangle$ being specified by the names a system uses. For wf the session conjunct is untouched، ν moving no session، and the name match holds on one side exactly when on the other.

(ii) is the proof of Lemma 7.9 verbatim but for its one hard paragraph، which becomes trivial: line 3's first disjunct is literally unchanged، ν leav- ing every session alone، and its second is preserved by (i). Nothing needs excluding، which is why no session-uniformity appears here. Every other check reads a name for equality or a membership in a set that travels with the code، and Definition 7.4 together with the extra clause of Defi- nition 7.14 makes the substitution compute the relabelled behaviour as before.

(iii) is the proof of Proposition 7.10، which used of the closures only that $\nu(\langle\pi\rangle \cup \langle\varphi\rangle) = \langle\nu\pi\rangle \cup \langle\nu\varphi\rangle$ and that they never stay put. What is

class environment the :7.5 Proposition after recorded reading the is lost
than rather hypothesis the of one the to corresponds conclusion the of
□ images. the name now must par since identically، constrained being

> and renaming a be ν Let .immaterial) is name caller's (The **نتیجه 7.16**
> $\nu\gamma =$ Then .(5.1 (Definition $\mathcal{F}$ for system game a be $\gamma = \{\mathcal{G}\mathfrak{m}\} \cup \sigma$ let
> every for and ،$\nu\mathcal{F}$ for game a is $\nu\mathcal{G}\mathfrak{m}$ system، game a is $\{\nu\mathcal{G}\mathfrak{m}\} \cup \nu\sigma$
> $\mathcal{G}\mathfrak{m}$ of Fin finalization
>
> $$\mathsf{Adv}^{\mathsf{Fin}}_{\nu\mathcal{G}\mathfrak{m},\nu\sigma,\nu\mathcal{Z},\nu\mathcal{X}} = \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z},\mathcal{X}}$$
>
> Since slot. adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every for
> reads form abbreviated the ،$\nu\mathcal{D} = \mathcal{D}$
>
> $$\mathsf{Adv}^{\mathsf{Fin}}_{\nu\mathcal{G}\mathfrak{m},\nu\sigma,\nu\mathcal{Z}} = \mathsf{Adv}^{\mathsf{Fin}}_{\mathcal{G}\mathfrak{m},\sigma,\mathcal{Z}}.$$
>
> environments of class a against ε within Fin property the has σ So
> class. that of image the against ε within it has $\nu\sigma$ when exactly

check game، a being $\nu\mathcal{G}\mathfrak{m}$ For (i). is wf$(\nu\gamma)$ with system a is $\nu\gamma$ That .اثبات
moving ν ،$(\bar{\nu}(\mathcal{G}\mathfrak{m}.\mathrm{F}), \mathbb{0}, \mathbb{0})$ is id process its :5.1 Definition of fields four the
is it so ،$\nu\mathcal{F}$.**P** $= \mathcal{F}$.**P** and untouched is **P** its instance؛ nor session neither
،$(\nu\mathcal{F}, \text{serves}_{\mathcal{G}\mathfrak{m}})$ is entry its (i)؛ by $\nu(\mathbf{Z}) = \mathbf{Z}$ is **N** its required؛ as $\nu\mathcal{F}$.**P** $\cup \{Z\}$
still 8 Line .**N** of membership a and **P** reading equivariant، is serves$_{\mathcal{G}\mathfrak{m}}$ and
id، process a of part no being ids party root، the at finalization every serves
on one same the names Fin so name، interface no moves renaming a and
FullFin interface the that event the is Win$_{\mathsf{Fin}}$ advantage، the For sides. both
(ii) by ؛$(\nu\mathcal{G}\mathfrak{m}.\text{pid}, Z)$ to identity that carries ν and ،1 returns $(\mathcal{G}\mathfrak{m}.\text{pid}, Z)$ at
correspondence، the under distributed identically are executions two the
5.8 Definition and sides both on probability same the has event the so
the For (iii). is $\nu\gamma$ for admissible is $\nu\mathcal{Z}$ That number. same the gives
of sets and pid $:= A$ are fields dummy's the :$\nu\mathcal{D} = \mathcal{D}$ form، abbreviated
call the places and $\mathrm{id}'.\mathrm{F} = Z$ on dispatches core its and fixes، (i) that ids
□ moves. mentions it name no so names، payload its

and use، not does γ name a with $\mathcal{G}\mathfrak{m}.\mathrm{F}$ of transposition the at Read
.5.1 Definition of wants one what says this alone، $\mathcal{G}\mathfrak{m}$ by carried $\mathcal{G}\mathfrak{m}.\mathrm{F}$ with
other no match؛ to renamed $\mathcal{F}$.**N** and name any given be may game The
property A either. does code of line no 7.14 Definition by and moves، field

property whose $\mathcal{F}$ the and called، is game its what to tied not therefore is use. will deployment the name caller whatever admit may proves one

a carries It programme. caller different a is gives renaming no What protocol the to game a never protocol، a to protocol a and game a to game the to answers 's$\mathcal{F}$ hands that protocol a and — earnest in $\mathcal{F}$ use will that 7.7 Proposition could. nothing so ،$\mathcal{F}$ of property no satisfies adversary inside but nowhere read being **N** side، other the from gap the narrows control access the and code 's$\mathcal{F}$ about statement a is property a :Guard$_{\mathcal{F}}$ rest the Crossing carry. could field that else nothing about and it، around Relocations crossing. the is 5.10 Proposition and for، is emulation what is so executions، of isomorphism an being each compose، renamings and once. at instance and session name، in moved be may system a

نکته 7.17 an not is Session-uniformity .distinguished) is 0 (Session artefact the of proofs، and the functionality it excludes is a real one. Line 3 compares the caller's session، and $\mathcal{A}$ and $\mathcal{Z}$ sit in session 0، so a functionality that is local in the sense of Chapter 1 yet admits **A** is addressable from the adversary slot exactly when its own session is 0: with $\mathcal{F}.s \neq 0$ the conjunct $\mathtt{id}'.s = \mathtt{id}.s$ fails and global($\mathcal{F}$) does not save it. The difference is observable — it is how $\mathcal{D}$ carries out an instruction of Section 4.4 at a corrupt party — so for such a pair no relocation moving session 0 is an isomorphism، and Definition 7.2 is necessary rather than convenient. It is also the least one can exclude: for every other callee the supplied machines are waived by global or refused at line 2، so no smaller condition would do and no larger one is needed.

The shape it excludes is not exotic، and one place the paper itself reaches for it should be said plainly. Remark 6.5 offers، as a way of closing the hosting route to an interface member، that such a member admit only $\mathbf{Z} \cup \mathbf{A}$ — the environment and the adversary and no one else. A member of that shape is not global، lying outside its **N**، so it is not session-uniform. That is no defect of Definition 7.2 but the artefact showing through at the place it costs most: line 3 lets the environment reach such a member in session 0 and in no other، so a protocol whose interface members admit exactly $\mathbf{Z} \cup \mathbf{A}$ is drivable only in session 0، whatever relocation has to say about it. The two routes an admissible environment has into a protocol divide accordingly. It may claim a fresh name، which line 2 admits only at a member admitting all of **Std**

one of run one not hence sessions، across shared hence global، —
a at admits 2 line which -identity،Z own its claim may it Or protocol.
.$\mathcal{O}$ session to member that pins 3 line then and — **Z** admitting member
session a in interface protocol session-local a reaches route Neither
choosing. own its of
re- worth is all at -identityZ own its claim can environment the That
name the available: route second the makes what is it since calling،
Defi- and $\langle\pi\rangle \cup \langle\varphi\rangle$ outside lie -identitiesZ so system، no by used is Z
par- its only is **N** whose member A claimable. them leaves 4.10 nition
artefact the there and route، neither by reachable is class name ent's
alone. slot adversary's the through shows
as 3 line Reading

$$\big(\mathrm{id}'.s = \mathrm{id}.s \ \vee\ \text{global}(\mathcal{F}) \ \vee\ \mathrm{id}'.F \in \{A, Z\}\big)$$

confer then would $\mathcal{O}$ Session condition. the with dispense would
reach- be would adversary the admitting functionality local a nothing،
be could 7.2 Definition and alike، session every in slot the from able
all of hold would results the above: statement every from dropped
the in else Nothing ones. session-uniform the than rather systems
being ids reserved the stays، (iii) condition — change would chapter
con- tuple the and is، rule session the whatever code in outright named
widening The .$\mathcal{O}$ session of free already are 7.11 Proposition of ditions
are names it machines two the rest: the of spirit the in and small is
them، of tested is global wherever disjunct second 'sglobal by global
$\mathcal{Z}$ and parties، corrupt to $\mathcal{A}$ confines still 3 line of gate corruption the
than more is buy would it What .par own its by and **N** by out stays
and session its to local be protocol a let would what is it relocation:
the pairing the is which time، same the at environment the by driven
as- silently suggestion 's6.5 Remark which and deny above routes two
record and ،[7] Canetti's is which rule، narrower the keep We sumes.
here. price its and option the both

نکته 7.18 (Relocation is not a many-instance theorem). Proposi-
tion 7.10 moves one copy and charges nothing. It does not follow، and
is not true، that n copies at n indices cost ε rather than $n\varepsilon$: the execu-
tions of $\pi(\vec{q}_1), \dots, \pi(\vec{q}_n)$ and of $\varphi(\vec{q}_1), \dots, \varphi(\vec{q}_n)$ are not each other's

— itself of copies n to system a carries relocation no — relabellings
،4.24 Theorem of chain hybrid the is them between route only the and
one statement the is together give results two the What adds. which
Corol- ؛$\vec{q}$ one at $\varphi(\vec{q})$ emulates $\pi(\vec{q})$ prove proof: single a from wants
whose ،4.24 Theorem and ؛ε within other every at it gives 7.12 lary
and disjoint pairwise being $\vec{q}_k$ the to come hypotheses disjointness
at ،$n\varepsilon$ within once at n all replaces shell، surrounding the of clear
the in functorial also is Relocation charges. 8 Chapter budgets the
only reads which of each ،6 Chapter and 4.1 Section of constructions
in- an since $\theta\big(\rho[\pi/\varphi]\big) = (\theta\rho)[\theta\pi/\theta\varphi]$ sets: identity and ids names،
right- the — $\theta\overline{\pi} = \overline{\theta\pi}$ difference، set the with commutes action jective
of system blocked the as requires، 6.2 Definition as read، side hand
relabels θ fields two the copies blocker a since — $(\theta\pi, \theta\varphi)$ pair the
Defini- since $\theta\big(\mathcal{Z}^{\rho\setminus\varphi}\big) = (\theta\mathcal{Z})^{\theta\rho\setminus\theta\varphi}$ and ،$I_{\theta\varphi} \setminus I_{\theta\pi} = \theta(I_\varphi \setminus I_\pi)$ and
half reserved whose set، outward declared bundle's a relabels 7.3 tion
relocated a So .7.5 Proposition of (ii) by fixed is $\{id : id.F \in \{A, C\}\}$
proof. relocated the of composition the is composition of proof

فصل 8

هزینه‌های عینی

collected are and line main the outside sit cost concrete of threads Two the cost the and members، its off read itself، in system a of cost the here: of budgets the on arithmetic charge، 4.5 Section of results composition .4.31 Definition

whole، a as cost a has also system a member، by member Budgeted up. level one read cost functionality’s a is it and

call a is π system a of invocation An .system) a of (Cost **8.1 تعریف** cas- a by it services π ؛π outside from members its of one reaching cost The outside. returns control before activations internal of cade total largest the ،T_π time per-invocation the :(T_π, R_π) pair the is π of activations the over summed — cascade such one in steps of number count invocation the and — time a at running machine one triggers، it $T_\pi R_\pi$ most at in runs π So answers. π invocations of number the ،R_π steps.

with $\pi = \{\mathcal{F}_1, \ldots, \mathcal{F}_n\}$ Let .costs) member from cost (System **8.2 گزاره** within $\mathcal{F}_i$ of activations the bound k_i let and ،(t_i, r_i) cost of $\mathcal{F}_i$ each Then .π of invocation one

$$T_\pi \leqslant \sum_i k_i t_i, \qquad R_\pi \leqslant \sum_i r_i,$$

each activates invocation every If steps. $\sum_i t_i r_i$ most at in runs π and $T_\pi \leqslant$ then — $k_i \leqslant 1$ subroutine-respecting، π — once most at member .$R_\pi \leqslant \min_i r_i$ members، all drives invocation each moreover if ؛$\sum_i t_i$

t_i most at for each times، k_i most at $\mathcal{F}_i$ activates invocation An .اثبات

whole the over $\mathcal{F}_i$ of activations The .$\sum_i k_i t_i$ most at spends it so steps، ac- one least at is invocation every and ،r_i most at number execution are steps total the ؛$\sum_i r_i$ most at number invocations the so tivation، mem- all drives invocation each When .$\sum_i($ of activations$\mathcal{F}_i) t_i \leqslant \sum_i r_i t_i$ for $R_\pi \leqslant r_i$ so times، R_π exactly $\mathcal{F}_i$ each activate invocations R_π bers، .i every ☐

pro- disjoint with systems be π_2 and π_1 Let .union) a of (Cost **8.3 نتیجه** invokes one of member no If .(T_2, R_2) and (T_1, R_1) costs and ids cess then other، the of member a

$$\begin{aligned}\left(T_{\pi_1\cup\pi_2},\ R_{\pi_1\cup\pi_2}\right) &= (T_1, R_1) \oplus (T_2, R_2)\\ &= \left(\max(T_1, T_2),\ R_1 + R_2\right).\end{aligned}$$

the and π_2 into calls k most at makes π_1 of invocation each instead If and entry، π_1 a from $T_{\pi_1\cup\pi_2} \leqslant T_1 + k\,T_2$ then acyclic، is graph cross-call unbounded cross-calls the with ؛$R_{\pi_1\cup\pi_2} \leqslant R_1 + R_2$ with symmetrically، union's. the determine longer no costs summary the

and π_2 or π_1 enters $\pi_1 \cup \pi_2$ outside from call a cross-calls no With .اثبات that to belonging and T_2 or T_1 most at costing there، stays cascade its $\max(T_1, T_2)$ is time per-invocation the so count؛ invocation subsystem's union the to internal is π_2 into π_1 from cross-call A add. counts the and at steps؛ T_2 most at ،π_2 of invocation one by cascade the extends and π_2 to external once call a while ،$T_1 + kT_2$ give acyclically، them، of k most number still invocations the so counted، longer no is π_1 from placed but .$R_1 + R_2$ most at ☐

can call one it without idle: not is proviso subroutine-respecting The towards degenerates T_π and end، without system the around driven be member's private a keeps that discipline same the — $\sum_i t_i r_i$ of whole the Def- of $\oplus$ the to dual is cost system This .6.5 Remark in pinned callers activation، per member one bundle، a as system a hosting :4.30 inition the in pays $\mathbf{c}_{\rho\setminus\varphi}$ cost the ،$(\max_i t_i,\ \sum_i r_i)$ into $\oplus$ by members its combines over times the sums subsystem a as it calling below؛ costs composition synchronous the in $(\sum_i t_i,\ \min_i r_i)$ calls، external counts and cascade a when tightly more second the ،$\sum_i t_i r_i$ total same the bound Both case. by itself: level system the at recurs $\oplus$ And pass. full a is invocation every at operator one it، by compose subsystems non-interacting 8.3 Corollary scales. both

turns time per-invocation the another one call π of members the When
clean. is case one and calling، that of shape the on

$\pi = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$ of graph call the Suppose .systems) (DAG **8.4 گزاره**
at $\mathcal{F}_i$ member internal each activates invocation every and acyclic is
the among time per-invocation largest the — times $\max_{j:\mathcal{F}_j\to\mathcal{F}_i} t_j$ most
Then it. call that members

$$T_\pi \leqslant \sum_i \Big(\max_{j:\mathcal{F}_j\to\mathcal{F}_i} t_j\Big) t_i \leqslant \Big(\max_i t_i\Big)\sum_i t_i,$$

graph. the of depth the of independent time per-invocation a

times a_i some $\mathcal{F}_i$ each activates source a entering invocation An .اثبات
at $\mathcal{F}_i$ internal each and once runs source A steps. $\sum_i a_i t_i$ spends and
whence throughout، $a_i \leqslant \max_i t_i$ so hypothesis، by times $\max_{j\to i} t_j$ most
□ .$T_\pi \leqslant \sum_i (\max_{j\to i} t_j)\, t_i \leqslant (\max_i t_i) \sum_i t_i$

bound not does alone acyclicity for place، its earns hypothesis The
per- the activation، one in $\mathcal{F}_i$ to makes $\mathcal{F}_j$ calls the for n_{ji} Writing .T_π
the counts a_i so ،$a_i = \sum_{j\to i} n_{ji}\, a_j$ solve counts activation invocation
them: multiplies graph reconvergent a and paths، $\mathcal{F}_i$source-to- weighted
$s \to \{A_k, B_k\} \to C_k \to \{A_{k+1}, B_{k+1}\} \to \cdots$ — diamonds N of chain a
per- the and 2^N is foot the at a so level، each at count the doubles —
the by node each Capping depth. the in exponential is time invocation
stops what exactly is sum their by than rather callers its over maximum
and more، once theme -over-summation⊕ the — accumulating paths the
Propo- of $k_i \leqslant 1$ case subroutine-respecting the of counterpart DAG the
bound caps invocation The .$\max_{j\to i} t_j$ to relaxed activation one ،8.2 sition
indepen- alone، times the in bound a is here point the case؛ any in $a_i \leqslant r_i$
depth. the of and caps the of dent

alone caps the statement: general the of whole the is remark last That
structure. call the on condition no with class، cost the close already

mem- whose $\pi = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$ system A .composition) (Cost **8.5 نتیجه**
most at over steps $\sum_i t_i r_i$ most at in runs (t_i, r_i) costs have bers
call its whatever $R_\pi \leqslant \sum_i r_i$ and $T_\pi \leqslant \sum_i t_i r_i$ so invocations، $\sum_i r_i$
and λ parameter a in polynomial is r_i and t_i each if Hence structure.
polynomially the polynomial: is (T_π, R_π) bounded، polynomially is n

formation. under closed are systems bounded

اثبات. The step and invocation bounds are Proposition 8.2، and one invo-
cation's cascade is part of the whole execution، so $T_\pi \leqslant \sum_i t_i r_i$ too. A
sum of polynomially many polynomials is a polynomial. □

No condition on the call graph is needed here، unlike the depth-free
time bound of Proposition 8.4، because the caps carry it. A reconvergent
cascade that would activate a member 2^N times can do so only if that
member answers 2^N calls، so $r_i \geqslant 2^N$ — a member already outside the
polynomial class; within the class the cap refuses the excess and trun-
cates the cascade at $\sum_i r_i$ activations. This is the efficiency counterpart
of Theorem 4.20: where composition preserves security — $\rho[\pi/\varphi]$ em-
ulates ρ once π emulates φ — cost composition preserves efficiency،
so a protocol assembled from polynomial-time functionalities is itself
polynomial-time. Together the two are what the concrete apparatus is
for: polynomial-time UC-secure protocols compose into polynomial-time
UC-secure protocols، which is what lets the asymptotic reading below be
a corollary rather than a theory of its own.

The same closure survives the operation the composition theorem
runs.

گزاره 8.6 (Replacement preserves polynomial boundedness). Let ρ and
π be systems whose members are polynomial and polynomially many،
and let the replacement of φ by π in ρ be well-formed. Then $\rho[\pi/\varphi]$
is polynomially bounded.

اثبات. The members of $\rho[\pi/\varphi] = (\rho \setminus \varphi) \cup \pi$ (Definition 4.1) are those
of ρ outside φ، each polynomial as a member of ρ، together with those
of π، again polynomial؛ they number at most $|\rho| + |\pi|$، polynomially many.
Corollary 8.5 applies. □

So the system that runs after a replacement is efficient whenever the
specification ρ and the realization π are — Theorem 4.20 read for cost
rather than security، and on the same operation.

Composition، numerically. In Theorem 4.20 the adversaries and the sim-
ulators carry over unchanged، so do their budgets. The environments are
the one place something is computed، and what absorption costs is the
cost of hosting. Systems are budgeted member by member، so the cost

of budgets the of -combination$\oplus$ the for $\mathbf{c}_{\rho\setminus\varphi}$ write up: looked be can the times، per-invocation their of largest the — $\rho \setminus \varphi$ of members the called- the to dual cost hosted-bundle the counts، invocation their of sum execution، any in them bounds which — 8.2 Proposition of cost subsystem 4.15 Definition of environment absorbed The them. driving is whoever so else، nothing and functionalities those and $\mathcal{Z}$ runs

$$\mathcal{Z} \in \mathbb{Z}_{\rho[\pi/\varphi],\,\rho}(\mathbf{z}) \qquad \Longrightarrow \qquad \mathcal{Z}^{\rho\setminus\varphi} \in \mathbb{Z}_{\pi,\varphi}\big(\mathbf{z} \oplus \mathbf{c}_{\rho\setminus\varphi}\big),$$

the- the into this Feeding .4.16 Proposition being $\mathbb{Z}_{\pi,\varphi}$ in containment the is conclusion the orem،

$$\mathrm{Adv}^{\mathrm{uc}}_{\rho[\pi/\varphi],\,\rho}[\mathbf{a},\mathbf{s},\mathbf{z}] \;\leqslant\; \mathrm{Adv}^{\mathrm{uc}}_{\pi,\varphi}\big[\mathbf{a},\ \mathbf{s},\ \mathbf{z} \oplus \mathbf{c}_{\rho\setminus\varphi}\big].$$

the costs and simulator، the in and adversary the in free is Composition side. environment the on once، system surrounding

sim- step's first the that asks 4.12 Proposition **numerically. Transitivity،** in- an becomes which class، adversary step's second the be class ulator A .$\mathbf{s} \leqslant \mathbf{a}'$ as soon as holds $\mathbb{S}(\mathbf{s}) \subseteq \mathbb{A}(\mathbf{a}')$ budgets: between equality will- is step next the adversary an be must build to willing is one simulator both for admissible class one for asks also proposition The face. to ing ،$\mathbb{Z} \subseteq \mathbb{Z}_{\pi,\varphi} \cap \mathbb{Z}_{\varphi,\psi}$ for here: carried be must hypothesis that and steps، well، as $\mathbb{Z}_{\pi,\psi}$ in lies 4.12 Proposition of proof the by which

$$\mathrm{Adv}^{\mathrm{uc}}_{\pi,\psi}[\mathbf{a},\mathbf{s}',\mathbb{Z}] \;\leqslant\; \mathrm{Adv}^{\mathrm{uc}}_{\pi,\varphi}[\mathbf{a},\mathbf{s},\mathbb{Z}] \;+\; \mathrm{Adv}^{\mathrm{uc}}_{\varphi,\psi}[\mathbf{s},\mathbf{s}',\mathbb{Z}].$$

brack- the places three all in $\mathbf{z}$ with Written here. do not will alone Budgets con- is first the and ،$\mathbb{Z}_{\varphi,\psi}(\mathbf{z})$ and $\mathbb{Z}_{\pi,\varphi}(\mathbf{z})$ ،$\mathbb{Z}_{\pi,\psi}(\mathbf{z})$ over range would ets outer the for admissible environment an — others the of neither in tained mid- the cover would hypothesis no so — IDs(φ) about nothing told is pair dis- composition The cancels. inequality triangle the that probability dle the into class one its maps absorption trouble: such no has above play of chain the Along together. put displays two the is 4.21 Corollary other. hop each add، errors the accumulate: effects two same the 4.24 Theorem at absorbs it what hosting of cost the by budget environment the widens inequalities the become 4.25 Remark of relations class the and step، that .$\mathbf{s}_{k+1} \leqslant \mathbf{a}_k$

emulation an that Say **up. blows simulator the whether and Nesting،** every for $\big(\mathbb{A}(\mathbf{a}), \mathbb{S}(f(\mathbf{a})), \cdot\big)$ against holds it if f overhead has statement one simulator the given، is adversary the whatever :$\mathbf{a}$ budget adversary builds costs f of it. In the per-invocation budget the two components behave differently،

$$f(t,r) \;=\; \big(\max(t,\tau),\;\; \gamma r+\delta\big),$$

the query count affine — γ the calls the simulator places for each call of the adversary it runs، δ its own — and the per-invocation time a maximum، τ the simulator's own per-activation work، since one machine runs per activation and the simulator only adds bounded work to each.

Theorem 4.20 does not touch f at all. Its adversary and simulator classes are those of the hypothesis، so a single replacement returns the very machine the hypothesis supplied and the whole price is paid on the environment side. Overheads compose in one place only: where a simulator is re-read as an adversary، which is Proposition 4.12. There $\mathcal{S}'$ is built from $\mathcal{S}$، itself built from $\mathcal{A}$، so the composite has overhead $f_2 \circ f_1$. Applying the results k times، the query budget obeys $r_j = \gamma\, r_{j-1} + \delta$، that is

$$r_k \;=\; \gamma^k r_0 \;+\; \delta\,\frac{\gamma^k-1}{\gamma-1} \quad (\gamma\neq 1), \qquad r_k \;=\; r_0 + k\delta \quad (\gamma = 1),$$

while the per-invocation time saturates، $t_k = \max(t_0,\tau_1,\dots,\tau_k)$، bounded by the largest level's own work and constant in depth.

The blow-up therefore lives entirely in the query count، a dichotomy turning on γ alone. A simulator that must place more than one call for some call of the adversary it runs has $\gamma > 1$، and k applications cost γ^k: genuinely exponential، and at $\gamma = 2$ a depth of 20 already costs a factor of 2^{20}، about a million، on the query count. A simulator that relays each call once and adds bounded traffic of its own has $\gamma = 1$، and k applications cost $r_0 + k\delta$: linear، δ paid once per level. This is the property to check when a simulator is built؛ nothing proved here implies it. It is also why straight-line simulators matter beyond the usual reasons، rewinding being the standard way to acquire $\gamma > 1$: one rewinding simulator anywhere in the tower puts a factor γ on every level below it.

The same formula governs breadth as well as depth. Theorem 4.24 builds $\mathcal{S}_1,\dots,\mathcal{S}_n$ by handing each simulator to the next step as its adversary، so n parallel instances compose n overheads exactly as an n-deep tower does، and Remark 4.25 read with budgets asks $\mathbf{a}_k \geqslant f_{k+1}(\mathbf{a}_{k+1})$ at every link. Composing n instances of a $\gamma = 1$ protocol costs $n\delta$ extra queries؛ composing n instances of a $\gamma = 2$ one is hopeless for even moderate n.

per- the in or side، environment the on happens comparable Nothing
absorbs، it shell the hosting of cost the adds level Each time. invocation
inner- the reaches nesting $k$depth- a so disjoint، are shells tower's a and
run- of that count invocation its — $\mathbf{z} \oplus \bigoplus_{i \leqslant k} \mathbf{c}_i$ budget at statement most
shell's، largest the time per-invocation its once، system ambient the ning
of applications k likewise، behave errors The tower. the deep however
exponen- degrades claim security a Where .$\sum_i \varepsilon_i$ giving 4.12 Proposition
possi- only the is $\gamma > 1$ overhead query simulator's the nesting، under tially
per-invocation the neither so activation، per runs machine one source: ble
multiply. can budget environment the nor time

security a mentions above Nothing **reading. asymptotic the Recovering**
the let ،λ by everything index statement، familiar the get To parameter.
negligible be advantage the that ask and ،λ in polynomials be budgets
and machines bounded polynomially the become then classes the it؛ in
is form concrete The emulation. UC of definition usual the 4.11 Definition
is form asymptotic the and in، proved are results composition the one the
practice-oriented ordering the — round way other the not it، of corollary a
.[2] for argued long has security provable

فصل 9

فراخوان‌های پاسخگو

write to way no had far so has framework the promise a needs 15 Chapter back، one get must it signature a for adversary the asks Sign When down. of rest the drive and off go to free adversary the were :now back it get and mid- suspended left be would party the answering، before execution the asks UC synchronous immediacy same the — lost locality and signature the restricting for settles there text The .[11] environment responsive a of instead. code in thing the says chapter This class. simulator

it — helpfully answer adversary the that Not **forbidden. be to has What** to has What answer. bad a with deals 12 Chapter and anything، return may returning، and call the receiving between leaving: token the is forbidden be a calls the on restriction a is That all. at call no place must responder the that. exactly for wrapper a has already paper the and makes، machine

maxi- its at Silence is Respond .answering) (Responsive **9.1 تعریف** identities، all of set the $\mathcal{I}_{\mathrm{all}}$ with mum:

$$\begin{aligned}&\mathsf{Respond}\big(\mathtt{id}.\mathsf{Op}(\mathtt{in})\ \text{from}\ \mathtt{id}'\big)\\ :=\ &\mathsf{Silence}\big(\mathcal{I}_{\mathrm{all}},\ \mathtt{id}.\mathsf{Op}(\mathtt{in})\ \text{from}\ \mathtt{id}'\big).\end{aligned}$$

pro- wrapped the call every 2 Chapter by so ،$\mathcal{I}_{\mathrm{all}}$ in lies claim Every pro- The placed. being ever without rej with refused is issues gramme value؛ a return still must and likes it as long as compute may gramme on. token the hand is do cannot it what

clause last its with Silence is Respond out، Written **handler. a as Read** so permits، it claim whose call a forwards Silence wrapper The deleted. a only re-raise، no has Respond re-raise؛ a partly is clause operation its ef- call ambient the discharges it 2 Chapter of vocabulary the In constant.

Respond **Wrapper**
$\mathtt{id}'$ from $\mathtt{id}$.Op($\mathtt{in}$) programme

:Respond($\mathtt{id}$.Op($\mathtt{in}$) from $\mathtt{id}'$)

handle $\mathtt{id}$.Op($\mathtt{in}$) from $\mathtt{id}'$ **with**
  return $\mathtt{out} \mapsto \mathtt{out}$
  $\mathtt{id}''$.FullOp($\mathtt{in}''$) as $\mathtt{id}'''$; $\mathtt{k} \mapsto \mathtt{k}$(rej)

locally handled is effect the scope its inside it: forwarding than rather fect It it. to respect with pure is programme wrapped the so completely، and on call a wanted: 15 Chapter property the locality، of definition the also is never token the activation، own responder's the within served is slot the responsive a — responsiveness of statement crisp the is That it. leaving oper- the and — Exec to escape not does effect call whose one is core do that calls the exactly dispatches Exec since follows، reading ational more. no and responder the of activation one is call responsive A escape.

car- adversary The .call) responsive interface، (Responsive **9.2 تعریف** and guard same the with FullA$^!$ interface second a ،FullA beside ries، :Silence of place in Respond but core same the

$$\mathtt{id}.\text{FullA}^!(\mathtt{in}) \text{ from } \mathtt{id}' : \quad \textbf{require } \text{Guard}_{\mathcal{A}}(\mathtt{id}, \mathtt{id}'); \quad \textbf{return } \text{Respond}\big(\mathtt{id}.\text{A}(\mathtt{in}) \text{ from } \mathtt{id}'\big).$$

identity with interface an Inside .FullA$^!$ on call a is call responsive A abbreviates $\mathcal{A}(\mathtt{in})$ as ،$\mathtt{id}$ as $(A, \mathtt{id}.P)$.FullA$^!(\mathtt{in})$ for $\mathcal{A}^!(\mathtt{in})$ write we $\mathtt{id}$ one. ordinary the

price. the are they and follow، things Three calling by reached is **C** set The register. the read cannot responder The computed is answer responsive a so refused، is call that and ،FullStatus subrou- no delegate: cannot It knows. already responder the what from being Corrupt corrupt، cannot it And environment. no party، other no tine، Sec- because own، its on stating deserves last The other. any like call a is reads two the between gap the and twice register the reads 3.5 tion in. slip can corruption a where exactly

reg- the execution، any In .corrupt) cannot call responsive (A **9.3 گزاره** call the when state its is returns call responsive a when **C** state ister's placed. was

اثبات. By Section 3.1، **C** changes only on line 5 of FullCorrupt، and only when that interface is reached by a call. Between the placing and the re- turn of a responsive call the responder holds the token throughout، and by Definition 9.1 every call it issues is refused before being placed، so FullCorrupt is not reached. No other machine holds the token in that in- terval، so none can reach it either. □

So if a core reaches the adversary only through $\mathcal{A}^!$، the party it serves is corrupt at line 7 exactly when it was corrupt at line 3، and the second mediation hook cannot fire on account of that call. Remark 3.2 describes what happens when it does; responsiveness is the condition under which it does not.

This is what Chapter 15 wanted. Writing $\mathcal{A}^!$ in place of $\mathcal{A}$ in Gen، Sign and Verify makes locality a property of the code rather than a condition on the simulator class، and the wrapper enforces it، so it holds of every occupant of the adversary slot — the simulator included، Definition 4.7 giving a simulator the adversary's interfaces، FullA$^!$ among them. What it does not do is make the answer good: a responsive adversary may still return $\perp$، and San repairs that. Responsiveness buys the token back; sanitization buys the value.

نکته 9.4 (Responsiveness and the dummy). The two chapters pull against each other، and it is worth being explicit about where. The dummy of Section 4.4 answers a call from a functionality by relaying it to the environment on line 5 — and that is a call، which Respond refuses. So $\mathcal{D}$ cannot serve a responsive call: the relay is refused، and the rule of Section 3.6 delivers its answer as $\perp$ where a real ad- versary would answer with substance. The regrouping (R1) would therefore fail for a system whose cores use $\mathcal{A}^!$، at the first respon- sive call — which is why Theorem 4.29 hypothesises systems whose cores place none. Recovering completeness for responsive systems needs a matching notion on the environment's side، so that the relayed question can be answered without the token moving on; we leave that open.

فصل 10

Language Core

protocols functionalities، book's this of code the give 21 to 13 Chapters an as not program، a as read be to meant is code that and games، and convention stated a by kept be to has that promise a — one of illustration for promise this exactly keep [5] will. good reader's the by than rather language terminating typed، strongly small، a with proofs game-based short a and (15 Figure (their grammar context-free a by given ،$\mathcal{L}$ call they the takes chapter This it. of top on allow they sugars the of list named here from chapter every what for grammar closed a here: steps two same it for word own book's this already body، operation's an — core a calls on each grammar، that from departures of list short own book's this and — it against checked be can come to still core every that so once named faith. on taken than rather

،Guard so. deliberately and exempt، is chapter this before Everything adver- dummy the ،(3.6 (Section Exec ،(2 (Chapter Mediate and Silence they core: a inside run not do (9 (Chapter Respond and (4.4 (Section sary guards، evaluating calls، dispatching cores، of system a runs what are exe- of definition A next. the to activation one from token the handing as exactly — executes it language the in program a itself not is cution informal their in sits adversary an meeting game a of account own]'s5] no- one the and — 15 Figure their of grammar the outside ،3 Section form continuation the own، core's a for mistaken be could that tation be- ،k resumption unbound explicit، an carrying id''.FullOp(in'') as id''' ; k operation's an inside occurs once never it layer: outer that to entirely longs — constantly does، it of relative plainer A on. 13 Chapter from code own the under resumed operation، functionality's another to call ordinary an that and — continuation fresh a handed than rather identity own caller's below. right، own its in matter core-language a is one op- every and statements، of sequence a is core A **grammar. core The** book's this what to Scoped core. one exactly is writes book this eration grammar: full 's$\mathcal{L}$ to than rather use، actually cores

$$
\begin{array}{rcl}
\text{core} & ::= & \varepsilon \ \mid\ \text{stmt}\,;\,\text{core} \\
\text{stmt} & ::= & \text{lv} \leftarrow e \ \mid\ \text{lv} \leftarrow_{\$} S \\
 & \mid & \textbf{if } e \textbf{ then } \text{core}\ [\,\textbf{else } \text{core}\,] \\
 & \mid & \textbf{for } x \in S \textbf{ do } \text{core} \\
 & \mid & \textbf{return } e \\
\text{lv} & ::= & x \ \mid\ x[e] \\
e & ::= & \ldots \ \mid\ \text{call} \\
\text{call} & ::= & \mathrm{id}''.\mathsf{Op}(e_1, \ldots, e_k) \textbf{ as } \mathrm{id}
\end{array}
$$

،[5]) grammar 's$\mathcal{L}$ usage. own book's this to scoped grammar، core The :5 شكل mathematical ordinary to left expressions of sub-grammar the with (15 Figure their six the over standard already is writes book this expression every since notation، (below). restriction Adversary-only their past generalized own، 's$\mathcal{L}$ is call below؛ types (.10 (Chapter

's$\mathcal{L}$ — table set، string، boolean، integer، five: 's$\mathcal{L}$ not types، base Six reading $\mathtt{L} : \mathbb{N} \to \{0,1\}^n \cup \{\square\}$ throughout، type function's a as kept array، book's this $\square$ there، does $A : \{0,1\}^* \to \{0,1\}^* \cup \{\text{undefined}\}$ as exactly -**for** one Only below. addition، own book's this identity، and — undefined pair a binds it of instance one the and ،**for** $x \in S$ **do** attested، is form obvious an — (16 and 17 lines ،20 (Chapter **for** $(m, y) \in N$ **do** directly، integer- bounded The it. destructuring and variable one binding for sugar the licensed and far، so unused reserve، in is allows also $\mathcal{L}$ form range it. needs core a moment

،id' **from** $\mathrm{id}.\mathsf{Op}(in)$ headed is core operation's Every **implicit. is caller The** though core the in anywhere readable is — identity caller's the — id' and it؛ like nothing carries $\mathcal{L}$ in oracle plain A .Op of argument no names it is who exactly asking by begin many so because it need cores book's this representative being operations 's$\mathcal{G}_{\mathsf{Clock}}$ runs، else anything before calling does core the before corruption 's$\mathrm{id}.P$ and $\mathrm{id}'.F$ tests 13 line at guard (the else). anything

the for expression call the reserves $\mathcal{L}$ **core. another call may core A** one. place not may games their of one inside oracle an alone؛ Adversary of protocol the — does book this in game and protocol functionality، Every neighbours)، and 22 (lines mid-operation $\mathcal{F}_{\mathsf{Sig}}$ and $\mathcal{F}_{\mathsf{PKI}}$ reads 20 Chapter (5 (line itself $\mathcal{F}_{\mathsf{Rand}}$ into back calls game unpredictability own 's$\mathcal{F}_{\mathsf{Rand}}$ and

others from built are protocols and functionalities book's this because —
and ،(1 (Chapter call may core a what exactly naming **U** declaration، by
own، 's$\mathcal{L}$ is itself call The other. any like functionality a is shell game's a
re- and expression other any like evaluates id''.Op$(e_1, \dots)$ **as** id unrelaxed:
stays What identity. own its under line، same the at core calling the sumes
k continuation unbound an with form، general the is core a outside strictly
this of line next "the than rather point resumption arbitrary an for standing
call a what for notation own model's execution the is that — core" same
defined. is it where is 3.6 Section and core's، a never general، in means

the throughout used line، **require** e own core's A **sugar. is Require**
sugar is come، to cores the of plenty inside and chapters execution-model
guard failed a though even ،rej literally not — **not if** e **then return** $\perp$ for
no value one the rej makes 3.6 Section that. exactly returns elsewhere
be- $\perp$ into rej core-level a laundering output، own its as return may core
com- case common the names merely **require** own core's a delivery؛ fore
or- an with hand، by out guard same the spell operations 's$\mathcal{G}_{\text{Clock}}$ pactly.
core-language one the are both again): 13 (line **if … then return** dinary
not. or sugared statement،

string- is neither :$\mathtt{L}[\mathtt{ctr}]$ ،$\mathtt{reg}[\mathrm{id}.\mathrm{P}]$ **rule. new no need indices Table**
— ([5]) enhancement third own its answer، the has already $\mathcal{L}$ and indexed،
table Every silently. first، encode$(\cdot)$ through passed is index non-string a
is there way؛ this reads already string-indexed not is that writes book this
it. for them citing beyond here add to nothing

makes 1.1 Definition sugar. not is departure One **type. sixth a is Identity**
process a party، a and id process a of $\mathrm{id} = (\text{pid}, \mathrm{P})$ pair a identity an
once، at levels both through reads core a and — $(\mathrm{F}, s, \mathrm{i})$ triple a itself id
reads ever core no field one the $\mathrm{id}.\mathrm{i}$ projections، named $\mathrm{id}.\mathrm{P}$ ،$\mathrm{id}.s$ ،$\mathrm{id}.\mathrm{F}$
projection، a carries types five 's$\mathcal{L}$ of None outright). so says 1 (Chapter
of account own]'s5] by equality-preservation، only gives encode$(\cdot)$ and
identity So serve. to that for often too far apart read is identity an — it
with atomic، than rather structured type، base sixth a as list type the joins
one the — operations only its as equality componentwise and projection
have. already not does it something for $\mathcal{L}$ ask cores book's this place

how compute؛ may core a what is above grammar The **code. the Reading**
kept. mostly already one and promise، separate a is page the on set is it
،**return** ،**do** ،**for** ،**else** ،**then** ،**if** default: algorithmicx's bold، are Keywords
declared is it wherever link a carries name operation's An .**require** and

caps small accentsoft in rendered both to، back called is it wherever and ev- from it to back point more two declaration، the anchors macro one — mutable own core's a for reserved typewriter، ،x in lives State site. call ery the and state، than rather parameters bound italic، stay out ،in ،id fields؛ own core's a what deciding reader future a — together run be not must two Com- other. no and distinction this exactly needs like looks type record That load-bearing. never side، the to off italic، ،commenttint in print ments nota- new not done، is written already chapters the over pass copy-editing already-declared an of mention prose first a sweep: uniformity a but tion constant the and throughout، declaration its to back linked is operation stor- and tag trace every with together ،done ،false ،true ،rej ،ok — atoms op- an than rather commenttint carry — book this in elsewhere key age a is "this separate does now colour bare so colour، own name's eration value." a is "this from name"

sin- a is together، buy، above departures the and grammar the What own core's a is math book's this of line a on، 13 Chapter from test: gle depar- the plus above table the against type-checks it when exactly code property's a verdict، game's a — specification a is it and here، named tures to gap the Closing not. does it when exactly — pick own 'sSan quantifier، compiling of matter a then is otherwise، or OCaml language، actual an deciding، first of not once، forms statement of handful a and types six to claiming ever was symbols of display given a what chapter، by chapter compute.

فصل 11

همسان تا خرابی

same the are that games two :[13 ،5] hops by advance proofs Code-based probabil- the on bound a and raised، been has flag a where except program licence the is playing game of lemma fundamental The raising. the of ity probabil- with differ outcomes their ،`bad` until identical are games the if — the on games both run coupling: a is proof its and — Pr[Bad] most at ity this In up. goes flag the before company part cannot they and coins same a environment، an runs execution An program. one not is game a paper them؛ among wanders token the system؛ a and register a occupant، slot rest the drive and mid-operation token the handed be may occupant the dis- to exists 9 Chapter freedom the — answering before execution the of lemma the replays chapter This it. of all survive must hop a and — cipline the nothing: costs wandering the that is finding the and executions، over all. at hypothesis no has below lemma

ordinary an of variable state ordinary an is `bad` **lives. flag the Where** but value any assigned never ،Initialize in `bad` ← false declared machine: machine the by state machine's a of rest the like owned afterwards، true — it off hand other every keeps already framework The it. declares that calls provoke is own not does it state to do can machine a thing one the answers، replaces Mediate party corrupt a at and it، extend or read that or code own owner's its by raised is flag a so — (5.4 (Remark state never interface an through immaterial، is it read may anyone Whether all. at not two the raised is flag the until :Guard bypasses which ،Leak through or lemma the raised is it after and alike، read any answer below machines members — once at flags carry may machines Several paid. already has the is below Bad and — own its owning each included، name a sharing flag. some raises machine some that event

are $\mathcal{F}'$ and $\mathcal{F}$ functionalities Standard .bad) until (Identical **11.1 تعریف**
line agree sInitialize their agree؛ fields five their if bad until identical
among `bad` $\leftarrow$ false and sides both to common declarations line، for
bad-guarded inside except agrees code remaining their and them؛
،`bad` $\leftarrow$ true assignment an of continuation the shapes: two of blocks،
the — block enclosing its of end the to assignment the past just from
ev- that so in، stands assignment the body operation or loop branch،
— assignment the across only reached is continuation the in erything
bad- the Erasing .`bad` is test whose conditional a of body the and
operation functionality، one leaves sides both from blocks guarded
assign- the particular in included؛ Leak line، for line and operation for
divergent own its heading each code، common are themselves ments
Equal .true but value any `bad` assigns side neither and continuation،
occu- Two erase. to nothing with bad، until identical are functionalities
sInitialize their when bad until identical are slot adversary the of pants
until identical are π' and π Systems way. same the read are، cores and
the of member a with one of member every pairs bijection some if bad
it. with bad until identical other

by checked is Membership]'s.5] both shape، the about things Two
— behaviour about claim a not code، of comparison a is it — inspection
divergent a that so built is it And audit. to cheap hops keeps what is which
its are block bad-guarded a into ways only the early: run cannot statement
the of test a and crossed، is it as flag the raises that assignment an head،
while runs bad-guarded nothing so — monotone is flag the and itself، flag
the fields، equal carry members paired Because down. is flag owner's its
$\langle\pi\rangle =$ names، same the use and identities same the occupy systems two
.$\mathbb{Z}_{\pi,\pi'} = \mathbb{Z}_{\pi,\pi} = \mathbb{Z}_{\pi',\pi'}$ both: serves environments of class one and ،$\langle\pi'\rangle$

π' and π systems Let .systems) over lemma، fundamental (The **11.2 لم**
be slot adversary the of $\mathcal{X}'$ and $\mathcal{X}$ occupants let bad، until identical be
some that event the for Bad Write .$\mathcal{Z} \in \mathbb{Z}_{\pi,\pi'}$ let and bad، until identical
execution whichever in ،`bad` $\leftarrow$ true assignment an executes machine
Then named. is

$$\Pr\left[\text{Bad in } \mathsf{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})\right] = \Pr\left[\text{Bad in } \mathsf{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C})\right],$$

value، common the for Pr[Bad] writing and،

$$\Big|\,\mathsf{Pr}\big[\mathsf{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C}) = 1\big]$$
$$-\ \mathsf{Pr}\big[\mathsf{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C}) = 1\big]\,\Big| \ \leqslant\ \mathsf{Pr}[\mathsf{Bad}].$$

then pairing the — systems game are systems two the moreover If of holds same the — preserved being fields game، to game matches share: games two the which ،Fin finalization every

$$\Big|\,\mathsf{Pr}\big[\mathsf{Win}_{\mathsf{Fin}} \text{ in } \mathsf{Exec}(\pi, \mathcal{Z}, \mathcal{X}, \mathcal{C})\big]$$
$$-\ \mathsf{Pr}\big[\mathsf{Win}_{\mathsf{Fin}} \text{ in } \mathsf{Exec}(\pi', \mathcal{Z}, \mathcal{X}', \mathcal{C})\big]\,\Big|$$
$$\leqslant\ \mathsf{Pr}[\mathsf{Bad}].$$

execu- both fixes tape every Fixing pointwise. read coupling، One .اثبات tions outright، so the two may be taken from one sample and compared step by step. The induction shows that while no flag is up the two corre- spond strictly and the token stands on common code — which is exactly what Definition 11.1's two bad-guarded shapes buy، neither being reach- able except across the assignment that raises the flag or through a test that fails while it is down. The first raising is therefore the same step on both sides، and that is the first display: Pr[Bad] is one number rather than two that happen to share a bound. Off Bad the two runs are equal outright، so every fact of the run has a single indicator there، and each difference of probabilities is at most the measure of the set where indi- cators may differ. Two conventions of Chapters 1 and 3.6 are used، the ones the absorption lemma restates: each machine draws from its own coin tape، uniform and independent of every other's، read only when its code samples — an environment's hosted copies keeping tapes of their own، bundling merging nothing — and Initialize places no calls. Fixing ev- ery tape therefore fixes an execution outright: which machine holds the token، at which line of which operation، with what state، and what its next statement does are functions of the configuration، coins entering only as the next unread positions of the holder's own tape.

Couple the two executions. The pairing matches machine to machine — member to member، occupant to occupant، the environment and the register each to themselves — and paired machines occupy the same iden- tities، so give paired machines equal tapes. Each execution separately keeps its distribution، tapes on either side being uniform and indepen-

point- compared and tapes of sample one off read be may both so dent،
of (C1)–(C4) if configuration a at strictly correspond two the Say wise.
tapes، unread equal states، equal — pairing the under hold 4.17 Definition
sus- values، local same the with line same the at hands paired in token the
and anywhere relay no with — frame for frame matching callers pended
elided. nothing

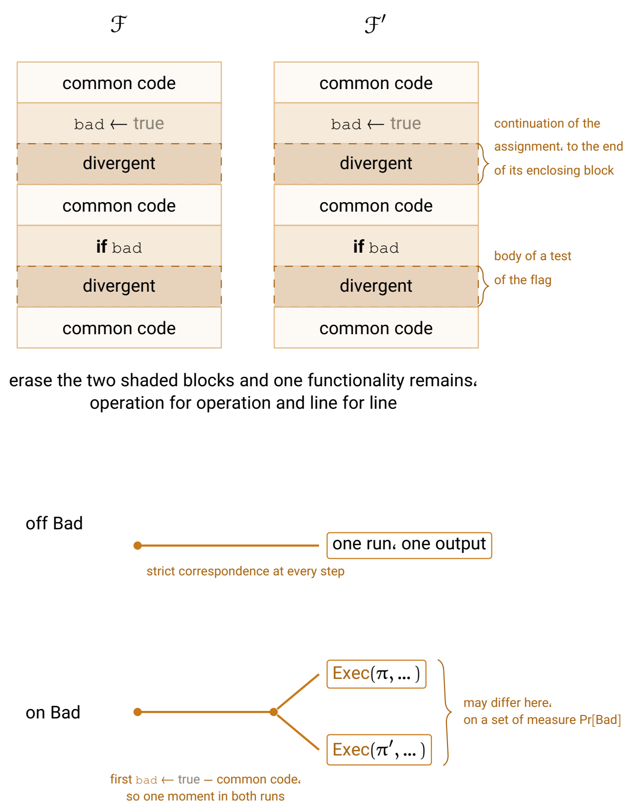
register the initialises 1 Line strictly. correspond they start the At
places Initialize — order whatever in sides، both on member every and
trace no leaves order the so owner's، its but state no touches and calls no
line، for line agree that sInitialize run members paired configuration؛ the in
draw، they where drawing، and calls no placing them، among `bad` $\leftarrow$ false
one the to tokens both hands 2 line and tapes؛ equal of positions same the
identity. same the at environment

side، either on raised yet flag no strictly، correspond two the Suppose
the at machines paired in stands token The run. statement one let and
is statement bad-guarded a code: common is line that and line، same
raises head the flag، the of test a through or head its through only reached
flag no and down، is flag the while fails test the crossed، is it as flag the
state same the over read statement، same the is statement the So up. is
If alike. tape for tape consumed having histories coins، next same the and
frame same the resume both returns، it if alike؛ write sides both writes، it
tar- — call same the place both call، a places it if value؛ same the with
equal over wrappers same the by met — identity claimed and payload get،
no at or machines، paired at landing state، register equal an and fields
rej — fixes 3.6 Section delivery every and alike؛ rej answered and machine
slot mediated a or rej return would core a where $\perp$ ،**require** refused a for
the survives correspondence Strict choice. a not rule، a is — one answers
code، common — `bad` $\leftarrow$ true assignment an is statement the if And step.
it، execute sides both — continuation divergent its of head the at standing
once. at executions both in raising first the is step the and

and raised ever is flag no either then: space، coupled the on Pointwise
exactly halts one so — step every at strictly correspond executions two the
one being environment the output، same the with does، other the when
and simultaneous، is it raising، first a is there or — alike resumed machine
has therefore Bad event The it. through and to up agree executions the
equality. with display، first the is which space، coupled the on indicator one
indicator one has run the of fact any so outright، equal are runs the Bad Off
the that $\mathrm{Win}_{\mathrm{Fin}}$ event the systems، game for and، output، the too: there
of difference Each .(5.8 (Definition 1 returns $(\mathcal{G}\mathfrak{m}.\mathrm{pid}, Z)$.FullFin interface

may indicators its where set the of measure the most at is probabilities .Pr[Bad] is which differ، □

worth are they and each، picture a have proof the and definition The Above consumes. other the what exactly is one since together، seeing then coupling the what is below code؛ the of asks 11.1 Definition what is earlier the of vocabulary box-and-wire the uses panel Neither it. with does to machine one from moved is here nothing point: the is that and figures، between. wire a draw to nothing is there so another،



شکل 6: Identical until bad، in two panels. Above، what Definition 11.1 asks of the code؛ below، what the coupling does with it. Neither panel uses the box-and-wire vocabulary of the earlier figures، because nothing here moves from one machine to another. (Definition 11.1 and Lemma 11.2.)

hop a keeps what is which inspection، by checkable is panel upper The behaviour. about claim a not code، of comparison a is it audit: to cheap they them between and allowed، ones only the are shapes shaded two The in ways only the — down is flag the while run can shaded nothing why are flag، the of test a and crossed، is it as flag the raises which head، the are runs what exactly is above shading the So down. is it while fails which else. nowhere and below، fork the beyond

The argument. the of whole the is that and point، single a is fork The both so code، common itself is block divergent each heading statement which — sample coupled same the of step same the at it execute runs two the in number same the is Pr[Bad] that display، first lemma's the gives no is there Bad Off bound. one with numbers two than rather executions indicator single a has it of fact every so run، one are runs two the all: at fork difference each and — $\mathrm{Win}_{\mathrm{Fin}}$ systems، game for and، output، the — there may indicators where set the of measure the most at is probabilities of .Pr[Bad] is measure its and branch، lower the is set That differ.

نکته 11.3 sys- the of nothing asks lemma The .assumed) is (Nothing refusal- not tightness، not well-formedness، not slot: the of or tems responsive of absence the not — savouring worth — and obliviousness، the for hypothesis a pays machine a moves that result Every calls. cores، responsive exclude 5.9 Proposition and 4.29 Theorem move: the and once، at sides both on exclusion the inherits 5.11 Corollary for three all outside sit channel the and network the clock، notifying respon- their — indifferent is above lemma The reason. that exactly nothing because — else anything as cheaply as hop notifications sive on resting runs، about identity an is it regrouping the Like moved. is fixed، are tapes its once execution an of determinism the but nothing for appears budget No achieve. runs two what of comparison a not enters 4.30 Definition and none، needs exactness reason: same the .Pr[Bad] bounding about sets application an when only

نکته 11.4 route: shorter tempting a is There .absorption) by not (Why exists 4.18 Lemma — environment the into execution whole the absorb [5] quote and program، single a is side each until — machines move to why see to instructive is it and works، it where works It pair. the on slot the along: come hypotheses Absorption's lemma. worse the is it re- a place may core no refusal-oblivious، be must moved occupant environment's the and ،(9.4 (Remark identity ceded a on call sponsive

> only lemma the proves reduction the so — -identityZ no hold must par notifying the and covers، already theorem dummy the systems the for than longer no is coupling direct The it. outside falls one، for clock، 4.17 Definition of machinery the nothing: carries and reduction the coupled executions two and rebracketed، execution one for built was — analysis case easier an with induction same the is tapes equal on over. smooth to refusal no claims، of rewriting no relay، no

between hop arguments Emulation concretely. for، is lemma the What simula- a variant، lazier a and functionality a objects: ideal neighbouring pair a is hop Each removed. check one with simulator same the and tor the and company، part two the where raised flags the bad، until identical the against execution، same the in — Pr[Bad] at hop the prices lemma proof a where alongside hopping occupants the with environment، same prop- a generally is that and ،Pr[Bad] bound to is remains What it. needs condition raising the that precisely is verdict whose finalization a :erty applies، transfer where inherited، and 5.8 Definition by priced occurred، func- signature the at sits instance canonical The .5.11 Corollary through honest- 'sVerify where `bad` raise to $\mathcal{F}_{\mathsf{Sig}}$ Instrument .15 Chapter of tionality and written being flag the see، can caller no addition an — fires test key stand. forgery the lets then that variant the it beside set and — read never proba- the costs them between hop the and bad، until identical are two The police. to exists FinUF quantity the — submitted ever is forgery a that bility

فصل 12

پاک‌سازی

af- not do attacks so outputs، simulator sanitize to need often will We fect ideal functionalities too adversely. In honest-majority settings، for instance، we typically sanitize a $\perp$ output to a non-$\perp$ one to preserve liveness. We write such a procedure $\mathsf{San}[\mathsf{Clean}](v; c)$، where Clean is a predicate on a value together with whatever context it consults، and re- place a simulator output v by $v' \leftarrow \mathsf{San}[\mathsf{Clean}](v; c)$، which operates as below unless stated otherwise. The context c after the semicolon lists the functionality state the predicate reads؛ it is read-only، and different predicates take different contexts.

Sanitizer $\mathsf{San}[\mathsf{Clean}]$
predicate Clean، value v، context c
:$\mathsf{San}[\mathsf{Clean}](v; c)$
:1 **if** $\mathsf{Clean}(v; c)$ **then**
:2 **return** v
:3 pick $v' \in \{ v' : \mathsf{Clean}(v'; c) \}$
:4 **return** v'

returns it what and fails، $\mathsf{Clean}(v; c)$ when only acts sanitization So access read-only than more no need sanitizers Our [1].Clean satisfies then specification the of part is picks 3 line How state. functionality the to observably are rules different so caller، the reaches value picked the — value good first lexicographically the fix we and — functionalities different only surfaces pick the looks: it than less matters choice The throughout.

it everything and infinite is from draws Clean space the whenever exists value good A[1] needed are halves Both .c context the in named value a is non-membership beyond excludes of conjunct first the is Σ or $\mathcal{K}$ infinite an in membership :15 Chapter of predicates the for 4.30 Definition Under .c in passed table a of entry an is exclude they else everything and each، many finitely records run bounded a — execution any in supported finitely is context the space. infinite an inside finite is set excluded the so — entries

some preferring adversary an and ،Clean fails answer own slot’s the when
it. submit simply may value good particular

فصل 13

تصادف

13.1 کارکرد

read to one right the and paper this in functionality smallest the is $\mathcal{F}_{\mathrm{Rand}}$
out، handed it what remembers strings، -bit$n$ uniform out hands It first.
1 Chapter remembers. still it what leaks and entry، an forget party a lets
named it and owed no code؛ here is the code.

It footnote. a than rather chapter own its worth it make things Two
is it so — adversary the on not clock، the on not — all at call no places
has 9 Chapter and slot، the to left is all at nothing where example one the
not is paper، this in other every unlike property، its And price. to nothing
bound the probability a to up unpredictable is randomness :0 at attained
decorative. being stops 5.8 Definition of ε the where is which name، to has

per instance one parameters، are **N** and **P** ،pid pattern. ’s$\mathcal{F}_{\mathrm{Sig}}$ **fields. The**
.1 Chapter of pattern local the in served protocol the naming **N** session،
strings، the of width the ،par $:= n$ make. to call no being there ،**U** $:= \emptyset$
.5 line at reads Rnd which

party’s corrupt a so entire، $\mathtt{L}$ returns Leak **for. is leak the what is Erasure**
the but defect a not is That adversary’s. the is draws of history whole
must it randomness draws that protocol a it: beside Erase having of point
Read leak. cannot erased been has what and entry، the erases retain not
and sampling، of as much as erasure secure of model a is $\mathcal{F}_{\mathrm{Rand}}$ way that
One way. same the shaped is 14 Chapter of store the why is pair the
wrongly: state to easy is it since plainly، stating worth is consequence
times of number the is $\mathtt{L}$ in entries of number the erased، is nothing while
erasing values؛ the as well as count the reports leak the so called، was Rnd
is exactly what takes an entry out of that count.

$\mathcal{F}_{\text{Rand}}$ Functionality
par := n ،**U** := $\emptyset$ ،**N** ،**P** ،pid

:Initialize()
:1 $\mathtt{ctr} \leftarrow 0$
:2 $\mathtt{L} : \mathbb{N} \to \{0,1\}^n \cup \{\square\}$
:3 $\mathtt{L}[*] \leftarrow \square$

$\mathtt{id}'$ from $\mathtt{id}$.Rnd()
:4 $\mathtt{ctr} \leftarrow \mathtt{ctr} + 1$
:5 $r \leftarrow_{\$} \{0,1\}^n$ / the one sampling in the paper
:6 $\mathtt{L}[\mathtt{ctr}] \leftarrow r$
:7 **return** $(\mathtt{ctr}, r)$ / the index، so a caller may erase

$\mathtt{id}'$ from $\mathtt{id}$.Erase(i)
:8 $\mathtt{L}[i] \leftarrow \square$
:9 **return** ok

$\mathtt{id}'$ from $\mathtt{id}$.Leak()
:10 **return** $\mathtt{L}$

index the with answers Rnd operation The code. the on remarks Three
be must draw a erase to means that caller A string. the as well as wrote it
so caller's، the not and state functionality's the is $\mathtt{ctr}$ and it، name to able
to willing caller a by used be only could it beside Erase the 7 line without
this and draws، it coin every erases 21.2 Section of protocol the — guess
so rewritten، never and counter a by indexed are Entries it. lets what is
never-written between distinction 's$\mathcal{F}_{\text{Store}}$ to companion no needs Erase
index an at $\square$ and reused، never is passed has Rnd index an erased: and
counter the says it above $\square$ while erased was entry the says $\mathtt{ctr}$ below
any as parties، across shared is counter the And there. reached not has
dis- draw instance one from drawing parties two is: state functionality's
per-party a than rather source a $\mathcal{F}_{\text{Rand}}$ makes what is which entries، tinct
generator.

13.2 ویژگی‌ها

advance in names environment no :unpredictability ،FinUnp property، One
.5.8 Definition of reading search the takes It returns. draw later a value a
$\mathcal{F}_{\text{Rand}}$.**N** := $\{\mathcal{G}m.\text{pid}\}$ with $\gamma := \{\mathcal{G}m\} \cup \{\mathcal{F}_{\text{Rand}}\}$ over played is game The
once argument tightness 's15.2 Section ،$\mathcal{F}_{\text{Rand}}$.**P** $\subseteq$ $\mathcal{G}m$.**P** and pinned
the draw a is see not does game the draw a reason: its for and more،
on. rule cannot verdict

Guess operation The relay. a not is that interface one carries shell The
environment the that so exists it all؛ at not $\mathcal{F}_{\text{Rand}}$ touches and call no places
verdict the lets what is sequence a being trace the and record، on go can
order، of use 's17.2 Section is That first. record on went it whether ask
answers، of order the read steadiness purpose: different a for borrowed
answer. an against claim a of order the reads unpredictability

follows finalization the :$\mathcal{F}_{\text{Rand}}$ over shell Game

،**U** $:= \{(\mathcal{F}_{\text{Rand}}, \text{serves}_{\mathcal{G}m})\}$ ،**N** $:=$ **Z** ،**P** $:= \mathcal{F}_{\text{Rand}}$.**P** $\cup \{Z\}$ ،pid $:= (\mathcal{G}m.F, 0, 0)$
par $:= \perp$

:Initialize()

:1 done $\leftarrow$ false
:2 $\mathtt{tr} \leftarrow ()$ / the trace، a sequence

id' from id.Rnd()

:3 **require** $\neg$done
:4 **require** $\mathrm{id}.P \neq Z$ / the root only finalizes
:5 $(i, r) \leftarrow \mathrm{id}_{\mathcal{F}_{\text{Rand}}}$.FullRnd() as id / the index is not recorded
:6 $\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{rnd}, \mathrm{id}.P, r)$
:7 **return** r

id' from id.Erase(i)

:8 **require** $\neg$done
:9 **require** $\mathrm{id}.P \neq Z$
:10 $\mathrm{id}_{\mathcal{F}_{\text{Rand}}}$.FullErase($i$) as id
:11 $\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{erase}, \mathrm{id}.P, i)$
:12 **return** ok

id' from id.Guess(y)

:13 **require** $\neg$done
:14 $\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{guess}, \mathrm{id}.P, y)$ / no call: the root may guess too
:15 **return** ok

id' from id.Leak()

:16 **return** $\perp$

shared is $\mathcal{F}_{\text{Rand}}$ over shell the :$\mathcal{G}m$ of finalization The

(unpredictability) id' from id.FinUnp()

:17 **require** $\neg$done
:18 **require** $\mathrm{id}.P = Z$
:19 done $\leftarrow$ true
:20 parse $\mathtt{tr}$ as $(e_j)_{j=1}^{m}$
:21 $w \leftarrow 1$ if $\exists j < k \ \exists y : \ e_j = (\text{guess}, \cdot, y) \wedge e_k = (\text{rnd}, \cdot, y)$
:22 else $w \leftarrow 0$
:23 **return** w

does Guess and root the refuse relays The code. the about things Two would root the at relay a 's:17.2 Section exactly is asymmetry the and not، ap- and ،1 line at rej answered be serve، not does $\mathcal{F}_{\text{Rand}}$ party a address guessed that environment an whereupon — trace the to (rnd, Z, rej) pend relay، the at Refused nothing. predicted having outright، win would rej actually $\mathcal{F}_{\text{Rand}}$ string a carries entry rnd every and appended، is nothing refused be to nothing has it so call، no places Guess operation The drew. hon- asks verdict the And included. root the everywhere، served is and for intercepts wrapper own game's the party corrupt a at nobody: of esty core's honest an is $\mathtt{tr}$ in entry every so ،(5.4 (Remark runs core the before .17.2 Section in as exactly already،

گزاره 13.1 ($\mathcal{F}_{\mathsf{Rand}}$ is unpredictable). Let $\mathcal{Z}$ place at most q_g calls on Guess and at most q_r on Rnd. Then for every occupant $\mathcal{X}$ of the adver- sary slot،

$$\Pr[\mathsf{Win}_{\mathsf{FinUnp}}] \leqslant q_g\, q_r\, 2^{-n}.$$

So $\{\mathcal{F}_{\mathsf{Rand}}\}$ has unpredictability within $q_g q_r 2^{-n}$ against $(\mathbb{Z}, \mathbb{X})$ for any $\mathbb{Z}$ class meeting those counts and $\mathbb{X}$ the class of all occupants.

اثبات. Fix k with $e_k = (\mathsf{rnd}, \cdot, r)$. By tightness no caller but $\mathcal{G}m$ reaches $\mathcal{F}_{\mathsf{Rand}}$، and $\mathcal{G}m$ reaches it only through the shell's Rnd relay (line 5)، so r is the value $\mathcal{F}_{\mathsf{Rand}}$'s own Rnd drew at line 5: uniform on $\{0,1\}^n$ and indepen- dent of everything the execution did before it، $\mathcal{F}_{\mathsf{Rand}}$ placing no call and reading no state to draw it.

Condition on the execution up to the moment before that draw. Every guess entry with $j < k$ is determined by that history، and there are at most q_g of them، so the probability that r equals any of their values is at most $q_g 2^{-n}$. There are at most q_r choices of k، and the verdict at line 21 returns 1 only if some pair (j, k) matches، so a union bound over k gives $q_g q_r 2^{-n}$.

Two details the bound rests on. The occupant of the slot does not ap- pear: $\mathcal{F}_{\mathsf{Rand}}$'s cores place no call، so nothing $\mathcal{X}$ does reaches the sampling، and the same bound holds against every occupant without an absorption argument. And a guess made after a draw is worthless by construction، the verdict asking $j < k$; an environment that reads r and then guesses it has predicted nothing، which is what makes the trace's order the content of this verdict. $\square$

The bound is the first in this paper that is not 0، and it is worth saying what that changes. Corollary 5.11 reads the same either way: a system π put in place of $\{\mathcal{F}_{\mathsf{Rand}}\}$ has unpredictability within $q_g q_r 2^{-n} + \varepsilon$ under the search reading، the ideal bound and the emulation error simply adding. What it changes is that the ideal side is no longer free — a realization inherits a bound that was already nontrivial، and the two terms are of dif- ferent kinds: one is the sampling's own، and no protocol does better than it، while the other is what the realization costs. Transfer itself is as cheap as it gets: $\mathcal{F}_{\mathsf{Rand}}$'s cores place no call، responsive or otherwise، so Re- mark 9.4's caveat never arises and Theorem 4.29 is untouched، exactly as for $\mathcal{G}_{\mathsf{PKI}}$ (Section 16.2).

فصل 14

ذخیره‌سازی

14.1 کارکرد

be them lets and values produces one where companion: ’s$\mathcal{F}_{\text{Rand}}$ is $\mathcal{F}_{\text{Store}}$ party A are. they until them holds and values takes other the forgotten، is what reports leak the it؛ erases and back، it gets index، an at value a puts keeping protocol a by means 1 Chapter what are they Together held. still persis- all ،$\mathcal{F}_{\text{Rand}}$ through randomness all — tape machine a on state no .Leak own its through holds it what exposing each ،$\mathcal{F}_{\text{Store}}$ through tence

parameters، **N** and **P** ،pid parameter: the save exactly، ’s$\mathcal{F}_{\text{Rand}}$ **fields. The** is state its ’s،$\mathcal{F}_{\text{Rand}}$ Like none. reading store a ،par $:= \perp$ and ،**U** $:= \emptyset$ put، another what get may party one — serves it parties the across shared what and tape، per-party a than rather store a it makes what is which point. the being sharing by means 1 Chapter

the and conditions، three of one in is index An **two. not states، Three** there؛ put ever was nothing if $\square$ is It purpose. on apart them keeps code was it if $\perp$ is it and erased؛ not and put was something if value a holds it empty either in index an accepts Put operation The erased. then and put get a so ،$\square$ only refuses Get reusable؛ is index erased an so condition، distinction The refused. than rather $\perp$ answered is index erased an on can caller a answers: two those between difference the in place its earns which gone، is and here was something from here ever was nothing tell rather it performed that protocol the to observable erasure makes what is alike both refuse would difference the hid that store A hole. silent a than happened. had erasure an whether asked be not could and

$\mathcal{F}_{\text{Store}}$ Functionality

pid, **P**, **N**, **U** $:= \emptyset$, par $:= \perp$

Initialize():
1: $\mathtt{L} : \mathbb{N} \to \{0,1\}^* \cup \{\perp, \square\}$
2: $\mathtt{L}[*] \leftarrow \square$

$\mathtt{id}.\mathtt{Put}(i, x)$ from $\mathtt{id}'$
3: **require** $\mathtt{L}[i] \in \{\perp, \square\}$ / an occupied index refuses
4: $\mathtt{L}[i] \leftarrow x$
5: **return** ok

$\mathtt{id}.\mathtt{Erase}(i)$ from $\mathtt{id}'$
6: $\mathtt{L}[i] \leftarrow \perp$
7: **return** ok

$\mathtt{id}.\mathtt{Get}(i)$ from $\mathtt{id}'$
8: **require** $\mathtt{L}[i] \neq \square$ / never written refuses
9: **return** $\mathtt{L}[i]$ / $\perp$ if erased

$\mathtt{id}.\mathtt{Leak}()$ from $\mathtt{id}'$
10: **return** $\mathtt{L}$

Two remarks on the code. The refusals are written as **require**s rather than as returns of rej, which is the same thing said in the framework's own words: Section 3.6 answers a refused **require** with rej, so lines 3 and 8 give exactly the two refusals a store owes, and no core of this paper returns rej by writing it. And Erase refuses nothing: erasing an index that holds nothing is a no-op that answers ok, because a protocol erasing defensively should not have to know whether anything was there.

14.2 ویژگی‌ها

Two properties, and they are the two halves of what a store is for. FinKeep, persistence: a value put and not erased is the value got back. FinGone, erasure: a value erased is not got back. Both take the search reading, and both are two finalizations of one game — they read the same trace of the same three relays and differ in the verdict alone, which is the test Section 17.2 gives for sharing a shell.

They are played over $\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \{\mathcal{F}_{\text{Store}}\}$ with $\mathcal{F}_{\text{Store}}.\mathbf{N} := \{\mathcal{G}\mathfrak{m}.\text{pid}\}$ and pinned $\mathcal{F}_{\text{Store}}.\mathbf{P} \subseteq \mathcal{G}\mathfrak{m}.\mathbf{P}$, tightness as before and for the same reason: a put the game does not see is a value the verdict cannot account for. The relays refuse the root, for Section 13.2's reason: both verdicts compare a value got against a put one, so a rej of the game's own making in the trace would be read as a value and could win either.

follow finalizations the :$\mathcal{F}_{\text{Store}}$ over shell Game

،**U** $:= \{(\mathcal{F}_{\text{Store}}, \text{serves}_{\mathcal{G}m})\}$ ،**N** $:=$ **Z** ،**P** $:= \mathcal{F}_{\text{Store}}$.**P** $\cup \{Z\}$ ،pid $:= (\mathcal{G}m.F, 0, 0)$
par $:= \perp$

:Initialize()

done $\leftarrow$ false
$\mathtt{tr} \leftarrow ()$

id′ from id.Put(i, x)

require ¬done
require $\text{id}.P \neq Z$
$a \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}$.FullPut$(i, x)$ as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{put}, \text{id}.P, i, x, a)$
return a

id′ from id.Erase(i)

require ¬done
require $\text{id}.P \neq Z$
$\text{id}_{\mathcal{F}_{\text{Store}}}$.FullErase$(i)$ as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{erase}, \text{id}.P, i)$
return ok

id′ from id.Get(i)

require ¬done
require $\text{id}.P \neq Z$
$y \leftarrow \text{id}_{\mathcal{F}_{\text{Store}}}$.FullGet$(i)$ as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{get}, \text{id}.P, i, y)$
return y

id′ from id.Leak()

return $\perp$

shared is $\mathcal{F}_{\text{Store}}$ over shell the :$\mathcal{G}m$ of finalizations The

(persistence) id′ from id.FinKeep()

require ¬done
require $\text{id}.P = Z$
done $\leftarrow$ true؛ parse $\mathtt{tr}$ as $(e_j)_{j=1}^{m}$
$w \leftarrow 1$ if $\exists j < k\ \exists i, x, y:\ e_j = (\text{put}, \cdot, i, x, \text{ok}) \wedge e_k = (\text{get}, \cdot, i, y)$
$\wedge\ y \neq x \wedge \neg\exists l \in (j, k) : e_l = (\text{erase}, \cdot, i)$؛
else $w \leftarrow 0$
return w

(erasure) id′ from id.FinGone()

require ¬done
require $\text{id}.P = Z$
done $\leftarrow$ true؛ parse $\mathtt{tr}$ as $(e_j)_{j=1}^{m}$
$w \leftarrow 1$ if $\exists j < k < l\ \exists i, x:\ e_j = (\text{put}, \cdot, i, x, \text{ok})$
$\wedge\ e_k = (\text{erase}, \cdot, i) \wedge e_l = (\text{get}, \cdot, i, x)$
$\wedge\ \neg\exists l' \in (k, l) : e_{l'} = (\text{put}, \cdot, i, \cdot, \text{ok})$؛
else $w \leftarrow 0$
return w

some if 1 returns FinKeep finalization The words. in verdicts، The index that at put successful last the value the than other answered get if 1 returns FinGone finalization The intervening. erasure no there، left have to supposed was erasure an value the exactly answered get some or- trace's the read Both between. in it restored having put no removed، is store a reason: same the for and does steadiness 's17.2 Section as der، between and next، the and call one between happens what about claim a ordering. an is

> and FinKeep of either Fin For .properties) both has $\mathcal{F}_{\mathsf{Store}}$) **14.1 گزاره** adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every for ،FinGone against 0 within property each has $\{\mathcal{F}_{\mathsf{Store}}\}$ So .$\Pr[\mathsf{Win}_{\mathsf{Fin}}] = 0$ slot، occupants. all of class the $\mathbb{X}$ for ،5.8 Definition of sense the in $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$

call a inside only changes $\mathtt{L}[i]$ properties: both carries invariant One .اثبات tight- from comes It names. call that value the to and records، trace the own game's the from guard)، the passes game the but caller (no ness from and runs، core any before party corrupt a at intercepting wrapper and call relay's a between runs nothing so call، no placing cores 's$\mathcal{F}_{\mathsf{Store}}$ then are properties Both order. execution in append entries and return its track and conjuncts، verdict's the assume move: same the by it off read Persis- named. entries the between $\mathtt{L}[i]$ written have could lines which can put later no so index، occupied an refusing put the on turns tence ،$\square$ nor $\perp$ neither containing $\{0,1\}^*$ on turns erasure unrecorded؛ intervene it. before put value the be cannot erase an after returns get a value the so trace the call a inside only changes $\mathtt{L}[i]$ invariant: one on rests Everything $\mathcal{G}m$ but caller no tightness By names. call that value the to and records، shell's the through only it reaches $\mathcal{G}m$ and ،$\mathcal{F}_{\mathsf{Store}}$ at 2 line 'sGuard passes wrapper own game's the party corrupt a at ؛(15 and 11 ،5 (lines relays three 's$\mathcal{F}_{\mathsf{Store}}$.(5.4 (Remark all at runs γ of core no and earlier step a intercepts call relay's a between runs γ of machine other no so call، no place cores below numbers Line order. execution in append entries and return، its and name 's$\mathcal{F}_{\mathsf{Store}}$ own cores unless said otherwise.

$e_k = (\mathsf{get}, \cdot, i, y)$ and $e_j = (\mathsf{put}, \cdot, i, x, \mathsf{ok})$ suppose persistence: For 'sPut so ،ok answered put The between. i at erase no and $j < k$ with 6 line 'sErase and line that Only .$\mathtt{L}[i] \leftarrow x$ set 4 line its and passed 3 line lies i at entry erase no recorded: is each invariant the by and ،$\mathtt{L}[i]$ write either، between lie cannot ok answering i at put further a and between، So on. j from occupied being $\mathtt{L}[i]$ and index occupied an refusing 3 line hence ؛x returns 9 line its and passes، 8 line 'sGet ،k at still $\mathtt{L}[i] = x \neq \square$ .0 gives 22 line at verdict the and $y = x$

$e_l =$ and $e_k = (\mathsf{erase}, \cdot, i)$ ،$e_j = (\mathsf{put}, \cdot, i, x, \mathsf{ok})$ suppose erasure: For .l and k between i at put successful no and $j < k < l$ with $(\mathsf{get}, \cdot, i, x)$ that line only the invariant the by and ،k at $\mathtt{L}[i] \leftarrow \perp$ set 6 line 'sErase answering i at put recorded a inside 4 line 'sPut is value a restore could

9 line ’sGet and ،l at $\mathtt{L}[i] = \perp$ So excludes. hypothesis the which ،ok argument، ’sPut a from 4 line by written was x :$x \neq \perp$ Now .$\perp$ returns ،x answer not did get the Hence .$\square$ nor $\perp$ neither contains $\{0,1\}^*$ and $\square$ .0 gives 29 line at verdict the and ،e_1 to contrary

beyond discharge to nothing with 5.11 Corollary by transfer Both the so all، at call no place cores ’s$\mathcal{F}_{\text{Store}}$ conditions: standing four its complete- dummy and way cheap the met is hypothesis responsive-call functionalities three The .$\mathcal{G}_{\text{PKI}}$ and $\mathcal{F}_{\text{Rand}}$ for as exactly untouched، is ness pat- the and nothing، pay that three the are nothing adversary the ask that the is clock the charges 17.2 Section what — coincidence a not is tern bill. no has notification no with functionality a and notification،

فصل 15

امضاهای دیجیتال

15.1 کارکرد

own Canetti’s of tradition the in signatures، digital idealizes below $\mathcal{F}_{\mathsf{Sig}}$
not and parameter a is id process Its .[6] functionality signature ideal
subscript the what is that — fixed is component name The constant. a
and session the but — 1 Chapter of convention naming the by records،
several and session per copy a hold may system one so not، are instance
Def- one and handle، to exists composition situation the session، a within
The pinned. were id process whole the if outright forbid would 1.2 inition
any at read be it lets 7 Chapter so component، neither reads below code
global any which ،7.2 Definition of condition the meeting **N** an for — other
abbreviates $\mathcal{A}(x)$ that 3.4 Section from Recall does. one local purely or
ideal the in served؛ being party the of behalf on adversary the to call a
here them of one every and calls، those answers simulator the execution
use. before 12 Chapter of sense the in sanitized is

throughout. same the is style The **functionalities. ideal write we How**
pin not does specification the and produced be to has value a Wherever
fly. the on choice the sanitize and it choose simulator the let we down، it
goal security the as freedom much as have to meant then is simulator The
goal the and — theorem a not intention، design a — more no and permits
code. the in buried than rather predicates the in visible stays

correct- protect two first the and times، three happens this $\mathcal{F}_{\mathsf{Sig}}$ In
adversary، the from key verification the takes Gen operation The ness.
issued already key every from distinct key، actual an it makes $\mathsf{Clean}_{\mathsf{vk}}$ and
generation so it، under valid recorded already signature any of free and
oper- The pre-forged. arrive nor collide neither keys and succeeds always
not signature actual an it makes Clean_{σ} and signature، the takes Sign ation
been has key a whenever succeeds signing so invalid، recorded already

asks and that، for asks Locality verifies. produces it what and generated،
elsewhere، token the passing without answer simulator the that addition in
The code. the of property a than rather class simulator the on condition a
outside verdict a instead: unforgeability for sanitized is Verify operation
key a under triple fresh a on 1 of verdict any does so and ،0 becomes $\{0, 1\}$
unforgeable. strongly signatures makes which generated، party honest an
rather inline written is sanitization the so unique، is value good the There
so authenticated، assumed not are keys Verification .San through than
own. its of vk a takes Verify

ad- the hands functionality، ideal an of requires 3.2 Section which ،Leak
recorded signatures the with together key verification party's the versary
fixes Initialize operation The state. the of else nothing and it، under valid
`Ver` and party served per key one holds `VK` types: their and tables two the
ev- starting both signature، and message key، of triple each on verdict the
recorded a from decided" yet "not separates that value the ،□ at erywhere
infinite :□ nor $\perp$ neither contain and infinite are Σ and $\mathcal{K}$ Both .1 or 0
two the of clean and values، good of out run never sanitizers the that so
them. refuses already membership that so markers

to restricted be must class simulator the wanted، is locality Where
placing without call simulator each answering those ones: responsive the
are calls simulator and once at returns token the so own، their of call any
enforces 9 Chapter discipline the — calls coroutine than rather subroutine
the nothing، costs this signatures stateless non-interactive، For code. in
already. responsive being simulators usual

Sec- re-entrance. against guards Verify and Gen in line further One
operations both and re-entered، be instance suspended a lets 3.6 tion
writing and entry table a testing between call adversary an at suspend
recorded value a and resumption، on entry the re-tests therefore each its
two issue could calls nested two re-test the Without wins. meanwhile
con- very the — triple one on verdicts different return or party، one to keys
record. to exists `Ver` sistency

$\mathcal{F}_{\mathsf{Sig}}$ Functionality

pid، P، N، $U := \{(\mathcal{A}, \text{serves})\}$، $\text{par} := \bot$

:Initialize()

$\mathtt{VK} : \mathcal{F}_{\mathsf{Sig}}.\mathbf{P} \to \mathcal{K} \cup \{\square\}$
$\mathtt{VK}[*] \leftarrow \square$
$\mathtt{Ver} : \mathcal{K} \times \mathcal{M} \times \Sigma \to \{0,1\} \cup \{\square\}$
$\mathtt{Ver}[*,*,*] \leftarrow \square$

$\mathtt{id}'$ from $\mathtt{id}$.Gen()

if $\mathtt{VK}[\mathtt{id.P}] \neq \square$ **then**
  return $\mathtt{VK}[\mathtt{id.P}]$
$\mathsf{vk} \leftarrow \mathcal{A}(\mathtt{id}.\mathsf{Gen})$
if $\mathtt{VK}[\mathtt{id.P}] \neq \square$ **then**
  return $\mathtt{VK}[\mathtt{id.P}]$ / recorded while suspended
$\mathsf{vk} \leftarrow \mathsf{San}[\mathsf{Clean}_{\mathsf{vk}}](\mathsf{vk}; \mathtt{VK}, \mathtt{Ver})$
$\mathtt{VK}[\mathtt{id.P}] \leftarrow \mathsf{vk}$
return vk

$\mathtt{id}'$ from $\mathtt{id}$.Sign(msg)

$\mathsf{vk} \leftarrow \mathtt{VK}[\mathtt{id.P}]$
if $\mathsf{vk} = \square$ **then**
  return $\bot$
$\sigma \leftarrow \mathcal{A}(\mathtt{id}.\mathsf{Sign}, \mathsf{msg})$
$\sigma \leftarrow \mathsf{San}[\mathsf{Clean}_\sigma](\sigma; \mathsf{msg}, \mathsf{vk}, \mathtt{Ver})$
$\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] \leftarrow 1$
return σ

$\mathtt{id}'$ from $\mathtt{id}$.Verify(vk, msg, σ)

if $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] \neq \square$ **then**
  return $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma]$
$b \leftarrow \mathcal{A}(\mathtt{id}.\mathsf{Verify}, \mathsf{vk}, \mathsf{msg}, \sigma)$
if $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] \neq \square$ **then**
  return $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma]$ / recorded while suspended
if $b \notin \{0,1\}$ **then**
  $b \leftarrow 0$
if $\exists\, P' \notin \mathbf{C} : \mathsf{vk} = \mathtt{VK}[P']$ **then**
  $b \leftarrow 0$ / no forgery under an honest key
$\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] \leftarrow b$
return b

$\mathtt{id}'$ from $\mathtt{id}$.Leak()

if $\mathtt{VK}[\mathtt{id.P}] = \square$ **then**
  return $(\bot, \emptyset)$
return $(\mathtt{VK}[\mathtt{id.P}], \{(\mathsf{msg}, \sigma) : \mathtt{Ver}[\mathtt{VK}[\mathtt{id.P}], \mathsf{msg}, \sigma] = 1\})$

:$\mathsf{Clean}_{\mathsf{vk}}(\mathsf{vk}; \mathtt{VK}, \mathtt{Ver})$

return $\mathsf{vk} \in \mathcal{K} \wedge \neg\exists P : \mathtt{VK}[P] = \mathsf{vk}$
  $\wedge\ \neg\exists\, \mathsf{msg}, \sigma : \mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] = 1$

:$\mathsf{Clean}_\sigma(\sigma; \mathsf{msg}, \mathsf{vk}, \mathtt{Ver})$

return $\sigma \in \Sigma \wedge \mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] \neq 0$

15.2 ویژگی‌ها

one give We instance. an worth is shape a and shape، a is 5.1 Definition the what that saying ،FinCor properties: two carrying $\mathcal{F}_{\mathsf{Sig}}$ over $\mathcal{Gm}$ game else nothing that saying ،FinUF and verifies، produces interface signing which games، two than rather shell one of finalizations two are They does. read they — for ask two these what and allows now shape the what is The alone. verdict the in differ and oracles، same the of trace، same the and call، to which choosing by on judged is it which chooses environment other. the against game the closes one calling

Both the take search the reading of Definition 5.8،

$$\mathsf{Adv}^{\mathsf{FinCor}}_{\mathcal{Gm},\mathcal{F}_{\mathsf{Sig}},\mathcal{Z},\mathcal{X}} := \Pr[\,\mathsf{Win}_{\mathsf{FinCor}}\,],$$

$$\mathsf{Adv}^{\mathsf{FinUF}}_{\mathcal{Gm},\mathcal{F}_{\mathsf{Sig}},\mathcal{Z},\mathcal{X}} := \Pr[\,\mathsf{Win}_{\mathsf{FinUF}}\,],$$

better no merely not and impossible be to meant is either winning because is time the half succeed forgeries whose scheme signature a coin: a than

average. not broken،
with ،γ := $\{\mathcal{G}m\} \cup \{\mathcal{F}_{\mathsf{Sig}}\}$ system game the over played are They
and — name its not pinned، id، process game's the — $\mathcal{F}_{\mathsf{Sig}}$.**N** := $\{\mathcal{G}m.\mathsf{pid}\}$
the at and، serves $\mathcal{F}_{\mathsf{Sig}}$ party every at exist relays the so ،$\mathcal{F}_{\mathsf{Sig}}$.**P** $\subseteq$ $\mathcal{G}m$.**P**
the — 1 line at rej answered are and serve not does it party a address root،
interface an Inside plays. it where not finalizes، game the where being root
party the at $\mathcal{F}_{\mathsf{Sig}}$ of interface the ،$(\mathcal{F}_{\mathsf{Sig}}.\mathsf{pid}, \mathtt{id}.\mathrm{P})$ for $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$ write we id at
5.1 Definition which ،$\{\mathcal{G}m\}$ at tightness is That at. called was game the
game legal a γ makes It over. twice keep its earns it here and requires،
2 line passes $\mathcal{G}m$ but caller No complete. trace the makes it and system؛
out- lie names fresh and -identities$Z$ whose environment، the not :$\mathcal{F}_{\mathsf{Sig}}$ at
not and -identity؛A an is claim whose adversary، the not ؛**N** pinned a side
game's the of instance invented an at hosts environment the machine a
every So refuses. name the than rather id the pinning what is which name،
A recorded. is and below 8 line through passes issues ever $\mathcal{F}_{\mathsf{Sig}}$ signature
interface" signing the from obtained not was "this is verdict whose game
know. to way other no has

follow finalizations the :$\mathcal{F}_{\mathsf{Sig}}$ over shell Game

par := $\perp$ ،**U** := $\{(\mathcal{F}_{\mathsf{Sig}}, \mathsf{serves}_{\mathcal{G}m})\}$ ،**N** := **Z** ،**P** := $\mathcal{F}_{\mathsf{Sig}}$.**P**$\cup\{Z\}$ ،pid := $(\mathcal{G}m.\mathrm{F}, 0, 0)$

:Initialize()
:1 done ← false
:2 tr ← ( ) // the trace، a sequence

id′ from id.Gen()
:3 **require** ¬done
:4 vk ← $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.FullGen() as id
:5 tr ← tr · (gen, id.P, vk)
:6 **return** vk

id′ from id.Sign(msg)
:7 **require** ¬done
:8 σ ← $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.FullSign(msg) as id
:9 tr ← tr · (sign, id.P, msg, σ)
:10 **return** σ

id′ from id.Verify(vk, msg, σ)
:11 **require** ¬done
:12 b ← $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.FullVerify(vk, msg, σ) asid
:13 tr ← tr · (ver, id.P, vk, msg, σ, b)
:14 **return** b

id′ from id.Leak()
:15 **return** $\perp$

shared is $\mathcal{F}_{\mathsf{Sig}}$ over shell the ؛$\mathfrak{Gm}$ of finalizations The

(correctness) $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinCor}()}$

:16 **require** $\neg$done
:17 **require** $\mathtt{id}.\mathrm{P} = \mathrm{Z}$
:18 done $\leftarrow$ true
:19 **C** $\leftarrow (\mathrm{C}, \mathrm{Z})$.FullStatus() as $\mathtt{id}$
:20 $w \leftarrow 1$ if $\exists \mathrm{P}, \mathrm{Q}, \mathsf{vk}, \mathsf{msg}, \sigma :\ \mathrm{P} \notin \mathbf{C} \wedge \mathrm{Q} \notin \mathbf{C} \wedge \sigma \in \Sigma$
:21 $\wedge$ $(\mathsf{gen}, \mathrm{P}, \mathsf{vk}) \in \mathtt{tr} \wedge (\mathsf{sign}, \mathrm{P}, \mathsf{msg}, \sigma) \in \mathtt{tr} \wedge (\mathsf{ver}, \mathrm{Q}, \mathsf{vk}, \mathsf{msg}, \sigma, 0) \in \mathtt{tr}$؛
:22 else $w \leftarrow 0$
:23 **return** w

(unforgeability) $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinUF}()}$

:24 **require** $\neg$done
:25 **require** $\mathtt{id}.\mathrm{P} = \mathrm{Z}$
:26 done $\leftarrow$ true
:27 **C** $\leftarrow (\mathrm{C}, \mathrm{Z})$.FullStatus() as $\mathtt{id}$
:28 $w \leftarrow 1$ if $\exists \mathrm{P}, \mathrm{Q}, \mathsf{vk}, \mathsf{msg}, \sigma :\ \mathrm{P} \notin \mathbf{C} \wedge \mathrm{Q} \notin \mathbf{C}$
:29 $\wedge$ $(\mathsf{gen}, \mathrm{P}, \mathsf{vk}) \in \mathtt{tr} \wedge (\mathsf{ver}, \mathrm{Q}, \mathsf{vk}, \mathsf{msg}, \sigma, 1) \in \mathtt{tr} \wedge (\mathsf{sign}, \mathrm{P}, \mathsf{msg}, \sigma) \notin \mathtt{tr}$؛
:30 else $w \leftarrow 0$
:31 **return** w

oc- means $\in$ sequence، a is trace The code. the about things Four and $\mathtt{tr}$ read verdicts The order. the reads verdict no and it، in currence serves$_{\mathfrak{Gm}}$ in exemption root the why is which nothing، call and register the rather end the at $\mathrm{P} \notin$ **C** ask Both .Z serve not need $\mathcal{F}_{\mathsf{Sig}}$ and suffices that makes (3.5 (Section monotonicity and signing، of time the at than honest was closes game the when honest party a reading: stronger the no where $\perp$ answers Sign because $\sigma \in \Sigma$ asks Correctness throughout. func- the against hold to signature no is $\perp$ and generated، been has key the not triple، the excludes 28 line :strong is unforgeability And tionality. forgery a as counts message signed a on signature second a so message، give. to written was sanitizer 's15 Chapter notion the is which —

گزاره 15.1 $\mathcal{F}_{\mathsf{Sig}}$) has both properties). For Fin either of FinCor and FinUF، for every $\mathbb{Z}_{\gamma,\gamma} \ni \mathcal{Z}$ and every occupant $\mathcal{X}$ of the adversary slot، $\Pr[\mathsf{Win}_{\mathsf{Fin}}] = 0$. So $\{\mathcal{F}_{\mathsf{Sig}}\}$ has each property within 0 against $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ in the sense of Definition 5.8، for the $\mathbb{X}$ class of all occupants.

اثبات. One preliminary and two verdicts. The preliminary is that both par- ties named by a verdict must be honest، and it is not bookkeeping: at a corrupt party the core does not run and the answer is the adversary's، so a mediated Verify could report a 0 on a good signature or a 1 on a forgery، Cor- wrong. nothing done having $\mathcal{F}_{\text{Sig}}$ with fire would finalization either and Verify and $\texttt{Ver}[\text{vk}, \text{msg}, \sigma] \leftarrow 1$ recording Sign from follows then rectness whether on splits Unforgeability line. first its at entry recorded a returning under 1 a computed have cannot it one: computed or entry an read Verify records Sign only and recorded، was entry the so key، party's honest an honest them makes Monotonicity close. the at honest parties Take one. firing 4 line mediated، is them at places game the call no so throughout، ver- Both .$\{0, 1\}^*$ in lying name own game's the and party corrupt a at only both: of it need both and ،Q verifier the of and P signer the of this ask dicts adversary's، the is answer the and run not does core the party corrupt a at signature، good a on 0 a — likes it whatever reports Verify mediated a so having $\mathcal{F}_{\text{Sig}}$ with 1 return would finalization either and — forgery a on 1 a or wrong. nothing done

.$\sigma \in \Sigma$ with $(\text{sign}, P, \text{msg}, \sigma) \in \texttt{tr}$ suppose correctness، For set then and Clean_σ against σ sanitized it so ran، Sign of core The never and once fixed Gen which ،$\text{vk} = \texttt{VK}[P]$ for $\texttt{Ver}[\text{vk}, \text{msg}, \sigma] \leftarrow 1$ at it returns and set entry that finds $\text{Verify}(\text{vk}, \text{msg}, \sigma)$ later Any changes. ،Q every for $(\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 0) \notin \texttt{tr}$ thus ؛1 returns it so line، first its .0 gives 20 line and

$\text{vk} =$ with $(\text{ver}, Q, \text{vk}, \text{msg}, \sigma, 1) \in \texttt{tr}$ suppose unforgeability، For computed it or entry recorded a read Verify Either .P honest an for $\texttt{VK}[P]$ sets $\texttt{Ver}$ writing before test last the :1 a computed have cannot It one. So witnesses. P which ،$P' \notin$ **C** some for $\text{vk} = \texttt{VK}[P']$ whenever $b \leftarrow 0$ whose party the at — 1 a records Sign only and recorded، was entry the Hence twice. key no issues Clean_{vk} since P being party that ،vk is key ☐ .0 gives 28 line and $(\text{sign}, P, \text{msg}, \sigma) \in \texttt{tr}$

ideal the says merely 0 is it that part؛ interesting the not is bound The functional- ideal an as properties، these have to written was functionality the and it، with does then 5.11 Corollary what is content The be. should ity $\{\mathcal{F}_{\text{Sig}}\}$of place in put π system a useful: that makes what is reading search adversary every against ،ε most at probability with finalization either wins is probability that reading search the under and dummy، the only not and reading decision the Had itself. property the bounds ε so advantage، the which ،$2\Pr[\text{Win}_{\text{Fin}}] - 1$ on 2ε read would conclusion same the taken been suc- forger a — all at constraint no is unwinnable be to meant game a for ap- 5 Chapter of rule calibration the is That it. meets time the half ceeding each displays: both carries 5.10 Proposition why and bites، it where plied inherit two these and fixes، reading own its scale the at inherits finalization .ε at

by discharged is none and corollary the with come conditions Four well-formed؛ be must π by $\{\mathcal{F}_{\mathsf{Sig}}\}$ of replacement The alone. emulation pin to members 'sπ asks which system، game a be again must $\{\mathfrak{Gm}\} \cup \pi$ simu- the ؛5.1 Definition of part is tightness since did، $\mathcal{F}_{\mathsf{Sig}}$ where $\mathfrak{Gm}$.pid environments the and refusal-oblivious؛ be must adversary the and lator argued separately never thus is realization A -identity.Z no silence must emulation one the from both inherits it — unforgeable or correct be to above finalizations two the and checks، four those with together proof، of. is inheritance the what are

5.11 Corollary bites. it where is this and it، with travels caveat One $\mathcal{F}_{\mathsf{Sig}}$ and call، responsive no place system game the of cores the that asks Should applies. corollary the so ،$\mathcal{A}(\cdot)$ through slot the reaches written as which — Verify and Sign ،Gen in $\mathcal{A}^{!}$ write and advice 's9 Chapter adopt one reason the for instead، directly argued be must transfer the — locality buys priced stand، things as are، inheritance and Locality gives. 9.4 Remark other. each against

فصل 16

زیرساختِ کلید عمومی

16.1 کارکرد

Verify — out hands it key the authenticate not does it that plainly said $\mathcal{F}_{\text{Sig}}$
keys verification because caller، any to unconnected own، its of vk a takes
be- $\mathcal{G}_{\text{PKI}}$ functionality The .(15.1 (Section authenticated assumed not are
cer- own Canetti's of shape the in gap، the closes that setup the is low
the to key a binding directory shared single، a :[6] functionality tification
because trusted be can Verify to handed vk a that so it، claims that party
believed. simply was it because than rather here from came it

only and key، one register may party Each said. quickly is does it What
one conjures which ،Gen unlike — input as key the takes Register one:
first the keeps and — factory a not and directory a is this slot، the from
two no that so file on already key every against sanitized given، is it key
anyone answers Retrieve operation The one. same the hold ever parties
has one no if nothing or name، they party the for file on is whatever with
registered.

id process The reason. same the for and 's،$\mathcal{G}_{\text{Clock}}$ are They **fields. The**
copy: one at system the caps 1.2 Definition so ،$(\text{GPKI}, 0, 0)$ constant، a is
key، which owns who of notion second a be would directory second a
global$(\mathcal{G}_{\text{PKI}})$ makes $\mathbf{N} := \textbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$ out. rule to exists PKI a what exactly
the universe the again is **P** and clock's، the as disjunct same the by hold
in state، is registration — it used has who than rather covers directory
The .$\mathbf{U} := \emptyset$ clock's: the than simpler is field One membership. not ،VK
with arriving key the all، at call no place Retrieve and Register operations
on depends $\mathcal{G}_{\text{PKI}}$ nothing is there so slot، the from than rather request the
has. already functionality every machinery ambient the beyond

ex- two the of each from line one takes Register **guards. borrowed Two**
for acting Z or A named caller a 's:$\mathcal{G}_{\text{Clock}}$ is first The it. before amples

nothing، changes and key own party’s that answered is party honest an honest an on tick to allowed than rather time the answered is it as exactly goes what decide not file، on is what ask may outsider an — behalf party’s the — proceeds call the so fire، not does check the party corrupt a At there. and one، ticks it as freely as behalf party’s corrupt a on registers adversary letting mediation show، will directory the what is submits it key whatever sends corrupt ’s$\mathcal{F}_{AC}$ for does it as unmediated through call this exactly later wins، succeed to call first the ’s:$\mathcal{F}_{Sig}$ is second The .(19.1 (Section as exactly it، overwriting than rather file on already key the returning ones from differs here Sanitizing exists. one once key a re-issues never Gen a fix each Clean_σ and Clean_{vk} repairs: it what in predicates signature both the is which supplied، caller the one fixes Clean_{reg} proposed؛ slot the value already presumably is key own party’s honest an — position exposed more one the and in، wrote adversary the whatever is one’s corrupt a but good، the key a claiming party second a is through let cannot directory a thing reason same the for here exists always value good A holds. already first those exactly are excluded keys the and infinite is $\mathcal{K}$:Clean_{vk} for does it .(12 Chapter of (footnote many finitely recorded، have can run bounded a

$\mathcal{G}_{PKI}$ Functionality

par := $\perp$ ،**U** := $\emptyset$ ،**N** := **Std** $\cup$ **A** $\cup$ **Z** ،**P** ،pid := (GPKI, 0, 0)

:Initialize()

:1 $\mathrm{VK} : \mathcal{G}_{PKI}.\mathbf{P} \to \mathcal{K} \cup \{\square\}$
:2 $\mathrm{VK}[*] \leftarrow \square$

id′ from id.Register(vk)

:3 **if** $\mathtt{id}'.\mathrm{F} \in \{A, Z\} \wedge \mathtt{id}.\mathrm{P} \notin \mathbf{C}$ **then**
:4 **return** $\mathrm{VK}[\mathtt{id}.\mathrm{P}]$ / a peek، not a write
:5 **if** $\mathrm{VK}[\mathtt{id}.\mathrm{P}] \neq \square$ **then**
:6 **return** $\mathrm{VK}[\mathtt{id}.\mathrm{P}]$
:7 vk $\leftarrow$ San[Clean_{reg}](vk; VK)
:8 $\mathrm{VK}[\mathtt{id}.\mathrm{P}] \leftarrow$ vk
:9 **return** vk

id′ from id.Retrieve(P)

:10 **return** $\mathrm{VK}[\mathrm{P}]$

id′ from id.Leak()

:11 **return** VK

:Clean_{reg}(vk; VK)

:12 **return** vk $\in \mathcal{K} \wedge \neg\exists\, \mathrm{P} : \mathrm{VK}[\mathrm{P}] =$ vk

party، the against recorded is Registration code. the on remarks Three the tied already has Guard are: ticks reason same the for caller، the not reg- P when token the holds machine whichever so ،2 line by $\mathtt{id}.\mathrm{P}$ to call none and ،Deregister no is There entry. own ’sP on lands key the isters، the is bound once key a but revocation، support can PKI real a added: is for left addition genuine a is revocation and is، there directory simplest the with answers Leak And over. papered gap a than rather elsewhere at entry one out gives already Retrieve than more no is which table، whole this only not interfaces، its of one every at public is directory the — time a times، enough asked ،Retrieve nothing adversary the tells Leak so one، not. would

No either. than less needs $\mathcal{G}_{\mathsf{PKI}}$ it، before examples both against Set key the — Gen in as nothing، of out choose to simulator's the is value than rather input caller's a repairs here sanitizing so — call the with arrives no And it. before San every from differs this place one the output، slot's a about responsive be to notification no is there so all، at core the leaves call as transfer below properties two the settle: to 9 Chapter for price no and can. property a as cheaply

16.2 ویژگی‌ها

local a — promise of kinds two having for properties two carried $\mathcal{G}_{\mathsf{Clock}}$ moves it that one، global a and coded، as moves counter the that one، same the splits $\mathcal{G}_{\mathsf{PKI}}$ directory The agreed. has honest everyone when only regis- a says fixity، ،FinFix game. one of finalizations two into and way، entry. own party's one about promise، local the — changes never key tered global the — key same the hold ever parties two no says binding، ،FinBind of reading search the take Both once. at $\mathcal{G}_{\mathsf{PKI}}.\mathbf{P}$ of all over quantified one، .5.8 Definition

$\mathcal{G}_{\mathsf{PKI}}$ with ،$\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \{\mathcal{G}_{\mathsf{PKI}}\}$ system game the over played are Both Re- — pinned ،$\mathbf{N} := \{\mathcal{G}\mathfrak{m}.\mathsf{pid}\}$ field: one save 16.1 Section in as taken iden- the for and ،(17.2 (Section clock the for as exactly forcing، 's6.5 mark adversary an against property a is $\mathcal{F}$ global a of property a reason: tical there key every about claims both are binding and fixity and it، shares that is. there all is trace game's the where only certifiable is،

session one for asks 3 line so global، longer no is directory the Pinned، passes $\mathcal{G}\mathfrak{m}$ but caller No it. meets 0 session own experiment's the and adversary's the not and hosts، it machine a not environment، the not :2 line ever $\mathcal{G}_{\mathsf{PKI}}$ key every So apply. even could 4 line before refused -identities،$\mathcal{A}$ whose game a — recorded is and below relay the through enters holds an Inside it. know to way other no has key" a share parties "two is verdict .$(\mathcal{G}_{\mathsf{PKI}}.\mathsf{pid}, \mathtt{id}.\mathrm{P})$ for $\mathtt{id}_{\mathcal{G}_{\mathsf{PKI}}}$ write we $\mathtt{id}$ at interface

follow finalizations the :$\mathcal{G}_{\mathsf{PKI}}$ over shell Game

par := $\bot$ ،**U** := $\{(\mathcal{G}_{\mathsf{PKI}}, \mathsf{serves}_{\mathcal{G}\mathfrak{m}})\}$ ،**N** := **Z** ،**P** := $\mathcal{G}_{\mathsf{PKI}}$.**P**$\cup\{\mathrm{Z}\}$ ،pid := $(\mathcal{G}\mathfrak{m}.\mathrm{F}, 0, 0)$

:Initialize()

done $\leftarrow$ false
$\mathtt{tr} \leftarrow ()$ / the trace، a sequence

id′ from id.Register(vk)

require $\neg$done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$
$\mathsf{vk}' \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{PKI}}}$.FullRegister(vk) as $\mathtt{id}$
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{reg}, \mathtt{id}.\mathrm{P}, \mathsf{vk}')$
return vk′

id′ from id.Leak()

return $\bot$

id′ from id.Retrieve(P)

require $\neg$done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$
$\mathsf{vk}' \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{PKI}}}$.FullRetrieve(P) as $\mathtt{id}$
return vk′

shared is $\mathcal{G}_{\mathsf{PKI}}$ over shell the :$\mathcal{G}\mathfrak{m}$ of finalizations The

(fixity) id′ from id.FinFix()

require $\neg$done
require $\mathtt{id}.\mathrm{P} = \mathrm{Z}$
done $\leftarrow$ true
C $\leftarrow$ (C, Z).FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists \mathrm{P}, \mathsf{vk}_1, \mathsf{vk}_2 :\ \mathrm{P} \notin \mathbf{C} \wedge \mathsf{vk}_1 \neq \mathsf{vk}_2$
$\wedge (\mathsf{reg}, \mathrm{P}, \mathsf{vk}_1) \in \mathtt{tr} \wedge (\mathsf{reg}, \mathrm{P}, \mathsf{vk}_2) \in \mathtt{tr}$؛
else $w \leftarrow 0$
return w

(binding) id′ from id.FinBind()

require $\neg$done
require $\mathtt{id}.\mathrm{P} = \mathrm{Z}$
done $\leftarrow$ true
C $\leftarrow$ (C, Z).FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists \mathrm{P}, \mathrm{Q}, \mathsf{vk} :\ \mathrm{P} \notin \mathbf{C} \wedge \mathrm{Q} \notin \mathbf{C} \wedge \mathrm{P} \neq \mathrm{Q}$
$\wedge (\mathsf{reg}, \mathrm{P}, \mathsf{vk}) \in \mathtt{tr} \wedge (\mathsf{reg}, \mathrm{Q}, \mathsf{vk}) \in \mathtt{tr}$؛
else $w \leftarrow 0$
return w

and relayed is Retrieve operation The code. the about things Two look not could that game a trace: its reads verdict neither though logged inter- the leaving and all، at $\mathcal{G}_{\mathsf{PKI}}$ exercising be not would up key anyone's games for Read relayed shell 's$\mathcal{G}_{\mathsf{Clock}}$ as exactly nothing، costs whole face and vk_1 need not does verdict 'sFinFix And mutators. the only read that means idempotence trace: its of reading steadiness's unlike ordered، vk_2 the write، the be can them produced that calls two the of one most at a identically، game the loses order either and it، of read later a only other first. came one whichever key changed a being key changed

گزاره 16.1 ($\mathcal{G}_{\mathsf{PKI}}$ has fixity). For every $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ and every occupant $\mathcal{X}$ of the adversary slot، $\Pr[\mathrm{Win}_{\mathsf{FinFix}}] = 0$. So $\{\mathcal{G}_{\mathsf{PKI}}\}$ has fixity within 0 against $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ in the sense of Definition 5.8، for $\mathbb{X}$ the class of all occupants.

اثبات. Suppose $(\mathsf{reg}, \mathrm{P}, \mathsf{vk}_1), (\mathsf{reg}, \mathrm{P}, \mathsf{vk}_2) \in \mathtt{tr}$ with $\mathbf{C} \notin \mathrm{P}$. By tightness both were logged at line 5، each running $\mathcal{G}_{\mathsf{PKI}}$'s Register at own core $\mathrm{P} = \mathtt{id}$. the relay's own name is $\mathcal{G}\mathtt{m}$'s، not $\mathcal{A}$ or $\mathcal{Z}$، so line 4 never intercepts it. The core's only write is line 8، reached only while $\mathtt{VK}[\mathrm{P}] = \square$، and once reached it fixes $\mathtt{VK}[\mathrm{P}]$ to some sanitized value for good — every later call at P finds $\mathtt{VK}[\mathrm{P}] \neq \square$ and returns line 6's value، the same one، unchanged. So the two calls producing vk_1 and vk_2 agree: whichever ran first wrote the value، and the other only read it back. The verdict at line 17 gives 0. □

گزاره 16.2 ($\mathcal{G}_{\mathsf{PKI}}$ has binding). For every $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ and every occupant $\mathcal{X}$ of the adversary slot، $\Pr[\mathrm{Win}_{\mathsf{FinBind}}] = 0$. So $\{\mathcal{G}_{\mathsf{PKI}}\}$ has binding within 0 against $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ in the sense of Definition 5.8، for $\mathbb{X}$ the class of all occupants.

اثبات. Suppose $(\mathsf{reg}, \mathrm{P}, \mathsf{vk}), (\mathsf{reg}, \mathrm{Q}, \mathsf{vk}) \in \mathtt{tr}$ with $\mathbf{C} \notin \mathrm{P}, \mathrm{Q}$ and $\mathrm{P} \neq \mathrm{Q}$. By Proposition 16.1's argument $\mathsf{vk} = \mathtt{VK}[\mathrm{P}]$ and $\mathsf{vk} = \mathtt{VK}[\mathrm{Q}]$ from the moment each was first written onward. Say Q's write came second، the case P's symmetric: at that moment Register's line 8 ran $\mathsf{vk} \leftarrow \mathsf{San}[\mathsf{Clean}_{\mathsf{reg}}](\cdot; \mathtt{VK})$ with $\mathtt{VK}[\mathrm{P}]$ already set، and San's guarantee (Chapter 12) is that its output satisfies whatever $\mathsf{Clean}_{\mathsf{reg}}$ input — membership in $\mathcal{K}$ and freshness against every key already on file، P's among them. So Q's write cannot equal $\mathtt{VK}[\mathrm{P}]$، contradicting $\mathtt{VK}[\mathrm{Q}] = \mathtt{VK}[\mathrm{P}] = \mathsf{vk}$. The verdict at line 25 gives 0. □

Both transfer by Corollary 5.11 with nothing extra to argue. The hypothesis it asks — that the game system's cores place no responsive call — is met the cheap way: $\mathcal{G}_{\mathsf{PKI}}$'s cores place no call of any kind، responsive or not، so Remark 9.4's caveat never arises and dummy completeness (Section 4.4) is untouched. Unlike $\mathcal{G}_{\mathsf{Clock}}$، whose notification bought locality at inheritance's expense، and unlike $\mathcal{F}_{\mathsf{Net}}$ and $\mathcal{F}_{\mathsf{AC}}$، whose slot calls priced their own transfer، $\mathcal{G}_{\mathsf{PKI}}$ pays nothing: a system π put in place of $\{\mathcal{G}_{\mathsf{PKI}}\}$

’s15.2 Section of conditions four the directly، binding and fixity inherits that directory A discharge. to left ones only the being paragraph closing spend. to nothing has nothing adversary the asks

فصل 17

ساعتِ جهانی

17.1 کارکرد

con- it about said was what of most and randomized، and local is $\mathcal{F}_{\text{Sig}}$ counts. both on opposite its is example second The simulator. the cerned and Tackmann Maurer، Katz، by introduced was functionality clock A Maurer، Badertscher، by recast ؛[11] UC in synchrony model to Zikas it ،[3] counter monotone a around setup shared a as Zikas and Tschudi composable the one the functionality، global canonical the become has be- $\mathcal{G}_{\text{Clock}}$ clock The by. time keep successors its and Bitcoin of analyses low is that functionality the in language the of this paper.

only and 0 at starts `now` counter The said. quickly is does it What enter Parties it. read may serves clock the Anyone forward. steps ever the and ،Deregister and Register through structure round the leave and Update to: it asked has party registered honest every when steps counter registered honest the once and party، calling the against tick a records them. clears and counter the advances 17 line on test the in، all are ticks the and done، is it in party honest last the when exactly ends round A code the how — early it close nor round the stall neither can adversary adversary the to reported is tick recorded Each below. is each refuses the with and (9 (Chapter responsively ،20 line on returns، Update before at settled is costs it what and buys notification the what discarded؛ answer stands it as counter the answers operation Every .17.2 Section of end the ticks the and time new the learns round a closes that tick the so return، on again. reads stands it where know must that party a old؛ the it before

a was 's$\mathcal{F}_{\text{Sig}}$ where ،(GClock, $0, 0$) constant، a is id process The **fields. The** which system، any in copy one most at puts then 1.2 Definition parameter. would clock second a and shared، instance، one means global point: the is first its by hold global($\mathcal{G}_{\text{Clock}}$) makes $\mathbf{N} := \mathbf{Std} \cup \mathbf{A} \cup \mathbf{Z}$ time. second a be session every and check session the waives Guard of 3 line so disjunct، component session the separation very the — instance one the reaches time keeps that protocol A here. made not deliberately make، to exists 1.3 Definition of disjunct global the and ،**U** its in $(\mathcal{G}_{\text{Clock}},$ serves$)$ declares copy. shared the at entry the meet session every of members lets what is notification the places، code the call one the for **U** $:= \{(\mathcal{A},$ serves$)\}$ And the than rather universe the as but parameter a stays **P** And .20 line of in is who while for، exists interface an parties the fixes it membership: through left and entered state، is moment given a at structure round the table، more one is membership dynamic — [3] in as Deregister and Register field. of kind new a not

operation، an of not functionality، the of field a is **N** **ticking. and Reading** :Read for one right the is above set the and core، every governs set one so and both، lets [3] — time the ask may environment the and adversary the global a makes what of much is clock shared the reading environment the free adversary an where mutators، three the for wide too is It global. setup it drop or finished، not had party the round a close could party a for act to of line first the as sits refinement The altogether. structure round the from answered is party honest an for acting $\mathcal{Z}$ or $\mathcal{A}$ named caller a core: each the — passes it party corrupt a at while nothing، changes and time the ،[3] in as owns، it parties the for ticks and deregisters registers، adversary corrupt a reading never 17 line matters، it where inert are ticks those and is party honest no while naming: deserves consequence One flag. party's a register may adversary the so vacuous، is 17 line of test the registered is time registration، honest first the until — freely it tick and party corrupt .[3] in as too there spend، to adversary's the

is it and party"، registered honest "every reads 17 line of quantifier The stan- a party corrupt a At itself. enforces $\mathcal{G}_{\text{Clock}}$ it of little how seeing worth inter- full the of 4 line by adversary the to handed is Update caller's dard party same the at call own adversary's the and runs، core the before face adversary's the are ticks party's corrupt a way either ؛13 line by admitted is conven- the by register the reads which — $\forall$ the and withhold، or give to therefore parties Corrupt regardless. them ignores — 3.1 Section of tion both party honest an At one. close help nor open round a hold neither ticks observers' the turns 13 line and fire، not does Mediate shut: routes tick- after corrupted party a piece: last the adds Monotonicity reads. into on، then from quantifier the outside is it but behind، flag stale a leaves ing it. clears advance next the and

$\mathcal{G}_{\text{Clock}}$ **Functionality**

par := $\perp$ ،**U** := $\{(\mathcal{A}, \text{serves})\}$ ،**N** := **Std** $\cup$ **A** $\cup$ **Z** ،**P** ،pid := (GClock, 0, 0)

:Initialize()

:1 `now` $\leftarrow 0$
:2 `reg`, `tick` : $\mathcal{G}_{\text{Clock}}$.**P** $\to \{0, 1\}$
:3 `reg`[∗] $\leftarrow 0$؛ `tick`[∗] $\leftarrow 0$

id′ from id.Register()

:4 **if** id′.F $\in \{A, Z\} \wedge$ id.P $\notin$ **C then**
:5 **return** `now`
:6 `reg`[id.P] $\leftarrow 1$؛ `tick`[id.P] $\leftarrow 0$ / a fresh member owes a tick
:7 **return** `now`

id′ from id.Deregister()

:8 **if** id′.F $\in \{A, Z\} \wedge$ id.P $\notin$ **C then**
:9 **return** `now`
:10 `reg`[id.P] $\leftarrow 0$
:11 **return** `now`

id′ from id.Update()

:12 **if** id′.F $\in \{A, Z\} \wedge$ id.P $\notin$ **C then**
:13 **return** `now` / reading، not ticking
:14 **if** `reg`[id.P] $= 0$ **then**
:15 **return** `now` / not in the round structure
:16 `tick`[id.P] $\leftarrow 1$
:17 **if** $\forall$ P $\in \mathcal{G}_{\text{Clock}}$.**P** $\setminus$ **C** : `reg`[P] $= 1 \Rightarrow$ `tick`[P] $= 1$ **then**
:18 `now` $\leftarrow$ `now` $+ 1$
:19 `tick`[∗] $\leftarrow 0$ / the round is over
:20 $\mathcal{A}^{!}$(id.Update, `now`) / notify؛ the answer is discarded
:21 **return** `now`

id′ from id.Read()

:22 **return** `now`

id′ from id.Leak()

:23 **return** (`now`, `reg`, `tick`)

recorded are ticks and Registrations code. the on remarks Three
register may P for acting machine any caller: the not party، the against
in- [3] .2 line on party the to call the tied already having Guard ،P tick or
before separately tick functionality and party registered every has stead
identity by indexed tables is wanted، if discipline، that closes؛ round the
’sUpdate is notification The changes. else nothing and party، than rather
and too، departures and registrations about adversary the tells [3] alone؛
And mechanical. is mutators two other the to pattern ’s20 line extending
is counter the secrets: no keeping clock a state، whole the returns Leak
and round the in is who only say tables the and anyway، read to anyone’s
it. with done is who

still is needed example signature the what of most ،$\mathcal{F}_{\text{Sig}}$ against Set
to simulator’s the is value No code. the in legible absence each absent،
to nothing is there so — discarded is answer notification’s the — choose
its re-tested and $\mathcal{A}(\cdot)$ through slot the reached $\mathcal{F}_{\text{Sig}}$ where And sanitize.
re- is call one clock’s the hung، it while happen could what against tables
and ،20 line precedes Update of mutation every construction: by sponsive
re-entered ever is instance no so leaving، from token the keeps Respond
oc- every against enforced — re-test to nothing is there and mid-operation
class. simulator a advise only could 15 Chapter where slot، the of cupant
plays 17.2 Section bites: it where recorded is costs notification the What
there. it prices and clock the over properties two

17.2 ویژگی‌ها

second، a worth is clock the once؛ 5.1 Definition instanced 15.2 Section is order whose paper this in thing one the is time because least not properties: two carrying $\mathcal{G}_{\text{Clock}}$ over $\mathcal{G}\mathfrak{m}$ game one give We content. the promises code the as moves counter the that saying steadiness، ،FinStead saying unanimity، ،FinUna and — pushed when only and one، by forward، — Both pushed. has party registered honest every when only moves it that be to meant being either winning ،5.8 Definition of reading search the take hard. than rather impossible

the with ،$\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \{\mathcal{G}_{\text{Clock}}\}$ system game the over played are They The pinned. ،$\mathbf{N} := \{\mathcal{G}\mathfrak{m}.\text{pid}\}$ field: one save 17 Chapter in as taken clock asks requires، 5.1 Definition which ،$\{\mathcal{G}\mathfrak{m}\}$ at Tightness forced. is change is **Std** bare a and ids، process whole admit to test under members the operational in why says 5.7 Definition refuses؛ 6.5 Remark what exactly any- as it touching functionality، global a shares environment an — terms adversary an against property a is $\mathcal{F}$ global a of property a and may، one the but nuisance a not is sharing properties two these For it. shares that all about assertions are unanimity and steadiness claim: the of negation all is trace its where only them certify can game a and are، there ticks the are certified properties the and privately، played is clock the So is. there 15.2 Section of situation the — change not does sharing which code، its of under $\mathcal{F}_{\text{Sig}}$ the and protocol its names ships that $\mathcal{F}_{\text{Sig}}$ the where exactly، game. the names test

one for asks Guard of 3 line so global، longer no is clock the Pinned، carries id process clock's the and 0 session in sits experiment the session؛ at 2 line passes now $\mathcal{G}\mathfrak{m}$ but caller no And passes. check the so ،$s = 0$ lie claims whose hosts، it machine a or environment the not clock: the refused are -identities$A$ whose adversary، the not and ،$\mathbf{N}$ pinned a outside changes counter the So time. the them answer even could 13 line before Register the inside only membership the and below 15 line inside only game a — recorded is change every and it، beside relays Deregister and way other no has pushed" one no when moved time "the is verdict whose and ،$(\mathcal{G}_{\text{Clock}}.\text{pid}, \mathtt{id}.\mathrm{P})$ for $\mathtt{id}_{\mathcal{G}_{\text{Clock}}}$ write we $\mathtt{id}$ at interface an Inside know. to gave. clock the τ answer the carries entry trace every

root. the refuse relays the :15.2 Section in analogue no has line One were and serve not does $\mathcal{F}_{\text{Sig}}$ party a addressed root the at relays the There well-shaped over existential verdicts، the which ،1 line at rej answered game's the of rej a so entry، every reads Steadiness read. never tuples،

nothing relay، the at refused trace؛ the into allowed be cannot making own
gave. actually clock the answer an is $\mathtt{tr}$ in entry every and appended، is
honest an answering π a rightly: and bites، this replacement a Against
that clock a spot، the on steadiness loses number a but anything read
moves. it however broken being time the say cannot

prop- single a that for asks shell shared a thing first the is line That
like existential، is verdict its — it need not does Unanimity not. would erty
steadi- is refusal the so — did those as rej a past read would and ’s،$\mathcal{F}_{\mathsf{Sig}}$
finalizations both because it carries shell the and alone، demand ness’s
serves، it verdicts the of strictest the to built is shell A shell. the off hang
trace a on played is unanimity others: the by paid is that of cost the and
thing one the is it because naming، worth cost a is It needs. it than cleaner
by weakened be would property a Where away. take can shell a sharing
second a is answer the over-served، merely than rather demand another’s
finalization. second a not and game

follow finalizations the :$\mathcal{G}_{\mathsf{Clock}}$ over shell Game

par := ،**U** := $\{(\mathcal{G}_{\mathsf{Clock}}, \mathsf{serves}_{\mathcal{G}m})\}$ ،**N** := **Z** ،**P** := $\mathcal{G}_{\mathsf{Clock}}.\mathbf{P} \cup \{Z\}$ ،pid := $(\mathcal{G}m.F, 0, 0)$
$\perp$

:Initialize()
done ← false
$\mathtt{tr} \leftarrow ()$ / the trace، a sequence

$\mathtt{id}'$ from $\mathtt{id}$.Register()
require ¬done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$ / the root only finalizes
$\tau \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{Clock}}}$.FullRegister() as $\mathtt{id}$
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{reg}, \mathtt{id}.\mathrm{P}, \tau)$
return τ

$\mathtt{id}'$ from $\mathtt{id}$.Deregister()
require ¬done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$
$\tau \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{Clock}}}$.FullDeregister() as $\mathtt{id}$
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{dereg}, \mathtt{id}.\mathrm{P}, \tau)$
return τ

$\mathtt{id}'$ from $\mathtt{id}$.Tick()
require ¬done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$
$\tau \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{Clock}}}$.FullUpdate() as $\mathtt{id}$
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{tick}, \mathtt{id}.\mathrm{P}, \tau)$
return τ

$\mathtt{id}'$ from $\mathtt{id}$.Read()
require ¬done
require $\mathtt{id}.\mathrm{P} \neq \mathrm{Z}$
$\tau \leftarrow \mathtt{id}_{\mathcal{G}_{\mathsf{Clock}}}$.FullRead() as $\mathtt{id}$
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{read}, \mathtt{id}.\mathrm{P}, \tau)$
return τ

$\mathtt{id}'$ from $\mathtt{id}$.Leak()
return $\perp$

shared is $\mathcal{G}_{\mathsf{Clock}}$ over shell the ؛$\mathcal{G}m$ of finalizations The

(steadiness) $\mathtt{id}'$ from <u>$\mathtt{id}$.FinStead()</u>

require $\neg$done
require $\mathtt{id}.P = Z$
done $\leftarrow$ true
parse $\mathtt{tr}$ as $\big((\mathsf{op}_j, P_j, \tau_j)\big)_{j=1}^{n}$؛ $\tau_0 := 0$
$w \leftarrow 1$ if $\exists j \in \{1, \dots, n\}:\ \ \tau_j \notin \{\tau_{j-1},\ \tau_{j-1}+1\}$
$\quad \vee \ \big(\tau_j = \tau_{j-1}+1 \ \wedge \ \mathsf{op}_j \neq \mathsf{tick}\big)$؛
else $w \leftarrow 0$
return w

(unanimity) $\mathtt{id}'$ from <u>$\mathtt{id}$.FinUna()</u>

require $\neg$done
require $\mathtt{id}.P = Z$
done $\leftarrow$ true
$\mathbf{C} \leftarrow (C, Z)$.FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists t \geqslant 0 \ \exists P \in \mathcal{G}_{\mathsf{Clock}}.\mathbf{P} \setminus \mathbf{C}$:
$\quad \big(\exists \tau_r \leqslant t : (\mathsf{reg}, P, \tau_r) \in \mathtt{tr}\big) \wedge (\mathsf{dereg}, P, \cdot) \notin \mathtt{tr}$
$\quad \wedge \ \big(\exists \tau > t : (\cdot, \cdot, \tau) \in \mathtt{tr}\big) \wedge \big(\forall \tau' \in \{t,\ t+1\} : (\mathsf{tick}, P, \tau') \notin \mathtt{tr}\big)$؛
else $w \leftarrow 0$
return w

an- an on 1 returns FinStead finalization The words. in verdicts، The
an or it، above one than more answer an predecessor، its below swer
must departures and registrations reads، tick: a at but anywhere increase
is $\tau_{j-1}+1$ answered tick A it. left answer last the where time the find
'sInitialize is $\tau_0 := 0$ and — round the closed that tick the is it — allowed
already is 1 answered read first a so observation، first the as taken word
re- FinUna finalization The pushed. anyone before moved that counter a
the at honest — party some although t some passed counter the if 1 turns
t answered tick no has — released never and t round by registered close،
covering $t+1$ answer the is، that ،t round during placed tick no :$t+1$ or
the not trace، the read conjuncts registration The it. closed that one the
did. itself game the what only name too they so tables، clock's

was trace the There instructive. both ،15.2 Section with contrasts Two
else، little reads FinStead here read؛ verdict no order whose sequence a
had verdicts both there And besides. much not and order an being time
any- of nothing asks FinStead named؛ they party every of $P \notin \mathbf{C}$ ask to
be- intercepts wrapper own game's the party corrupt a at because one،
an is $\mathtt{tr}$ in entry every so ،(5.4 (Remark records and runs core the fore
is honesty name must that verdict one The already. answer core's honest
it names it and call، not did that parties concerns claim whose 's،FinUna

stronger the that making again (3.5 (Section monotonicity — close the at not؛ other the register، the reading one — so differ two the That reading. for: is shell shared a what is — membership other the order، reading one them made that verdicts، the not common، in have they oracles the is it game. one

نکته 17.1 property The .parties) across read steadiness is (Agreement hold parties honest :agreement is for wanted usually is clock shared a corol- a but game third a not is it Here time. the of view common a the over quantifies 28 Line design. by and above، verdict the of lary en- tick consecutive two between party: by party not interleaved، trace two so asked، whoever predecessor، its equal must answer every tries the answered are ticks two same the between reading parties honest own party's each — party per Stated won. is game the or value same agree- and weaker، strictly be would verdict the — steady subsequence 5 with P answer then could replacement a property: separate a ment What property. neither fail and activations adjacent in 3 with Q and to- single a model: execution the is claims both carry verdict one lets and exist not do reads" "simultaneous so trace، the orders totally ken Two entries. adjacent of consistency the to reduces view common a a only، parties honest among is Agreement naming. worth are limits says it and ؛(5.4 (Remark adversary's the being answers party's corrupt while stale grow may view whose ask، not does that party a of nothing the of disagreement not party، the of staleness — ahead runs clock the never that caller a promise could functionality a nothing and clock، syn- loosely the ،Δ some within agreeing views — variant drift A calls. ،28 line of weakening genuine a be would — [11] of clocks chronized third a than rather finalization third a is it shape present the under and the in differs and relays same the of trace same the reads it game: shell. this off it hanging for test the exactly is which alone، verdict

گزاره 17.2 and FinStead of either Fin For .properties) both has $\mathcal{G}_{\mathsf{Clock}}$) slot، adversary the of $\mathcal{X}$ occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every for ،FinUna $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ against 0 within property each has $\{\mathcal{G}_{\mathsf{Clock}}\}$ So .$\Pr[\mathsf{Win}_{\mathsf{Fin}}] = 0$ occupants. all of class the $\mathbb{X}$ for ،5.8 Definition of sense the in

اثبات. One invariant carries both properties: the counter changes only in- side a tick the trace records and by exactly one، and membership only first the is it Establishing departures. and registrations recorded inside caller no so tightness، — things three on turns it and work، paragraph’s only them reaching game the cores؛ clock’s the reaches game the but handing γ of core no and mediated؛ is answer no so parties، honest at cores clock’s the call single the one، run could that anything to token the each order، execution in append therefore Entries responsive. being place com- by follows then Steadiness returned. call its as counter the carrying Unanimity runs. nothing them between since entries، consecutive paring de- and verdict the of conjuncts four the assumes it half: longer the is corruption of monotonicity using invariant، the from contradiction a rives invariant: one on rests Everything throughout. honest party the keep to exactly by and records، trace the tick a inside only changes counter the departures. and registrations recorded inside only membership the one؛ at counter the — 17 Chapter of cores the in only written are tables Both `reg` passes، 17 line of test the when one by stepping ،Update of line one passing calls on only run cores those and — Deregister and Register in only calls $\mathcal{G}\mathrm{m}$ and ،2 line passes $\mathcal{G}\mathrm{m}$ but caller no pinned **N** with :Guard wrapper own its one corrupt a at — parties honest at relays، the through mediated is call no so — ran γ of core no and earlier step a intercepted γ of core no Moreover recorded. answer the is answer core’s each and clock’s the call one the one: run could that anything to token the hands Respond — responsive is which ،20 line of notification the is place cores slot، the occupies whatever place، would answerer its call every refuses relays’ the and — case any in it precedes Update of mutation every and γ of machine other no with returning chain that clock، the on is call one each order، execution in append therefore Entries between. in run having returned. call its as counter the carrying

$.0 = \tau_0$ ’sInitialize is counter the entry first the before steadiness: For entry at counter the so runs، γ of core no entries consecutive Between own ’sj entry inside taken steps the by $\mathrm{j} - 1$ entry at that from differs j tick. a for one or none departure، a or registration a read، a for none call: tick، a at only increment the with throughout، $\tau_j \in \{\tau_{j-1}, \tau_{j-1} + 1\}$ Hence $.0$ gives 28 line and

some of hold 36 line of conjuncts four the suppose unanimity: For mono- by throughout، honest — close the at $\mathrm{P} \in \mathcal{G}_{\text{Clock}}.\mathbf{P} \setminus \mathbf{C}$ some and t the only ؛$\texttt{reg}[\mathrm{P}] \leftarrow 1$ set $\tau_r \leqslant \mathrm{t}$ answered registration recorded A tonicity. Deregister no placed game the ،`reg` write Deregister and Register of cores honest an reach cannot observers the and — entry dereg no is there — P at ،$\tau > \mathrm{t}$ answers entry Some on. moment that from $\texttt{reg}[\mathrm{P}] = 1$ so party،

stepped it so took، counter the values are answers the invariant the by and
ev- of flags the read 17 line its tick؛ recorded some inside $t+1$ to t from
them، among P — then registered and moment that at honest party ery
flag That .$\texttt{tick}[P] = 1$ so — $\tau_r \leqslant t$ and growing ever only register the
the of step every at cleared and P at Update of core the by only written is
stood counter the while set was it so re-registration، every at and counter
itself was it if $t+1$ or ،t answered tick that ؛P at game the of tick a by ،t at
$(\text{tick}, P, \tau') \in \texttt{tr}$ So recorded. was it and round، the closed that one the
36 line and conjunct، fourth the contradicting ،$\tau' \in \{t, t+1\}$ some for
.0 gives □

was specification the that only says 0 being bound the ،$\mathcal{F}_{\text{Sig}}$ with As
no- the here and transfer، the is content the properties؛ its have to written
cores the that asks 5.11 Corollary price. its collects 20 line of tification
the so one، is line that and call، responsive no place system game the of
for directly، argued be must $\{\mathcal{G}_{\text{Clock}}\}$ of place in put π system a to transfer
Theo- guards hypothesis same the and — gives 9.4 Remark reason the
com- dummy forgoes clock notifying the over built system a so ،4.29 rem
and 20 line delete — notification the without clock A it. with pleteness
them inherits and proof same the by properties both has — entry **U** the
discharge. to nothing with corollary the through

notification A whether. not price، which only is note، choice، The
to token the hand would took، $\mathcal{F}_{\text{Sig}}$ route the ،$\mathcal{A}(\cdot)$ ordinary the through
would answers answering؛ before ticks nested drive to free adversary an
ideal the of fail would steadiness and order، trace of out append then
because only $\mathcal{G}_{\text{Clock}}$ of properties are above verdicts two the — itself clock
chose $\mathcal{F}_{\text{Sig}}$ where So returns. Update before on acted be cannot 20 line
locality choose must clock notifying a locality، with paid and inheritance
tells that one the is neither pays that clock the inheritance؛ with pay and
nothing. adversary the

فصل 18

Network –DelayedΔ

18.1 کارکرد

Communica- network. a is reader first the and read، be to exists clock The
measured delays form، diffusion the]'s؛11] is delay bounded a with tion
analyses blockchain composable the network the is clock، shared a on
language the in functionality that is below $\mathcal{F}_{\text{Net}}$ network The .[3] over run
it. be than rather clock the use to example first the and — paper this of
is $\mathcal{F}_{\text{Net}}$ code؛ no owed and $\mathcal{F}_{\text{Diffuse}}$ channel diffusion a named 1 Chapter
attached. time with like looks channel a such what

mes- the timestamps Send operation The said. quickly is does it What
inter- clock's the ،$\mathtt{id}_{\mathcal{G}_{\text{Clock}}} := (\mathcal{G}_{\text{Clock}}.\text{pid}, \mathtt{id}.\mathrm{P})$ through — clock the by sage
tells it، stores — 17.2 Section of notation the served، being party the at face
The id. message a answers and — public is network the — adversary the
ev- answers then early، reveal to what adversary the asks Fetch operation
The old. Δ everything with together party fetching the to released erything
and ،Δ within when chooses adversary the point: the is power of division
with- is do cannot it what own؛ its of traffic add may and first، whom to
last 'sFetch in sits bound that and time، its past message honest an hold
is τ time at sent message A it. reaches slot the of answer no where line،
first. favoured adversary the whoever on، $\tau + \Delta$ time from fetch every in

are **N** and **P** ،pid clock's. the not pattern، 's$\mathcal{F}_{\text{Sig}}$ follow They **fields. The**
naming **N** its session، per instance one local، is network a parameters:
in- every and — 1 Chapter of pattern local the in serves it protocol the
com- "τ "time makes what is which clock، shared one the reads stance
en- first the :**U** $:= \{(\mathcal{G}_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$ sessions. across parable
met، is and does network the party every serve to clock the requires try
global the by copy shared the at in، sits network the session whatever
only well-formed is $\mathcal{F}_{\text{Net}}$ containing system a — 1.3 Definition of disjunct

be- calls slot two the covers second the and — alongside clock the with
is n 's$\mathcal{F}_{\mathsf{Rand}}$ as reads، actually code of line a parameter a ،par $:=$ Δ low.
.(13 (Chapter

mes- a deliver adversary the lets [3] **called. not answered، is Scheduling**
lit- Transcribed accord. own its of network the on acting by early sage
A containing **N** local a so calls، adversary the interface an is that erally
shape one the other، no in and 0 session in slot the from reachable —
asked is slot the So carry. cannot relocation and excludes 7.2 Definition
(S, I) pair a back takes and time the adversary the hands Fetch instead:
in- sanitized — traffic fresh of injections ids، stored the among releases —
are calls slot Both .15 Chapter in as canonical، being repair good the line،
it before mutates Send operation The discipline. clock's the responsive،
responsive- and answer، the reading mutations its cannot، Fetch notifies؛
straight comes token the immaterial: difference the makes what is ness
re-tests re-entrance 's$\mathcal{F}_{\mathsf{Sig}}$ and call، either inside runs else nothing back،
too. here unnecessary stay

are Fetch and Send party's corrupt A **code. of line a without Corruption،**
buffer the so run، cores the before interface full the of 4 line by intercepted
— mediated never are reads clock cores' the and only، sends honest holds
"sends"، party corrupt a What word. own clock's the always is timestamp a
of injections the instead: side receiving the from supplies adversary the
the to released ،∞ timestamp and $\perp$ sender with buffer the enter 16 line
guarantee، 's20 line into ripening no so and age، no — alone party fetching
traffic. own adversary's the conscripts never and honest the for is which

$\mathcal{F}_{\text{Net}}$ Functionality

par := Δ ،**U** := $\{(\mathcal{G}_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$ ،**N** ،**P** ،pid

:Initialize()
:1 $\texttt{ctr} \leftarrow 0$
:2 $\texttt{buf} \leftarrow \emptyset$ / entries (m, P, msg, τ)
:3 $\texttt{rel} : \mathbb{N} \times \mathcal{F}_{\text{Net}}.\textbf{P} \to \{0, 1\}$
:4 $\texttt{rel}[*, *] \leftarrow 0$ / what is released to whom

id′ from id.Send(msg)
:5 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
:6 $m \leftarrow \texttt{ctr}$؛ $\texttt{ctr} \leftarrow \texttt{ctr} + 1$
:7 $\texttt{buf} \leftarrow \texttt{buf} \cup \{(m, \text{id}.P, \text{msg}, \tau)\}$
:8 $\mathcal{A}^!(\text{id}.\text{Send},\ m,\ \text{msg})$ / the network is public
:9 **return** m

id′ from id.Leak()
:10 **return** $(\texttt{buf},\ \texttt{rel})$

id′ from id.Fetch()
:11 $\tau \leftarrow \text{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
:12 $(S, I) \leftarrow \mathcal{A}^!(\text{id}.\text{Fetch},\ \tau)$ / a non-pair reads as $(\emptyset, \emptyset)$
:13 $S \leftarrow S \cap \{m : (m, \cdot, \cdot, \cdot) \in \texttt{buf}\}$؛
:14 $I \leftarrow I \cap \mathcal{M}$ / sanitized inline
:15 $\texttt{rel}[m, \text{id}.P] \leftarrow 1$ for each $m \in S$
:16 for each msg $\in I$:
:17 $\quad \texttt{buf} \leftarrow \texttt{buf} \cup \{(\texttt{ctr}, \bot, \text{msg}, \infty)\}$؛
:18 $\quad \texttt{rel}[\texttt{ctr}, \text{id}.P] \leftarrow 1$؛ $\texttt{ctr} \leftarrow \texttt{ctr} + 1$
:19 **return** $\{(m, \text{msg}) : (m, P', \text{msg}, \tau_s) \in \texttt{buf}$
:20 $\quad \wedge\ (\texttt{rel}[m, \text{id}.P] = 1\ \vee\ \tau_s + \Delta \leqslant \tau)\}$

net- the sender: no carry pairs Fetched code. the on remarks Three what are injections the and]’s،3] diffusion، unauthenticated an is work sent، one no messages contain may fetch a — means unauthenticated one is diffusion authenticated An write. to adversary’s the being wire the drop keeps، already buffer the sender the return — away line one and field dif- a and step further a is addressing with Authentication injections. the fetch one what sticky: is `rel` table The it. gives 19 Chapter object؛ ferent grows only network the of view party’s a so hide، may fetch later no reveals Δ past silence adversary’s the so deadline، the or `rel` reads 20 line and — state، whole the returns Leak And punished. than rather ignored simply is adversary the to told were contents its secrets: no keeping network the doing. own slot’s the is table release the and arrived، they as 8 line at

18.2 ویژگی‌ها

monotone ،Δ by delivery unformulated: properties three left 18.1 Section all: at game no needs last The sending. before delivery no and views، `buf` with intersecting 20 line name، can S nothing is `buf` in yet not id an release ever can slot the of occupant no so runs، else anything before unformulated، leave still we views Monotone sent. been yet not has what formalize we What timing. to orthogonal and alone `rel` of property a being by ripened message a :liveness ،FinLive name the under first، the is here property a formalize we And fetch. honest later no from missing is Δ an message a :authentication ،FinAuth have، to claimed never was $\mathcal{F}_{\text{Net}}$

reading search the take Both someone. by sent was fetches party honest
.5.8 Definition of

shape that where is this and game، one of finalizations two are They unau- an being and، outright first the has $\mathcal{F}_{\text{Net}}$ network The most. the earns injec- the on remark own 's18.1 Section — design by diffusion thenticated oracles، of set one shell، one second: the lacks provably — 16 line of tions the and unwinnable is one which of it off read verdicts two and trace، one be would that games separate two Under .1 probability with won is other twice. measured object one is it here objects؛ two about statements two en- an flag: single 's5.3 Remark is apart measurements the keeps What no so ،FinLive against game the closed has FinAuth calls that vironment conjunction. a is pair the about nothing and both، on judged is run

with ،$\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ system game the over played are Both tight- 's15.2 Section — $\mathcal{F}_{\text{Net}}.\mathbf{P} \subseteq \mathcal{G}\mathfrak{m}.\mathbf{P}$ and pinned $\mathcal{F}_{\text{Net}}.\mathbf{N} := \{\mathcal{G}\mathfrak{m}.\text{pid}\}$ follow already fields own 's$\mathcal{F}_{\text{Net}}$ since whole، transplanted argument ness be- box interface The .(18.1 (Section clock's the than rather pattern 's$\mathcal{F}_{\text{Sig}}$ the not :$\mathcal{F}_{\text{Net}}$ at Guard passes $\mathcal{G}\mathfrak{m}$ but caller No full. in $\mathcal{G}\mathfrak{m}.\mathbf{U}$ states low in- invented an at hosted machine a not adversary، the not environment، stores، ever core 's$\mathcal{F}_{\text{Net}}$ message every So name. game's the of stance be- relays the through passes ،Fetch a to returns ever it answer every and someone" by "sent names verdict whose game a — recorded is and low and clock the on dependence own 's$\mathcal{F}_{\text{Net}}$ it. know to way other no has $\mathcal{F}_{\text{Net}}$ for was it as exactly met is and this of any by unaffected is slot the .1.3 Definition of disjunct global the by copy shared the at clock the itself، for $\mathrm{id}_{\mathcal{G}_{\text{Clock}}}$ and $(\mathcal{F}_{\text{Net}}.\text{pid}, \mathrm{id}.\mathrm{P})$ for $\mathrm{id}_{\mathcal{F}_{\text{Net}}}$ write we id at interface an Inside
.17.2 Section of notation the second the ،$(\mathcal{G}_{\text{Clock}}.\text{pid}, \mathrm{id}.\mathrm{P})$

follow finalizations the :$\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{Net}}$ over shell Game

U := ،**N** := **Z** ،**P** := $\mathcal{F}_{\text{Net}}.\mathbf{P} \cup \{Z\}$ ،pid := $(\mathcal{G}\mathfrak{m}.\mathrm{F}, 0, 0)$
par := $\perp$ ،$\{(\mathcal{F}_{\text{Net}}, \text{serves}_{\mathcal{G}\mathfrak{m}}), (\mathcal{G}_{\text{Clock}}, \text{serves})\}$

:Initialize()
done ← false
$\mathtt{tr} \leftarrow ()$ / the trace، a sequence

id′ from id.Send(msg)
require ¬done
$\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
$\mathfrak{m} \leftarrow \mathrm{id}_{\mathcal{F}_{\text{Net}}}$.FullSend(msg) as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{send}, \mathrm{id}.\mathrm{P}, \mathfrak{m}, \tau, \text{msg})$
return $\mathfrak{m}$

id′ from id.Fetch()
require ¬done
$\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
$M \leftarrow \mathrm{id}_{\mathcal{F}_{\text{Net}}}$.FullFetch() as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{fetch}, \mathrm{id}.\mathrm{P}, \tau, M)$
return M

id′ from id.Leak()
return $\perp$

shared is $\mathcal{G}_{\mathsf{Clock}}$ and $\mathcal{F}_{\mathsf{Net}}$ over shell the ؛$\mathcal{G}\mathfrak{m}$ of finalizations The

(liveness) $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinLive}()}$

require $\neg$done
require $\mathtt{id}.P = Z$
done $\leftarrow$ true
C $\leftarrow (C, Z)$.FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists P, Q, m, \tau_s, \mathsf{msg}, \tau, M :\; P \notin \mathbf{C} \wedge Q \notin \mathbf{C}$
$\quad \wedge$ (send, P, m, τ_s, msg) $\in \mathtt{tr} \wedge$ (fetch, $Q, \tau, M) \in \mathtt{tr}$
$\quad \wedge\; \tau \geqslant \tau_s + \Delta \wedge (m, \mathsf{msg}) \notin M$؛
else $w \leftarrow 0$
return w

(authentication) $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinAuth}()}$

require $\neg$done
require $\mathtt{id}.P = Z$
done $\leftarrow$ true
C $\leftarrow (C, Z)$.FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists Q, \tau, M, m, \mathsf{msg}$:
$\quad Q \notin \mathbf{C} \wedge$ (fetch, $Q, \tau, M) \in \mathtt{tr} \wedge (m, \mathsf{msg}) \in M$
$\quad \wedge \neg\exists P, \tau_s$: (send, P, m, τ_s, msg) $\in \mathtt{tr}$؛
else $w \leftarrow 0$
return w

'sFetch or 'sSend reads relay Neither code. the about things Three read both so — read to none is there — timestamp a for value return own readings two the ؛$\mathcal{F}_{\mathsf{Net}}$ to call the above line one themselves، clock the guard own 's$\mathcal{F}_{\mathsf{Net}}$ not them، between runs γ of core no because coincide relay the honest، $\mathtt{id}.P$ finds and **C** reads only which check، corruption and .(17.1 (Section call no places which itself، Read not and all، at run having follow fields 's$\mathcal{F}_{\mathsf{Net}}$'s:17.2 Section unlike ،$\mathtt{id}.P = Z$ excludes relay Neither worse the any reads verdict neither and clock's، the not and pattern 's$\mathcal{F}_{\mathsf{Sig}}$ demand the spared is shell the so — too fetching or sending root the for they shell the get finalizations both and clock's، the of made steadiness of nothing asks verdict authentication's And alone. for asked have would is pair fetched the — about ask to none returns Fetch because sender، a exactly is whom، by sent، ever was m some whether and alone، (m, msg) whole the what is which recover، otherwise can $\mathcal{F}_{\mathsf{Net}}$ of caller no fact the testing. is game

occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every For .liveness) has $\mathcal{F}_{\mathsf{Net}}$) **18.1 گزاره** liveness has $\{\mathcal{F}_{\mathsf{Net}}, \mathcal{G}_{\mathsf{Clock}}\}$ So .$\Pr[\mathsf{Win}_{\mathsf{FinLive}}] = 0$ slot، adversary the of $\mathcal{X}$ class the $\mathbb{X}$ for ،5.8 Definition of sense the in $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ against 0 within occupants. all of

of activation some at written ،$(\text{send}, P, m, \tau_s, \text{msg}) \in \mathtt{tr}$ Suppose .اثبات (Sec- one internal 's$\mathcal{F}_{\text{Net}}$.Send and read clock own Its relay. Send the core the so given، just argument the by value، same the answer (18.1 tion .buf in $(m, P, \text{msg}, \tau_s)$ exactly stores

's$\mathcal{F}_{\text{Net}}$.Fetch .$\tau \geqslant \tau_s + \Delta$ with $(\text{fetch}, Q, \tau, M) \in \mathtt{tr}$ also suppose Now and $(m', P', \text{msg}', \tau_{s'}) \in \mathtt{buf}$ with (m', msg') pair every returns core clock the of reading own its τ ،(20 (line $\mathtt{rel}[m', Q] = 1 \vee \tau_{s'} + \Delta \leqslant \tau$ $(m, P, \text{msg}, \tau_s)$ is entry buffer the :$m' = m$ Taking relay's. the again، and، what- met is disjunct second the so hypothesis، by holds $\tau_s + \Delta \leqslant \tau$ and settles which ،12 line at answered slot the whatever and is $\mathtt{rel}[m, Q]$ ever gives 18 line and ،$(m, \text{msg}) \in M$ Hence disjunct. neither and I and S only .0 □

گزاره 18.2 ($\mathcal{F}_{\text{Net}}$ does not have authentication). There are $\mathcal{Z}^* \in \mathbb{Z}_{\gamma,\gamma}$ and an occupant $\mathcal{X}^*$ of the adversary slot with $\Pr[\text{Win}_{\text{FinAuth}}] = 1$: no $\varepsilon < 1$ bounds $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$'s authentication against any class containing them.

answers $\mathcal{X}^*$ occupant The machines. two in counterexample A .اثبات ex- is which — call no places and injection one with query fetch 's$\mathcal{F}_{\text{Net}}$ the to relay to is answer to way only whose dummy، the not is it why actly point whole the is difference That refuses. Respond call a environment، the than rather slot the of occupant every over properties these stating of and satisfied، out come would FinAuth dummy، the at read alone: dummy on finalizes and once fetches then $\mathcal{Z}^*$ environment The vacuously. only the reaches sanitizing، survives injection The else. nothing and FinAuth .1 advantage with fires verdict the so ran، ever relay Send no and trace، $\mathrm{id}.A(\mathrm{id}''.\text{Fetch}, \tau)$ query the On call. no place but fields 's$\mathcal{A}$ have $\mathcal{X}^*$ Let for $(\emptyset, \{\text{msg}^*\})$ returns it — 12 line of shape the — $\mathrm{id}''.F = \mathcal{F}_{\text{Net}}.\text{pid}$ with is it call، no Placing .$\perp$ returns it else anything on ؛$\text{msg}^* \in \mathcal{M}$ fixed one dummy، the unlike calls، slot 's$\mathcal{F}_{\text{Net}}$ of either at Respond by refused never does Respond call a ،(5 (line $\mathcal{Z}$ to relay to is answer to way only whose exactly and ،$\mathcal{D}$ from differs $\mathcal{X}^*$ where exactly is This .(9.4 (Remark refuse of occupant every over stated are 18.1 and 17.2 ،15.1 Propositions why ،5.8 Definition of case dummy's the — alone dummy the not and slot the and satisfied، as FinAuth read would subscript، the from dropped $\mathcal{X}$ with only vacuously، $\mathcal{D}$ being unable to inject at all.

claim then once، Fetch() call ،$Q \in \mathcal{F}_{\text{Net}}$.**P** for $(\mathcal{G}m.\text{pid}, Q)$ claim $\mathcal{Z}^*$ Let the is which other، no and FinAuth and — FinAuth() call and $(\mathcal{G}m.\text{pid}, Z)$ beside one the not and property this on judged be to takes it what of whole 's$\mathcal{G}m$ of identities only claims and nothing، hosts one، no corrupts It it. .$\mathcal{Z}^* \in \mathbb{Z}_{\gamma,\gamma}$ so name، own

of loop the so ،$(I \leftarrow I \cap \mathcal{M})$ sanitizing survives $I = \{\text{msg}^*\}$ fetch، that At ؛$\texttt{rel}[m, Q] \leftarrow 1$ sets and $\texttt{buf}$ to $(m, \perp, \text{msg}^*, \infty)$ fresh some adds 16 line No .$(m, \text{msg}^*) \in M$ with $(\text{fetch}, Q, \tau, M) \in \texttt{tr}$ and returned، is (m, msg^*) indeed — $(\text{send}, \cdot, m, \cdot, \text{msg}^*)$ entry no carries $\texttt{tr}$ so ran، ever relay Send $Q \notin$ **C** So injection. the to fresh being m all، at m naming entry send no .1 gives 27 line and throughout،

$\mathcal{Z}^*$ environment The .18.1 Proposition touches run that about Nothing $\text{Win}_{\text{FinLive}}$ and answered never was FinLive so ،FinAuth at game the closed same the and ،5.3 Remark by disjoint are events two the occur: not did always it one beside keeps always it property a host therefore can shell □ other. each about anything saying two the without loses

of cores the if only 5.11 Corollary by transfers liveness ،$\mathcal{G}_{\text{Clock}}$ with As 8 lines do، 's$\mathcal{F}_{\text{Net}}$ of both and call، responsive no place system game the transfer the have then must $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ of place in put π system a ؛12 and clock's the as exactly gives، 9.4 Remark reason the for directly، argued no did note، Tightness، thing. same the 17.2 Section cost notification own unlike system: game a as γ licensing beyond above proof either in work a out rule to needs here verdict neither unforgeability، and correctness what off read to only sees، never $\mathcal{G}m$ route some by $\texttt{buf}$ entering message a survive would propositions both so — placed itself $\mathcal{G}m$ what to happens it. for asks above nothing though wanted، one were dependence، weaker Proposi- transfer: to nothing having nothing، transfers Authentication exactly specification، the about fact a but inherit to bound a not is 18.2 tion than rather now proved — prose in asserted already 18.1 Section fact the its ignores that one than stranger no adversary an by and said، merely does What it. on improve to asked is $\{\mathcal{F}_{\text{Net}}, \mathcal{G}_{\text{Clock}}\}$ of replacement No mail. traffic sender's corrupt a :$\mathcal{F}_{\text{AC}}$ 's19 Chapter is property analogous the have through than rather identity، own its under itself، Send through enters there adver- the is "Authentication ،19 (Chapter answer slot unaccountable an verdict a for sender a carries pair fetched the so interface")، own sary's addressed، fetched، — condition win transplanted the there and name، to met. not is — sent never and

فصل 19

Authenticated -DelayedΔ Channel

19.1 کارکرد

one the is object complementary The signs. nobody and diffuses $\mathcal{F}_{\text{Net}}$ arrives none and attached sender its with arrives message every where and Δ same the given ،[7] of channel authenticated the — nobody from two the and functionality، that is below $\mathcal{F}_{\text{AC}}$ channel The clock. same the the than more deal good a in differ they side: by side reading worth are lives. actually authentication where is difference the and field، sender

as recipient a takes Send operation The said. quickly is does it What fresh a under it files clock، the by pair the stamps message، a as well this and secrecy، not is authentication — adversary the tells and index، ad- everything gathers Fetch operation The all. at nothing hides channel adversary the asks age، of come has that party fetching the to dressed the empties and union، the answers early، release to rest the of which sender، record: whole the is gets receiver the What answered. it entries ad- the — 's$\mathcal{F}_{\text{Net}}$ is power of division The message. time، send recipient، bound: the is so and — order what in and Δ within when chooses versary slot the whatever names، it party the of fetch next the in is old Δ entry an answers.

The fields. They are $\mathcal{F}_{\text{Net}}$'s، for $\mathcal{F}_{\text{Net}}$'s reasons. pid، **P** and **N** are pa- rameters، one instance per session، with **N** naming the protocol served in the local pattern of Chapter 1، and $\Delta := \text{par}$. The **U** field is $\{(\mathcal{G}_{\text{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$، its clock entry met at the shared copy by the global disjunct of Definition 1.3. Both conjuncts of serves are needed and neither is a formality: the first asks the clock to serve every party the chan- nel does، and without the second the channel's own name would not be

.4 line on read every refuse would Guard so clock، the at admitted

be- 16 line needs $\mathcal{F}_{\mathsf{Net}}$ **interface. own adversary's the is Authentication**
under: enter to name no traffic adversarial gives channel diffusion a cause
it on write slot the letting by modelled is forgery so anonymous، is wire the
At it. delivers already interface full the and name، a is there Here directly.
$\mathcal{A}$ admit 4 and 3 lines — core the reaches adversary the sender corrupt a
test- both calls، its on fires hook mediation neither and party corrupt a at
sender's the under Send through enters traffic corrupt so — $\mathtt{id}'.\mathsf{F} \neq \mathsf{A}$ ing
$\mathcal{F}_{\mathsf{AC}}$ channel The anyone's. like clock the by timestamped identity، own
an of field sender the and none، needs and injections no has therefore
the interface which about fact a but wire the about report a not is entry
is forgery is: channel authenticated an what is That through. came call
from. forge to nowhere is there because impossible

adversary the message A ripens. traffic adversarial that is price The
12 line so timestamp، real a with list the in is party corrupt a from sends
$\mathcal{F}_{\mathsf{Net}}$ network The it. retract can answer later no and Δ by it deliver will
the for is guarantee the so ∞ at injections filing deliberately، this refuses
con- a is difference The traffic. own slot's the conscripts never and honest
message a it: against made choice a than rather addressing of sequence
is sender the message a is recipient named a and sender named a with
accepted it deadline the to adversary the holding and for، record the on
field sender the keeps that reading the is party corrupt a as spoke it when
says. it what meaning

dis- it answering so recipient، one names entry Each **empties. Fetching**
them between slot the and 12 line what clears 16 line and it، charges
owed is message diffused a — this do cannot $\mathcal{F}_{\mathsf{Net}}$ network The selected.
and instead table `rel` sticky the carries it why is which — everyone to
differ- are guarantees two The answered. already has it what re-answers
only view party's a that is 's$\mathcal{F}_{\mathsf{Net}}$ other: the implies neither and kind in ent
over protocol A once. exactly delivered is message a that 's$\mathcal{F}_{\mathsf{AC}}$ grows،
fetched، it what lose not must $\mathcal{F}_{\mathsf{AC}}$ over one and deduplicate must $\mathcal{F}_{\mathsf{Net}}$
absence. other's the is obligation each and

release to indices of set a is answer slot's The **release. the Sanitizing**
are that entries naming set a is be can it wrong thing only the and early،
procedure 's12 Chapter all. at entries not or — party's fetching the not
it through answer the runs 14 line so fault، of kind this exactly repairs

under

$$\mathsf{Clean}_{\mathsf{addr}}(S;\ \mathtt{list}, P)\ :\Longleftrightarrow\quad S \subseteq \{\, i : \mathtt{list}[i] = (\cdot, P, \cdot, \cdot)\,\},$$

things Two context. read-only the as party fetching the and list the with right the San makes what are they because saying، worth are that follow so exists، always value good A assertion. an than rather here instrument satisfies $\emptyset$ case: this in prove to nothing has 12 Chapter of footnote the lex- the also is it set empty the being and holds، list the whatever $\mathsf{Clean}_{\mathsf{addr}}$ release early no to repaired is answer failing a so such، first icographically had، have not could it nothing adversary the costs repair that And all. at may released subset good particular some wanting adversary an since 'sSan in assertion An untouched. it returned be and subset that submit of denial a slot the hand and answers good on same the read would place dies fetch party's honest the and index stray one — ones bad on service out. rule to exists 12 Chapter failure the is which —

load- is it here and discipline، clock's The **responsive. are calls slot Both** time the reads Fetch operation The out. spelling worth way a in bearing responsive not call that Were .13 line on slot the asks and 10 line on clock the drive could and between in token the hold would adversary the used 12 line time the by stale be then would τ answering؛ before forward deadline its past message a and ،$\tau_i + \Delta \leqslant \tau$ clear would entries fewer it، and read the makes what is Responsiveness withheld. lawfully be would and notifies it before mutates Send operation The moment. one test the nothing reason: 's$\mathcal{F}_{\mathsf{Net}}$ for too، responsive is but argument، such no needs call. either inside run may else

$\mathcal{F}_{\mathsf{AC}}$ Functionality

par := Δ ،**U** := $\{(\mathcal{G}_{\mathsf{Clock}}, \text{serves}), (\mathcal{A}, \text{serves})\}$ ،**N** ،**P** ،pid

:Initialize()

:1 $\mathtt{ctr} \leftarrow 0$
:2 $\mathtt{list} : \mathbb{N} \to (\mathcal{F}_{\mathsf{AC}}.\mathbf{P}^2 \times \mathbb{N} \times \mathcal{M}) \cup \{\perp\}$
:3 $\mathtt{list}[*] \leftarrow \perp$ / entries (P, Q, τ, msg)

id′ from id.Send(Q, msg)

:4 $\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\mathsf{Clock}}}.\mathsf{FullRead}()$ as id
:5 $\mathtt{ctr} \leftarrow \mathtt{ctr} + 1$
:6 $\mathtt{list}[\mathtt{ctr}] \leftarrow (\mathrm{id}.P, Q, \tau, \text{msg})$
:7 $\mathcal{A}^!(\mathrm{id}.\mathsf{Send}, \mathtt{ctr}, Q, \tau, \text{msg})$
:8 **return** ok

id′ from id.Leak()

:9 **return** $\mathtt{list}$

id′ from id.Fetch()

:10 $\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\mathsf{Clock}}}.\mathsf{FullRead}()$ as id
:11 $R \leftarrow \{m : \mathtt{list}[m] = (\cdot, \mathrm{id}.P, \tau_m, \cdot)$
:12 $\quad \wedge\ \tau_m + \Delta \leqslant \tau\}$ / come of age
:13 $S \leftarrow \mathcal{A}^!(\mathrm{id}.\mathsf{Fetch}, \tau)$ / released early
:14 $S \leftarrow \mathsf{San}[\mathsf{Clean}_{\mathsf{addr}}](S;\ \mathtt{list}, \mathrm{id}.P)$
:15 $M \leftarrow \{\mathtt{list}[m] : m \in R \cup S\}$
:16 $\mathtt{list}[m] \leftarrow \perp$ for each $m \in R \cup S$
:17 **return** M

P against checked not is recipient The code. the on remarks Three and filed is parties channel's the outside addressed message a :Send on fetch to declines who party a silence same the is which fetched، never entries about statement a is below guarantee the and produce، would receiver a so message، the with travels time send The fetches. someone re- fetch 's$\mathcal{F}_{\mathsf{Net}}$ — itself deadline the check and receives it what date can auditable locally is Δ 's$\mathcal{F}_{\mathsf{AC}}$ that is difference the and thing، such no turns the returns Leak And functionality. the about fact a only is 's$\mathcal{F}_{\mathsf{Net}}$ where the to told were contents its secrets: no keeping channel the entire، list is drained been since has what and arrived، they as 7 line at adversary leave. watched itself slot the what

19.2 ویژگی‌ها

$\mathcal{F}_{\mathsf{Net}}$ one the and ،Δ by delivery claims: two named above paragraph The A names. it party the by sent، was fetched message every — make cannot in yet not index an game: no needs again sending، before delivery no third، nothing and $\emptyset$ to S repairs San so ،Clean$_{\mathsf{addr}}$ fails drained، already or ،`list` for point same the made 18.2 (Section either early released is real yet not the formalize below FinAuth and FinLive finalizations The buffer). 's$\mathcal{F}_{\mathsf{Net}}$ the — over starting than rather pair 's18.2 Section transplanting two، other channel diffusion a between change not does finalize log، relay، of shell same the being these names، the do neither and one، addressed an and reading. search the take Both channel. different a of asked properties two the and time، this both، keep: can $\mathcal{F}_{\mathsf{AC}}$ two the of which is changes What for name is transplant the That design. whole the of point the is second name is the sharpest way to put the difference between the two channels differently. answered ،$\mathcal{F}_{\mathsf{AC}}$ of and $\mathcal{F}_{\mathsf{Net}}$ of asked game، one — with ،$\gamma := \{\mathcal{G}\mathfrak{m}\} \cup \{\mathcal{F}_{\mathsf{AC}}, \mathcal{G}_{\mathsf{Clock}}\}$ system game the over played are Both ar- tightness same the — $\mathcal{F}_{\mathsf{AC}}.\mathbf{P} \subseteq \mathcal{G}\mathfrak{m}.\mathbf{P}$ and pinned $\mathcal{F}_{\mathsf{AC}}.\mathbf{N} := \{\mathcal{G}\mathfrak{m}.\mathsf{pid}\}$ already fields 's$\mathcal{F}_{\mathsf{AC}}$ reason، same the for and ،18.2 Section as gument Guard passes $\mathcal{G}\mathfrak{m}$ but caller No .(19.1 (Section pattern 's$\mathcal{F}_{\mathsf{Net}}$ following Fetch record every and ،`list` in places ever Send entry every so ،$\mathcal{F}_{\mathsf{AC}}$ at inter- The recorded. is and below relays the through passes returns، ever for $\mathtt{id}_{\mathcal{F}_{\mathsf{AC}}}$ write we $\mathtt{id}$ at interface an Inside full. in $\mathcal{G}\mathfrak{m}.\mathbf{U}$ states box face of notation the second the ،$(\mathcal{G}_{\mathsf{Clock}}.\mathsf{pid}, \mathtt{id}.\mathrm{P})$ for $\mathtt{id}_{\mathcal{G}_{\mathsf{Clock}}}$ and $(\mathcal{F}_{\mathsf{AC}}.\mathsf{pid}, \mathtt{id}.\mathrm{P})$ .17.2 Section

follow finalizations the ؛$\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{AC}}$ over shell Game

U $:=$ ،**N** $:=$ **Z** ،**P** $:=$ $\mathcal{F}_{\text{AC}}$.**P** $\cup$ $\{Z\}$ ،pid $:=$ $(\mathcal{G}\mathrm{m}.\mathrm{F}, 0, 0)$
par $:= \perp$ ،$\{(\mathcal{F}_{\text{AC}}, \text{serves}_{\mathcal{G}\mathrm{m}}), (\mathcal{G}_{\text{Clock}}, \text{serves})\}$

:Initialize()

done $\leftarrow$ false
$\mathtt{tr} \leftarrow ()$ / the trace، a sequence

id' from id.Send(Q, msg)

require $\neg$done
$\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
$r \leftarrow \mathrm{id}_{\mathcal{F}_{\text{AC}}}$.FullSend(Q, msg) as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{send}, \mathrm{id}.P, Q, \tau, \text{msg})$
return r

id' from id.Fetch()

require $\neg$done
$\tau \leftarrow \mathrm{id}_{\mathcal{G}_{\text{Clock}}}$.FullRead() as id
$M \leftarrow \mathrm{id}_{\mathcal{F}_{\text{AC}}}$.FullFetch() as id
$\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\text{fetch}, \mathrm{id}.P, \tau, M)$
return M

id' from id.Leak()

return $\perp$

shared is $\mathcal{G}_{\text{Clock}}$ and $\mathcal{F}_{\text{AC}}$ over shell the ؛$\mathcal{G}\mathrm{m}$ of finalizations The

(liveness) id' from id.FinLive()

require $\neg$done
require $\mathrm{id}.P = Z$
done $\leftarrow$ true
C $\leftarrow (C, Z)$.FullStatus() as id
$w \leftarrow 1$ if $\exists P, Q, \tau_s, \text{msg}, \tau, M :\ P \notin \mathbf{C} \wedge Q \notin \mathbf{C}$
$\wedge\ (\text{send}, P, Q, \tau_s, \text{msg}) \in \mathtt{tr} \wedge (\text{fetch}, Q, \tau, M) \in \mathtt{tr}$
$\wedge\ \tau \geqslant \tau_s + \Delta \wedge \neg\exists \tau', M' :$
$(\text{fetch}, Q, \tau', M') \in \mathtt{tr} \wedge (P, Q, \tau_s, \text{msg}) \in M'$؛
else $w \leftarrow 0$
return w

(authentication) id' from id.FinAuth()

require $\neg$done
require $\mathrm{id}.P = Z$
done $\leftarrow$ true
C $\leftarrow (C, Z)$.FullStatus() as id
$w \leftarrow 1$ if $\exists Q, \tau, M, P, \tau_s, \text{msg} :$
$Q \notin \mathbf{C} \wedge (\text{fetch}, Q, \tau, M) \in \mathtt{tr} \wedge (P, Q, \tau_s, \text{msg}) \in M$
$\wedge\ (\text{send}, P, Q, \tau_s, \text{msg}) \notin \mathtt{tr}$؛
else $w \leftarrow 0$
return w

the read relays Both familiar. them of two code، the about things Three readings two the ’s:18.2 Section as reason same the for themselves clock call no placing Read them، between runs γ of core no because coincide nothing having ،ok return، own ’s$\mathcal{F}_{\text{AC}}$.Send keeps relay ’sSend own. its of and back pass to id message a had which ’s،$\mathcal{F}_{\text{Net}}$ unlike — report to else over search no needs verdict authentication’s And out. hands never $\mathcal{F}_{\text{AC}}$ names already record fetched a namesake’s: its unlike times، or senders

was tuple exact the that check direct a is condition win the so ،τ_s and P
the — it produced have might what over one existential an not sent، never
the at does it what exactly verdict، the in doing، is carries $\mathcal{F}_{\mathsf{AC}}$ address
interface.

occupant every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every For .liveness) has $\mathcal{F}_{\mathsf{AC}}$) **19.1 گزاره**
liveness has $\{\mathcal{F}_{\mathsf{AC}}, \mathcal{G}_{\mathsf{Clock}}\}$ So .$\Pr[\mathsf{Win}_{\mathsf{FinLive}}] = 0$ slot، adversary the of $\mathcal{X}$
class the $\mathbb{X}$ for ،5.8 Definition of sense the in $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ against 0 within
occupants. all of

of activation some at written ،$(\mathsf{send}, P, Q, \tau_s, \mathsf{msg}) \in \mathtt{tr}$ Suppose .اثبات
one internal 's$\mathcal{F}_{\mathsf{AC}}$.Send and read clock own Its .(5 (line relay Send the
between runs γ of core no :18.2 Section of argument the by agree، (4 (line
relay the since honest $\mathtt{id}.P$ finding check corruption and guard 's$\mathcal{F}_{\mathsf{AC}}$ them،
fresh some wrote core the So call. no placing itself Read and all، at ran
moment. that at $\mathtt{list}[m] = (P, Q, \tau_s, \mathsf{msg})$ with m
$\mathtt{list}[m]$ If .$\tau \geqslant \tau_s + \Delta$ with $(\mathsf{fetch}, Q, \tau, M) \in \mathtt{tr}$ too Suppose
by $m \in R$ then runs، core fetch's that when $(P, Q, \tau_s, \mathsf{msg})$ holds still
its τ for hypothesis، by $\tau_s + \Delta \leqslant \tau$ and ،Q names entry the — 12 line
— relay's the again، argument same the by and، clock the of reading own
whatever and 13 line at answers slot the whatever $(P, Q, \tau_s, \mathsf{msg}) \in M$ so
been already had $\mathtt{list}[m]$ instead If .R touching neither it، with does San
the — Q by fetch earlier some of 16 line at only happened that cleared،
in placed already indices on only and — entry an clears ever that line one
.M own fetch's earlier that in present already hence ،$R \cup S$ own fetch's that
verdict the and 's،Q of fetch some in appears $(P, Q, \tau_s, \mathsf{msg})$ way Either
□ .0 gives 18 line at

oc- every and $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every For .authentication) has $\mathcal{F}_{\mathsf{AC}}$) **19.2 گزاره**
has $\{\mathcal{F}_{\mathsf{AC}}, \mathcal{G}_{\mathsf{Clock}}\}$ So .$\Pr[\mathsf{Win}_{\mathsf{FinAuth}}] = 0$ slot، adversary the of $\mathcal{X}$ cupant
،5.8 Definition of sense the in $(\mathbb{Z}_{\gamma,\gamma}, \mathbb{X})$ against 0 within authentication
$\mathcal{F}_{\mathsf{Net}}$ proved 18.2 Section property the — occupants all of class the $\mathbb{X}$ for
instead. here held lacks،

$(P, Q, \tau_s, \mathsf{msg}) \in$ and $Q \notin \mathbf{C}$ with $(\mathsf{fetch}, Q, \tau, M) \in \mathtt{tr}$ Suppose .اثبات
index both and ،$M = \{\mathtt{list}[m] : m \in R \cup S\}$ code، 's$\mathcal{F}_{\mathsf{AC}}$.Fetch By .$M$
match pattern own its by R stands: already it as $\mathtt{list}$ on only draw sets
with i indices only admits $\mathsf{Clean}_{\mathsf{addr}}$ since sanitizing، after S and ،(12 (line
values among picks (12 (Chapter 3 line and already $\mathtt{list}[i] = (\cdot, Q, \cdot, \cdot)$

not does `list` index an manufactures neither — predicate that meeting some for $\texttt{list}[m] = (P, Q, \tau_s, \text{msg})$ if only $(P, Q, \tau_s, \text{msg}) \in \mathcal{M}$ So hold. point. some at m

is $\perp$ than other value a to $\texttt{list}[m]$ writes ever that line only The that — pinned $\mathcal{F}_{\text{AC}}.\mathbf{N} = \{\mathcal{G}m.\text{pid}\}$ — tightness by and core، own 'sSend the not environment، the not places: relay 's$\mathcal{G}m$ calls on only runs core game's the of instance invented an at hosted machine a not adversary، activa- some at happened $(P, Q, \tau_s, \text{msg})$ producing write the So name. pre- the of argument coinciding-read the by and relay، Send the of tion Hence .$(\text{send}, P, Q, \tau_s, \text{msg})$ exactly as there logged was it proof vious $\square$ .0 gives 28 line at verdict the and ،$(\text{send}, P, Q, \tau_s, \text{msg}) \in \texttt{tr}$

only 5.11 Corollary by transfer properties both ،$\mathcal{F}_{\text{Net}}$ and $\mathcal{G}_{\text{Clock}}$ with As there، as here، and — call responsive no place cores system's game the if 'sSend covers responsive" are calls slot "Both 's19.1 Section does: neither $\{\mathcal{F}_{\text{AC}}, \mathcal{G}_{\text{Clock}}\}$of place in put π system A alike. query 'sFetch and notification 9.4 Remark reason the for directly، argued transfers both have then must thing. same the 18.2 Section cost notification own 's$\mathcal{F}_{\text{Net}}$ as exactly gives، re- negative a as only standing left is here nothing network، the Unlike from transplanted 18.2 Section condition win the is 19.2 Proposition sult. laid proofs، two the and — instead here closed close، not could and $\mathcal{F}_{\text{Net}}$ tightness whether fact: one in difference whole the locate side، by side it $\mathcal{F}_{\text{AC}}$ For it. produced that Send a to back record fetched a trace can cannot، it $\mathcal{F}_{\text{Net}}$ for entry؛ an gain to way other no having `list` can، always trans- game one chapters، Two in. way other the exactly being 16 line them. between difference entire the is address the and twice، planted

comparison the makes what is names the keeps transplant the That resem- that properties two not is FinAuth finalization The all. at sayable — sent never addressed، fetched، — condition win one but other each ble Under .$\mathcal{F}_{\text{AC}}$ for 0 and $\mathcal{F}_{\text{Net}}$ for 1 is answer the and channels، two of asked game، own its carried property each where replaced، 5 Chapter shape the would sameness the and $\mathcal{G}m'_{\text{Auth}}$ and $\mathcal{G}m_{\text{Auth}}$ been have would two the notation. the is it here remark؛ a been have

فصل 20

تحققِ کانالِ اصالت‌دار

proto- a it: earns chapter This postulated. was 19 Chapter of channel The unauthen- the directory، public-key a — subroutines ideal three over ρ col that — 15 Chapter of signatures the and ،18 Chapter of network ticated ev- ،0 within exactly، so does and channel، authenticated an UC-realizes information-theoretic. reduction whole the and ideal being subroutine ery the exercises that one the and paper، the in construction largest the is It sanitization، discipline، responsive the composition، once: at it of most do- each appear، all 11 Chapter of lemma fundamental the and tightness، could. other no job a ing

from key signing a draws time first the for sending party A **does. ρ What** the reads it Q to msg send To .$\mathcal{F}_{\mathsf{PKI}}$ directory the in it registers and $\mathcal{F}_{\mathsf{Sig}}$ signs ،s number sequence per-party a increments ،τ timestamp a for clock over signature its with tuple the diffuses and ،$(\P, Q, \tau, s, \mathsf{msg})$ tuple the item diffused a keeps and delivers، $\mathcal{F}_{\mathsf{Net}}$ what collects it fetch، To .$\mathcal{F}_{\mathsf{Net}}$ un- verifies that signature a carries fetcher، the to addressed is it if only seen not pair (sender, s) a has and key، directory sender's claimed the der carries Diffusion .$(\P, Q, \tau, \mathsf{msg})$ as up hands it survives What before. and second the supplies signature the authenticity؛ no and addresses no a trusts receiver a that so first، the tuple signed the inside address the sig- own sender's the under is name that because exactly name sender's nature.

and obstruction، the is sender corrupt A **.$\mathcal{F}_{\mathsf{AC}}$ not ،$\mathcal{F}^{\circ}_{\mathsf{AC}}$ is target the Why** ad- the network diffusion a Over incidental. than rather instructive is it — fetch a during particular in likes، it whenever wire the on writes versary the of window responsive the inside chosen are (16 (line injections 's$\mathcal{F}_{\mathsf{Net}}$ the key a under signed injected، so message A them. carries that fetch and receiver the by accepted is party، corrupt a for registered adversary only delivers which ،$\mathcal{F}_{\mathsf{AC}}$ so it، placed ever Send prior no but delivered؛

call can simulator no and it، produce cannot stored، Send own its what
Worse، gap. the repair to window responsive fetch's the inside $\mathcal{F}_{\mathsf{AC}}$.Send
val- signature the — choose to own its is timestamp sender's corrupt the
time true the stamps $\mathcal{F}_{\mathsf{AC}}$ where — carries payload the τ whatever idates
au- an what are both ؛ρ in defect a is gap Neither witnessed. it call a of
20 Chapter and party، corrupt a gives unavoidably diffusion thenticated
does subroutines these over protocol no that argument the with closes
traffic sender's corrupt a with $\mathcal{F}_{\mathsf{AC}}$:$\mathcal{F}^{\circ}_{\mathsf{AC}}$ is object realized the So better.
exactly is which timestamped، freely and unstored fetch، the at admitted
honest- The less. no and channel authenticated]'s7] of level guarantee the
party's honest an that authenticity، and ،Δ by delivery — guarantees sender
survive — names it party honest the by sent genuinely was record fetched
objects two the one no corrupts that adversary an against And whole.
a for Pr[Bad] at difference the prices lemma fundamental the coincide:
exactly، itself، $\mathcal{F}_{\mathsf{AC}}$ realizes ρ so raise، can sender corrupt a only that `bad`
.(20.2 (Corollary corrupt is nobody whenever

re- the are signatures The **shelf. the off quite not are subroutines Two**
and — Verify and Sign ،Gen in $\mathcal{A}^!$ — 9 Chapter of $\mathcal{F}^!_{\mathsf{Sig}}$ variant sponsive
respon- is call slot one its atomic: is fetch 's$\mathcal{F}_{\mathsf{AC}}$ token. the is reason the
answer. its and call the between execution the drives occupant no so sive،
would verification a ،$\mathcal{A}(\cdot)$ ordinary the through $\mathcal{F}_{\mathsf{Sig}}$ reach to fetch 'sρ Were
re- before elsewhere act it let and mid-fetch token the adversary the hand
simulator no and show cannot fetch atomic 's$\mathcal{F}^{\circ}_{\mathsf{AC}}$ interleaving an turning،
's$\mathcal{F}_{\mathsf{Net}}$ escape؛ the removes $\mathcal{F}^!_{\mathsf{Sig}}$ variant responsive The reproduce. could
is fetch every and send every so responsive، already are calls 's$\mathcal{F}^{\circ}_{\mathsf{AC}}$ and
of $\mathcal{G}_{\mathsf{PKI}}$ global the not local، is directory the And sides. both on atomic
real the in ρ.Send honest an by written is directory shared a :16 Chapter
where ،Retrieve through environment the by back straight read and world
peek the only holds simulator the and it writes nothing world ideal the in
commit- the difference، unsimulatable an — parties honest at grants $\mathcal{G}_{\mathsf{PKI}}$
،$\mathcal{F}_{\mathsf{PKI}}$ local A coordinates. paper's this in setup global a of problem ment
out kept is Z it؛ closes hybrid، any like session the to private identities its
a being environment the by retrieval direct a reason، same the for **N** its of
serve. to machine no has world ideal the call

key registered a sanitizes $\mathcal{F}_{\mathsf{PKI}}$ **purpose. on bindings forgets directory The**
fresh- cross-party 's$\mathcal{G}_{\mathsf{PKI}}$ dropping slot، per — more nothing and key a be to
ad- the weapon: a be would freshness The .(Clean$_{\mathsf{reg}}$ ،16 (Chapter ness
$\mathcal{F}_{\mathsf{Sig}}$.Gen value the being it key، signing party's honest an chooses versary

the have and slot corrupt a at key that pre-register could it so returns، direc- leaving one، different a to repaired registration own party's honest rejected message every party's the and disagreeing key signing and tory Per-slot unconditionally. promises $\mathcal{F}^{\circ}_{\mathrm{AC}}$ liveness the of denial a — ever for sender's the binding: on rested never attribution because enough is fixity and up، name that looks receiver the tuple، signed the inside travels name different a naming tuple a on key party's honest an under valid signature a costs and here nothing buys Binding suppresses. $\mathcal{F}_{\mathrm{Sig}}$ forgery a is sender goes. it so liveness،

$\mathcal{F}_{\mathrm{PKI}}$ directory Local

par := $\perp$ ،**U** := $\emptyset$ ،**N** ،**P** ،pid

:Initialize()

$\mathrm{VK} : \mathcal{F}_{\mathrm{PKI}}.\mathbf{P} \to \mathcal{K} \cup \{\square\}$

$\mathrm{VK}[*] \leftarrow \square$

id$'$ from id.Register(vk)

if $\mathtt{id}'.\mathrm{F} \in \{A, Z\} \wedge \mathtt{id}.\mathrm{P} \notin$ **C then**

  return $\mathrm{VK}[\mathtt{id}.\mathrm{P}]$ / a peek، not a write

if $\mathrm{VK}[\mathtt{id}.\mathrm{P}] \neq \square$ **then**

  return $\mathrm{VK}[\mathtt{id}.\mathrm{P}]$

vk $\leftarrow$ San[$\mathrm{Clean}_{\mathrm{key}}$](vk)

$\mathrm{VK}[\mathtt{id}.\mathrm{P}] \leftarrow$ vk

return vk

id$'$ from id.Retrieve(P)

return $\mathrm{VK}[\mathrm{P}]$

id$'$ from id.Leak()

return VK

:$\mathrm{Clean}_{\mathrm{key}}$(vk)

return vk $\in \mathcal{K}$

(S, I) pair a back takes Fetch changed: operation one with $\mathcal{F}_{\mathrm{AC}}$ is $\mathcal{F}^{\circ}_{\mathrm{AC}}$ of set a I before، as ids stored among releases early the S slot، the from injection each pins Sanitizing unstored. delivers slot the records injected ever only can injection an so party، fetching the to and sender corrupt a to ،Send own 's$\mathcal{F}_{\mathrm{AC}}$ it؛ for asked that fetch the into word party's corrupt a put unchanged. are Initialize and Leak

replaced ،$\mathcal{F}_{AC}$ of fetch the :$\mathcal{F}^{\circ}_{AC}$ Functionality
(19.1 (Section $\mathcal{F}_{AC}$ as fields

id' from $\underline{id.Fetch()}$

$\tau \leftarrow id_{\mathcal{G}_{Clock}}$.FullRead() as id
C $\leftarrow (C, id.P)$.FullStatus() as id
$R \leftarrow \{m : \mathtt{list}[m] = (\cdot, id.P, \tau_m, \cdot) \wedge \tau_m + \Delta \leqslant \tau\}$
$(S, I) \leftarrow \mathcal{A}^!(id.\text{Fetch}, \tau)$ / releases and injections
$S \leftarrow$ San[Clean$_{addr}$]$(S;\ \mathtt{list}, id.P)$
$I \leftarrow$ San[Clean$_{inj}$]$(I;$ **C**$, id.P)$
$M \leftarrow \{\mathtt{list}[m] : m \in R \cup S\} \cup I$
$\mathtt{list}[m] \leftarrow \perp$ for each $m \in R \cup S$
return M

:$\underline{\text{Clean}_{inj}(I;\ \mathbf{C}, P_f)}$

return $I \subseteq \{(P, Q, \tau, \text{msg}) : P \in \mathbf{C} \wedge Q = P_f\}$

the at interfaces the through subroutines four its reads protocol The **U** Its .18 Chapter of $id_{\mathcal{F}} := (\mathcal{F}.\text{pid}, id.P)$ notation the served، being party payloads own its stamps ρ since them، among clock the four، all names returns. never $\mathcal{F}_{Net}$ which stamp، ’s$\mathcal{F}_{Net}$ recover not does and

$\{\mathcal{F}_{PKI}, \mathcal{F}_{Net}, \mathcal{F}^!_{Sig}\}$ over ρ Protocol
،**U** $:=$ $\{(\mathcal{F}_{PKI}, \text{serves}), (\mathcal{F}_{Net}, \text{serves}), (\mathcal{F}^!_{Sig}, \text{serves}), (\mathcal{G}_{Clock}, \text{serves})\}$ ،**N** ،**P** ،pid
par := ⊥

:$\underline{\text{Initialize()}}$

$\mathtt{gen} : \mathbf{P} \to \{0,1\}$؛ $\mathtt{gen}[*] \leftarrow 0$
$\mathtt{seq} : \mathbf{P} \to \mathbb{N}$؛ $\mathtt{seq}[*] \leftarrow 0$
$\mathtt{seen} : \mathbf{P} \to \mathcal{P}(\mathbf{P} \times \mathbb{N})$
$\mathtt{seen}[*] \leftarrow \emptyset$

id' from $\underline{id.\text{Send}(Q, \text{msg})}$

if $\mathtt{gen}[id.P] = 0$ **then**
  vk $\leftarrow id_{\mathcal{F}_{Sig}}$.FullGen() as id
  $id_{\mathcal{F}_{PKI}}$.FullRegister(vk) as id
  $\mathtt{gen}[id.P] \leftarrow 1$
$\tau \leftarrow id_{\mathcal{G}_{Clock}}$.FullRead() as id
$\mathtt{seq}[id.P] \leftarrow \mathtt{seq}[id.P] + 1$
$x \leftarrow (id.P, Q, \tau, \mathtt{seq}[id.P], \text{msg})$
$\sigma \leftarrow id_{\mathcal{F}_{Sig}}$.FullSign$(x)$ as id
$id_{\mathcal{F}_{Net}}$.FullSend$((x, \sigma))$ as id
return ok

id' from $\underline{id.\text{Fetch()}}$

$N \leftarrow id_{\mathcal{F}_{Net}}$.FullFetch() as id
$M \leftarrow \emptyset$
for $(m, y) \in N$ **do**
  parse y as (x, σ)، $x = (P, Q, \tau, s, \text{msg})$ / else skip
  if $Q \neq id.P$ **then** skip
  vk $\leftarrow id_{\mathcal{F}_{PKI}}$.FullRetrieve$(P)$ as id
  if vk $= \square$ **then** skip
  $b \leftarrow id_{\mathcal{F}_{Sig}}$.FullVerify$(\text{vk}, x, \sigma)$ asid
  if $b \neq 1$ **then** skip
  if $(P, s) \in \mathtt{seen}[id.P]$ **then** skip
  $\mathtt{seen}[id.P] \leftarrow \mathtt{seen}[id.P] \cup \{(P, s)\}$
  $M \leftarrow M \cup \{(P, Q, \tau, \text{msg})\}$
end for
return M

world real the In slot. adversary the of $\mathcal{A}$ occupant an Fix **simulator. The** must $\mathcal{S}_{\mathcal{A}}$ and gone، are they world ideal the in hybrids؛ three the faces $\mathcal{A}$ ،4.26 (Definition absorption an is It itself. inside from faces their present each copy internal one with $\mathcal{A}$ bundles $\mathcal{S}_{\mathcal{A}}$ side): adversary the on read execution real the as exactly them against $\mathcal{A}$ runs ،$\mathcal{F}^!_{Sig}$ and $\mathcal{F}_{Net}$ ،$\mathcal{F}_{PKI}$ of no- Send honest an On it. tells $\mathcal{F}^{\circ}_{\mathsf{AC}}$ what from them drives and would، ،$\P$ for `seq` internal the advances it ؛$(\P, \mathsf{Q}, \tau, \mathsf{msg})$ it hands $\mathcal{F}^{\circ}_{\mathsf{AC}}$ tification internal the into steps own ’sρ replays and ،$x = (\P, \mathsf{Q}, \tau, s, \mathsf{msg})$ forms answers، $\mathcal{A}$ hosted the query $\mathcal{A}^!$ whose ،$\mathcal{F}^!_{\mathsf{Sig}}$.Sign internal an — hybrids hosted the so — delivers it notification whose ،$\mathcal{F}_{\mathsf{Net}}$.Send internal an and makes $\mathcal{F}^{\circ}_{\mathsf{AC}}$ When place. would world real the queries the precisely sees $\mathcal{A}$ ’s$\mathcal{F}_{\mathsf{Net}}$ internal the over filter ’sρ.Fetch runs $\mathcal{S}_{\mathcal{A}}$ query، fetch responsive its hand would ρ records which exactly learns ،$\mathcal{A}$ hosted the against delivery their by entries honest early-released or ripe the :(S, I) answers and up، key a under verifying those — records corrupt-sender the and ،S in ids $\mathcal{F}^{\circ}_{\mathsf{AC}}$ through، straight passes it Corruption .I in — slot corrupt a at registered does. ’s3.2 Remark as bundle the surviving Corrupt

passes above paragraph the state of pieces two explicit: Made chosen. than rather it above just code the by forced both silently، over with slot the notifies (19.1 Section ،$\mathcal{F}_{\mathsf{AC}}$ from (inherited Send own ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ the alone، $(\P, \mathsf{Q}, \tau, \mathsf{msg})$ not — index list own its m ،$(\mathsf{Send}, m, \mathsf{Q}, \tau, \mathsf{msg})$ The .(9.2 (Definition at arrives query the identity the off read $\P$ sender it later؛ entry the name to below، `fid` index، that keep must $\mathcal{S}_{\mathcal{A}}$ simulator And itself. $\mathcal{F}^{\circ}_{\mathsf{AC}}$`.list` to access no having time، fetch at it recover cannot reads test freshness its `seen` the and ،`seq` ،5 line at `gen` — tables own ’sρ world ideal the of part no being ρ host، not does $\mathcal{S}_{\mathcal{A}}$ tables are — 24 line at shad- needs filter its reproducing so here، internal subroutines its only and reading by than rather construction by ’sρ to equal kept own، its of ows them.

whether decide both and open، leaves prose the questions routing Two $\mathcal{A}$ hosted the call a of becomes what is first The all. at works simulator the neither and party corrupt a at $\mathcal{A}$ admits 3 Line identity. own ’sρ on places reaches call a such world real the in so calls، its on fires hook mediation mediated is places then core that call subroutine every and — core ’sρ so corrupt)، party the and ρ being caller the ،4 (line $\mathcal{A}$ to back straight instead would it outward Relayed buffer. ’s$\mathcal{F}_{\mathsf{Net}}$ reaches ever it of nothing Δ by delivered is and ripens that record a files Send whose ،$\mathcal{F}^{\circ}_{\mathsf{AC}}$ reach real-world no with delivery a — (19.1 (Section answers slot the whatever rout- explicit ’s4.13 Definition relayed: not are calls these So counterpart. against core ’sρ runs below handler last the where inside، them keeps ing declared is set outward The really. does Mediate as exactly $\mathcal{A}$ hosted the accordingly.

handlers Both reaches. answer responsive a far how is second The call every refuses (9.1 (Definition Respond and call، responsive a answer placed not are machines hosted two between calls ؛places responder the nothing but reachable، are copies internal the so ،(4.13 (Definition all at reached، be otherwise would it outside things Two is. bundle the outside each. avoids code ’s$\mathcal{S}_{\mathcal{A}}$ and

register، corruption the reading by opens interface full copy’s hosted A computed is answer responsive a — outright refuses 9.2 Definition which copies the drive handlers the So knows. already responder the what from nothing and above، FullGen than rather Gen written cores، their through before mediated is fetch the or send the at corrupt party a it: by lost is and behalf its on issued ever is query slot no so runs، core own ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ adds interface full the party honest an at and one، serves ever handler no opens $\mathcal{F}_{\mathsf{Net}}$ And fires. hook neither and passes، gate the — silencer its only hosted. than rather shared is which ،$\mathcal{G}_{\mathsf{Clock}}$ reading by Fetch and Send both carries، already query or notification the τ the with answered is read That it send the on one: convenient a merely not and value right the is which the on and share، reads own ’s$\mathcal{F}_{\mathsf{Net}}$ and ’sρ shows proof the stamp the is past advanced have can call responsive no which read، ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ is it fetch .(9.3 (Proposition

$\mathcal{S}_{\mathcal{A}}$ Simulator

،4.26 (Definition $\mathcal{F}^{!}_{\mathsf{Sig}}$ ،$\mathcal{F}_{\mathsf{Net}}$ ،$\mathcal{F}_{\mathsf{PKI}}$ of each copy internal one and $\mathcal{A}$ hosts ؛IDs($\mathcal{A}$) occupies serves، still execution ideal the identities the set outward side)؛ adversary the on read (4.13 (Definition hand by routes handler last the which own، ’sρ less

:Initialize()

:1 $\mathtt{gen} : \mathbf{P} \to \{0,1\}$؛ $\mathtt{gen}[*] \leftarrow 0$
:2 $\mathtt{seq} : \mathbf{P} \to \mathbb{N}$؛ $\mathtt{seq}[*] \leftarrow 0$
:3 $\mathtt{seen} : \mathbf{P} \to \mathcal{P}(\mathbf{P} \times \mathbb{N})$؛ $\mathtt{seen}[*] \leftarrow \emptyset$
:4 $\mathtt{fid} : \mathbf{P} \times \mathbb{N} \to \mathbb{N} \cup \{\square\}$؛ $\mathtt{fid}[*] \leftarrow \square$ / own record $\mapsto$ $\mathcal{F}^{\circ}_{\mathsf{AC}}$’s index

¶ at $(m, Q, \tau, \mathsf{msg})$ note Send ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ on

:5 **if** $\mathtt{gen}[¶] = 0$ **then**
:6 vk $\leftarrow$ internal $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.Gen()
:7 internal $\mathtt{id}_{\mathcal{F}_{\mathsf{PKI}}}$.Register(vk)
:8 $\mathtt{gen}[¶] \leftarrow 1$
:9 $\mathtt{seq}[¶] \leftarrow \mathtt{seq}[¶] + 1$؛ $s \leftarrow \mathtt{seq}[¶]$
:10 $x \leftarrow (¶, Q, \tau, s, \mathsf{msg})$
:11 $\sigma \leftarrow$ internal $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.Sign($x$) / hosted $\mathcal{A}$ answers the query
:12 internal $\mathtt{id}_{\mathcal{F}_{\mathsf{Net}}}$.Send($(x, \sigma)$) / its clock read answered τ
:13 $\mathtt{fid}[¶, s] \leftarrow m$

Q at (τ) query Fetch responsive ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ on

:14 N $\leftarrow$ internal $\mathtt{id}_{\mathcal{F}_{\mathsf{Net}}}$.Fetch() / clock answered τ؛ $\mathcal{A}$ answers the release query
:15 $S, I \leftarrow \emptyset, \emptyset$
:16 **for** $(m, y) \in N$ **do**
:17 parse y as (x, σ)، $x = (¶, Q', \tau_x, s, \mathsf{msg})$ / else skip
:18 **if** $Q' \neq Q$ **then** skip
:19 vk $\leftarrow$ internal $\mathtt{id}_{\mathcal{F}_{\mathsf{PKI}}}$.Retrieve(¶)
:20 **if** vk $= \square$ **then** skip
:21 $b \leftarrow$ internal $\mathtt{id}_{\mathcal{F}_{\mathsf{Sig}}}$.Verify(vk, x, σ)
:22 **if** $b \neq 1$ **then** skip
:23 **if** $(¶, s) \in \mathtt{seen}[Q]$ **then** skip
:24 $\mathtt{seen}[Q] \leftarrow \mathtt{seen}[Q] \cup \{(¶, s)\}$
:25 **if** $\mathtt{fid}[¶, s] \neq \square$
:26 **then** $S \leftarrow S \cup \{\mathtt{fid}[¶, s]\}$
:27 **else** $I \leftarrow I \cup \{(¶, Q, \tau_x, \mathsf{msg})\}$
:28 **end for**
:29 **return** (S, I)

¶ corrupt a at Fetch or Send ’s$\mathcal{A}$ on

:30 run ρ’s own core for that operation (the protocol box above) on gen، seq، seen، with every subroutine call it places — to $\mathcal{F}^{!}_{\mathsf{Sig}}$، $\mathcal{F}_{\mathsf{PKI}}$، $\mathcal{F}_{\mathsf{Net}}$ or $\mathcal{G}_{\mathsf{Clock}}$ — handed to the hosted $\mathcal{A}$ rather than placed، as the full interface’s line 4 hands it really؛ return the core’s own value
:31 never relayed outward: no $\mathcal{F}^{\circ}_{\mathsf{AC}}$.list entry، and none drained

Corrupt and Routing،

:32 any other call between the hosted $\mathcal{A}$ and an internal hybrid: served inside، the bundle’s ordinary routing (Definition 4.13.(A Corrupt leaves under the hosted $\mathcal{A}$’s own claim، the bundle occupying IDs($\mathcal{A}$)، so Corrupt’s own gate $\mathtt{id}'.F = A$ (its line 3(is met without rewriting

read to easy but proof the by used all code، ’s$\mathcal{S}_{\mathcal{A}}$ about things Three entry list undrained “an exactly is 26 line at $\mathtt{fid}[¶, s] \neq \square$ First، past. record a recognizes $\mathcal{S}_{\mathcal{A}}$ words: own proof’s the in record” the matches is which it، produced that Send honest the relayed once it when precisely all. at fid into gets entry an way only the

in- the of reading a not and flag own ’sρ of shadow a is gen Second، Register ’s$\mathcal{F}_{\mathsf{PKI}}$ — agree two the party honest an at though ،VK ’s$\mathcal{F}_{\mathsf{PKI}}$ ternal

ρ only so ،4 line its at peek a into there call -caller's$\mathcal{Z}$ or -$\mathcal{A}$ an turns
is which one، corrupt a at company part They entry. that writes ever
before away mediated is Register 'sρ there taken: not is shortcut the why
regardless، $\mathtt{gen}[\P] \leftarrow 1$ sets ρ while unset $\mathtt{VK}[\P]$ leaving runs، core 's$\mathcal{F}_{\mathsf{PKI}}$
re- would $\mathtt{VK}$ testing simulator A answer. the reading not 5 line 'sρ.Send
hosted the show and send later every party's the on Register and Gen run
sees. never $\mathcal{A}$ real the queries of pair a $\mathcal{A}$

in- ones ripe ،$\mathcal{S}$ into record honest matching every puts 26 line Third،
own. its on R through ones ripe the taken have would $\mathcal{F}^{\circ}_{\mathsf{AC}}$ where cluded،
entry undrained any admits $\mathsf{Clean}_{\mathsf{addr}}$ nothing: costs and deliberate is This
their drains and 7 line its at $\mathcal{S}$ with R unions $\mathcal{F}^{\circ}_{\mathsf{AC}}$.Fetch and ،$\mathsf{Q}$ to addressed
it Naming once. it drains and delivers twice entry ripe a naming so union،
with query the inside own its of read clock no having do، can $\mathcal{S}_{\mathcal{A}}$ all also is
early. from ripe tell to which

قضیه 20.1 ρ) realizes the authenticated channel.) Let $\pi :=$
well- is $\pi \cup \{\mathcal{G}_{\mathsf{Clock}}\}$ that so matched parameters with $\{\rho, \mathcal{F}_{\mathsf{PKI}}, \mathcal{F}_{\mathsf{Net}}, \mathcal{F}^{!}_{\mathsf{Sig}}\}$
UC- π Then .$\mathcal{F}_{\mathsf{Net}}$.par $= \mathcal{F}^{\circ}_{\mathsf{AC}}$.par $= \Delta$ and ،ρ.**N** $= \mathcal{F}^{\circ}_{\mathsf{AC}}$.**N** formed،
slot adversary the of $\mathcal{A}$ occupant every for :0 within $\{\mathcal{F}^{\circ}_{\mathsf{AC}}\}$ emulates
،$\mathcal{Z} \in \mathbb{Z}_{\pi,\{\mathcal{F}^{\circ}_{\mathsf{AC}}\}}$ every for gives، above $\mathcal{S}_{\mathcal{A}}$ simulator the

$$\mathsf{Exec}\big(\pi, \mathcal{Z}, \mathcal{A}, \mathcal{C}\big) \equiv \mathsf{Exec}\big(\{\mathcal{F}^{\circ}_{\mathsf{AC}}\}, \mathcal{Z}, \mathcal{S}_{\mathcal{A}}, \mathcal{C}\big),$$

the not adversary، every over is quantifier The distributions. as equal
Theo- so responsive، is world either in call slot every alone: dummy
against exhibited is simulator the and this reach not does 4.29 rem
directly. $\mathcal{A}$ each

اثبات. One induction on activations، carrying an invariant in three parts.
routing، the so ،4.18 Lemma in as machine to machine pairs coupling The
state the is new is what lemma's؛ that are arguments refusal and claims
the of whole the is it on invariant the and cut، $\mathcal{F}^{\circ}_{\mathsf{AC}}/\rho$ the across carried
hold hybrids internal the parts: three the states invariant The argument.
$\mathcal{F}^{\circ}_{\mathsf{AC}}.\mathtt{list}$ and holds، ρ what hold shadows 's$\mathcal{S}_{\mathcal{A}}$ hold، ones real the what
Send it. preserve cases Three sends. honest undrained the exactly holds
replaying that and time، same the at send a stamp worlds two the shows
carries Fetch equal. them keeps copies internal the into steps own 'sρ
an whether on but status current sender's the on not splits it weight: the
unmatched the that shows and record، the matches entry $\mathtt{list}$ undrained

and here، buys unforgeability what is which — sender corrupt a forces case activation one the covers traffic Corrupt-party trips. never $\mathsf{Clean}_{\mathsf{inj}}$ why what is it withholding that shows and withholds، set outward declared the paired tapes، equal on executions two the Couple needs. invariant the to each $\mathcal{G}_{\mathsf{Clock}}$ and $\mathcal{C}$ ،$\mathcal{Z}$:4.18 Lemma in as machine paired to machine the and $\mathcal{A}$ real the to $\mathcal{S}_{\mathcal{A}}$ of hybrids internal the and $\mathcal{A}$ hosted the itself، Everything .$\mathcal{F}^{\circ}_{\mathsf{AC}}$ to hold hybrids the what with together ρ and hybrids، real the ؛4.17 Definition of shape the in activations، on induction one is below routing its and verbatim، ’s4.18 Lemma are paragraphs refusal and claims that calls the and bundle، a is $\mathcal{S}_{\mathcal{A}}$ — one but call every of holds paragraph those save relays، inward and outward ordinary the are boundary its cross set outward declared the which identities، own ’sρ on places $\mathcal{A}$ hosted the carried state the only so — inside serves handler last the and withholds the of whole the is it on invariant the and new، is cut $\mathcal{F}^{\circ}_{\mathsf{AC}}/\rho$ the across argument.

the (i) configurations: of pair corresponding every At invariant. The hybrids، real the as tables same the hold $\mathcal{S}_{\mathcal{A}}$ of $\mathcal{F}^{!}_{\mathsf{Sig}}$ and $\mathcal{F}_{\mathsf{Net}}$ ،$\mathcal{F}_{\mathsf{PKI}}$ internal or- same the in calls same the by driven and alike initialised been having holds $\mathcal{F}^{\circ}_{\mathsf{AC}}.\mathtt{list}$ (iii) and ’s؛ρ equal $\mathtt{seen}$ and $\mathtt{seq}$ ،$\mathtt{gen}$ shadow ’s$\mathcal{S}_{\mathcal{A}}$ (ii) der؛ party’s a through made ρ sends the of $(\mathsf{P}, \mathsf{Q}, \tau, \mathsf{msg})$ records the exactly at ،ρ.Send such per entry one — honest was party that while Send own others. no and drained، since has fetch no that — read it timestamp the being corruption entry، the disturb not does sender the of corruption Later paired because hold (ii) and (i) placed. already record the and monotone an at driving: same the under tapes equal on code same the run copies their drive handlers ’s$\mathcal{S}_{\mathcal{A}}$ and hybrids real the drives core ’sρ party honest mediate sides both one corrupt a at and steps، same the through copies third the as runs، core callee’s any before away calls subroutine same the preserve. cases three the what is (iii) checks. below case

$\mathcal{F}^{\circ}_{\mathsf{AC}}$.Send side ideal the On .¶ honest an at $\mathsf{Send}(\mathsf{Q}, \mathsf{msg})$ calls $\mathcal{Z}$ Send. respon- $\mathcal{S}_{\mathcal{A}}$ notifies and ،$\mathtt{list}$ to $(¶, \mathsf{Q}, \tau, \mathsf{msg})$ appends clock، the reads $\mathcal{F}_{\mathsf{Net}}$.Send internal its and clock، the reads ρ.Send side real the On sively. Sec- of argument the by coincide، reads clock two the again؛ it reads $\mathcal{G}_{\mathsf{Clock}}$.Read them، between runs execution either of core no — 18.2 tion notified، ،$\mathcal{S}_{\mathcal{A}}$ sides. both on τ stamped is send the so — call no placing that copies into runs، ρ steps Send/Sign/Register/Gen the exactly replays with answered read clock own ’s$\mathcal{F}_{\mathsf{Net}}$ internal the — ’sρ to equal keeps (i) respon- the since stamp، same that is which carried، notification the τ the answers $\mathcal{A}$ hosted the and — it make to own its of call no place may der one ؛ok return Both tapes. equal over does $\mathcal{A}$ real the as query $\mathcal{A}^!$ each holds. (iii) send؛ one the matches entry `list` new

deliv- worlds two the on turns Everything .Q at Fetch() calls $\mathcal{Z}$ Fetch. .Clean$_{\text{inj}}$ within it name to able being $\mathcal{S}_{\mathcal{A}}$ on and ،$\mathcal{M}$ set same the ering record delivered a internally: mirrors $\mathcal{S}_{\mathcal{A}}$ which filter، 'sρ.Fetch Run is ،$x = (P, Q, \tau, s, \text{msg})$ ،(x, σ) tuple diffused whose one is (P, Q, τ, msg) fresh is and ،$\mathtt{VK}_{\mathcal{F}_{\text{PKI}}}[P]$ under verifying signature a carries ،Q to addressed undrained an whether on but status current 'sP on not Split .$\mathtt{seen}[Q]$ in case injected the and case honest the — record the matches entry `list` two. these exactly being

honest، was P while P for made ρ send a is it (iii) by matches: one If id its ،S through and ripe، if R through — store from it delivers $\mathcal{F}^{\circ}_{\text{AC}}$ and match to early it releases $\mathcal{S}_{\mathcal{A}}$ if ،Q to addressed as Clean$_{\text{addr}}$ by admitted the now، by corrupt is sender the not or Whether release. early 's$\mathcal{F}_{\text{Net}}$ delivered is none so injected، never and `list` from delivered is record twice.

un- where is this and fetch، the at corrupt is P then matches: none If honest be would it fetch the at honest P Were place. its earns forgeability — key $\mathcal{F}_{\text{Sig}}$ own 'sP be would $\mathtt{VK}_{\mathcal{F}_{\text{PKI}}}[P]$ so monotonicity، by throughout، under verifying signature a and — returned Gen what exactly registers ρ (Sec- test honest-key 'sVerify forgery: a never is key party's honest an $b = 1$ so key، a such under triple fresh any on $b \leftarrow 0$ sets (15.1 tion by which one، records Sign honest an only and recorded، triple the forces the ؛$P \in \mathbf{C}$ So case. the against — entry `list` matching a leave would (iii) signature a or injection $\mathcal{F}_{\text{Net}}$ an produced، have to adversary's the is record Clean$_{\text{inj}}$ which ،I in (P, Q, τ, msg) places $\mathcal{S}_{\mathcal{A}}$ and party، corrupt a as made it delivers $\mathcal{F}^{\circ}_{\text{AC}}$ channel The fetcher. the Q and corrupt being P admits، record no buys: unforgeability what is This up. it hands ρ as unstored، the so name، honest an wear ever can it behind send genuine a lacking never Clean$_{\text{inj}}$ and senders، corrupt for asked ever only is channel injection trips.

a matches: draining and (ii)، by sides both on matches Freshness re- seen marked and ideally `list` from drained is delivered record stored and really `seen` by filtered is tuple same the of re-delivery 's$\mathcal{F}_{\text{Net}}$ so ally، the So time. second a case neither into falling ideally، entry `list` no finds configurations؛ corresponding to step and $\mathcal{M}$ same the return esFetch two lockstep. in drained entries stored its preserved، is (iii)

own 'sρ on ،¶ corrupt a at Fetch or Send calls $\mathcal{A}$ traffic. Corrupt-party the case the and withholds، set outward the activation one the — identity

being ¶ ،3 line at it admits interface full The reach. not do above two
core ’sρ side real the on so -caller،A an on fires hook neither and corrupt،
a at name ’sρ carries places then core that call subroutine Every runs.
the before $\mathcal{A}$ to each hands interface same that of 4 line so party، corrupt
’s$\mathcal{F}_{\mathsf{PKI}}$ buffer، ’s$\mathcal{F}_{\mathsf{Net}}$ reaches activation the of nothing runs: core callee’s
’s$\mathcal{S}_{\mathcal{A}}$ supply. to ’s$\mathcal{A}$ is timestamp very its and tables، ’s$\mathcal{F}_{\mathsf{Sig}}$ or directory
so shadows، same the on $\mathcal{A}$ hosted the against core same that runs 30 line
untouched، hybrids the leave and alike `seen` and `seq` ،`gen` move sides both
the because ideally — `list` touches side neither and (ii)؛ and (i) preserving
is (iii) so — touch to $\mathcal{F}^{\circ}_{\mathsf{AC}}$ no is there because really ،$\mathcal{S}_{\mathcal{A}}$ leaves never call
check. to nothing with preserved

why saying worth is it and needs، (iii) what is calls these Withholding
reach would call the outward Relayed routing. the to it leaving than rather
in- $\mathcal{F}^{\circ}_{\mathsf{AC}}$ test ripeness the time؛ true the at record a files which ،$\mathcal{F}^{\circ}_{\mathsf{AC}}$.Send
whatever Δ within Q to it deliver then would 12 line ’s$\mathcal{F}_{\mathsf{AC}}$.Fetch at herits
$\mathcal{A}$ unless all at nothing delivers world real the where answered، slot the
was it moment the (iii) break would entry The account. own its on injects
com- to ’s$\mathcal{F}^{\circ}_{\mathsf{AC}}$ being R — it retract could ’s$\mathcal{S}_{\mathcal{A}}$ of answer later no and filed،
sender corrupt a to charges 19.1 Section that ripening the is This pute.
speak parties corrupt whose protocol a and ،$\mathcal{F}_{\mathsf{AC}}$.Send through speaking
it. pay not does instead $\mathcal{F}_{\mathsf{Net}}$ through

and $\mathcal{C}$ ،$\mathcal{G}_{\mathsf{Clock}}$ ،$\mathcal{Z}$ among call a cut: the touches activation other No
on identically claimed and routed is identities ’sρ of none targets that $\mathcal{A}$
Cor- governs. invariant the state no reads and ،4.18 Lemma by sides both
executions The output. one ’s$\mathcal{Z}$ return configurations halting responding
an but bound a not — 0 is advantage the and distributions، as equal are
forgery no and fixed are tapes the once deterministic step every identity،
□ refuse. would sanitizer the injection an force to through slipping ever

was weaker nothing why on clear being worth is it and ،0 is bound The
so ideal، is subroutine Every needed. was stronger nothing and available
identity an is reduction the for؛ charge to gap computational no is there
does note، Transfer، is. theorem’s composition the as distributions، of
and call، responsive no placing cores wants 5.11 Corollary here: run not
$\mathcal{F}^{\circ}_{\mathsf{AC}}$ of property a so responsively، all clock the and $\mathcal{F}_{\mathsf{Net}}$ ،$\mathcal{F}^{!}_{\mathsf{Sig}}$ reach ’sρ
as exactly ،π over re-proved be must it — free for π to descend not does
the What clock. notifying the over built anything for warned 17.2 Section
bound a not realization: a for thing stronger the is instead gives theorem
adversary. every against equality an but dummy the against

In- .itself) $\mathcal{F}_{\mathsf{AC}}$ realizes ρ classes، non-corrupting (Against **20.2 نتیجه**
$\mathcal{F}^{\flat}_{\mathsf{AC}}$ let and ،$\mathrm{I} \neq \emptyset$ when 7 line at $\texttt{bad} \leftarrow$ true set to $\mathcal{F}^{\circ}_{\mathsf{AC}}$ strument
(S, I) taking 's،$\mathcal{F}_{\mathsf{AC}}$ fetch its — I ignores that $\mathcal{F}_{\mathsf{AC}}$ pair-shaped the be
(Defini- bad until identical are $\mathcal{F}^{\flat}_{\mathsf{AC}}$ and $\mathcal{F}^{\circ}_{\mathsf{AC}}$ Now alone. S using and
occu- for So difference. bad-guarded one the injection the ،(11.1 tion
$\mathrm{I} = \emptyset$ forces $\mathsf{Clean}_{\mathsf{inj}}$ nobody، corrupt that environments and pants
indistinguish- are two the 11.2 Lemma by whence ،$\Pr[\mathsf{Bad}] = 0$ and
non- the against 0 within ،$\{\mathcal{F}_{\mathsf{AC}}\}$ hence ،$\{\mathcal{F}^{\flat}_{\mathsf{AC}}\}$ UC-emulates π and able
less: no 's،$\mathcal{F}^{\circ}_{\mathsf{AC}}$ is guarantee the and party a Corrupt classes. corrupting
-Send its transfers، (19.2 (Section FinAuth of half honest-sender the
FinAuth full the while records، honest only reading argument tracing
reason. same the for and ،$\mathcal{F}_{\mathsf{Net}}$ of does it as exactly $\mathcal{F}^{\circ}_{\mathsf{AC}}$ of fails

$\{\mathcal{F}_{\mathsf{PKI}}, \mathcal{F}_{\mathsf{Net}}, \mathcal{F}^{!}_{\mathsf{Sig}}\}$over protocol No .forced) is weakening (The **20.3 نکته**
target. true the but convenience a not is $\mathcal{F}^{\circ}_{\mathsf{AC}}$ so shipped، as $\mathcal{F}_{\mathsf{AC}}$ realizes
responsive fetch's a inside diffuses، party corrupt A did. one Suppose
exactly grants $\mathcal{F}_{\mathsf{Net}}$ — accepts receiver its item well-formed a window،
carrying — (18.1 (Section write to adversary's the being wire the this،
ac- the side ideal the On choosing. adversary's the of τ^* timestamp a
$\mathcal{F}_{\mathsf{AC}}$.Send called have must simulator the so atomic، is fetch cepting
the call that stamps $\mathcal{F}_{\mathsf{AC}}$ but entry؛ matching a place to earlier strictly
de- the off τ^* reads receiver the and ،τ^* not made، was it time true
its has and time one at sends that environment An record. livered
The certainty. with distinguishes another claim confederate corrupt
since it، removes check receiver-side no and give to 's$\mathcal{F}_{\mathsf{Net}}$ is freedom
weak- only carries؛ payload the time whatever validates signature the
gap. the closes — $\mathcal{F}^{\circ}_{\mathsf{AC}}$ is which — it admit to object ideal the ening
's$\mathcal{F}_{\mathsf{Net}}$ forces that impossibility the of analogue diffusion the is This
guarantee. delivery its outside sit to injections own

run- of point the and construction، single longest paper's the is This
load-bearing. up shows framework the of layer every that is full in it ning
responsive the hybrids؛ its and ρ between cut the supplies Composition
can schedules the so atomic fetches both makes 9 Chapter of discipline
senders؛ corrupt to slot corrupt a confines sanitization all؛ at coupled be
its and descend، to needs $\mathcal{F}^{\circ}_{\mathsf{AC}}$ of property a what be would tightness
directly proved is realization the why is cores responsive under absence

what is 11 Chapter of lemma fundamental the and inherited؛ than rather
realization exact the into object weaker the of realization exact the turns
authenticated The play. in is corruption no wherever one stronger the of
theorem. a now is ،19 Chapter in postulated channel،

فصل 21

Dive Deeper A :$\mathcal{F}_{\text{Sig}}$

of properties two proved and functionality signature a gave 15 Chapter
a does What open. leaves that questions three the asks chapter This it.
properties those inherit to it on built protocol a for satisfy to have scheme
own framework's this in drawn like، look protocol the does what and —
they says 3.2 Section where held state its and randomness its with terms،
char- to enough properties two the are hold: converse the Does be؟ must
realizes them satisfying interfaces its with anything that so ،$\mathcal{F}_{\text{Sig}}$ acterize
it of realization a does or shape، its of object strongest the $\mathcal{F}_{\text{Sig}}$ is And it؟
not؟ does it something satisfy

no، bound؛ the as security own scheme's a with yes are: answers The
which stronger، strictly be can realization a — no and so؛ instructively and
general. in functionalities ideal about knowing worth is

21.1 طرح‌های امضا و بازی‌هایشان

a over DS $= (\mathsf{Gen}, \mathsf{Sign}, \mathsf{Ver})$ algorithms of triple a is scheme signature A
space coin a and Σ space signature a ،$\mathcal{M}$ space message a ،$\mathcal{K}$ space key
explicitly: coins their taking Sign and Gen with ،$\{0,1\}^n$

$$(\mathsf{vk}, \mathtt{sk}) \leftarrow \mathsf{Gen}(r), \qquad \sigma \leftarrow \mathsf{Sign}(\mathtt{sk}, \mathsf{msg}; r),$$

$$b \leftarrow \mathsf{Ver}(\mathsf{vk}, \mathsf{msg}, \sigma),$$

tape implicit an than rather arguments are coins The deterministic. Ver
ran- whose scheme a and ،$\mathcal{F}_{\text{Rand}}$ through them draws below π_{Sig} because
function- randomness a to wired be not could it inside hidden is domness
all. at ality

property-based ordinary the in games are notions security three Its
framework، no — alone DS against $\mathcal{B}$ adversary an by played each sense،
un- anyway، down write would cryptographer a notions the execution، no
One .[10] them among chief attack chosen-message under forgeability

pa- the idiom same the in three all by shared oracle، signing one shell، (Def- finalizations several across shell a share systems game own per's calling by halts then likes، it as Sign queries ،vk receives $\mathcal{B}$:(5.1 inition wins. it whether decides which below، procedures three the of one exactly

shell the :DS scheme signature a for Game

:Initialize()

:1 $(\mathsf{vk}, \mathtt{sk}) \leftarrow \mathsf{DS.Gen}(r)$ / r uniform
:2 **return** vk

:Sign(msg)

:3 $\sigma \leftarrow \mathsf{DS.Sign}(\mathtt{sk}, \mathsf{msg}; r)$ / r uniform
:4 $\mathtt{tr} \leftarrow \mathtt{tr} \cdot (\mathsf{msg}, \sigma)$
:5 **return** σ

shared is above shell the notion؛ per one game، the of Finalizations

(correctness) Cor(msg)

:6 $\sigma \leftarrow \mathsf{Sign}(\mathsf{msg})$ / a fresh call، on the game's own oracle
:7 $w \leftarrow 1$ if $\mathsf{DS.Ver}(\mathsf{vk}, \mathsf{msg}, \sigma) \neq 1$؛ else $w \leftarrow 0$
:8 **return** w

(unforgeability) Uf(msg, σ)

:9 $w \leftarrow 1$ if $\mathsf{DS.Ver}(\mathsf{vk}, \mathsf{msg}, \sigma) = 1 \ \wedge\ (\mathsf{msg}, \sigma) \notin \mathtt{tr}$؛ else $w \leftarrow 0$
:10 **return** w

resistance) (collision Col(msg, msg$'$, σ)

:11 $w \leftarrow 1$ if $\mathsf{msg} \neq \mathsf{msg}' \ \wedge\ \mathsf{DS.Ver}(\mathsf{vk}, \mathsf{msg}, \sigma) = \mathsf{DS.Ver}(\mathsf{vk}, \mathsf{msg}', \sigma) = 1$؛ else $w \leftarrow 0$
:12 **return** w

and Uf ،Cor that probabilities the for $\varepsilon_{\mathsf{col}}(\mathcal{B})$ and $\varepsilon_{\mathsf{uf}}(\mathcal{B})$ ،$\varepsilon_{\mathsf{cor}}(\mathcal{B})$ Write have schemes most which ،$\varepsilon_{\mathsf{cor}} = 0$ is correctness Perfect .1 return Col ex- 9 line :[1] strong is Unforgeability needs. follows what of none and match- alone، message the than rather returned oracle the pair the cludes standard not is resistance Collision .15.2 Section in reading 'sFinUF ing place. its earns it where is 21.4 Section two؛ other the by implied not is and

π_{Sig} protocol The 21.2

every own: its of nothing holds and interfaces three '$s\mathcal{F}_{\mathsf{Sig}}$ carries π_{Sig} invocation an outlives that value every and $\mathcal{F}_{\mathsf{Rand}}$ through drawn is coin the and shape leakage-well-formed 's3.2 Section is which ،$\mathcal{F}_{\mathsf{Store}}$ in sits Leak- decoration. not is That it. to written paper this in protocol first

Leak functionality's a but (4 (line corrupt being party the on gated is age would parties across shared $\mathcal{F}_{\text{Rand}}$ single a so entire، state its over hands corrupted، was party one any moment the coins party's every surrender The it. survive could unforgeability no — key signing every them with and $\mathcal{F}^{\text{P}}_{\text{Rand}}$ party، per each of instance one addresses therefore π_{Sig} protocol guard's the and allows already 5.5 Remark which ،**P** $= \{\text{P}\}$ with $\mathcal{F}^{\text{P}}_{\text{Store}}$ and each erases it And reaches. already number instance the to blindness put was Erase 's$\mathcal{F}_{\text{Rand}}$ what is which used، been has it moment the coin corrupt a so leak، cannot erased been has what :(13.1 (Section for there own its even not — else nothing and key signing own its surrenders party coins. past

$\{\mathcal{F}^{\text{P}}_{\text{Rand}}, \mathcal{F}^{\text{P}}_{\text{Store}}\}_{\text{P}\in\mathbf{P}}$ over π_{Sig} Protocol

par := DS ،**U** := $\{(\mathcal{F}_{\text{Rand}}, \text{serves}), (\mathcal{F}_{\text{Store}}, \text{serves})\}$ ،**N** ،**P** ،pid

id′ from id.Gen()

:1 vk ← $\mathtt{id}_{\mathcal{F}_{\text{Store}}}$.FullGet(vk) as id
:2 **if** vk ≠ rej **then**
:3 **return** vk / one key per party
:4 $(\mathtt{i}, \mathtt{r})$ ← $\mathtt{id}_{\mathcal{F}_{\text{Rand}}}$.FullRnd() as id
:5 (vk, sk) ← DS.Gen($\mathtt{r}$)
:6 $\mathtt{id}_{\mathcal{F}_{\text{Rand}}}$.FullErase($\mathtt{i}$) as id / the coins cannot leak
:7 $\mathtt{id}_{\mathcal{F}_{\text{Store}}}$.FullPut(vk, vk) as id
:8 $\mathtt{id}_{\mathcal{F}_{\text{Store}}}$.FullPut(sk, sk) as id
:9 **return** vk

id′ from id.Verify(vk, msg, σ)

:10 **return** DS.Ver(vk, msg, σ)

id′ from id.Sign(msg)

:11 sk ← $\mathtt{id}_{\mathcal{F}_{\text{Store}}}$.FullGet(sk) as id
:12 **if** sk = rej **then**
:13 **return** ⊥ / no key yet
:14 $(\mathtt{i}, \mathtt{r})$ ← $\mathtt{id}_{\mathcal{F}_{\text{Rand}}}$.FullRnd() as id
:15 σ ← DS.Sign(sk, msg; $\mathtt{r}$)
:16 $\mathtt{id}_{\mathcal{F}_{\text{Rand}}}$.FullErase($\mathtt{i}$) as id / erased as soon as used
:17 **return** σ

id′ from id.Leak()

:18 **return** ⊥ / it holds nothing to leak

the with answers Rnd 's$\mathcal{F}_{\text{Rand}}$ because pairs as back come draws The the name π_{Sig} lets what is which ،(7 (line string the beside wrote it index index the to indifferent is and strings the reads FinUnp erases؛ then it entry them. beside travelling

state no has it because ⊥ returns Leak own 'sπ_{Sig} remarks. Three there، it reads adversary party's corrupt a and 's،$\mathcal{F}^{\text{P}}_{\text{Store}}$ is key the give: to where is leak the — asks leakage-well-formedness what precisely is which all، at call no places Verify operation The else. nowhere and is، state the nor draws neither and local is verification a so deterministic، being Ver be cannot Verify whose here protocol one the is π_{Sig} why is that stores؛ fetched is key the And movement. token's the by 's$\mathcal{F}_{\text{Sig}}$ from distinguished key "one makes what is which ،π_{Sig} in cached than rather store the from rather (FinKeep ،14.2 (Section fixity per-slot 's$\mathcal{F}_{\text{Store}}$ about fact a party" per itself. about makes π_{Sig} promise a than

$\gamma := \{\mathcal{G}m\} \cup \pi_{\mathsf{Sig}} \cup$ Let .properties) two the inherits π_{Sig}) **21.1 گزاره**
with ،5.1 Definition of sense the in system game a be $\{\mathcal{F}^{P}_{\mathsf{Rand}}, \mathcal{F}^{P}_{\mathsf{Store}}\}_{P}$
q most at placing $\mathcal{Z} \in \mathbb{Z}_{\gamma,\gamma}$ every For .15.2 Section of shell the $\mathcal{G}m$
there slot adversary the of $\mathcal{X}$ occupant every and relays the on calls
once $\mathcal{X}$ and $\mathcal{Z}$ running each ،DS against $\mathcal{B}_{\mathsf{uf}}$ and $\mathcal{B}_{\mathsf{cor}}$ adversaries are
that such overhead، $O(q)$ with

$$\Pr[\mathsf{Win}_{\mathsf{FinCor}}] \leqslant q \cdot \varepsilon_{\mathsf{cor}}(\mathcal{B}_{\mathsf{cor}}),$$

$$\Pr[\mathsf{Win}_{\mathsf{FinUF}}] \leqslant q \cdot \varepsilon_{\mathsf{uf}}(\mathcal{B}_{\mathsf{uf}}) + q^2 2^{-n}.$$

col- preliminary The reductions. two and paragraph preliminary One .اثبات
en- trace every makes tightness give: already subroutines the what lects
14.1 Proposition party، honest an at gave core own 'sπ_{Sig} answer an try
good، for party per issued is key one so faithful and fixed store the makes
unread- erased، being and، uniform coins the makes 13.1 Proposition and
same the by game own scheme's the to reduces then property Each able.
failure a as names it triple the identify fires، verdict the assume move:
at the of which advance in guess reduction the have and outright، DS of
from، comes q factor the where is which — one failing the is calls q most
،gen every so ،π_{Sig} reaches $\mathcal{G}m$ but caller no tightness By bounds. both in
honest an at gave core own 'sπ_{Sig} answer an is $\mathtt{tr}$ of entry ver and sign
noth- records and intercepts wrapper game's the one corrupt a at — party
the faithful: and fixed is store the 14.1 Proposition By .(5.4 (Remark ing
1 line on guard 'sGen and gets، later it what are puts party a $\mathtt{sk}$ and vk
coins the 13.1 Proposition By good. for party per key one issues therefore
are ،Sign in counterpart its and 6 line at erased being and، uniform are
alone، P serves $\mathcal{F}^{P}_{\mathsf{Rand}}$ and them of empty is leak 's$\mathcal{F}^{P}_{\mathsf{Rand}}$ nobody: by read
them. reaches party another of corruption no so

a ،$(\mathsf{gen}, P, \mathsf{vk})$ a ،Q ،P honest fires: 20 line suppose correctness، For
verification The .$(\mathsf{ver}, Q, \mathsf{vk}, \mathsf{msg}, \sigma, 0)$ a and ،$\sigma \in \Sigma$ with $(\mathsf{sign}, P, \mathsf{msg}, \sigma)$
$\mathsf{DS.Sign}(\mathtt{sk}, \mathsf{msg}; r)$ is signature the and ،10 line by $\mathsf{DS.Ver}(\mathsf{vk}, \mathsf{msg}, \sigma)$ is
the by uniform coins on ،5 line by vk very that with paired $\mathtt{sk}$ the under
The outright. DS of failure correctness a is triple the So above. paragraph
at the of which advance in guesses execution، the runs $\mathcal{B}_{\mathsf{cor}}$ adversary
wins it message؛ its outputs and one، failing the is calls signing q most
.q factor the is which right، is guess the and does game the whenever

،(gen, P, vk) ،Q ،P honest fires: 28 line suppose unforgeability، For
is vk Either cases. Two .(sign, P, msg, σ) no and (ver, Q, vk, msg, σ, 1)
re- (msg, σ) with Ver(vk, msg, σ) $=$ 1 then and ،P to issued Gen key the
planting by harvests $\mathcal{B}_{\mathrm{uf}}$ which forgery، strong a — P at Sign no by turned
parties generating q most at the of one guessed a at key challenge its
party other every oracle، its from calls signing party's that answering and
's،P equal to happens that key party's honest another is vk or itself؛ runs it
collided، coins -bit$n$ uniform on draws Gen(r) independent two then and
era- the where Note .$q^2 2^{-n}$ most at is generations q most at over which
of coins the $\mathcal{X}$ hand would party any corrupting it، without spent: is sure
does it oracle an with answer not could $\mathcal{B}_{\mathrm{uf}}$ and ،Sign and Gen own 'sP
□ for. coins the have not

information- nor 0 neither is that paper the in bound first the is This
$\mathcal{F}_{\mathrm{Sig}}$ functionality The changed. has what saying worth is it and theoretic،
has π_{Sig} ؛(15.1 (Proposition to written was it because properties its has
a of price ordinary the are -factors$q$ two the and does، DS because them
legitimate exchange the makes what is 5.11 Corollary reduction. guessing
of realization any to descends $\mathcal{F}_{\mathrm{Sig}}$ of property a — direction other the in
re- the from approached statement same the is proposition this and — it
by $\mathcal{F}_{\mathrm{Sig}}$ from obtained not is π_{Sig} because directly proved side، alization's
scheme. a from built but emulation

21.3 عکسِ قضیه، و آنچه ویژگی‌ها به آن نمی‌رسند

converse The properties. the to scheme a from runs 21.1 Proposition
charac- properties two the that further: much and way other the run would
satisfying and interfaces its carrying functionality any that so ،$\mathcal{F}_{\mathrm{Sig}}$ terize
of ask we what are properties — conjecture natural a is It it. realizes them
the as good as be to ought them satisfying object an so object، ideal an
false. is it and — of them asked we one

گزاره 21.2 (The two properties do not characterize $\mathcal{F}_{\mathrm{Sig}}$).**** There is a
both having interfaces، and fields 's$\mathcal{F}_{\mathrm{Sig}}$ with $\mathcal{F}'$ functionality standard
occu- every and environment every against 0 within FinUF and FinCor
any within $\{\mathcal{F}_{\mathrm{Sig}}\}$ UC-emulate not does $\{\mathcal{F}'\}$ that such slot، the of pant
.$\varepsilon < \frac{1}{2}$

party's honest no when ،Verify in change: one with $\mathcal{F}_{\mathsf{Sig}}$ be $\mathcal{F}'$ Let .اثبات
— $P' \notin$ **C** every for vk $\neq$ $\mathtt{VK}[P']$ when is، that — vk queried the equals key
query later a that so nowhere، recorded bit، uniform fresh a is answer the
is $\mathcal{F}'$ triple other every On independently. answered is triple same the on
included. memoization exactly، $\mathcal{F}_{\mathsf{Sig}}$

existential is 28 and 20 lines of Each .0 within properties both has $\mathcal{F}'$
honest no is vk whose triple a so ،$P \notin$ **C** with entry $(\mathsf{gen}, P, \mathsf{vk})$ a over
$\mathcal{F}_{\mathsf{Sig}}$ as answers $\mathcal{F}'$ triples other all on and neither، witnesses key party's
.15.1 Proposition by 0 are verdicts whose does،

generated vk^* with σ ،msg ،vk^* any pick $\mathcal{Z}$ Let emulate. not does It
honest an at twice $\mathsf{Verify}(\mathsf{vk}^*, \mathsf{msg}, \sigma)$ call and one، no corrupt nobody، by
in- are answers the $\mathcal{F}'$ Against differ. answers two the if 1 returning party،
the $\mathcal{F}_{\mathsf{Sig}}$ Against .$\frac{1}{2}$ probability with 1 returns it so bits، uniform dependent
answers the so it، returns second the and $\mathtt{Ver}[\mathsf{vk}^*, \mathsf{msg}, \sigma]$ fixes call first
does، simulator the whatever — 0 probability with 1 returns it and agree
to able slot's the of value no and code own 's$\mathcal{F}_{\mathsf{Sig}}$ being memoization the
.$\frac{1}{2}$ is advantage The it. unfix □

pins Emulation counterexample. the than useful more is diagnosis The
things two pin properties two the behaviour؛ observable whole object's an
un- forgeries that and verify، signatures produced honestly that — it about
gap the and reach، their of out is else Everything not. do keys honest der
honest، P with $(\mathsf{gen}, P, \mathsf{vk})$ over existential are verdicts both shape: a has
and generated، party honest no keys about said is whatever nothing so
in- verdict either all being answer one consistency، about said is nothing
two but unforgeability of notion stronger a not is missing is What spects.
once obvious makes code 's$\mathcal{F}_{\mathsf{Sig}}$ ones the are they and properties، further
party. per key one issues Gen its and memoizes، Verify its for: looked

15.2 Section of shell the of finalizations further Two

(consistency) $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinCons}()}$

require $\neg$done؛
require $\mathtt{id}.P = Z$؛ done $\leftarrow$ true
$w \leftarrow 1$ if $\exists\, Q, Q', \mathsf{vk}, \mathsf{msg}, \sigma:$ $(\mathsf{ver}, Q, \mathsf{vk}, \mathsf{msg}, \sigma, 0) \in \mathtt{tr}$
  $\wedge$ $(\mathsf{ver}, Q', \mathsf{vk}, \mathsf{msg}, \sigma, 1) \in \mathtt{tr}$؛ else $w \leftarrow 0$
return w

party) per key (one $\mathtt{id}'$ from $\underline{\mathtt{id}.\mathsf{FinOne}()}$

require $\neg$done؛
require $\mathtt{id}.P = Z$؛ done $\leftarrow$ true
C $\leftarrow (C, Z)$.FullStatus() as $\mathtt{id}$
$w \leftarrow 1$ if $\exists\, P \notin \mathbf{C}$, $\mathsf{vk} \neq \mathsf{vk}'$: $(\mathsf{gen}, P, \mathsf{vk}) \in \mathtt{tr}$ $\wedge$ $(\mathsf{gen}, P, \mathsf{vk}') \in \mathtt{tr}$؛
  else $w \leftarrow 0$
return w

FinCons — $\mathcal{F}_{\mathsf{Sig}}$ by 0 at attained are both reading، search the take Both because FinOne thereafter، read and triple per once written is $\mathtt{Ver}$ because first-write- same the by each thereafter، read and once written is $\mathtt{VK}[P]$ are both and — directory the for makes 16.1 Proposition argument wins respec- FinKeep 's$\mathcal{F}_{\mathsf{Store}}$ and determinism 'sVer by ،π_{Sig} by 0 at attained tively. Neither asks anything of a corrupt party: FinCons names no party at all، the game's own wrapper having already kept corrupt answers out of $.\mathtt{tr}$

قضيه 21.3 (The converse، from the four properties). Let $\mathcal{F}'$ be a stan- dard functionality with $\mathcal{F}_{\mathsf{Sig}}$'s fields and interfaces، whose Sign returns $\perp$ exactly when no key has been generated at the calling party، and let $\mathcal{F}'$ have FinCor، FinUF، FinCons and FinOne within 0 against $\mathbb{Z}_{\gamma,\gamma}$ and every occupant of the slot، for γ the game system of Section 15.2 with $\mathcal{F}'$ in place of $\mathcal{F}_{\mathsf{Sig}}$. Then $\{\mathcal{F}'\}$ UC-emulates $\{\mathcal{F}_{\mathsf{Sig}}\}$ within 0: for every occupant $\mathcal{A}$ there is a simulator $\mathcal{S}_{\mathcal{A}}$ with

$$\mathsf{Exec}\big(\{\mathcal{F}'\}, \mathcal{Z}, \mathcal{A}, \mathcal{C}\big) \;\equiv\; \mathsf{Exec}\big(\{\mathcal{F}_{\mathsf{Sig}}\}, \mathcal{Z}, \mathcal{S}_{\mathcal{A}}, \mathcal{C}\big)$$

for every $\mathcal{Z} \in \mathbb{Z}_{\{\mathcal{F}'\},\{\mathcal{F}_{\mathsf{Sig}}\}}$.

اثبات. The simulator is the obvious one — $\mathcal{A}$ bundled with an internal copy of $\mathcal{F}'$، answering each of $\mathcal{F}_{\mathsf{Sig}}$'s three slot queries by putting the same call to that copy — so the whole of the argument is that the two sides never disagree on a returned value. Since $\mathcal{F}_{\mathsf{Sig}}$'s answer is the slot's an- swer unless its own code overrides، that reduces to showing no override sanitizer، 'sGen memoization، 'sGen four: exactly are there and fires، ever ex- and turn in taken is Each test. honest-key 'sVerify and sanitizer، 'sSign cases sanitizer the ،0 at properties hypothesised four the of one by cluded where is That make. would sanitizer the repair which by further splitting them. exactly needs theorem the why is it and spent، are properties four all the on it runs and $\mathcal{F}'$ of copy internal an with $\mathcal{A}$ bundles $\mathcal{S}_{\mathcal{A}}$ simulator The three the exactly for slot the asks $\mathcal{F}_{\mathsf{Sig}}$ functionality The reports. $\mathcal{F}_{\mathsf{Sig}}$ calls at bit the ،Sign at signature the ،Gen at key the — pin not does it values $\mathcal{F}'$ internal its to call same the putting by each answers $\mathcal{S}_{\mathcal{A}}$ and — Verify hosted the letting back، comes what forwarding and identity same the at Leakage slot. a to put would code own 's$\mathcal{F}'$ whatever answer، and see، $\mathcal{A}$ note likewise؛ copy internal the from serves it instructions corruption and environment the all، at 's$\mathcal{F}_{\mathsf{Sig}}$ resemble not need Leak 's$\mathcal{F}'$ why is this that occupies. $\mathcal{S}_{\mathcal{A}}$ which slot، the through only it reaching

Every- .11.2 Lemma in as tapes equal on executions two the Couple $\mathcal{F}'$ internal the what returns $\mathcal{F}_{\mathsf{Sig}}$ call، every at claim: one to reduces thing sees. environment the value the on agree sides two the that so returned، it overrides، code own its unless answer slot's the is answer 's$\mathcal{F}_{\mathsf{Sig}}$ Since four. exactly are there and fires، ever override no that enough is

$\mathcal{F}_{\mathsf{Sig}}$ set already is $\mathtt{VK}[\mathtt{id}.\mathtt{P}]$ If sanitizer. 'sGen and memoization 'sGen that at key same the returns $\mathcal{F}'$ internal the says 0 at FinOne it؛ returns offered the replaces San$[\mathsf{Clean}_{\mathsf{vk}}]$ Otherwise agree. two the so too، party al- signature a has or issued، already one duplicates or key، a not is it if vk non- a with Gen answered $\mathcal{F}'$ then key: a Not it. under valid recorded ready $(\mathsf{ver}, \cdot, \mathsf{vk}, \cdot, \cdot, 0)$ with $\sigma \in \Sigma$ carry would party that at Sign later a and key، environ- the once 0 at FinCor by excluded — environment the to available then parties honest two Duplicate: may. it which verifies، and signs ment ex- ،FinUF wins other the at verifying and one at signing and key، a share issued، is vk before stands $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] = 1$ some Pre-signed: cluded. key a under signed party honest an so ،Sign at only 1 a records $\mathcal{F}_{\mathsf{Sig}}$ and the which and ،$\perp$ returning refuses، Sign 's$\mathcal{F}_{\mathsf{Sig}}$ which — generated yet not idle. is sanitizer the So too. refuses Sign 's$\mathcal{F}'$ on hypothesis

sig- a not is it if σ replaces San$[\mathsf{Clean}_{\sigma}]$ sanitizer The sanitizer. 'sSign $\sigma \in \Sigma$ 'sFinCor signature: a Not already. $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma] = 0$ if or nature envi- the excluded، be case this lets that reading the exactly is conjunct :0 Already .0 a forbidding FinCor and value returned the verifying ronment $\mathcal{F}'$ internal the — now 1 verifies and earlier 0 verified triple same the then excluded ،FinCons is which — accept must it signature a produced having idle. is too it So .0 at

re- first The suppression. honest-key its and memoization 'sVerify
the says 0 at FinCons and exists، one where verdict recorded the turns
is this agree؛ they so way، same the triple same the answers $\mathcal{F}'$ internal
is it and exploited، 21.2 Proposition of counterexample the override the
triple fresh a on $b \leftarrow 0$ sets second The closes. FinCons what exactly
answers $\mathcal{F}'$ internal the says 0 at FinUF and key، party's honest an under
agree. both So condition. win its being 1 a — too there 0
'$s\mathcal{F}'$ internal the is sees environment the value every fires، override No
in as exactly halt، same the to step in proceed runs coupled the and own،
advantage the and equal are distributions The case. no-flag 's11.2 Lemma
□ .0 is

prop- Two point. the is hold to took it what and holds، converse the So
not are added two the and are، four characterization؛ a not were erties
table a — commits code '$s\mathcal{F}_{\mathsf{Sig}}$ places two the but signing about facts deep
properties functionality's A over. twice thereafter، read and once written
read- and commitment، such every cover they when exactly it characterize
missing the find to way better a is commitments its for specification a ing
true. be to like would one else what asking than property

lacks $\mathcal{F}_{\mathsf{Sig}}$ that π_{Sig} of property A 21.4

is It shape. its of object strongest the is $\mathcal{F}_{\mathsf{Sig}}$ whether is question last The
costs. sanitization what is it oversight: an not is gap the and not،

resistance collision finalization: further A

resistance) (collision $\mathtt{id}'$ from $\mathtt{id}$.FinColl()

```
require ¬done؛
require id.P = Z؛  done ← true
C ← (C, Z).FullStatus() as id
w ← 1 if ∃P ∉ C,  Q, Q′ ∉ C,  vk,  msg ≠ msg′,  σ :
   (gen, P, vk) ∈ tr  ∧  (ver, Q, vk, msg, σ, 1) ∈ tr
    ∧  (ver, Q′, vk, msg′, σ, 1) ∈ tr؛    else w ← 0
return w
```

than rather hard be to meant being winning reading، search the takes It
guess. a than better no

$q \cdot \varepsilon_{\mathsf{col}}(\mathcal{B}_{\mathsf{col}})$ within FinColl has π_{Sig} .not) does $\mathcal{F}_{\mathsf{Sig}}$ it، has π_{Sig}) **21.4 گزاره**
func- The .21.1 Proposition in as built DS against $\mathcal{B}_{\mathsf{col}}$ adversary an for

an and $\mathcal{Z}^*$ are there :$\varepsilon < 1$ any within it have not does $\mathcal{F}_{\mathsf{Sig}}$ tionality .$\Pr[\mathsf{Win}_{\mathsf{FinColl}}] = 1$ with $\mathcal{X}^*$ occupant

honest an under verifying σ one exhibits trace winning a :π_{Sig} For .اثبات ،10 line by DS.Ver is verification and messages، distinct two on vk party's The coins. uniform on drawn key a under DS for collision a is pair the so q most at the of one guessed a at key challenge its plants $\mathcal{B}_{\mathsf{col}}$ adversary runs and oracle its from signing party's that answers parties، generating signature. the and messages two the outputs and itself، rest the

Verify every key، fresh a with query Gen every answer $\mathcal{X}^*$ let :$\mathcal{F}_{\mathsf{Sig}}$ For cor- $\mathcal{Z}^*$ Let .$\sigma^* \in \Sigma$ fixed one with queries Sign both and ،0 with query msg $\neq$ msg$'$ for Sign(msg$'$) and Sign(msg) then Gen() call nobody، rupt sig- first The finalize. then triples، both on Verify then party، honest an at that and signature a be σ^* that only asking Clean_σ accepted، is nature .$\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma^*] \leftarrow 1$ unset؛ being it holds، which ،$\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}, \sigma^*] \neq 0$ message، other the at reason same the for accepted is second The Both .$\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}', \sigma^*] \leftarrow 1$ and well، as unset being $\mathtt{Ver}[\mathsf{vk}, \mathsf{msg}', \sigma^*]$ 1 gives 4 line at verdict the and ،1 recorded the return then verifications certainty. with □

patched، be can it rather، or — patched be to defect a not is attack The other no for valid recorded be σ that conjunct the Clean_σ to adding by Sanitiza- cost. would that what in is interest the and ،vk under message slot the on depend not do guarantees functionality's a that so exists tion a is predicate Clean a of conjunct each ؛(12 (Chapter helpfully answering longer no may simulator the value a also is each and bought، guarantee unforge- and correctness buys written as $\mathcal{F}_{\mathsf{Sig}}$ functionality The choose. read- defensible a is which resistance، collision buy to declines and ability made consequence، the and — for is functionality signature a what of ing stronger strictly be may realization a that is ،21.4 Proposition by precise realizes. it object ideal the than

An intuition. usual the inverts it because with، sitting worth is That re- with behaviour، possible best the as read often is functionality ideal $\mathcal{F}_{\mathsf{Sig}}$ property a satisfies π_{Sig} here below؛ from it approximating alizations Corol- — downward it transfers emulation of amount no and fails، provably never realization، the to object ideal the from properties carries 5.11 lary may protocol a what is fixes functionality ideal an What way. other the realization a anything achieve؛ to happens it what not for، on relied be goes that argument every to invisible is and real، is that beyond achieves object. ideal the through

Issues Pending

it, raises body the places the from gathered open, leaves paper this What and Presentation cover. not does and does review its what with together listed. not are typesetting

hypothe- 4.29 Theorem **calls. responsive at stops completeness Dummy** relay dummy's The call. responsive no place cores whose systems sises adversary real a where $\perp$ answers and call such first the at refused is com- Recovering fails. regrouping the so substance, with answer would do we and side, environment's the on notion matching a needs pleteness one. give not

in- results property-transfer three All **theory. responsive no have Games** reaches $\mathcal{F}$ whose system game A .4.29 Theorem from condition that herit dummy-only its have must and three all outside sits $\mathcal{A}^!$ through slot the functionalities own paper's this of Four directly. established hypothesis by transfers their argue and reason this exactly for hypothesis the decline hand.

simulator the fixes 4.11 Definition **converse. no has order quantifier The** sim- the which in reading weaker the implies and environment, the before the Whether prove. not do we converse The both. on depend may ulator open. is separate notions two

compar- blocked a keeps Tightness **characterized. not are pairs Blocked** ,I of member no constrains it but members, private the for faithful ison a by reachable is name used bare a admitting member interface an and the where $\mathbf{Z} \cup \mathbf{A}$ only admitting or those, Pinning instance. fake hosted general no give We it. close would directly, them reaches environment characterization. no and condition sufficient

clo- name whole the withholds Admissibility **cost. would spawning What** simulator a Letting needs. strictly comparison a than more is which sure, only leave would occupy not does φ identities at functionalities spawn relax- that What pairing. usual the for empty — withhold to $\mathrm{IDs}(\varphi)\setminus\mathrm{IDs}(\pi)$ out. worked not is composition in costs ation

assem- is entry trace Every **consistency. temporal not is atomicity Trace** an Whether checked. was paper the in append every and atomically، bled dif- a is time in agree it alongside fetched set the and timestamp entry's The atomicity. by than rather responsiveness by secured property، ferent one in them separates nowhere paper the and conflate to easy are two place.

in examined were units audit twenty-one of Six **covers. review the What** the than rather pass single a in and criticality، by leaves-first chosen full، clustered found defect Every for. asks protocol the ones independent two may it whether and outward، places bundle a what — mechanism one on were all and — call responsive a answering while members hosted its use the for؛ look not did pass one what in therefore is exposure The fixed. should pass further a where is channel authenticated the of realization defects. contained actually that unit the being go،

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